

PROFESSIONAL INVESTMENT PORTFOLIO MANAGEMENT

Boosting Performance with Machine-Made Portfolios and Stock Market Evidence

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Preface

This book is intended for investment professionals. It applies a new asset asset pricing model dubbed the *ZCAPM* to the problem of building high-performing stock portfolios. In a recent book by Kolari, Liu, and Huang (2021), extensive empirical tests demonstrated that the ZCAPM provides a better out-of-sample fit to the cross section of average stock returns than well-known multifactor asset pricing models.

From 2013 to 2015, we worked with the Teachers Retirement System of Texas (TRS) and Texas A&M University to manage a research and development fund. The ZCAPM was used to build a diversified stock portfolio. In crossing the bridge between academic research and professional practice, we learned a number of important investment lessons. Unfortunately, our joint activities were cut short due to management issues.

The purpose of this book is to apply the ZCAPM to build efficient stock portfolios. The first chapter lays the foundation by covering modern investment principles invented by Harry Markowitz. In Chapter 2, we show how William Sharpe used Markowitz's mean-variance investment parabola to derive the now famous Capital Asset Pricing Model (CAPM). A number of CAPM variants emerged on conditional, consumption, international, intertemporal, production, and zero-beta portfolio asset pricing. Unfortunately, as discussed in Chapter 3, shortly after Professors Markowitz and Sharpe won the Nobel Prize in Economic Sciences in 1990, Fama and French published a series of papers that found beta risk in the CAPM was not related to average returns in the cross section of stocks in the market. Concluding the

CAPM was *dead*, the authors proposed an innovative three-factor model to supplant the CAPM. Other researchers quickly proposed many new factors and different multifactor models. The field of asset pricing was in a state of confusion with no clear direction about which factors and models to use.

Chapters 4 and 5 review the theoretical and empirical ZCAPM, respectively. In their 2021 book, Kolari, Liu, and Huang derived the ZCAPM from the Markowitz investment parabola as well as the zero-beta CAPM of Black (a close relative of the CAPM). The ZCAPM has two factors—namely, average market returns related to beta risk and the cross-sectional return dispersion of stocks in the market associated with zeta risk. Of these two factors, zeta risk was consistently more significantly priced than popular multifactors, such as the size, value, profit, capital investment, and momentum factors. The dominance of the ZCAPM over multifactor models across different test asset portfolios, individual stocks, and time periods simplifies the problem of which asset pricing model to use. Also, the authors conducted some experiments that suggested the model could be applied to real world investments. We close Chapter 5 by discussing resistance in the academic community to recognize the empirical success of the ZCAPM.

Chapter 6 reviews portfolio performance metrics, including return computations and different risk measures to evaluate portfolio performance over time.

Chapter 7 makes a major departure from previous literature, including Kolari, Liu, and Huang. We employ the ZCAPM to construct a proxy for the global minimum variance portfolio G. This portfolio is important in the context of the ZCAPM because it has beta risk but no zeta risk. It represents the vertex of the Markowitz investment parabola. In later chapters, using the G portfolio as a starting point, we progressively add risk to stock portfolios and observe their out-of-sample performance on a return/risk basis.

Chapters 8–10 take a deep dive into the application of the ZCAPM to a variety of stock portfolio investments. All experiments are run on an out-of-sample basis to mimic an investable strategy by a real world investor. That is, build a portfolio, observe its return in the next month, and then rebalance the portfolio at the end of each month over time. Our analysis period is from July 1964 to December 2022. Net long portfolios are constructed by combining portfolio G with long-short portfolios that have increasing levels of zeta risk. Average annual returns and Sharpe ratios prove that ZCAPM-based stock portfolios outperform the CRSP market index by substantial amounts in a number of cases. They show that CRSP as a general market index is not an efficient portfolio as many researchers have believed in the past. Since most professional investment managers cannot consistently beat general stock

market indexes over time, these findings bode well for our ZCAPM technology. Further analyses of long only portfolios sorted on different levels of zeta risk and beta risk corroborate these findings. Returns are particularly sensitive to zeta risk, which can be used to build relatively high-performing stock portfolios.

Chapter 11 provides an interesting look inside Markowitz's mean-variance investment parabola. We create stock portfolios sorted on beta and zeta risks and then plot their locations in return/risk space. A definite risk structure or architecture emerges within the parabola. As zeta risk increases, investment returns increase vertically or upward toward the efficient frontier that is the upper boundary of the parabola. Given some zeta risk level, as beta risk increases, average returns and total risk move horizontally in the parabola. This beta-zeta risk architecture represents an important tool for professional managers to evaluate investment portfolio performance and make future investment decisions.

Chapter 12 takes our analyses one step closer to real world practice by applying the ZCAPM to mutual fund investments. Many workers invest in mutual funds to save for the future in their retirement pension accounts. But is there some optimal way to manage their mutual fund investments? We employ the ZCAPM to build optimized portfolios containing 10 mutual funds that are rebalanced monthly over time. Our results show that average annual returns increase, in particular, with respect to zeta risk and to a lesser extent beta risk. We conclude that professional mutual fund managers and investors can use ZCAPM-based investment technology to boost long-run average stock returns.

Lastly, Chapter 13 gives a summary of the book with some conclusions drawn from our empirical stock portfolio findings. The continued growth of artificial intelligence and machine learning (AIML) models in investment practice nowadays bodes well for our ZCAPM-based investment methods. The authors seek to collaborate with the professional investment community to implement ZCAPM-AIML methods in the real world.

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Our research on the ZCAPM has been ongoing for more than a decade. During that time, we have been fortunate to have useful conversations with many researchers at academic institutions and professional managers

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Part I

Introduction



1

Portfolio Theory and Practice

1.1 The Two-Step Portfolio Process

From a theoretical perspective, Markowitz viewed portfolio selection as a two-step process. In the first step, investors assess the potential future performance of securities. Because it is expected or anticipated in the future, we refer to this analysis as *ex ante*. Of course, nobody knows the future per se. Investors can only make probabilistic assessments of future performances of securities. In the second step, using this information, investors choose a portfolio of securities.

But what rule should an investor use to guide their selection of securities in their portfolio? A longtime workhorse for this purpose is the present value formula that seeks to maximize the discounted value of future investment returns. For example, assuming a three-year investment, the present value for a risky security can be computed as follows:

$$V = \frac{\$10}{(1+0.10)} + \frac{\$12}{(1+0.10)^2} + \frac{\$8}{(1+0.10)^3} \approx \$25, \tag{1.1}$$

where the numerator shows risky cash flows in future years, and the denominator sets the discount rate at 10%. The discount rate can be interpreted as the required rate of return of investors. Rewriting the above formula in general ex ante (or future) terms, we have:

$$V = \frac{E(CF_1)}{[1+E(R)]} + \frac{E(CF_2)}{[1+E(R)]^2} + \frac{E(CF_3)}{[1+E(R)]^3}, \tag{1.2}$$

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where $E(CF_t)$ = expected cash flow at time t, and $E(R)$ = discount rate or expected rate of return in the three-year investment period. Because future cash flows are risky and not known with certainty, we use the expected value operator. Based on estimates of expected cash flows, discount rate $E(R)$ sets their present value equal to the current price of the investment. To estimate uncertain cash flows for a security, investors utilize available information gathered about the firm, industry, and general economy.

Notice that the discount rate takes into account the risk of uncertain cash flows. Under the reasonable assumption that investors do not like risk (i.e., are risk averse), the required discount rate by investors increases as risk increases, which lowers the present value or asset price. That is, investors demand a higher expected return as compensation for the risk that they bear. If there is no risk of future cash flows, the discount rate is the riskless rate of interest. U.S. government bond rates are often used to proxy the riskless rate. Thus, the expected risky discount rate has two components:

$$E(R) = R_f + RP, \tag{1.3}$$

where R_f is the riskless rate, and RP is the risk premium. The latter risk premium increases as: (1) investors become more risk averse; and (2) the risk of the security increases.

How can investors evaluate expected rates of return? Even though the future is unknown, such that future cash flows are not known with certainty, investors can estimate probabilities of different possible outcomes or states. Table 1.1 gives a simple example using probabilities. Let's assume that there are three possible future states: 1. pessimistic, 2. average, and 3. optimistic. These states can be denoted $s = 1, 2, 3$ with assigned probabilities $p_s = 0.25, 0.50, 0.25$ and possible returns $R_{sj} = 5\%, 10\%, 15\%$, respectively. Assuming that we hold the security for one year, the expected annual return would be:

$$E(R_j) = \sum_{s=1}^{S} p_s R_{sj}, \tag{1.4}$$

which equals 20% in Table 1.1.

If might seem logical that investors choose securities that maximize expected returns, but Markowitz argued against this investment approach. In his words: "The hypothesis (or maxim) that the investor does (or should) maximize discounted return must be rejected" Markowitz (1952, p. 77).

Table 1.1 Expected rate of return calculation (in percent terms)

State	p_s = probability	R_{sj}	$p_s R_{sj}$	$R_{sj} - E(R_j)$	$[R_{sj} - E(R_j)]^2$	$p_s[R_{sj} - E(R_j)]^2$
1	0.25	10	2.50	−10	100	25.00
2	0.50	20	10.00	0	0	0.00
3	$\frac{0.25}{100}$	30	$\frac{7.50}{E(R_j)=20}$	10	100	$\frac{25.00}{\sigma^2(R_j)=50.00}$ $\sigma(R_j) = 7.0711$

Instead, investors should maximize expected returns *per unit risk*. Using statistical methods, he proposed that total risk can be measured as the variance of returns.

Variance is a statistical measure of the dispersion, volatility, or variability of returns. We can compute the variance of returns as:

$$\sigma^2(R_j) = \sum_{s=1}^{S} p_s[R_{sj} - E(R_j)]^2. \tag{1.5}$$

It is important to keep track of the dimension or units of measurement of terms. Unlike expected returns, the dimension of the variance of returns is not in percent terms. But if we compute the standard deviation of returns, its dimension is in percent similar to expected returns. To get the standard deviation of returns, we take the square root of their variance:

$$\sigma(R_j) = \sqrt{\sigma^2(R_j)}. \tag{1.6}$$

Going back to the data in Table 1.1, the variance of returns is 50.00, such that the standard deviation of returns is $\sigma(R_j) = \sqrt{50.00} \approx 7.07\%$. We can interpret these results to mean that, for a one standard deviation change in returns due to volatility, returns will vary around the expected return from a low of 12.93% (= 20% − 7.07%) to a high of 27.07% (= 20% + 7.07%). A two standard deviation change gives a range from 5.86% to 30.14%. The latter range of future returns accounts for up to 95% of all possible outcomes. Because it is fairly wide, we can infer security j is quite risky, which is consistent with the high expected return equal to 20%.

Markowitz applied these return and risk concepts for individual securities to portfolios. For example, the expected return of a portfolio is:

$$E(R_P) = w_1 E(R_1) + w_2 E(R_2), \tag{1.7}$$

where w_1 is the weight (or proportion) invested in security 1, and w_2 is the weight (or proportion) invested in security 2. Since weights must add to one, or $w_1 + w_2 = 1$, we know that $w_2 = (1 - w_1)$.

To compute the variance of portfolio returns is more complicated. From statistics, we know that:

$$\sigma^2(R_P) = w_1^2\sigma^2(R_1) + 2w_1w_2Cov(R_1, R_2) + w_2^2\sigma^2(R_2), \tag{1.8}$$

where $Cov(R_1, R_2)$ = the covariance of returns between security 1 and security 2 over time. In practical terms, covariance simply measures the extent to which the returns for security 1 and 2 move together over time. The definition of covariance is:

$$Cov(R_1, R_2) = \rho_{12}\sigma(R_1)\sigma(R_2), \tag{1.9}$$

where ρ_{12} = the correlation of returns between security 1 and security 2 over time. Here is where an investment manager can make a difference. By choosing securities with low correlations (or ρs) with other securities in a portfolio, the total risk of the portfolio as measured by its variance of returns in Eq. (1.8) can be reduced. In so doing, *diversification benefits* can be achieved.

For two securities, using simpler notation wherein $Cov(R_j, R_k)$ is written as σ_{jk}, return variance in Eq. (1.8) can be expressed in matrix form as:

$$\sigma_P^2 = (w_1, w_2)\begin{pmatrix}\sigma_{11} & \sigma_{12}\\ \sigma_{21} & \sigma_{22}\end{pmatrix}\begin{pmatrix}w_1\\ w_2\end{pmatrix} = w_1^2\sigma_1^2 + w_1w_2(\sigma_{12} + \sigma_{21}) + w_2^2\sigma_2^2, \tag{1.10}$$

where $w_1 + w_2 = 1$, $\sigma^2(R_1) = \sigma_{11} = \sigma_1^2$, and $\sigma^2(R_2) = \sigma_{22} = \sigma_2^2$. Because the covariance terms σ_{12} and σ_{21} are equal to one another, their sum equals $2\sigma_{12}$, which reduces Eq. (1.8) to:

$$\sigma_P^2 = w_1^2\sigma_1^2 + 2w_1w_2\sigma_{12} + w_2^2\sigma_2^2, \tag{1.11}$$

With these preliminary statistical concepts in hand, we can analyze real world investment portfolios.

1.2 Real World Portfolio Analyses

Can we apply Markowitz's statistical concepts to actual investment portfolios? For this purpose, instead of using future expected returns, we need to use actual returns—that is, *ex post* or historical returns. Table 1.2 gives an example for two securities with previous actual or realized returns over the past five years.

The sample mean or average return for security j is:

$$\bar{R}_j = \frac{\sum_{t=1}^{T} R_{jt}}{T}, \tag{1.12}$$

Table 1.2 Actual or realized (ex post) returns and risk over time (in percent terms)

Years	R_1	R_2	$R_1 - E(R_1)$	$[R_1 - E(R_1)]^2$	$R_2 - E(R_2)$	$[R_2 - E(R_2)]^2$
1	14	14	−10.4	108.16	0.8	0.64
2	26	16	1.6	2.56	2.8	7.84
3	16	10	−8.4	70.56	−3.2	10.24
4	32	6	7.6	57.76	−7.2	51.84
5	$\frac{34.4}{R_1=24.4}$	$\frac{20}{R_2=13.2}$	9.6	$\frac{92.16}{\sigma^2(R_1)-331.2/4=82.80}$	6.8	$\frac{46.24}{\sigma^2(R_2)=116.80/4=29.20}$
				$\sigma(R_1) = 9.08$		$\sigma(R_2) = 5.40$

where there are $t = 1, \ldots, T$ periods. To compute the variance of returns (i.e., total risk) for security j, we use the following statistical formula:

$$\sigma^2(R_j) = \frac{\sum_{t=1}^{T}(R_{jt} - \bar{R}_j)^2}{T-1}. \tag{1.13}$$

Using the returns over time in Table 1.2 for security 1 and 2, their respective average returns are $\bar{R}_1 = 24.4\%$ and $\bar{R}_2 = 13.2\%$. Their variance of returns are $\sigma^2(R_1) = 82.80$ with $\sigma(R_1) = 9.08\%$ and $\sigma^2(R_2) = 29.20$ with $\sigma(R_2) = 5.40\%$, respectively. The problem for the investment manager is to select the optimal weights w_1 and w_2 to minimize portfolio risk.

For two securities that have correlation $-1 < \rho < 1$, we can solve the optimal weights to minimize the portfolio variance. Let's define $w_1 = w$ and $w_2 = (1 - w)$. From Eq. (1.11), we have:

$$\sigma_P^2 = w^2\sigma_1^2 + 2w(1-w)\rho_{12}\sigma_1\sigma_2 + (1-w)^2\sigma_2^2. \tag{1.14}$$

Rearranging terms, we get:

$$\sigma_P^2 = w^2\sigma_1^2 + 2w\rho_{12}\sigma_1\sigma_2 - 2w^2\rho_{12}\sigma_1\sigma_2 + (1-w)^2\sigma_2^2. \tag{1.15}$$

The change in return variance per unit change in w is:

$$\frac{\partial\sigma_P^2}{\partial w} = 2w\sigma_1^2 + 2(1-2w)\rho_{12}\sigma_1\sigma_2 - 2(1-w)\sigma_2^2. \tag{1.16}$$

When this relation reaches 0, at which point return variance can no longer be reduced for any small increase in w, security 1 has optimal weight:

$$w^* = \frac{\sigma_1^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 - 2\rho_{12}\sigma_1\sigma_2 + \sigma_2^2}, \tag{1.17}$$

and security 2 has optimal weight $(1 - w^*)$. In the special case that the two securities have no correlation (i.e., $\rho_{12} = 0$), the optimal weights are simplified to $w^* = \frac{\sigma_1}{(\sigma_1+\sigma_2)}$ and $(1 - w^*) = \frac{\sigma_2}{(\sigma_1+\sigma_2)}$. Of course, if $\rho_{12} \neq 0$, the optimal weights are affected.

In our example in Table 1.2, securities 1 and 2 have correlation $\rho_{12} = 0.170833$. Now we can compute the optimal weight for security 1 to be $w^* = 74.42/95.25 = 0.78$, such that optimal weight for security 2 is $(1 - w^*) = 0.22$. Based on these optimal weights, the expected return and return

variance on this two-security portfolio are:

$$E(R_P) = w^* E(R_1) + \left(1 - w^*\right) E(R_2) = 0.78(24.4\%) + 0.22(13.2\%) = 21.94\%.$$

$$\begin{aligned}\sigma_P^2 &= w_1^2\sigma_1^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2 + w_2^2\sigma_2^2 \\ &= \left(0.78^2\right)(82.80) + 2(0.78)(0.22)(0.170833)(9.08)(5.40) \\ &\quad + \left(0.22^2\right)(29.20) = 54.66\end{aligned}$$

Using these results, we can compute $\sigma_P = \sqrt{\sigma_P^2)} = \sqrt{54.66} = 7.39\%$. Finally, the ratio $E(R_P)/\sigma_P = 21.94\%/7.39\% = 2.97$ provides a measure of return/risk for comparison to other portfolios. This portfolio is widely known as the *global minimum variance portfolio* G.

The above analyses show that the correlation ρ between securities can be used to reduce the total risk of a portfolio. Figure 1.1 illustrates three different cases for a two-security portfolio: (1) perfect positive correlation ($\rho_{12} = 1$), (2) perfect negative correlation ($\rho_{12} = -1$), and (3) some positive correlation ($-1 < \rho_{12} < 1$). No change in risk occurs in the first case. Conversely, risk is reduced to zero in the second case. The third case is more likely in the real world wherein two securities typically have some degree of positive (but not perfect) correlation. As Markowitz proved, *diversification* is achieved by reducing the volatility of returns of the portfolio.

Figure 1.2 graphically summarizes Markowitz diversification when $-1 < \rho_{12} < 1$. We plot the average returns for a two-security portfolio on the Y-axis given the standard deviations of returns for different combinations of these securities on the X-axis. The efficient frontier contains the most desirable portfolios with the highest return per unit risk. The curved bold line depicts the *mean-variance investment parabola*. Portfolio G is shown to have minimum variance. The dashed line between security 1 and 2 contains the portfolios in the case of perfectly positive correlated returns, or $\rho_{12} = 1$. The other dashed lines plot portfolios in the case of perfectly negative correlation. If all risk is eliminated, we obtain portfolio X with zero standard deviation of returns (i.e., see Fig. 1.1).

In the real world, the number of assets from which to choose is large. Focusing on the U.S. stock market, there are over 8,000 different common stocks. Due to diversification benefits, the mean-variance parabola will gradually shift to the left in return/risk space. Figure 1.3 illustrates the parabola in

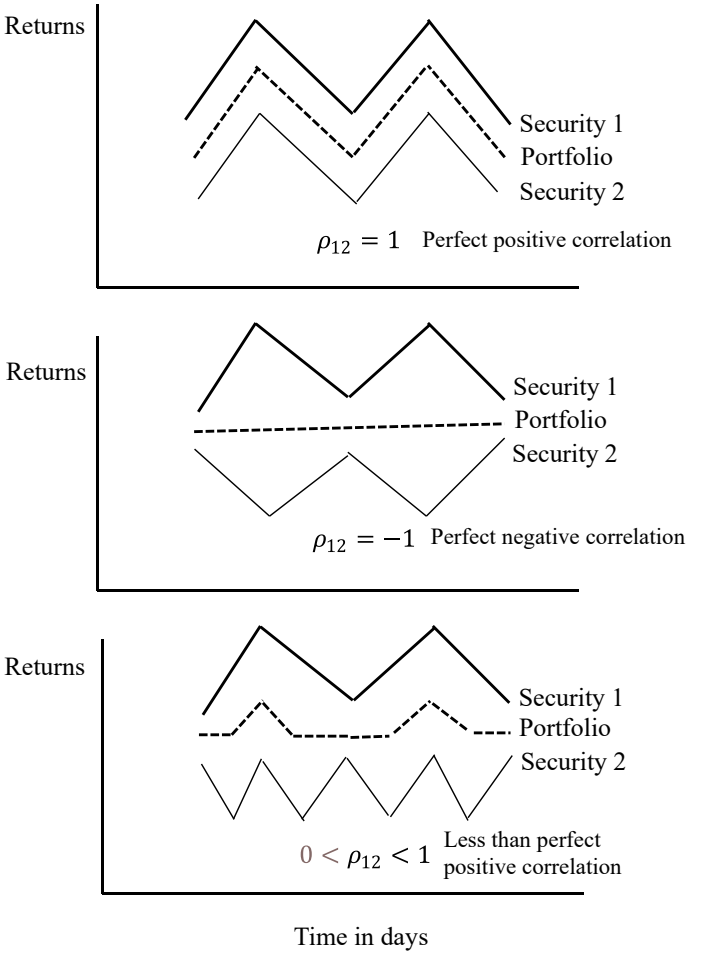


Fig. 1.1 Correlation of security returns and effects on portfolio returns

the case of many assets. Inefficient assets or portfolios lie within the opportunity set outlined by the boundary of the parabola. The parabola is split into two halves by the axis of symmetry shown by the dashed line that horizontally runs through portfolio G.

For N assets, the average portfolio return is:

$$\mu_P = \sum_{j=1}^{N} w_j R_j, \tag{1.18}$$

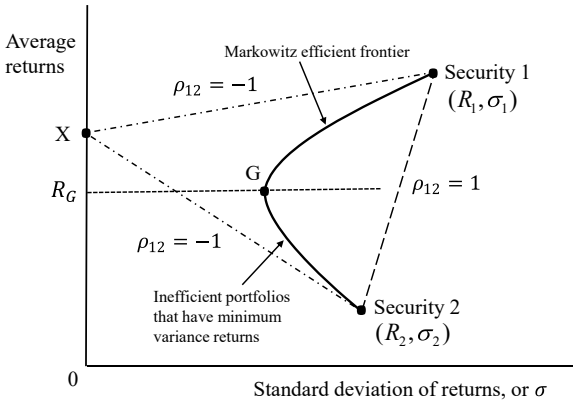


Fig. 1.2 Combinations of two securities to form a portfolio

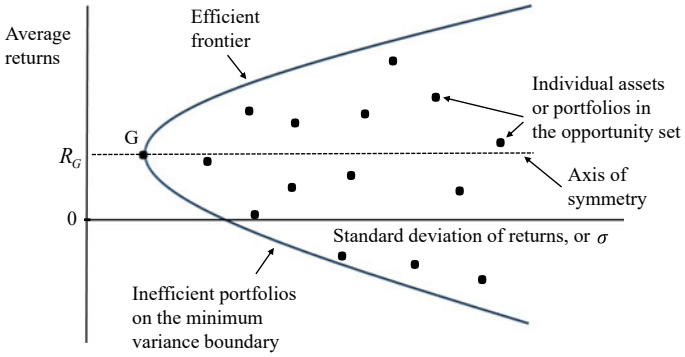


Fig. 1.3 The minimum variance boundary of portfolios and opportunity set of all assets

where μ_P denotes the population mean return, w_j is the weight for the jth security, $\sum_{j=1}^{N} w_j = 1$, and other notation is as before. The variance of returns for this portfolio is:

$$\sigma_P^2 = \sum_{j=1}^{N} w_j^2 \sigma_j^2 + \sum_{j=1}^{N} \sum_{k \neq 1}^{N} w_j w_k \rho_{ji} \sigma_j \sigma_k, \tag{1.19}$$

where $Cov(R_j, R_k) = \sigma_{jk} = \rho_{jk} \sigma_j \sigma_k$. Using this covariance definition, we can re-write the variance of returns for the portfolio as:

$$\sigma_P^2 = \sum_{j=1}^{N} \sum_{k=1}^{N} w_j w_k Cov(R_j, R_k) = \sum_{j=1}^{N} \sum_{k=1}^{N} w_j w_k \sigma_{jk}. \tag{1.20}$$

Now the computation of optimal weights is much more complicated than for only two securities. Appendix reviews Markowitz's derivation of optimal weights for many assets. While a mathematical solution can be achieved, implementation of these equations using computer programs in the real world is very difficult. For say 500 stocks, there would be $N(N-1))/2$ correlations, or almost 25,000 correlations! For 8,000 stocks in the U.S. stock market, the number of correlations is almost 32 million ... a formidable task. Of course, computers can crank out these correlations fairly rapidly, but then the covariance matrix needs to be estimated. To do this, a very large number of observations spanning over 30 years of daily returns would be required. Since old stock market data may well have little to do with the market in recent years or months, the analysis would become pointless. Also, the equations in Appendix show that the covariance matrix needs to be inverted (i.e., Σ^{-1}); however, this computation can fail due to matrix singularity (due to high correlations between some securities at times). These computational problems make it virtually impossible to build highly diversified portfolios with many securities.

Even if you could estimate optimal weights for diversified portfolios that lie on the efficient frontier, an even more daunting task remains. Your analyses were based on historical returns. According to the Randow Walk Hypothesis of Malkiel (2003), asset prices in the market move randomly over time and are independent of one another. Hence, all stock returns in an efficient portfolio will randomly change in the next out-of-sample period (e.g., one-month-ahead). These random movements not only change the expected returns of diversified portfolios but also their variances, correlations, and covariances. All of the thousands of estimates in the covariance matrix represent moving targets; consequently, measurement errors in the inputs for optimization multiply as assets are added to the analyses. Additionally, each month the entire analyses would need to be replicated, and the portfolio rebalanced to the new optimal weights. So estimation errors would multiply further as each set of optimal weights is estimated. Given all of these issues, it is not surprising that Michaud (1989) found that equal-weighted portfolio returns outperformed optimally weighted efficient portfolios in many out-of-sample tests. Alas, optimized efficient portfolios are an impossible dream.

1.3 New Investment Parabola Insights

Figures 1.2 and 1.3 depict the mean-variance investment parabola at a snapshot in time from a static perspective. But in the real world, the parabola is continuously moving and changing its shape over time. To illustrate, Fig. 1.4 shows the mean-variance parabola dynamically moving over time. The vertical width of the parabola changes say from day-to-day in response to increases or decreases in the cross-sectional return dispersion of all assets in the market. Also, the level of the whole parabola shifts as its axis of symmetry moves over time. Note that the axis of symmetry is approximated by the average market return on a general stock market index (e.g., the S &P 500 index). These two kinds of dynamic changes transform the shape of the parabola and, subsequently, move returns on assets up and down over time.

From time $t = 1$ to $t = 2$ in Fig. 1.4, average market returns do not change but return dispersion increases the width of the parabola. Asset A in the upper half of the parabola experiences increasing returns in this case. In this scenario, assets in the upper half experience increasing returns, whereas assets in the lower half experience decreasing returns. Notice here that, depending on where assets are located in the parabola, increasing return dispersion can have opposite return effects on different assets. At time $t = 3$, both average market returns and return dispersion decrease moving the return

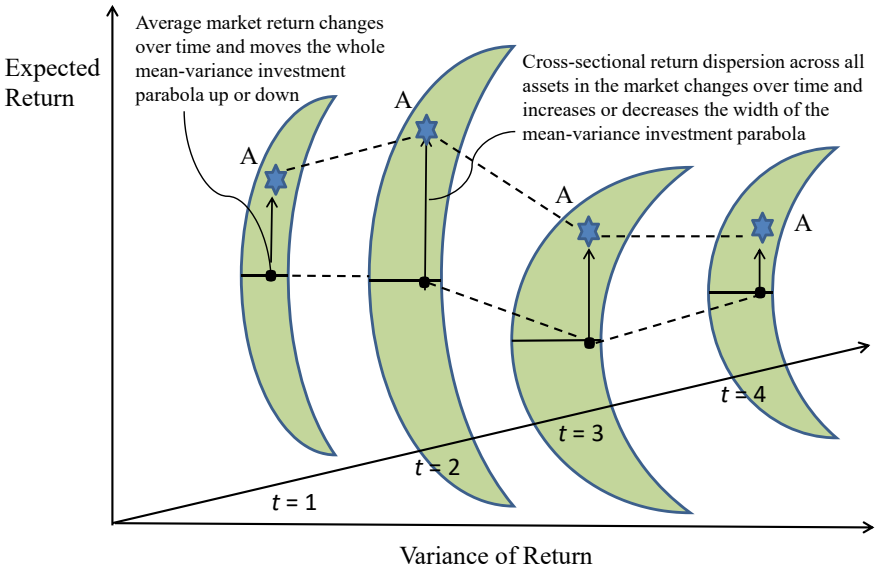


Fig. 1.4 The minimum variance boundary of portfolios and opportunity set of all assets

on asset A down. Finally, at time $t = 4$, the average market return increases but return dispersion decreases, which together offset one another, such that the return on asset A is unchanged. These market forces that impact the investment parabola have powerful effects on all asset returns.

A skeptic might ask: Is it really true that the width of the parabola is determined by return dispersion across assets in the market. Not previously known, a recent book by Kolari et al. (2021) provided mathematical proofs that return dispersion plays a major role in determining the width of the parabola. Let's review one of their proofs. They define return dispersion as the cross-sectional standard deviation of returns for assets in the market. For instance, on any day in the stock market, there is a mean return among all stocks (e.g., S &P 500 index return) as well as a standard deviation returns across all stocks on that day (or return dispersion).

Kolari, Liu, and Huang begin by writing the well-known variance of portfolio returns (see Eq. 1.39 in Appendix):

$$\sigma_P^2 = \frac{1}{C} + \frac{C(\bar{R}_P - \frac{A}{C})^2}{BC - A^2}, \tag{1.21}$$

where $A \equiv \bar{R}'\Sigma^{-1}e$, $B \equiv \bar{R}'\Sigma^{-1}\bar{R}$, $C \equiv e'\Sigma^{-1}e$ with e defined as an $n \times 1$ vector of ones, Σ is the covariance matrix, σ_P is the volatility of an efficient portfolio on the efficient frontier, and $\bar{R}_P$ is the expected return of the efficient portfolio.

Assume that there are three stocks in the market which are independent of one another (i.e., uncorrelated). Also assume that there are three risk factors in the market. The stocks have expected returns defined as $\bar{R}' = (\bar{R}_1, \bar{R}_2, \bar{R}_3)$, which yields covariance matrix $Diag(\sigma_1^2, \sigma_2^2, \sigma_3^2)$. To simplify the derivation, assume that $\sigma_1 = \sigma_2 = \sigma_3 = \sigma$. Given these starting conditions, we have:

$$\begin{aligned} C &= e'\Sigma^{-1}e = \frac{3}{\sigma^2} \\ A &= \bar{R}'\Sigma^{-1}e = \frac{\bar{R}_1 + \bar{R}_2 + \bar{R}_3}{\sigma^2} \\ B &= \bar{R}'\Sigma^{-1}\bar{R} = \frac{\bar{R}_1^2 + \bar{R}_2^2 + \bar{R}}{\sigma^2}. \end{aligned} \tag{1.22}$$

Using these values, we can write the denominator of Eq. (1.21) as:

$$BC - A^2 = \frac{3\left(\bar{R}_1^2 + \bar{R}_2^2 + \bar{R}_3^2\right) - \left(\bar{R}_1 + \bar{R}_2 + \bar{R}_3\right)^2}{\sigma^4}$$

$$= \frac{2\left(\bar{R}_1^2 + \bar{R}_2^2 + \bar{R}_3^2\right) - 2\left(\bar{R}_1\bar{R}_2 + \bar{R}_1\bar{R}_3 + \bar{R}_2\bar{R}_3\right)}{\sigma^4}$$
$$= \frac{\left(\bar{R}_1 - \bar{R}_2\right)^2 + \left(\bar{R}_1 - \bar{R}_3\right)^2 + \left(\bar{R}_2 - \bar{R}_3\right)^2}{\sigma^4}. \quad (1.23)$$

Our main interest is the cross-sectional variance σ_{CS}^2, which can be defined as:

$$\sigma_{CS}^2 = \frac{\begin{array}{l}\left[\bar{R}_1 - \left(\bar{R}_1 + \bar{R}_2 + \bar{R}_3\right)/3\right]^2 + \left[\bar{R}_2 - \left(\bar{R}_1 + \bar{R}_2 + \bar{R}_3\right)/3\right]^2 \\ + \left[\bar{R}_3 - \left(\bar{R}_1 + \bar{R}_2 + \bar{R}_3\right)/3\right]^2\end{array}}{3-1}$$
$$= \frac{\begin{array}{l}\left[\left(\bar{R}_1 - \bar{R}_2\right) + \left(\bar{R}_1 - \bar{R}_3\right)\right]^2 + \left[\left(\bar{R}_2 - \bar{R}_1\right) + \left(\bar{R}_2 - \bar{R}_3\right)\right]^2 \\ + \left[\left(\bar{R}_3 - \bar{R}_1\right) + \left(\bar{R}_3 - \bar{R}_2\right)\right]^2\end{array}}{18}$$
$$= \frac{\left(\bar{R}_1 - \bar{R}_2\right)^2 + \left(\bar{R}_2 - \bar{R}_3\right)^2 + \left(\bar{R}_3 - \bar{R}_1\right)H2}{6} \quad (1.24)$$

Using Eq. (1.23) and (1.24), we get $BC - A^2 = 6\sigma_{CS}^2/\sigma^4$. As such, we can write Eq. (1.21) as:

$$\sigma_P^2 = \frac{\sigma^2}{3} + \frac{\sigma^2}{2\sigma_{CS}^2}(\bar{R}_P - (\bar{R}_1 + \bar{R}_2 + \bar{R}_3)/3)^2. \quad (1.25)$$

The above Eq. (1.25) can be used to construct the mean-variance parabola. If we vary the cross-sectional variance σ_{CS}^2 from 1 to 8, Fig. 1.5 shows that the width of the mean-variance parabola becomes larger. Hence, we can infer that the width or span of the parabola is determined in large part by return dispersion.[1]

An important implication of this new finding is that the average market return lies somewhere in the middle of the parabola along the axis of symmetry. How do we know this? If the width of the parabola approximately encompasses the cross-sectional distribution of returns among all assets in the market, the mean or average return should be in the middle of this distribution. For example, the return on the S &P 500 index on

[1] Kolari et al. (2021) provided another more sophisticated mathematical proof of this result for a large number of N assets using random matrix theory.

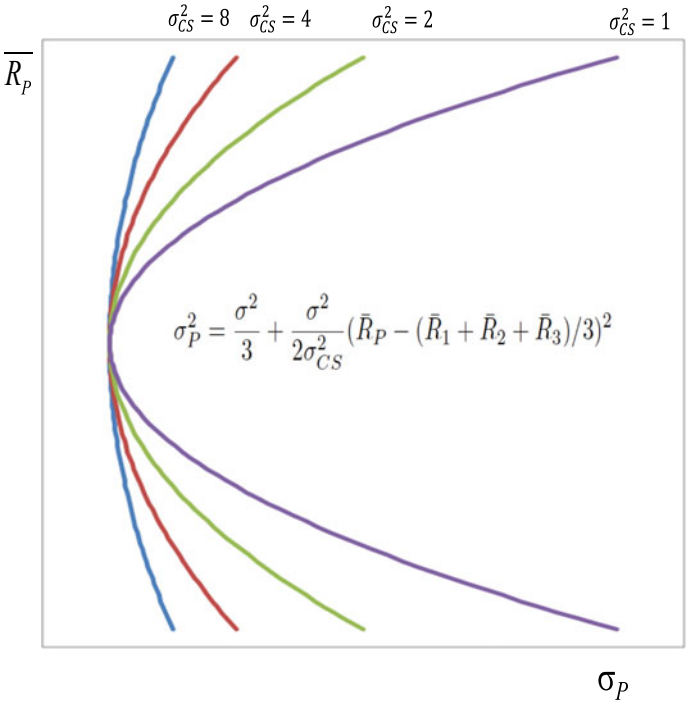


Fig. 1.5 Cross-sectional return variance (denoted σ^2_{CS}) directly affects the width of the mean-variance investment parabola

any given day lies somewhere in the middle of the parabola. This implication is a second insight about the parabola that was not previously known. Most finance researchers and professionals believe that the value-weighted average of returns in the stock market is representative of a fairly efficient portfolio located in the vicinity of the efficient frontier. However, according to Kolari, Liu, and Huang, because general stock market indexes lie along the axis of symmetry of the parabola, their returns are far from efficient! The same holds for other asset classes, including bond market indexes, real estate market indexes, commodities indexes, etc. Each of these asset classes can be graphed in a separate mean-variance parabola or combined into one aggregate investment parabola.

1.4 A New Approach to Finding Efficient Portfolios

With these new insights into the geometry of the mean-variance parabola in hand, how can we use this information to find a more efficient portfolios? As already discussed, optimization methods proposed by Markowitz are a mathematical tour de force but fall down in the real world due to problems related to estimating a massive number of parameters as well as estimation of expected returns in the future. In this book we propose a new approach. To identify more efficient portfolios, we take advantage of the architecture of the mean-variance investment parabola.

Figure 1.6 gives a diagram that communicates the intuition of our portfolio approach. To reach efficient portfolios, an investment manager needs to move along the axis of symmetry using the average market return and then vertically upward using cross-sectional return dispersion. In this setup, it is critical to control risk. On the X-axis is the standard deviation of returns measured over some time period (e.g., one day where general market returns are computed every 10 minutes during the trading day). By fixing the standard deviation of returns, or total risk, at different levels, a series of efficient portfolios can be obtained via the upward arrows in the diagram. Controlling risk related to average market returns is beta risk. In upcoming Chapter 2, we discuss beta risk in the context of well-known asset pricing models. Lastly, controlling risk related to cross-sectional return dispersion is zeta risk. In Chapter 3, we cover this important risk, which is based on the recently invented ZCAPM asset pricing model by Kolari et al. (2021).

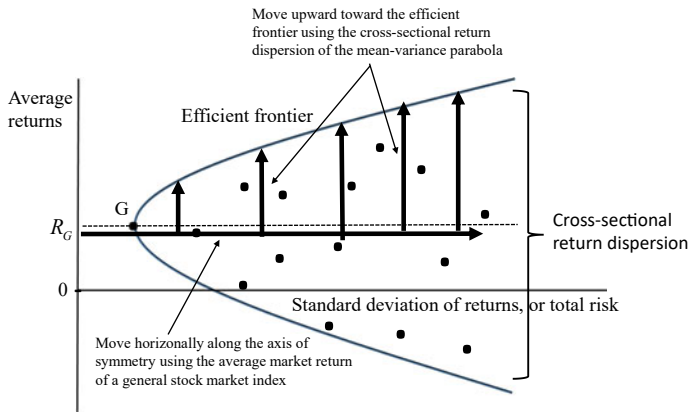


Fig. 1.6 How to reach the efficient frontier using average market returns and the cross-sectional standard deviation of returns of assets in the market

The initial step in our investment approach is identifying the global minimum variance portfolio G. What are the weights for individual assets that should be used to estimate G? As you can see in Fig. 1.6, the answer to this question is fundamental to the parabola. Portfolio G pins the level of the entire parabola. Notice that it has no zeta risk related to cross-sectional return dispersion. It gives the best estimate of the average market return and, therefore, beta risk. Also, it can be used to compute a value-weighted measure of cross-sectional return dispersion. These are the key ingredients needed to control risk and build more efficient portfolios.

1.5 Summary

Harry Markowitz (1952, 1959) received the Nobel Prize in Economics for inventing modern portfolio selection methods. He proposed the mean-variance investment parabola with an efficient frontier of portfolios that maximize returns per unit total risk as measured by the standard deviation of returns. Unfortunately, the dream of building efficient portfolios for investors has proven elusive. One problem is that sophisticated computer techniques must be used to estimate large numbers of statistical parameters that are constantly changing over time. A second, and bigger, problem is that efficient portfolios computed in-sample using ex post historical returns do not stay efficient in out-of-sample (e.g., one-month-ahead) tests. In this regard, Michaud (1989) found that efficient portfolios using optimal weights for securities cannot outperform equal-weighted portfolios in the future.

In this book, we propose a new approach to building efficient portfolios. More specifically, we utilize a recent asset pricing dubbed the *ZCAPM* by Kolari et al. (2021) to control risk in portfolio formation. These authors posited two new insights into the mean-variance investment parabola: (1) the width of the parabola is determined in large part by the cross-sectional return dispersion among all assets in the market; and (2) the average market return for a general market index lies approximately on the axis of symmetry that splits the parabola into two symmetric halves. Using these insights, an investment manager can move horizontally along the axis of symmetry and then vertically up to reach efficient portfolios. Previous researchers and professionals have commonly believed that average market returns lie in the vicinity of the efficient frontier. According to Kolari, Liu, and Huang, general market indexes are far from efficient as they lie in the middle of the mean-variance parabola in the vicinity of its axis of symmetry.

To implement this portfolio strategy, we employ the ZCAPM to control beta risk related to average market returns and zeta risk associated with return dispersion. We provided a graph of the investment parabola to communicate the intuition of this portfolio approach. In forthcoming chapters, we put together the pieces needed to understand and build efficient portfolios. Later chapters document evidence based on U.S. stock returns over many years.

As we will see, controlling risk is the key to constructing efficient portfolios that outperform general market indexes on an out-of-sample basis. Indexes such as the S &P 500 index in the U.S. stock market are good benchmarks for comparison. More than 95% of active investment managers cannot consistently outperform this index over 5 year investment horizons. If we can outperform the S &P 500 index in out-of-sample tests over a long period of time, the validity of our new investment approach is verified. We encourage readers to not only learn our new methods but conduct tests themselves. Investment managers interested in working jointly on efficient portfolios are invited to contact us. We own and operate an investment securities firm for this purpose.

Appendix: Optimal Weights for Many Assets

The appendix reviews the estimation of optimal weights in the case of many assets. Here we assume that all information about securities is based on historical returns. For a given set of N securities that comprises some portfolio P, based on their past returns and covariances of returns, we seek to find the minimum variance portfolio I that has optimal weights for these securities. By varying the portfolio return R_I, a mean-variance investment parabola can be traced out with the full range of minimum variance portfolios. We caution that this mean-variance parabola is in-sample in the sense that is constructed using historical returns for securities.

Following Markowitz (1959), the variance of portfolio I is minimized by means of a Lagrangian function subject to the constraints that the portfolio return equals R_I and securities' weights sum to one (i.e., Lagrangian multipliers η and κ):

$$\underset{\omega,\eta,\kappa}{\text{Min}} L = \frac{1}{2}\omega'\Sigma\omega + \eta[R_I - \omega' R_P] + \kappa(1 - \omega' e), \tag{1.26}$$

where ω is the vector of weights (with dimension $N \times 1$), e is a vector of ones (with dimension $N \times 1$), $R_P = [R_1, R_2, \ldots, R_N]'$ is the expected one-period return vector for N assets (with dimension $N \times 1$), and Σ is the

covariance matrix for N assets (with dimension $N \times N$). The latter return covariance matrix is:

$$\Sigma = \begin{vmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \cdots & \sigma_{1N} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \sigma_{N1} & \cdots & \sigma_{N3} & \cdots & \sigma_{NN} \end{vmatrix}. \tag{1.27}$$

Optimizing function L, the following first derivatives for each asset weight in the vector ω are computed to obtain $N + 2$ first order conditions:

$$\frac{\partial L}{\partial \omega} = \Sigma\omega - \eta R_P - \kappa e = 0 \tag{1.28}$$

$$\frac{\partial L}{\partial \eta} = R_P - \omega' R_P = 0 \tag{1.29}$$

$$\frac{\partial L}{\partial \kappa} = e - \omega' e = 0. \tag{1.30}$$

Based on Eq. (1.28), we have:

$$\begin{aligned} w &= \Sigma^{-1}(\eta R_P + \kappa e) \\ &= \eta\Sigma^{-1} R_P + \kappa\Sigma^{-1} e. \end{aligned} \tag{1.31}$$

Multiplying this equation by R'_P:

$$R'_P\omega = \eta R'_P\Sigma^{-1} R_P + \kappa R'_P\Sigma^{-1} e = R_I. \tag{1.32}$$

Multiplying the same equation by e':

$$e'\omega = \eta e'\Sigma^{-1} R_P + \kappa e'\Sigma^{-1} e = 1. \tag{1.33}$$

Upon setting

$$\begin{aligned} A &\equiv R'_P\Sigma^{-1} e \\ B &\equiv R'_P\Sigma^{-1} R_P \\ C &\equiv e'\Sigma^{-1} e, \end{aligned} \tag{1.34}$$

we obtain the known scalars for η and κ:

$$\eta = \frac{CR_I - A}{D} \tag{1.35}$$

$$\kappa = \frac{B - AR_I}{D}, \tag{1.36}$$

where $D = BC - A^2$.

Lastly, we can solve for the optimal weights ω^* as follows:

$$\begin{aligned} w^* &= \frac{B - AR_I}{BC - A^2}\Sigma^{-1}e + \frac{CR_I - A}{BC - A^2}\Sigma^{-1}R_P \\ &\equiv \phi + \psi R_I, \end{aligned} \tag{1.37}$$

where

$$\begin{aligned} \phi &\equiv \frac{B\Sigma^{-1}e - A\Sigma^{-1}R_P}{BC - A^2} \\ \psi &\equiv \frac{C\Sigma^{-1}R_P - A\Sigma^{-1}e}{BC - A^2} \end{aligned} \tag{1.38}$$

are known $N \times 1$ vectors. These equations yield the familiar definition of the variance of efficient portfolio I:

$$\sigma_I^2 = \frac{1}{C} + \frac{C\left[R_I - \frac{A}{C}\right]^2}{BC - A^2}. \tag{1.39}$$

Quadratic computer programs can provide exact solutions for the estimation of the above equations. Approximate solutions are available from some programs. These programs can be purchased on the internet.

References

Kolari, J.W., W. Liu, and J. Huang. 2021. *A new capital asset pricing model: Theory and evidence*. New York, NY: Palgrave Macmillan.

Malkiel, B.G. 2003. *A random walk down Wall Street: The time-tested strategy for successful investing*. New York, NY: W. W. Norton.

Markowitz, H.M. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.
Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: Wiley.
Michaud, R.O. 1989. The Markowitz optimization enigma: Is 'optimized' optimal? *Financial Analysts Journal* 45: 31–42.

Part II

Previous Asset Pricing Models



2

General Equilibrium Asset Pricing Models

2.1 The Present Value Formula

In theory, asset valuation can be expressed using the well-known present value formula. The value or price of an asset P at time 0 is:

$$P_0 = \sum_{t=1}^{T} \frac{E(CF_t)}{[1 + E(R)]^t}, \tag{2.1}$$

where $E(CF_t)$ are expected cash flows generated by the asset over time $t = 1, \ldots, T$, and $E(R)$ is the expected discount rate. For risky investments, the expected cash flows are not known for certainty. Even so, some estimates of future cash flows can be reasonably assessed based on good, bad, and average scenarios. In this regard, information about the asset, firm, industry, and economy is useful.[1]

[1] We should mention that many researchers re-write the present value formula in a one-period context as:

$$P_t = E_t(m_{t+1}x_{t+1}),$$

where P_t is the beginning period price, x_{t+1} represents cash flows or payoffs in the period from t to $t+1$ equal to $1+R_{t+1}$, E_t is the expectations operator conditional on market information at time t, and m_{t+1} is the *stochastic discount factor* (SDF). The variable m is sometimes referred to as the *asset pricing kernel* or *marginal rate of substitution*. The SDF, which is defined as $m_{t+1} = 1/(1 + R_{t+1})$, discounts expected cash flows to present value prices. We can rearrange this equation to express returns as $R_{t+1} = x_{t+1}/P_t$, such that $E_t[m_{t+1}(1 + R_{t+1})] = 1$. These equations represent the most general forms of asset pricing models. Since all models are special cases of these two equations,

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The main problem in the present value formula is: How can we estimate the discount rate for these cash flows. Of course, as the risk of future cash flows increases, a higher expected rate of return is required by investors, which increases the discount rate and lowers the asset price. But what risks are relevant to the discount rate and how can we measure them?

Asset pricing models seek to answer these questions. In so doing, they provide an estimate of the discount rate in Eq. (2.1). Given an expected discount rate, the present value of cash flows, or price, can be computed. Also, in the context of the present book, relevant risks used to determine the discount rate can be employed to control portfolio risk.

2.2 The CAPM

Sharpe (1964) and others cited earlier developed the CAPM in a market equilibrium setting. He assumed the following about the capital market for long-term assets: (1) perfect competition; (2) homogeneous expectations of investors about market information; (3) investors maximize expected utility of wealth and are risk averse with quadratic utility functions; (5) borrowing and lending by investors at the riskless rate is possible; and (6) no short selling is allowed by investors who can only hold long positions in assets.

2.2.1 Existence of a Riskless Asset

Extending Markowitz (1952, 1959), Sharpe introduced a riskless asset in the market. The upshot of this possibility is the Capital Market Line (CML) displayed in Fig. 2.1. Equilibrium asset prices trade off the expected rates of return of assets against their risks. Risk here is measured by the standard deviation of returns, or *total risk*. If there is zero risk, the investor earns the riskless rate R_f, which is based on the price of time or pure interest rate.[2] Given some expected rate of return per unit risk, under conditions of risk, the investor demands increasing risk premiums as risk increases.

they fall under the general heading of "*m*-talk." See Cochrane (2005) and Ferson (2019) for further discussion and applications of m-talk in asset pricing models.

[2] For example, assuming no risk, if you are indifferent between $100 today or $101 at the end of a year, your rate of time preference is 1%. This rate can be attributed to your utility preference of present consumption over future consumption.

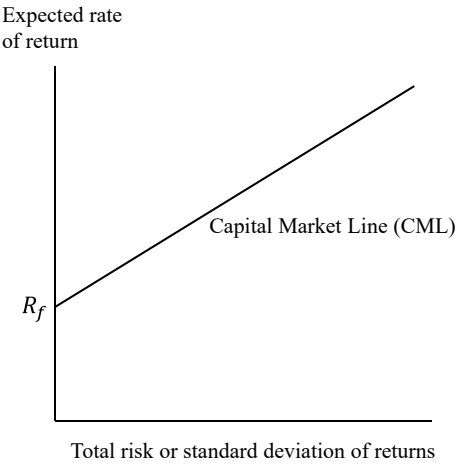


Fig. 2.1 Equilibrium relationship between expected rates of return and risk

2.2.2 Capital Market Line

What if the investor can hold both riskless and risky assets? This portfolio combination (C) of assets has expected return as follows:

$$E(R_C) = aR_f + (1 - a)E(R_P), \tag{2.2}$$

where R_f = the riskless rate for asset f held in proportion a, and $E(R_P)$ = the expected return on portfolio P held in proportion $(1 - a)$. Following Markowitz as discussed in the previous chapter, the standard deviation of combined portfolio returns, or total risk is:

$$\sigma(R_C) = [a^2\sigma^2(R_f) + (1 - a)^2\sigma^2(R_P) + 2a(1 - a)Cov(R_f, R_P)]^{1/2}, \tag{2.3}$$

where $Cov(R_f, R_P) = \rho_{fP}\sigma(R_f)\sigma(R_P) = 0$. Since the riskless rate R_f is constant over time, its correlation with R_P is zero. This simplification reduces the above expression to:

$$\sigma(R_C) = (1 - a)^2\sigma^2(R_P). \tag{2.4}$$

Using these formulas, we can draw Fig. 2.2. There we see different combinations of the riskless asset and risky portfolios on the Markowitz efficient frontier. Combinations with tangent market portfolio M yield the highest returns at all levels of total risk. This line is known as the *Capital Market Line*

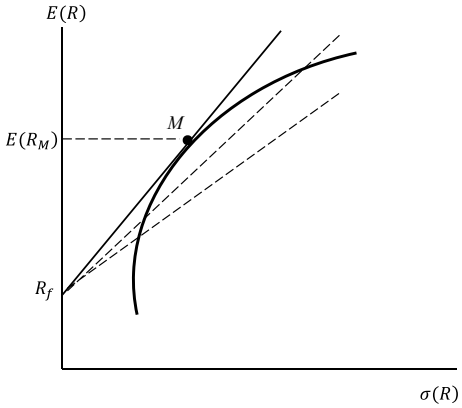


Fig. 2.2 Locating market portfolio as tangent point on ray from riskless rate

(CML) and represents the new efficient frontier for investors. Other lines below the CML are inefficient investments. If investors put all their funds into the market portfolio, they would be located at M. If they put some of their funds in the riskless rate (i.e., lending), they would be somewhere on the line between R_f and M. Finally, if they borrow funds at the rate R_f and invest in M, they would move up the CML to the right of M.

The obvious question is: How can an investor buy M? This market portfolio is believed to be a value-weighted average of all assets in the market—for example, in the stock market we could proxy it with the S &P 500 index. Historical evidence has shown that over 90% of investment managers cannot outperform this stock market index over three-to-five-year periods on a consistent basis over time. Hence, it would appear to be a potential proxy for M. However, as we will see in Chapter 4, this inference is incorrect.

2.2.3 Deriving the CAPM

The CML shows the general equilibrium relationship between returns and total risk for portfolios. Next, Sharpe considered the case of an individual asset. If we combine asset i with M, we get the following expected return:

$$E(R_C) = aE(R_i) + (1 - a)E(R_M). \tag{2.5}$$

The standard deviation of returns is:

$$\sigma(R_C) = [a^2\sigma^2(R_i) + (1 - a)^2\sigma^2(R_M) + 2a(1 - a)Cov(R_i, R_M)]^{1/2}, \tag{2.6}$$

where $Cov(R_i, R_M) = \rho_{iM}\sigma(R_i)\sigma(R_M)$ is the covariance between asset i returns and M returns.

To derive the CAPM, Sharpe set the slopes of the CML and efficient frontier equal to one another and solved for the expected return of asset i. Figure 2.2 shows the slope of the CML is $[E(R_M) - R_f]/\sigma(R_M)$. The slope of the efficient frontier is more difficult. Using Eq. (2.5), we can write $dE(R_C)/da = E(R_i) - E(R_M)$. Using Eq. (2.6), we have $d\sigma(R_C)/da = [a\sigma^2(R_i) - \sigma^2(R_M) + a\sigma^2(R_M) + Cov(R_i, R_M) - 2a\,Cov(R_i, R_M)]/\sigma(R_C)$. At the market portfolio M where $a = 1$, we can write the change in portfolio C returns per unit change in total risk, or slope of the CML, as:

$$\frac{dE(R_C)}{d\sigma(R_C)} = \frac{[E(R_i) - E(R_M)]}{[-\sigma^2(R_M) + Cov(R_i, R_M)]/\sigma(R_M)}. \tag{2.7}$$

Setting these slopes equal to one another, we get:

$$\frac{[E(R_M) - R_f]}{\sigma(R_M)} = \frac{[E(R_i) - E(R_M)]}{[-\sigma^2(R_M) + Cov(R_i, R_M)]/\sigma(R_M)}. \tag{2.8}$$

Isolating the expected return on asset i results in the asset pricing equation for the CAPM:

$$E(R_i) = R_f + [E(R_M) - R_f]\frac{Cov(R_i, R_M)}{\sigma^2(R_M)}. \tag{2.9}$$

Sharpe writes the CAPM more simply as:

$$E(R_i) = R_f + \beta_i[E(R_M) - R_f], \tag{2.10}$$

where

$$\beta_i = \frac{Cov(R_i, R_M)}{\sigma^2(R_M)}. \tag{2.11}$$

Here β_i represents *beta risk*. It measures the systematic risk of asset i with respect to the excess return on the market portfolio, or $E(R_M) - R_f$. Assuming asset i is the market portfolio, we know that beta is 1. So beta less or greater than one implies systematic risk less or greater that the general market.

Returning to Markowitz, we can now decompose total risk for asset i into beta risk and idiosyncratic risk, which are systemic and unsystematic risks.

Diversified investors need only pay attention to beta risk, as idiosyncratic risk is diversified away and therefore not priced in assets.

2.2.4 Security Market Line

We can draw a graph of the CAPM. Figure 2.3 shows that a linear relation between expected returns $E(R_i)$ and beta risk β_i for individual assets. Sharpe named this relation the *Security Market Line* (SML). Notice that the expected return on the market portfolio M, or $E(R_M)$, has beta risk $\beta = 1$. The market risk premium is the excess return over the riskless rate $E(R_M) - R_f$. The risk premium for an individual asset is $\beta_i[E(R_M) - R_f]$. This risk premium plus the riskless rate yields the expected return for an asset. Thus, the CAPM enables an estimate of the expected discount rate in Eq. (2.1). If an asset had an expected rate of return greater than (less than) this expected rate from the CAPM, according to Eq. (2.1), it would be underpriced (overpriced).

2.3 The Market Model

Researchers set out to empirically test the CAPM using stock return data. To do so, a *market model* form the CAPM was developed by Markowitz (1959), Sharpe (1963), and Fama (1968). The following simple regression model was proposed:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + e_{it}, \tag{2.12}$$

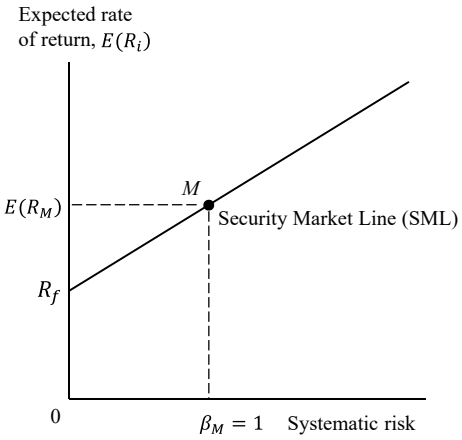


Fig. 2.3 Pricing capital assets using systematic or beta risk

where $R_{it} - R_{ft}$ is the realized excess rate of return on asset i over the riskless rate at time t (e.g., one day or month), $R_{mt} - R_{ft}$ is the realized excess average rate of return on the stock market as a whole, and e_{it} is a random error term that captures the scatter of assets' returns around the SML. The estimated parameters are the intercept, or α_i, and beta risk, or β_i. Looking at the CAPM in Eq. (2.10), it is clear that α_i should equal zero. If not, something is missing in the CAPM. We use lower case m here to denote a proxy for the true theoretical market portfolio M. Also, the riskless rate is commonly proxied with the U.S. Treasury bill rate.

2.3.1 Early CAPM Tests

An often-cited paper on early CAPM tests is by Black, Jensen, and Scholes (BJS) (1972). They employed monthly U.S. stock returns from 1926 to 1966. Individual stocks were grouped into 10 portfolios with different levels of beta. They found that the SML was flatter than expected. Stock portfolios that had betas greater (less) than one had negative (positive) intercepts. They conducted t-tests of the αs for the portfolios to determine whether or not they were equal to zero as posited by theory. Many researchers refer to the intercept as Jensen's (1968) alpha, who observed that α provides a way to test the CAPM. If $\alpha > 0$, we can infer that mispricing exists due to missing risk factors that are not in the model.

Black, Jensen, and Scholes found that, contrary to CAPM theory, portfolios with very high or low betas had significant αs. Further analyses of different subperiods showed that the SML was flatter later in their sample period. The average αs for the portfolios ranged from about -0.50% and $+0.35\%$ per month, which is -6% to $+4\%$ on an annualized basis. Given that excess market returns are around 6%, these results suggest that the CAPM is missing something. More than beta risk is needed to capture all of the risks inherent in stock returns. For this reason, the authors rejected the CAPM.

Another way to test the CAPM is to see how well it performs in out-of-sample tests. That is, estimate the parameters during some period of time, then see if beta risk is related to returns in the next period. No cheating can happen here as one-month-ahead returns (for example) are related to previously estimated beta risk. A two-step estimation process was developed by Fama and MacBeth (1973). First, estimate the market model over say 60

months (or 5 years) for 20 beta-sorted portfolios. Second, estimated betas for these portfolios are used in the following cross-sectional regression model[3]:

$$R_{pT+1} = \alpha_0 + \lambda_m \beta_{pt-1} + e_{pT+1}, \tag{2.13}$$

where R_{pT+1} is the portfolio return in the next month $T + 1$ after the previous 60-month estimation period, α_0 is the intercept term, β_{pt-1} is the portfolio beta estimated using monthly returns in the previous estimation period, and e_{pT+1} is the random error term. The main focus in this regression equation is on the estimated parameter λ. This parameter tells us if previously estimated beta risk is related to future returns across the 20 beta-sorted portfolios. More specifically, it measures the market price of risk of beta, or the estimated market risk premium associated with beta risk. In their tests covering the period 1935 to 1968, they found mixed results with λ significant in some subperiods but insignificant in others. While the evidence supported the CAPM to some extent, it was fairly weak in general.

Interestingly, Fama and MacBeth found that the estimated market price of risk λ was less than the average market risk premium $R_{mt} - R_{ft}$ used in step one with the market model. They inferred that the market risk premium should be defined as $E(R_{mt}) - E(R_{0t})$, where $E(R_{0t}) > R_{ft}$. The rate of return $E(R_{0t})$ obviously has higher risk than the riskless Treasury bill rate. Also, the rate $E(R_{0t}) > R_{ft}$ helps to explain the flatter than expected SML. Looking back at Fig. 2.3, the higher intercept implied by $E(R_{0t})$ would tilt the SML downward to make it flatter in shape.

2.3.2 Investment Portfolio Implications

The CAPM says that all investors should prefer market portfolio M. Researchers have argued that M contains all assets in the market in proportion to their market values with respect to the total or aggregate value of the entire market.[4] This portfolio approach excludes all other efficient portfolios from investor consideration. According to the CAPM, other efficient portfolios, in combination with the riskless asset, will yield inferior returns per unit total risk. Hence, investors do not need to be constantly evaluating all kinds of risks of firms, including credit risk, management risk, liquidity risk, legal risk, etc. to construct portfolios from individual stocks based on

[3] We simplify their analyses here to ease exposition and emphasize important findings in their study.
[4] For example, see Fama (1968).

accounting and financial fundamentals. All idiosyncratic risks are diversified away by M, and beta risk takes into account all relevant pricing risks.

In the CAPM world, only beta risk matters in pricing assets. If an investor buys a range of portfolios with betas from low (less than one) to high (greater than one), the future returns on these portfolios should be ordered in line with these beta risks from low to high. Nobody knows where general market returns will go in the future with certainty. Future returns are very difficult (if not impossible) to accurately predict. However, if investors hold portfolios with different levels of beta risk, their future returns will be predictable in the sense that the relative returns of these portfolios will be ordered according to their relative beta risks.

In the real world, two problems with these investment portfolio ideas gradually materialized. First, even with the invention of powerful computers in the 1970s and 1980s, for large numbers of individual stocks (for example), nobody could estimate efficient portfolios let alone M using sophisticated statistical software. We will discuss econometric problems in this estimation process in Chapter 7 in the context of finding minimum variance portfolio G. In the meantime, it is sufficient to observe that no investment company in the marketplace claims to use computerized portfolio techniques in constructing efficient portfolios that consistently outperform the S&P 500 index on a return/risk basis. Second, and particularly troubling, when we put together portfolios of individual stocks sorted on beta risk, as shown by Fama and French (1996) and citations therein, there appears to be no relation between their future returns and their previously estimated betas. These problems are major challenges to utilizing the CAPM in portfolio management.

2.4 The Zero-Beta CAPM

Fischer Black (1972) derived a more general equilibrium asset pricing model that, unlike the CAPM, allowed for short sales of assets and borrowing at a higher rate than the riskless rate. His so-called *zero-beta CAPM* is as follows:

$$E(R_i) = E(R_Z) + \beta_i[E(R_M) - E(R_Z)], \tag{2.14}$$

where β_i = beta risk, $E(R_Z)$ is the expected return on the zero-beta portfolio Z that is the borrowing rate, and other notation is the same as the CAPM. Since $E(R_Z) > R_f$, the slope of the SML is lower and intercept higher than the CAPM in line with empirical evidence discussed above.

The zero-beta portfolio is orthogonal to (or uncorrelated with) the market portfolio and therefore has a beta equal to zero. Figure 2.4 shows how to geometrically locate the zero-beta portfolio. A ray from the market portfolio M down through the minimum variance portfolio G intersects the Y-axis at the rate of return coinciding with the zero-beta portfolio Z.

In Fig. 2.4 notice that, if the investor could borrow at R_f, then we would get two efficient frontiers: (1) the ray from R_f to tangent portfolio T on the efficient frontier; and (2) the efficient frontier from portfolio T to portfolio S. Any portfolio between T and S can be constructed by using different weighted combinations of M and Z.

Interestingly, in the zero-beta CAPM, investors can hold any risky efficient portfolio I and its zero-beta counterpart ZI (i.e., an infinite number of such portfolio pairs). In this case, we have:

$$E(R_i) = E(R_{ZI}) + \beta_i[E(R_I) - E(R_{ZI}], \tag{2.15}$$

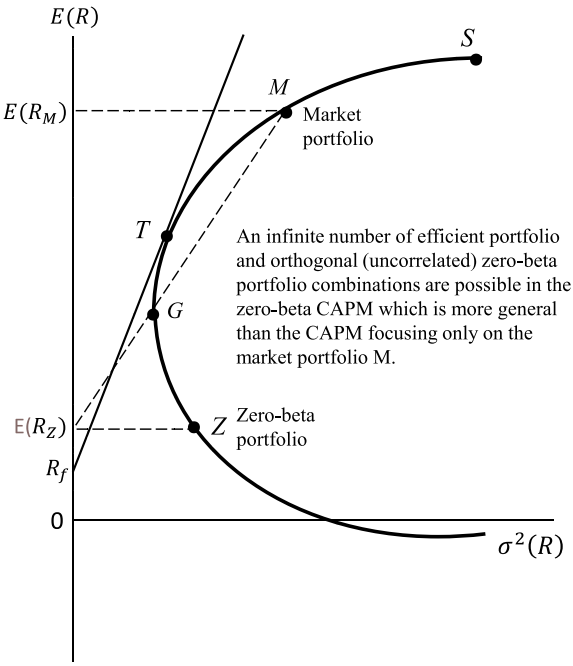


Fig. 2.4 Black's zero-beta CAPM

where β_i corresponds to beta risk related to risk premium defined as $E(R_I) - E(R_{ZI})$.[5] This specification using any efficient portfolio I and its zero-beta portfolio ZI is much more general than the CAPM that focuses only on the market portfolio M. Copeland and Weston (1980, p. 184) have an excellent discussion of this form of the zero-beta ZCAPM.

Lastly, if a riskless asset exists, the zero-beta CAPM becomes:[6]

$$E(R_i) - R_f = \beta_{i,I}[E(R_I) - R_f] + \beta_{i,ZI}[E(R_{ZI}) - R_f], \qquad (2.16)$$

where $\beta_{i,I}$ and $\beta_{i,ZI}$ become the beta risks of asset i with respect to efficient portfolio I and zero-beta portfolio ZI, respectively. Thus, we have a two-factor model with two beta risks related to the excess returns on the efficient portfolio and zero-beta portfolio.

2.4.1 Investment Portfolio Implications

In the real world, it is not clear how to find portfolios M and G and therefore portfolio Z. Without portfolio Z, it is difficult to empirically test the zero-beta CAPM with stock return data. Nonetheless, the zero-beta CAPM implies that investors should seek portfolios that lie on the efficient frontier, not just market portfolio M. Regrettably, Black only provided a theoretical analysis of general equilibrium asset pricing with no recommended market model for test purposes as in the case of the CAPM.

In forthcoming Chapters 4 and 5, to implement the zero-beta ZCAPM in the real world, we review a special case of the zero-beta CAPM named the *ZCAPM* by Kolari et al. (2021). This new asset pricing model can be estimated with available stock returns. The theoretical ZCAPM provides a novel way to reach a unique pair of efficient and orthogonal zero-beta portfolios. The empirical ZCAPM (akin to a market model form of the theoretical CAPM) enables the estimation of two market risks related to average market returns (beta risk) and cross-sectional return dispersion in the market (zeta risk), respectively. The latter return dispersion captures the difference in returns between portfolios I and ZI in Eq. (2.16), thereby avoiding the need to know their specific returns. This model is employed in forthcoming chapters to construct efficient portfolios for the U.S. stock returns.

[5] See Roll (1980) for this form of the zero-beta CAPM.

[6] See equation (40) in Black (1972, p. 454).

2.5 Alternative CAPM Forms

It is worthwhile mentioning that, in an effort to fix the CAPM in terms of empirical relevance, researchers have proposed a number of alternative forms of the CAPM. The most prominent among these alternatives is the intertemporal CAPM (ICAPM) by Merton (1973). Extending the CAPM from a single period to multiple periods, which was a major advancement, the ICAPM allows for underlying state variables that investors seek to hedge over time as the investment opportunity set changes over time. Hence, this model allows for multiple risk factors beyond only beta risk related to the market portfolio.

The ICAPM led to a number of other CAPM-based models. The international asset pricing model (IAPM) of Solnik (1974a, b) uses an international market factor to price assets and allows for different riskless rates in different countries. This model assumes that (for example) stock markets are integrated or inter-connected across different countries. Later studies found that stock market in different countries are not integrated for the most part, which means that pricing stocks tend to be local (domestic) rather than global (international) in nature.

Lucas (1978) and Breeden (1979) proposed the consumption CAPM (CCAPM). Investors smooth consumption by buying and selling financial assets. If times are bad (good), they sell (buy) assets to maximize their lifetime expected utility. An argument in favor of the CCAPM is that aggregate consumption is observable in the real world (unlike market portfolio M). Even so, empirical tests yielded weak support for the CCAPM akin to the CAPM.

The production CAPM (PCAPM) of Cochrane (1991) focuses on producers rather than consumers. Asset returns are explained by investment returns to capital investments related to the production function of the firm. Some evidence supports this model due to the relation between stock returns and the ratio of investment-to-capital over time.

In empirical tests using U.S. stock returns, time variation in beta risk associated with the investment/capital ratio appeared to improve the empirical results for the PCAPM. In his tests, Cochrane implemented the conditional CAPM by Hansen and Hodrick (1983), Ferson, Kandel, and Stambaugh (1987), and others, which allows for time-varying risk parameters in asset pricing models using instrumental variables (e.g., interest rates levels). Consistent with Merton's ICAPM, conditional models seek to take into account the business cycle with respect to economic expansions and recessions as well as other changing market conditions. An advantage of conditional models is

that they can be adapted to any asset pricing model to potentially increase its empirical fit to real world returns.

2.6 Road Map of General Equilibrium Models

Figure 2.5 gives a road map of general equilibrium asset pricing models. These theory-based models are grounded in Markowitz diversification, general equilibrium conditions, capital market assumptions, and risk averse investor behavior.

The first stop in our road trip is the CAPM of Sharpe with beta risk and the market factor. Due to early empirical evidence that suggested a flatter slope of the SML than expected, Black's zero-beta CAPM was proposed to allow borrowing at a higher rate than the riskless rate using the zero-beta portfolio rate of return. As will be seen in Chapter 4, the zero-beta portfolio is a key ingredient in the development of the more recent ZCAPM that we employ in this book for portfolio construction purposes.

CAPM theory was further developed by Merton's ICAPM, who greatly expanded the analyses to a multiperiod investment framework. A number of extensions of the ICAPM were developed, including the IAPM by Solnik, the CCAPM by Lucas and Breeden, the PCAPM by Cochrane, and conditional models by Hansen and Hodrick as well as Ferson, Kandel, and Stambaugh. These models showed that the CAPM is consistent with a variety of real world conditions, such as international markets, aggregate consumption and production in the economy, and time-varying business conditions.

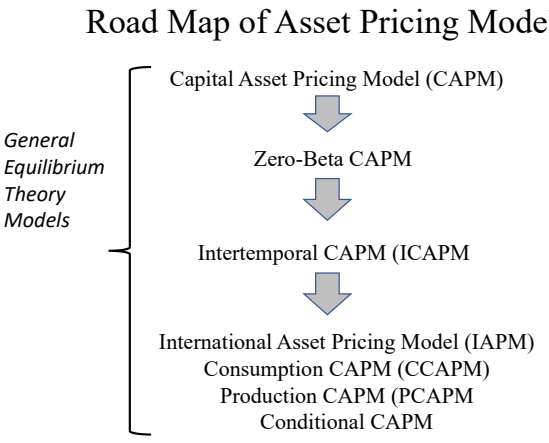


Fig. 2.5 CAPM-based models developed over time to encompass most real world market and economic conditions

2.7 Summary

This chapter has reviewed the development of asset pricing models in the financial economics literature. Based on Markowitz diversification principles, Sharpe and other authors created the CAPM in the 1960s. The CAPM proved to be a landmark model that created the field of asset pricing. While theoretically elegant based on Markowitz portfolio concepts and general equilibrium conditions, tests of the market model form of the CAPM suggested a weaker relation between returns and market beta risk than predicted by theory.

Subsequently, Black proposed the zero-beta CAPM that allowed a higher borrowing rate than the riskless rate in the CAPM. A zero-beta portfolio was identified on the investment parabola that is orthogonal (uncorrelated) to its counterpart efficient portfolio. Merton's ICAPM was a major advancement that derived the CAPM in a multiperiod framework. It led to other CAPM-based models, including the IAPM, CCAPM, and PCAPM. Lastly, conditional models allow the risk parameters (or betas) in these models to time vary with changing business conditions.

Markowitz diversification and general equilibrium asset pricing models laid the theoretical foundation for modern portfolio management. The pursuit of efficient portfolios had begun. After more than 60 years, the dream of building efficient portfolios has been elusive. In this book, we propose new and novel methods for estimating efficient portfolios using asset pricing models. It turns out that the critical ingredients are the measurement and control of market risks.

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Black, F., M.C. Jensen, and M. Scholes. 1972. The capital asset pricing model: Some empirical tests. In *Studies in the theory of capital markets*, ed. M.C. Jensen. New York, NY: Praeger.

Breeden, N.D. 1979. An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics* 7: 265–296.

Cochrane, J.H. 1991. Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance* 46: 209–237.

Cochrane, J.H. 2005. *Asset pricing*. Revised. Princeton, NJ: Princeton University Press.

Copeland, T. E., and J. F. Weston. 1980. *Financial theory and corporate policy.* Reading, MA: Addison-Wesley Publishing Company.

Fama, E.F. 1968. Risk, return, and equilibrium: Some clarifying comments. *Journal of Finance* 23: 29–40.

Fama, E.F., and K.R. French. 1996. The CAPM is wanted, dead or alive. *Journal of Finance* 51: 1947–1958.

Fama, E.F., and J.D. MacBeth. 1973. Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy* 81: 607–636.

Ferson, W.E. 2019. *Empirical asset pricing: Models and methods*. Cambridge, MA: The MIT Press.

Ferson, W. E., S. Kandel, and R. F. Stambaugh. 1987. Tests of asset pricing with time-varying expected risk premiums and market betas. *Journal of Finance* 42: 201–220.

Hansen, L.P., and R.J. Hodrick. 1983. Risk averse speculation in the forward foreign exchange market: An econometric analysis of linear models. In *Exchange rates and international macroeconomics*, ed. J.A. Frenkel. Chicago, IL: University of Chicago Press.

Jensen, M.C. 1968. The performance of mutual funds in the period 1945–1964. *Journal of Finance* 23: 389–416.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Lucas, R.E. 1978. Asset prices in an exchange economy. *Econometrica* 46: 1429–1445.

Markowitz, H. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: Wiley.

Merton, R.C. 1973. An intertemporal capital asset pricing model. *Econometrica* 41: 867–887.

Mossin, J. 1966. Equilibrium in a capital asset market. *Econometrica* 34: 768–783.

Roll, R. 1980. Orthogonal portfolios. *Journal of Financial and Quantitative Analysis* 15: 1005–1012.

Sharpe, W.F. 1963. A simplified model for portfolio analysis. *Management Science* 9: 277–293.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.

Solnik, B.H. 1974a. The international pricing of risk: An empirical investigation of the world capital market structure. *Journal of Finance* 29: 365–378.

Solnik, B.H. 1974b. An international market model of security price behavior. *Journal of Financial and Quantitative Analysis* 9: 537–554.

Treynor, J. L. 1961. Market value, time, and risk, Unpublished manuscript.

Treynor, J. L. 1962. Toward a theory of market value of risky assets, Unpublished manuscript.



3

Multifactor Asset Pricing Models

3.1 Arbitrage Pricing Theory

Stephen Ross (1976) broke away from the CAPM equilibrium asset pricing framework with his *arbitrage pricing theory* (APT). Assuming that arbitrage opportunities by investors determine asset prices, the APT is a linear asset pricing model with multiple factors. Arbitrage means to buy and sell an asset that is under- or overpriced until a single price is reached in the market. No wealth is needed as buying and selling of an asset can occur simultaneously. For example, arbitrage portfolios buy assets in a long position that are financed by selling short other assets (i.e., selling borrowed assets). The sum of the weights of this arbitrage portfolio equals zero, which results in the name *zero-investment portfolio*. The APT is theoretically foundational to later multifactor models that construct risk factors using zero-investment portfolios with little or no connection of the CAPM and its equilibrium framework. Some researchers also cite Merton's ICAPM for justification of multiple factors in multifactor models.

The APT with $k = 1, \ldots, K$ common factors, or $F_1, F_2, \ldots, F_K$, and beta risk coefficients $\beta_{i1}, \beta_{i2}, \ldots, \beta_{iK}$, can be written as a multifactor model:

$$R_i = E(R_i) + \beta_{i1} F_1 + \cdots + \beta_{iK} F_K + \epsilon_i. \tag{3.1}$$

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Given a riskless asset, it can be re-written in the context of beta risks and associated risk premiums (denoted γ_k):

$$E(R_i) - R_f = \gamma_1 \beta_{i1} + \cdots + \gamma_K \beta_{iK}. \tag{3.2}$$

Asset-specific idiosyncratic risk is diversified away. Arbitragers price assets using long/short, zero-investment factors that hedge various market risks. Unfortunately, the APT does not identify these factors in the real world. In effect, a kind of "invisible hand" exists in financial markets that linearly price assets.

3.2 Fama and French Three-Factor Model

What could the APT factors be in the real world? Fama and French (1992, 1993, 1995, 1996, 1998, 2004) published a series of papers that documented the failure of the CAPM to explain the cross section of average stock returns. Declaring the CAPM *dead*, they proposed the landmark three-factor model that included the market factor plus size and value multifactors. As mentioned earlier, it was not long before other researchers proposed other multifactors and resultant models. The race was on—may the best multifactor model win!

In the Fama and French three-factor model, the CAPM market factor is augmented with size and value factors. Following the APT by Ross, these factors are zero-investment portfolios. The *size factor* is long small stocks' returns minus (or short) big stocks' returns, where market capitalization is used to define small and big. The *value factor* is long value stocks' returns minus (or short) growth stocks' returns, where the ratio of book market value of equity to market value of equity is used to define value or growth. This long/short approach to factor construction was a major advance in asset pricing.

The three-factor model is written as:

$$\begin{aligned} R_{it} - R_{ft} = \alpha_i + \beta_{i,m}(R_{mt} - R_{ft}) + \beta_{i,S}\mathrm{SMB}_t \\ + \beta_{i,V}\mathrm{HML}_t + e_{it}, \end{aligned} \tag{3.3}$$

where $\beta_{i,S}$ and $\beta_{i,V}$ are the beta risk loadings for the size and value factors denoted SMB and HML, respectively. These loadings measure the sensitivity of asset returns to the new factors. This model can be depicted in four-dimensional space as shown in Fig. 3.1.

Three-Factor Model

$$R_{it} - R_{ft} = \alpha_i + \beta_{i,m}(R_{mt} - R_{ft}) + \beta_{i,S}SMB_t + \beta_{i,V}HML_t + \varepsilon_{it}$$

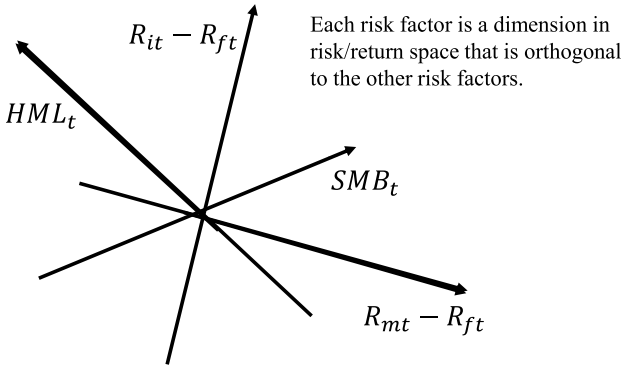


Fig. 3.1 The Fama and French three-factor model with excess returns a function of market, size, and value factors takes into account four dimensions in return/risk space

Using 25 stock portfolios sorted into size and value quintiles and U.S. stock returns over about 40 years, Fama and French reported convincing evidence that the size and value factors improved the market model version of the CAPM with only the market factor. The goodness-of-fit as measured by R^2 increased considerably, and the intercept term α_i noticeably decreased. The beta risk loadings for size and value factors appeared to have some statistical significance in cross-sectional regressions like Eq. (2.13) in Chapter 2. Hence, unlike beta risk of the market factor, investors' price beta risks related to the size and value factors to some extent.

Some researchers immediately criticized the three-factor model. Black (1993) argued that there was little or no theoretical foundation for the size and value factors. He believed that *data mining* was used to come up with these factors. Just keep running stock return data until something is statistically significant. No clear economic meaning can be given to explain small stock returns minus big stock returns. Many differences exist between small and big firms. It is impossible to ascribe what risk or risks are manifested within the long/short size factor. Similarly, value minus growth firms have many differences that disallow any clear definition of risk. We generally refer to unexplained returns as anomalies. That is, the size and value factors have anomalous returns not explained by the market factor or some other factor in the various CAPM, APT, and other theoretical asset pricing models.

Another study by Kothari et al. (1995) found some evidence to support the CAPM using annual rather than monthly returns as in Fama and French.

Upon estimating cross-sectional regression relation (2.13) in Chapter 2, they detected a significant market price of risk with respect to beta loadings, or λ_m. Also, tests of the Fama and French model using industry portfolios seemed to indicate that the size and value factors were much weaker than for portfolios sorted on size and value factors. Researchers consider the size and value portfolios as endogenous to the size and value factors. Normally, in regression analyses, it is desirable to use exogenous that are unrelated to the factors in a model, such as industry portfolios.[1] Responding to these issues, Fama and French (1996) noted that, even if market beta is significant, the CAPM needs to be augmented with the size and value factors to better explain average stock returns. Without them, the CAPM still has problems with its goodness-of-fit and Jensen's alpha.

3.3 The Factor Zoo and Multifactor Models

The success of the three-factor model led to boom in asset pricing factors. Researchers developed all kinds of long/short factors that had some significance in explaining average returns in time-series regression tests of alpha intercepts as well as cross-sectional regression tests of their market prices of risk. As noted above, Cochrane (2011) opined that a factor zoo problem was fogging up the asset pricing picture.

3.3.1 Carhart Four-Factor Model

The next popular multifactor model was developed by Carhart (1997), who augmented the three-factor model with a momentum factor. Jegadeesh and Titmen (1993) discovered that momentum portfolios, defined as long high performing stock in the last year and short low performing stocks in the last year, generated anomalous returns of 1.3–1.5% per month (or 16–17% per year). Momentum strategies for investment have long been known in practice. Carhart specified a four-factor model to take into account momentum as follows:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i,m}(R_{mt} - R_{ft}) + \beta_{i,S}\mathrm{SMB}_t + \beta_{i,V}\mathrm{HML}_t + \beta_{i,\mathrm{MOM}}\mathrm{MOM}_t + e_{it}, \tag{3.4}$$

[1] See studies by Lewellen et al. (2010), Daniel and Titman (2012), and others.

where MOM_t is the momentum factor. For U.S. mutual funds in the sample period 1963–1993, he found that R^2 estimates were very high in excess of 90%. Also, Jensen's α_i estimates were fairly low (i.e., mispricing error due to missing factors was small).

3.3.2 Hou, Xue, and Zhang Four-Factor Model

Another widely accepted four-factor model was created by Hou et al. (2015). It can be written as:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i,m}(R_{mt} - R_{ft}) + \beta_{i,S}\text{ME}_t + \beta_{i,R}\text{ROE}_t + \beta_{i,C}\text{IA}_t + e_{it}, \tag{3.5}$$

where ME is equity market capitalization, ROE is return on equity, and IA is the investment to assets ratio of the firm. These new factors were grounded in the production CAPM (PCAPM) of Cochrane (1991) as well as the q-theory of investment (i.e., firms with relatively high capital investment and profit earn higher stock returns than other firms). Using anomalous stock portfolios in their tests, the so-called *q-factor model* performed well in terms of lower Jensen's αs than the Fama and French three-factor model and Carhart four-factor model.

3.3.3 Stambaugh and Yuan Four-Factor Mispricing Model

A somewhat similar study by Stambaugh and Yuan (2017) developed another four-factor model is as follows:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i,m}(R_{mt} - R_{ft}) + \beta_{i,S}\text{SMB}_t + \beta_{i,MT}\text{MGMT}_t + \beta_{i,P}\text{PERF}_t + e_{it}, \tag{3.6}$$

where MGMT is a management factor, and PERF is a performance factor. These new factors were constructed from stocks' ranking with respect to 11 anomalous return portfolios—namely, net stock issues, equity issuance, accruals, net operating assets, asset growth, investment to assets, financial distress, O-score of distress, momentum, gross profitability, and return on assets. After forming clusters of stocks using these anomalies, they created long/short zero-investment portfolios for MGMT and PERF dubbed *mispricing factors*. Departing from other studies, the size factor is based on

small and big stocks which are not used in these mispricing factors. Also, no value factor (HML) is employed, which had some correlation with their new factors. Like Hou, Xue, and Zhang, anomalous portfolios were utilized in their model tests. They found that Jensen's α was lower than for the Fama and French three-factor model, Carhart four-factor model, and Hou et al. q-factor model. Relevant to the factor zoo problem, the authors concluded that more parsimonious asset pricing models with fewer factors can be developed by constructing mispricing factors from various anomalous return portfolios.

3.3.4 Fama and French Five-Factor Model

Extending their earlier model, Fama and French (2015) put forward the following five-factor model:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i,m}(R_{mt} - R_{ft}) + \beta_{i,S}\text{SMB}_t + \beta_{i,V}\text{HML}_t + \beta_{i,R}\text{RMW}_t + \beta_{i,C}\text{CMA}_t + e_{it}, \quad (3.7)$$

where RMW_t is long high profit firms' stock returns minus (short) low profit firms' stock returns, and CMA_t is long conservative capital investment firms' stock returns minus (short) aggressive capital investment firms' stock returns. Stock portfolios for testing the model were constructed using different sorts on size, value, operating profit, and capital investment (but industry portfolios were not tested). Fama and French interpreted these four factors as proxies for unknown state variables akin to Merton's ICAPM discussed earlier. Their empirical tests showed that Jensen's α was reduced in many cases to zero. However, there were some portfolios that exhibited significant αs leaving the possibility of a missing factor.

3.3.5 Fama and French Six-Factor Model

To better explain average stock returns, Fama and French (2018) offered a six-factor model:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i,m}(R_{mt} - R_{ft}) + \beta_{i,S}\text{SMB}_t + \beta_{i,V}\text{HML}_t + \beta_{i,R}\text{RMW}_t + \beta_{i,C}\text{CMA}_t + \beta_{i,M}\text{MOM}_t + e_{it}, \quad (3.8)$$

where the momentum factor (MOM) is added to their five-factor model. It is interesting that, except for the market and size factors, the factors only used small stocks in their construction. Thus, most of these factors are associated with risks inherent in small (as opposed to big) stocks. At the end of

their paper, Fama and French advised that researchers try to use theory to guide factor selection as well as out-of-sample empirical tests to check the robustness of multifactor models.

3.3.6 Machine Learning Models

Despite their admonition about the theory of factor selection, which echoes Black's (1993) earlier criticisms, Fama and French (2020) published a study that explores a wide variety of models and their specifications in search of the best fit to stock return data. In one model, they take advantage of machine learning to construct factors. To let the machine define factors via artificial intelligence (AI), they run the following cross-sectional regression equation:

$$R_{it} = R_{zt} + R_{\mathrm{MC}t}\mathrm{MC}_{it-1} + R_{\mathrm{BM}t}\mathrm{BM}_{it-1} + R_{\mathrm{OP}t}\mathrm{OP}_{it-1} + R_{\mathrm{INV}t}\mathrm{INV}_{it-1} + e_{it}, \tag{3.9}$$

where the independent variables are MC_{it-1}, BM_{it-1}, OP_{it-1}, and INV_{it-1} defined as the observed values of size, the book-to-market ratio, operating profitability, and the rate of growth of assets. The intercept term is denoted R_{zt}. The market prices of risk, or λ_k coefficients, are denoted $R_{\mathrm{MC}t}$, $R_{\mathrm{BM}t}$, $R_{\mathrm{OP}t}$, and $R_{\mathrm{INV}t}$, respectively. Fama and French use these λ_ks as the long/short, zero-investment factors in different specifications of their previous asset pricing models.[2] For simplicity, they refer to them as *cross section factors*.

In one model using cross section factors, they specify a mathematical four-factor model as follows:

$$R_{it} - R_{zt} = \mathrm{MC}_{it-1}R_{\mathrm{MC}t} + \mathrm{BM}_{it-1}R_{\mathrm{BM}t} + \mathrm{OP}_{it-1}R_{\mathrm{OP}t} + \mathrm{INV}_{it-1}R_{\mathrm{INV}t} + e_{it}, \tag{3.10}$$

where no intercept mispricing term exists, and beta risk loadings are the accounting variables lagged one period. However, the beta risk loadings are not coefficients estimated in a time-series regression model. Instead, in relation (3.10), the regression coefficients are replaced by time-varying accounting variables. The cross section factors are the multifactors. Since nothing is estimated via statistical regression methods, it is a mathematical model. In the absence of an intercept term, they computed the average of the

[2] This interpretation of λ_ks in cross-sectional asset pricing regression analyses as mimicking portfolios of factor returns is well known. See Ferson et al. (1999), Back et al. (2013, 2015) and Ferson (2019, p. 223).

error terms e_{it} over t time periods to proxy α_i in the model.[3] In empirical tests, this mathematical model produced lower Jensen αs than other model specifications, including their earlier five- and six-factor models.

Lastly, another example of machine learning models is by Lettau and Pelger (2020). They employed a statistical method called Principal Component Analysis (PCA) to derive latent asset pricing factors. PCA takes a large number of observations (i.e., stock returns for portfolios or individual stocks here) and then seeks to find how many dimensions are needed to summarize their returns. In a space reduction technique, the underlying dimensions are considered to be *latent* or *hidden*. The dimensions or components are orthogonal to one another (or uncorrelated). This setup is amenable to asset pricing models that require relatively uncorrelated factors. Their analyses of U.S. stock returns from 1963 to 2017 revealed that five components described returns. Using these components as factors, they tested a five-factor model. The α estimates for their PCA model outperformed the Fama and French three- and five-factor models. The authors conducted further tests to show that the latent factors in their model tended to be correlated with market, value, and momentum factors. Hence, they inferred that machines do a good job of capturing risk factors in stock returns.

3.4 Roadmap of Multifactor Models

Figure 3.2 gives a road map of multifactor models starting with *discretional multifactor models* based on the arbitrage pricing theory (APT) of Ross. We connect the APT to the three-factor model and other multifactor models due to the use of long/short zero-investment factors in these asset pricing models.[4] These models have become tremendously popular in academic research and investment practice in the real world. Their main advantage is that many long/short factors can be developed that fit return data. However, due to the resultant factor zoo, their principal disadvantage is that it is not clear what factors to choose in the asset pricing models. The theory could guide factor selection, but these models (with some exceptions) have loose theoretical foundations in most cases. Importantly, their primary justification is

[3] In regression analysis, the average of the residual error terms is zero, but in the mathematical model their average is not set to zero.

[4] We should mention that many authors of multifactor models point to Merton's ICAPM for theoretical support also. Like Ross, who proposed multiple unidentified zero-investment risk factors, Merton did not identify the multiple state variables in the ICAPM, only that they associated with the risks of unfavorable changes in future investment opportunities.

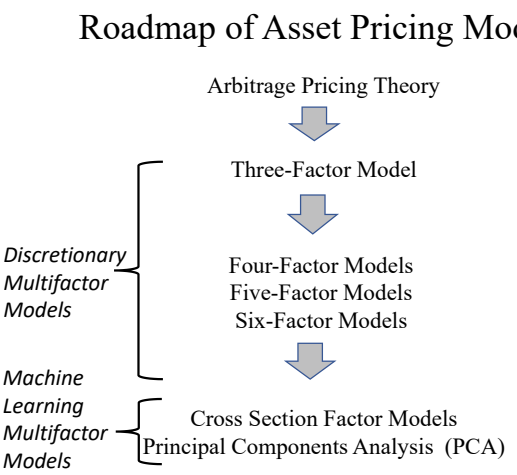


Fig. 3.2 Popular multifactor models have long/short factors related to firm and stock characteristics

empirical success in explaining stock returns. We refer to them as *discretionary models* due to the use of researcher judgment in constructing various multifactors.

Finally, *machine learning multifactor models* are starting to emerge that use computers and statistical methods to build multifactor models. In this approach, artificial intelligence (AI) removes researcher judgment (discretion) in model development. Instead, machine learning lets the data tell you what the factors are. This abstract approach avoids human error, which is a strength; however, it suffers from black box ambiguity. Inputs and outputs of analyses can be observed in these models, but their internal workings are difficult to discern. How exactly were the factors derived? How would they change using different computing algorithms and statistical methods? What do the factors mean? With no theoretical foundation, they are entirely mechanical in nature.

3.4.1 Investment Portfolio Implications

In the early 2000s, the consulting firm Towers Watson (later Willis Towers Watson) began using the term "smart beta" to describe investment portfolio strategies that do not simply weight stocks by the market values. Unlike value-weighted index funds, smart beta funds seek to form portfolios based on different risk factors in the market, such as liquidity risk, momentum risk,

size risk, value risk, volatility risk, and quality risk (related to profits, dividends, leverage, efficiency, etc.). Some investors refer to smart beta as strategic beta, alternative beta, or factor investing.

Over $1 trillion is invested in smart beta funds nowadays. According to Wikipedia, major institutional investors, such as Blackrock iShares, Vanguard, Invesco and First Trust, Legg Mason, Henderson Rowe, and Wisdom Tree, create and sell smart beta funds. Also, over 1,000 smart beta exchange-traded funds (ETFs) are available in the market. They comprise approximately 20% of the overall ETF marketplace. Some funds are multifactor small beta ETFs. Stocks selection and weighting methods vary across ETFs.

Why are smart beta portfolios so popular? The answer is that, in an effort to outperform general stock market indexes, they take advantage of factors developed in academic and practitioner research to build portfolios. Normally, smart beta portfolios are long only (no short positions) that emphasize portfolios contained in long/short multifactors. For example, small-cap funds invest in stocks with relatively small market capitalization. The Fama and French studies showed that small stocks tend to outperform large stocks over time, at least in rate of return terms. Anomalous returns (related to the returns of stocks with different firm characteristics) can be earned that are not explained by the general market index. These funds seek to outperform market indexes to earn "alpha" equal to the return difference between smart beta returns and general market index returns. Of course, the factor zoo provides a ready supply of anomalies to use in producing smart beta portfolios.

Black (1993) has criticized multifactor models due to potential *data mining*. This search method is marked by data dredging of historical stock returns to find past patterns in stocks that yield good return performance. By using computers to torture the data, researchers can find return anomalies that look like something significant but in actuality are not. Indeed, the results of data mining can be completely accidental or random. With reference to the size and value factors in the Fama and French three-factor model, Black cautions: "Lack of theory is a tipoff: watch out for data mining!"

Smart beta portfolios emphasize manager discretion and data mining in the construction of smart beta portfolios. Not surprisingly, their historical performance has not been good—they underperform value-weighted general market stock indexes. According to a recent study by Huang et al. (2021), a major reason for this underperformance is use of extensive backtests of investment portfolios over historical periods that have no predictive power in the future in live investment portfolio tests.

3.5 Summary

This chapter has reviewed the development of multifactor models in the asset pricing literature. Due to weaker than expected empirical results for the market model version of the theoretical CAPM by Sharpe and others, Fama and French proposed the three-factor model that augmented the market factor with novel long/short size and value factors. This new approach to constructing factors is closely related to the arbitrage pricing theory of Ross, who posited multiple long/short, zero-investment factors are used by investors to arbitrage away various market risks.

The empirical success of this model with stock returns led to an avalanche of new factors (or multifactors) by many researchers. These discretionary models are based on the researcher judgment. We discussed some of the more prominent multifactor models, including Carhart's four-factor model, the Hou, Xue, and Zhang four-factor *q*-model, Stambaugh and Yuan four-factor mispricing model, and Fama and French five- and six-factor models. Due to so many different models and factors being proposed by researchers, Cochrane observed a new problem—the factor zoo. What factors should be chosen to use in asset pricing models? With relatively weak theoretical foundations to help guide factor selection, multifactor models continue to be proposed by researchers. Which model should be used and why?

One answer to the problems of multifactor models is to let the data choose the factors using machine learning and artificial intelligence (AI). For example, Fama and French used cross-sectional regression analyses to create cross-section factor models. Also, Lettau and Pelger implemented the statistical method of Principal Components Analysis (PCA) to develop models. While these machine learning models take researchers' judgment and discretion out of the model development process and seem to do a good job of picking empirically significant factors, they are block box in nature. Even so, despite "flying blind" on autopilot based on computers and sophisticated statistical programs, these models give some promise that important asset pricing factors exist in the market.

In the next chapter, we go "back to the future" in a sense by reviewing a new version of the zero-beta CAPM named the *ZCAPM* by Kolari, Liu, and Huang (2021). In contrast to previous theoretical CAPM models, which did not fit stock return data for the most part and so motivated the development of multifactor models, the ZCAPM outperforms multifactor models. In effect, theoretical CAPM models are not dead anymore but alive and

well. Due to its exceptional empirical success, we utilize the ZCAPM in the forthcoming chapters to develop both minimum variance portfolio G and relatively efficient portfolios that outperform general stock market indexes.

References

Back, K., N. Kapadia, and B. Ostdiek. 2013. Slopes as factors: Characteristic pure plays. Working Paper, Rice University.

Back, K., N. Kapadia, and B. Ostdiek. 2015. Testing factor models on characteristic and covariance pure plays. Working Paper, Rice University.

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Black, F. 1993. Beta and return. *Journal of Portfolio Management* 20: 8–18.

Carhart, M.M. 1997. On persistence in mutual fund performance. *Journal of Finance* 52: 57–82.

Cochrane, J.H. 1991. Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance* 46: 209–237.

Cochrane, J.H. 2011. Presidential address: Discount rates. *Journal of Finance* 56: 1047–1108.

Daniel, K., and S. Titman. 2012. Testing factor-model explanations of market anomalies. *Critical Finance Review* 1 (103–139): 29–40.

Fama, E.F., and K.R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427–465.

Fama, E.F., and K.R. French. 1993. The cross-section of expected returns. *Journal of Financial Economics* 33: 3–56.

Fama, E.F., and K.R. French. 1995. Size and book-to-market factors in earnings and returns. *Journal of Finance* 50: 131–156.

Fama, E.F., and K.R. French. 1996. The CAPM is wanted, dead or alive. *Journal of Finance* 51: 1947–1958.

Fama, E.F., and K.R. French. 1998. Value versus growth: The international evidence. *Journal of Finance* 53: 1975–1999.

Fama, E.F., and K.R. French. 2004. The Capital Asset Pricing Model: Theory and evidence. *Journal of Economic Perspectives* 18: 25–46.

Fama, E.F., and K.R. French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116: 1–22.

Fama, E.F., and K.R. French. 2018. Choosing factors. *Journal of Financial Economics* 128: 234–252.

Fama, E.F., and K.R. French. 2020. Comparing cross-section and time-series factor models. *Review of Financial Studies* 33: 1892–1926.

Ferson, W.E. 2019. *Empirical asset pricing: Models and methods*. Cambridge, MA: The MIT Press.

Ferson, W.E., S. Sarkissian, and T. Simin. 1999. The alpha force asset pricing model: A parable. *Journal of Financial Markets* 2: 49–68.

Hou, K., C. Xue, and L. Zhang. 2015. Digesting anomalies: An investment approach. *Review of Financial Studies* 28: 650–705.

Huang, S., Y. Song, and H. Xiang. 2021. The smart beta mirage. Working Paper, The University of Hong Kong and University of Washington.

Jegadeesh, N., and S. Titman. 1993. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance* 48: 65–91.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Kothari, S.P., J. Shanken, and R.G. Sloan. 1995. Another look at the cross-section of expected stock returns. *Journal of Finance* 50: 185–224.

Lettau, M., and M. Pelger. 2020. Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies* 33: 2274–2325.

Lewellen, J., S. Nagel, and J.A. Shanken. 2010. A skeptical appraisal of asset pricing tests. *Journal of Financial Economics* 96: 175–194.

Markowitz, H. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: Wiley.

Merton, R.C. 1973. An intertemporal capital asset pricing model. *Econometrica* 41: 867–887.

Ross, S.A. 1976. The arbitrage theory of capital asset pricing. *Journal of Economic Theory* 13: 341–360.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.

Stambaugh, R.F., and Y. Yuan. 2017. Mispricing factors. *Review of Financial Studies* 30: 1270–1315.

Part III

The ZCAPM



4

A New Asset Pricing Model: The ZCAPM

4.1 Theoretical ZCAPM

The theoretical ZCAPM is grounded in Markowitz's mean-variance investment parabola. Here we review the investment parabola discussed in Chapter 1, the derivation of the ZCAPM equilibrium relation using the parabola, and the relation of the ZCAPM to the architecture of the parabola.

4.1.1 Markowitz Investment Parabola

In Fig. 4.1, we draw the Markowitz mean-variance investment parabola with mean returns on the Y-axis denoted as $E(R_P)$ and return variance on the X-axis denoted as σ_P^2. How can we compute these variables in the real world? As an example, return variance for one day could be computed using a series of 10-minute returns computed during the day when the market is open and trading. The mean 10-minute return is simply the average of this series. These mean return and variance of return estimates would determine the location of each asset within the parabola for the day.

As discussed in Chapter 1, the global minimum variance portfolio G pins the level of the parabola at some expected return denoted $E(R_G)$. The efficient frontier consists of the portfolios on the parabola to the right of G. These portfolios have the highest expected return per unit return variance (or total risk). Those on the bottom boundary of the parabola to the right of G are inefficient.

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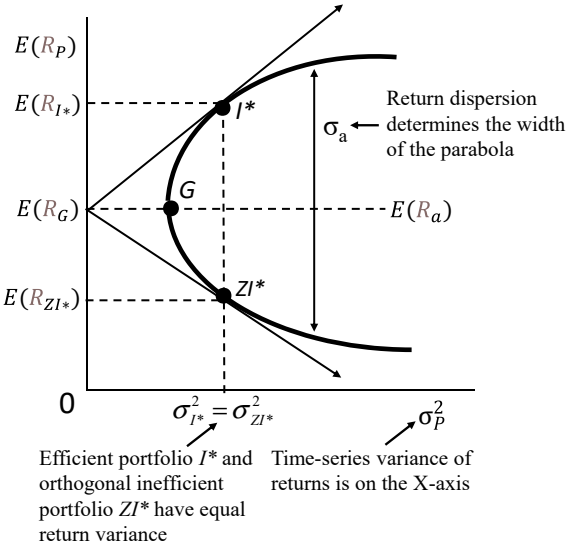


Fig. 4.1 Locating orthogonal portfolios I^* and ZI^* on the mean-variance parabola (*Source* Kolari et al. 2021, p. 59)

Let's put this picture of the parabola in the context of Black's (1972) zero-beta CAPM. Without a riskless asset, we can write Eq. (2.15) in Chapter 2 for his model as:

$$E(R_i) = E(R_{ZI}) + \beta_{i,I}[E(R_I) - E(R_{ZI})], \tag{4.1}$$

where $\beta_{i,I}$ is beta risk associated with the excess return of efficient portfolio I defined as $E(R_I) - E(R_{ZI})$. Here we use zero-beta portfolio return $E(R_{ZI})$ in place of the riskless rate R_f as the borrowing rate in the CAPM. Notice that Black's model is more general than the CAPM that relies solely on the market portfolio M, as opposed to any efficient portfolio I. If we could locate M, then the zero-beta portfolio would be ZM. If we locate portfolio I^*, the zero-beta portfolio would be ZI^*.

As shown in Chapter 1, KLH proved that the width of the parabola is largely determined by the cross-sectional standard deviation of returns for all assets. This market return dispersion is denoted σ_a in Fig. 4.1. Continuing our earlier example, on any given day, the return dispersion can be computed using the cross-sectional (not time-series) standard deviation of returns of all assets in the market. Looking at the parabola as if standing on the Y-axis, this result is intuitively obvious. The width of the parabola contains the distribution of expected returns for all assets in the market (i.e., the investment opportunity set). Naturally, the average market return must lie approximately

in the middle of this distribution on the axis of symmetry of the parabola that splits into two symmetric halves. KLH inferred that $E(R_{at}) \approx E(R_{Gt})$. This result is surprising! Most researchers believe that the average market return (e.g., the S &P 500 index return) is relatively efficient and lies in the vicinity of market portfolio M on the efficient frontier. When estimating the market model and multifactor models, a general market index is normally used to proxy the market portfolio return.

Roll (1977) is famous for criticizing tests of the CAPM using various general market indexes as proxies for the market portfolio. He argued that, if these proxy market indexes are inefficient portfolios, tests of the CAPM would be useless. Evidence against the CAPM using such proxies is unacceptable. From Fig. 4.1, we can see that, consistent with the Roll critique, such proxies are far from efficient. This problem explains the flatter than expected Security Market Line (SML) estimated by many previous empirical tests of the CAPM using the market model that we discussed in Chapter 2.

4.1.2 Derivation of the ZCAPM Equilibrium Relation

Referring to Fig. 4.1, two portfolios denoted I^* and ZI^* on the parabola are shown that have the same return variance, or $\sigma^2_{I^*} = \sigma^2_{ZI^*}$. Portfolio I^* is on the efficient frontier. Its zero-beta counterpart is ZI^*. These portfolios are orthogonal (uncorrelated) to one another. KLH proved this orthogonal relationship with tangent lines extending from $E(R_G) \approx E(R_a)$ to portfolios I^* and ZI^*. The Roll approach covered in Chapter 2 would yield the same zero-beta portfolio (i.e., extend a ray from I^* down through G to the Y-axis and then move horizontally to reach ZI^*).

The unique pair of portfolios I^* and ZI^* represent a special case of the zero-beta CAPM (that allows an infinite number of such portfolio pairs). Looking at Fig. 4.1, it is obvious that the expected returns of these two portfolios can be approximately defined as follows:

$$E(R_{I^*}) \approx E(R_a) + \sigma_a \tag{4.2}$$

$$E(R_{ZI^*}) \approx E(R_a) - \sigma_a. \tag{4.3}$$

In words, the geometry of the parabola shows that an investor can move along the axis of symmetry at $E(R_a)$ to the right of portfolio G until the point at which the time-series variance of portfolios I^* and ZI^* are equal to one another. Subsequently, by moving up or down by the cross-sectional standard

deviation of returns (return dispersion) σ_a, the investor can reach portfolios I^* and ZI^*, respectively.[1]

We are now ready to derive the theoretical ZCAPM. Expected returns $E(R_{I^*})$ and $E(R_{ZI^*})$ in Eqs. (4.2) and (4.3) can be substituted into zero-beta CAPM Eq. (4.1) as follows:

$$\begin{aligned} E(R_i) &= E(R_{ZI^*}) + \beta_{i,I^*}[E(R_{I^*}) - E(R_{ZI^*})] \\ &= E(R_a) - \sigma_a + \beta_{i,I^*}\{[E(R_a) + \sigma_a] - [E(R_a) - \sigma_a]\} \\ &= E(R_a) + (2\beta_{i,I^*} - 1)\sigma_a \\ E(R_i) &= E(R_a) + Z^*_{i,a}\sigma_a, \end{aligned} \tag{4.4}$$

where $Z^*_{i,a} = 2\beta_{i,I^*} - 1$. Here $Z^*_{i,a}$ represents zeta risk that captures the sensitivity of the return of asset i to market return dispersion.

Next, KLH assume that a riskless asset f exists with the rate of return R_f. Now we have three assets to consider: I^*, ZI^*, and f. Using these assets, we can define the expected return of asset i as:

$$E(R_i) = w_{I^*}E(R_{I^*}) + w_{ZI^*}E(R_{ZI^*}) + w_f R_f. \tag{4.5}$$

Where the sum of the weights $w_{I^*} + w_{ZI^*} + w_f = 1$. Using expected returns $E(R_{I^*})$ and $E(R_{ZI^*})$ defined above, we can re-write this expression as:

$$E(R_i) = (w_{I^*} + w_{ZI^*})E(R_a) + (w_{I^*} - w_{ZI^*})\sigma_a + w_f R_f. \tag{4.6}$$

Assuming that investors can hold long and short positions in assets, upon rearranging terms, we have:

$$E(R_i) - R_f = (w_{I^*} + w_{ZI^*})[E(R_a) - R_f] + (w_{I^*} - w_{ZI^*})\sigma_a. \tag{4.7}$$

Finally, based on Eq. (4.4), we write the theoretical ZCAPM as:

$$E(R_i) - R_f = \beta_{i,a}[E(R_a) - R_f] + Z^*_{i,a}\sigma_a, \tag{4.8}$$

where $\beta_{i,a} = w_{I^*} + w_{ZI^*}$ is the *beta risk* coefficient measuring the sensitivity of the ith asset's excess returns to average market excess returns of all assets, and $Z^*_{i,a} = w_{I^*} - w_{ZI^*}$ is the *zeta risk* coefficient measuring the sensitivity of an asset's excess returns to the market return dispersion of all assets.[2]

[1] KLH write Eqs. (4.2) and (4.3) more generally using $f(\theta)\sigma_a^2$ instead of σ_a^2, where they define $f(\theta) > 0$ to be a complex function of other terms. We simplify the analyses here for ease of exposition.

[2] The ZCAPM implies the following general equilibrium conditions:

The ZCAPM provides an empirically testable model. Average market return $E(R_a)$ and market return dispersion σ_a are common statistical measures that can be readily computed using market return data in the real world. There is no need to identify an unobservable market portfolio M per the CAPM. Also, there is no need to find orthogonal efficient and zero-beta portfolios in the zero-beta CAPM. Now estimates of beta risk and zeta risk can be reliably estimated.

The ZCAPM does present one new challenge. It would be more advantageous to have an estimate of minimum variance portfolio G. It is the only portfolio inside or on the parabola that has no zeta risk related to market return dispersion and, therefore, is a good benchmark for beta risk. If portfolio G could be constructed, it could be used to compute potentially better estimates of average market return and market return dispersion in the empirical ZCAPM. In forthcoming Chapter 7, we propose and implement methods for building portfolio G. In later chapters, we build efficient portfolios using the ZCAPM, which are improved by our new portfolio G.

As a final note, beta risk in the ZCAPM is related to average market returns, rather than market portfolio M returns as in the CAPM. Also notice that zeta risk coefficient $Z^*_{i,a}$ can be positive or negative sign in order to take into account portfolios I^* and ZI^*, respectively, on the parabola. Hence, zeta risk captures two-sided (or bi-directional) market volatility related to market return dispersion.

4.2 Graphical Depictions of the ZCAPM

The theoretical ZCAPM can be better understood with the aid of some pictures. In this section we provide graphical depictions of the ZCAPM. Also, we present some real world results that demonstrate how the ZCAPM can be applied to investment portfolio management.

(1) assuming all funds are invested in either I^* or ZI^*, then $\beta_{I^*,a} = \beta_{ZI^*,a} = 1$ and $Z^*_{I^*,a} = 1$ or $Z^*_{ZI^*,a} = -1$, respectively;
(2) assuming no riskless asset, Eq. (4.8) reduces to Eq. (4.4) (i.e., $\beta_{i,a} \equiv w_{I^*} + w_{ZI^*} = 1$); and
(3) assuming the restriction $w_f > 0$ (i.e., no borrowing at the riskless rate is allowed), then $\beta_{i,a} < 1$.

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4.2.1 Beta Risk and Zeta Risk in the ZCAPM

We can draw a graph of the ZCAPM as shown in Fig. 4.2. Beta risk gives the slope of the line between the excess returns of asset i denoted $R_i - R_f$ on the Y-axis and average excess market returns denoted $R_a - R_f$ on the X-axis. Zeta risk takes into account return dispersion around this line in either an upward direction $(+Z^*_{i,a}\sigma_a)$ or a downward direction $(-Z^*_{i,a}\sigma_a)$. The two-sided, opposite effects of zeta risk on asset returns are obvious in this graph.

Another way to understand the effects of market return dispersion on asset returns is shown in Fig. 4.3. There are two time periods. We consider the simple case of two assets B and C. At $t = 1$, market return dispersion is $\sigma_{a1} = 1\%$. At $t = 2$ market return dispersion increases to $\sigma_{a1} = 2\%$. From the diagram, we see that, as cross-sectional market volatility increases from $t = 1$ to $t = 2$, the expected return on asset B decreases, but the expected return on asset C increases. Market return dispersion has opposite effects on asset returns depending on whether their returns are in the lower or upper half of the return distribution. It is easy to see that, if market return dispersion decreased (instead of increased), then asset B's expected return would increase, but asset C's expected return would decrease. Thus, market return dispersion can have powerful effects on the returns of assets within the investment parabola.

In combination with beta risk, the dual systematic risk effects of return dispersion in the market give the mean-variance parabola its upper and lower boundaries. We can think of the upper (lower) boundary as having some

$$\text{ZCAPM}: R_{it} - R_{ft} = \beta_{i,a}(R_{at} - R_{ft}) + Z^*_{i,a}\sigma_{at} + u_{it}$$

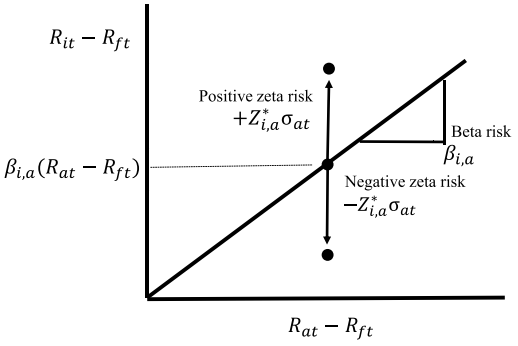


Fig. 4.2 The ZCAPM is depicted with beta risk related to average excess market returns and zeta risk associated with positive and negative sensitivity to return dispersion (*Source* Kolari et al. 2021, p. 72)

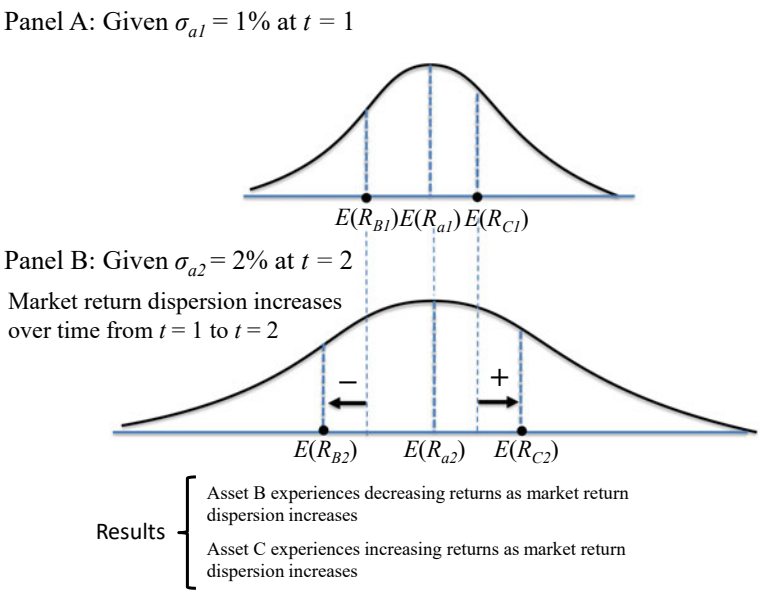


Fig. 4.3 Two-sided, opposite effects of return dispersion on the expected returns of two assets (*Source* Kolari et al. 2021, p. 94)

amount of positive (negative) zeta risk—for example, $Z^*_{i,a} = 0.8$ (-0.8). Along the upper (lower) boundary, assets will have increasing levels of beta risk. Hence, beta and zeta risk are important in defining the architecture of the mean-variance parabola.

4.2.2 Architecture of the Investment Parabola and the ZCAPM

How can beta risk and zeta risk be used to invest in portfolios? The answer to this question lies in the fact that the boundary of the parabola and its interior can be described by these risks. To illustrate the architecture of the parabola, KLH produced Fig. 4.4 using real world stock returns. The parabola shown there is constructed based on 25 U.S. stock portfolios formed from all common stocks in the period from 1965 to 2018.

The authors estimated the ZCAPM using daily returns in the initial period from January 1964 to December 1964 for all common stocks. To do this, the empirical form of the ZCAPM was employed, which we cover in the next chapter. Stocks were sorted into 25 portfolios based on quintiles of their estimated beta risks and zeta risks. Then portfolio returns in the out-of-sample month of January 1965 were computed. Next, the one-year estimation period

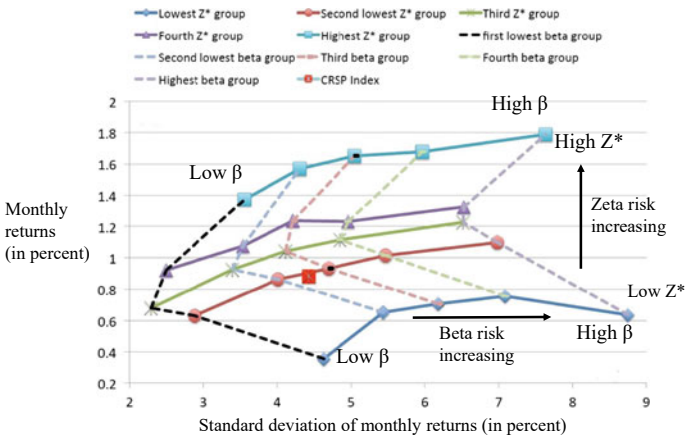


Fig. 4.4 This graph depicts the architecture of the investment parabola, including not only its boundary but internal structure. The investment opportunity set plots out-of-sample returns for portfolios formed from individual U.S. stocks using average beta risk and zeta risk over time. A total of 25 beta-zeta sorted portfolios are shown. Only long positions are allowed in the portfolios. The empirical ZCAPM is estimated using daily returns for January 1964 to December 1964 to obtain the beta and zeta risk parameters. One-month-ahead returns in January 1965 are computed for each portfolio. The analyses are rolled forward one month at a time until the last one-month-ahead return in December 2018. All returns are averaged for portfolios over these 636 out-of-sample months (*Source* Kolari et al. 2021, p. 272)

for estimating the ZCAPM was rolled forward one month and the analyses were repeated to get out-of-sample February 1965 portfolio returns. This process was repeated until the last out-of-sample month December 2018. The average returns for these 636 out-of-sample returns for each portfolio were computed as well as the average beta risk and zeta risk for each portfolio. Also, the standard deviations of returns (total risk) for these 636 returns were computed for each portfolio.

Figure 4.4 plots the average returns and standard deviations of returns for the 25 beta-zeta sorted portfolios. Interestingly, the portfolios form a beta risk/zeta risk grid that depicts the architecture of the investment parabola. Portfolios on the upper efficient boundary of the parabola have relatively high zeta risk. On this boundary, higher returns are earned as beta risk increases. The bottom inefficient boundary contains portfolios with relatively low zeta risk. Again, given zeta risk, higher returns are earned as beta risk increases. The performance of a general market index (viz., the CRSP index) is shown by the square in the diagram. Notice that, consistent with ZCAPM theory, the CRSP index lies approximately in the middle of the parabola in the vicinity of the axis of symmetry of the empirically constructed parabola. It

clearly is inefficient compared to higher zeta risk portfolios invoking the Roll criticism of its use in empirical tests of the CAPM.

An intersecting latticework of beta-zeta risk portfolios forms the investment parabola. Thus, assets are not randomly located within the parabola. There is a definite structure related to systematic beta risks and zeta risks that determine the architecture of the parabola. These new insights into the parabola enable new methods for controlling the risk of portfolios and thereby their relative returns and, ultimately, their locations within the parabola.

In forthcoming chapters, we apply these investment portfolio fundamentals to the construction of relatively efficient portfolios. As we will see, these novel portfolio techniques can be used to construct any portfolio, including empirically efficient portfolios desired by investment managers and their clients.

4.3 Summary

Kolari, Liu, and Huang (KLH) recently proposed a new theoretical CAPM dubbed the *ZCAPM*. Their new model has two factors: average market returns and the cross-sectional standard deviation of returns for all assets in the market (or *return dispersion*). These factors are associated with beta risk and zeta risk, respectively. Using two unique orthogonal portfolios on the mean-variance investment parabola, the ZCAPM is mathematically derived as a special case of Black's zero-beta CAPM.

In their derivation, KLH utilized two new insights about the investment parabola of Markowitz: (1) return dispersion plays an important role in determining the width or span of the parabola at any time t; and (2) by implication, the average market return lies approximately on the axis of symmetry in the middle of the parabola. Based on these insights, they located two orthogonal portfolios on the parabola with equal total risk that are denoted as I^* and ZI^*.

KLH substituted the expected returns of efficient portfolio I^* and zero-beta counterpart portfolio ZI^* into Black's specification of the zero-beta CAPM. The resultant theoretical ZCAPM is modified to include a riskless rate. Conveniently, the ZCAPM contains readily estimable factors in the real world—namely, the average market return and market return dispersion. To illustrate the ZCAPM in the context of the investment parabola, we reviewed previous empirical evidence by KLH that showed how beta risk and zeta risk can be used to shape the architecture of parabola. Using these two market

risks, all assets (including portfolios) on both the boundary of parabola and the interior of the parabola can be located.

In the next chapter, we specify the empirical ZCAPM, which contains a hidden signal variable to determine the direction of zeta risk. Later chapters employ the empirical ZCAPM to build high-performing portfolios based on beta and zeta risks that outperform the general stock market indexes in terms of their return and risk characteristics.

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Fama, E.F., and K.R. French. 2018. Choosing factors. *Journal of Financial Economics* 128: 234–252.

Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2022a. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*. Available online at https://www.mdpi.com/1911-8074/15/3/137. Reprinted in Kolari, J.W., and S. Pynnonen, eds. 2022. *Frontiers of asset pricing*. Basel, Switzerland: MDPI.

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022b. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. Testing for missing asset pricing factors. Paper presented at the Western Economic Association International, San Diego, CA.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Liu, W. 2013. A new asset pricing model based on the zero-beta CAPM: Theory and evidence. Doctoral dissertation, Texas A &M University.

Liu, W., J.W. Kolari, and J.Z. Huang. 2012. A new asset pricing model based on the zero-beta CAPM market model (CAPM). Presentation at the annual meetings of the Financial Management Association, October, Best Paper Award in Investments, Atlanta, GA.

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. Return dispersion and the cross-section of stock returns. Presentation at the annual meetings of the Southern Finance Association, October, Palm Springs, CA.

Markowitz, H.M. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: Wiley.

Roll, R. 1977. A critique of the asset pricing theory's tests, part I: On past and potential future testability of the theory. *Journal of Financial Economics* 4: 129–176.



5

The Empirical ZCAPM

5.1 Specification of the Empirical ZCAPM

KLH specified an innovative empirical form for the ZCAPM that takes into account positive and negative zeta risk. They proposed the following two-factor regression model:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{at} - R_{ft}) + Z_i D_{it}\sigma_{at} + u_{it}, \quad t = 1, \ldots, T, \tag{5.1}$$

where β_i measures beta risk sensitivity of asset i excess asset returns to excess average market returns, or $R_{at} - R_{ft}$, Z_i measures zeta risk sensitivity to cross-sectional return dispersion σ_{at} (or RD), D_{it} is a signal variable with values $+1$ and -1 to capture positive and negative return dispersion effects on stock returns at time t, respectively, and error term $u_{it} \sim (0, \sigma_i^2)$. Note that the dummy signal variableD_{it} determines the direction or sign of zeta risk. Unfortunately, this signal variable cannot be directly observed. We do not know at any time t if it will be positive or negative. Therefore, it is a latent or hidden variable. Commonly used ordinary least squares (OLS) regression methods cannot be adapted for this hidden variable.

This recent book gives the complete theoretical derivation as well as extensive empirical test results for the ZCAPM. See also numerous works by the authors, including Liu (2013), Liu et al. (2012, 2020), Kolari et al. (2022a, b, 2023), Kolari and Pynnonen (2023), and Kolari et al. (2024).

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Statisticians have developed methods to estimate hidden variables. A widely accepted approach is the expectation-maximization (EM) algorithm developed by Dempster et al. (1977). Studies by Jones and McLachlan (1990), McLachlan and Peel (2000), McLachlan and Krishnan (2008), and others further developed these methods. See Wikipedia for an excellent discussion of the EM algorithm and further citations to the statistics literature.

The EM algorithm begins by specifying the hidden variable D_{it} as an independent random variable with a two-point distribution:

$$D_{it} = \begin{cases} +1 \text{ with probability } p_i \\ -1 \text{ with probability } 1 - p_i, \end{cases} \tag{5.2}$$

where p_i (or $1 - p_i$) is the probability of a positive (or negative) return dispersion effect, and D_{it} is independent of u_{it}. No previous studies utilize the EM algorithm in an asset pricing regression model.

In empirical ZCAPM relation (5.1), $Z_{i,a}D_{i,t}$ is an interaction term. Depending on the sign of signal variable $D_{i,t}$, the interaction term can take on two possible values at time t: $+Z_{i,a}$ and $-Z_{i,a}$. Given that the mean of binary signal variable $D_{i,t}$ equals $2p_i - 1$, we can specify zeta risk as $Z^*_{i,a} = Z_{i,a}(2p_i - 1)$. Using this definition of zeta risk, the empirical ZCAPM becomes:

$$R_{it} - R_{ft} = \beta_{i,a}(R_{at} - R_{ft}) + Z^*_{i,a}\sigma_{at} + u_{it}, \quad t = 1, \ldots, T. \tag{5.3}$$

EM regression provides estimates of beta risk coefficient $\beta_{i,a}$ as well as zeta risk coefficient $Z^*_{i,a}$ that coincides with theoretical ZCAPM relation (4.8) in previous Chapter 4. Here we see that the probability p_i of signal variable D_{it} in period $t = 1, \ldots, T$ determines the positive or negative sign of $Z^*_{i,a}$. When $p_i > 1/2$, the sign of $Z^*_{i,a}$ will be positive; contrarily, when $p_i < 1/2$, $Z^*_{i,a}$ will have a negative sign. Hence, zeta risk coefficient $Z^*_{i,a}$ gives the positive or negative sensitivity of asset returns with respect to a one unit change in return dispersion σ_{at} (or *RD*).

Interestingly, the empirical ZCAPM does not have an intercept or alpha term. Normally, OLS models use an intercept term to increase the goodness-of-fit of the regression model to the data. KLH tried incorporating an intercept but the variance of residual error terms did not decrease, which means that goodness-of-fit is not improved by the intercept.

KLH provide free software programs on the internet to estimate the empirical ZCAPM. Matlab, R, and Python programs for estimation purposes are provided at the following website: GitHub (https://github.com/zcapm).

We have found that it takes less time to estimate the ZCAPM using the R programs. Also, they make available programs to estimate standard Fama-MacBeth cross-sectional regression tests.

5.2 Cross-Sectional Test Methodology

Standard cross-sectional tests of the empirical ZCAPM were conducted by KLH. They utilized daily stock returns for almost all U.S. stocks in the Center for Research and Security Prices (CRSP) database at the University of Chicago from 1964 to 2018. The average market return R_{at} is proxied using the value-weighted CRSP index return. To proxy value-weighted market return dispersion on each day t, they computed:

$$\sigma_{at} = \sqrt{\frac{n}{n-1}\sum_{i=1}^{n} w_{it-1}(R_{it} - R_{at})^2}, \tag{5.4}$$

where n is the total number of common stocks (numbering in the 1000s each year), w_{it-1} is the previous day's market value weight for the ith stock (i.e., market capitalization of the stock divided by the market capitalization of all n stocks), and R_{it} is the return of the ith stock on day t. They found that the average daily market return in the sample period was 0.04 average excess return per month of the general stock market return over the Treasury bill rate was 0.50 daily return dispersion was 0.99%. It is obvious from these average daily values that return dispersion is much larger than the daily market return. Additionally, the authors reported that the correlation between the two market variables R_{at} and σ_{at} was only 0.07 in the sample period. We can infer that these asset pricing factors are orthogonal to one another and, therefore, are different market variables containing different market information for the most part.

KLH compared the empirical ZCAPM to popular asset pricing models by performing standard cross-sectional Fama and MacBeth (1973) regression tests. The tests employed a variety of different test asset portfolios, different subperiods of analyses, individual stocks, and out-of-sample returns in the next month.

In the first step, using one year of daily returns, the empirical ZCAPM in Eq. (5.3) is estimated. In the second step, they estimated the following one-month-ahead cross-sectional regression test for $i = 1, \ldots, N$ assets in month $T + 1$:

$$R_{i,T+1} - R_{fT+1} = \lambda_0 + \lambda_\beta \hat{\beta}_i + \lambda_{Z^*} \hat{Z}_i^* + u_{it}, \tag{5.5}$$

where λ_β and λ_{Z^*} are coefficients that can be interpreted as the market price of beta risk (related to sensitivity to average market returns) and market price of zeta risk (associated with sensitivity to cross-sectional market volatility or return dispersion) in percent terms, respectively, and other notation is as before. Notice that this regression has observations across the $i = 1, \ldots, N$ assets, as opposed to time t as in the time-series ZCAPM regression (5.3). The estimated risk premiums λ_β and λ_{Z^*} are known to approximate mimicking portfolio returns. That is, λ_β returns are based on the returns of long stocks with relatively high betas and returns of short stocks with lower betas. Also, λ_{Z^*} returns are comprised of the returns on long stocks with relatively high zetas and the returns of short stocks with lower zetas.

Beta risk loadings ($\hat{\beta}_i$) are invariant to using daily, monthly, or annual returns in their time-series regression estimation. However, zeta risk loadings ($\hat{Z}_i^*$) are not time invariant. Because zeta risk loading is estimated using daily returns and then used in the above cross-sectional regression with monthly excess returns on the left-hand side, we need to rescale $\hat{Z}_i^*$ from a daily to monthly basis as follows:

$$R_{i,T+1} - R_{fT+1} = \lambda_0 + \lambda_\beta \hat{\beta}_i + \lambda_{Z^*} \hat{Z}_i^* N_{T+1} + u_{it}, \tag{5.6}$$

where N_{T+1} is the number of trading days in month $T + 1$ (i.e., 21 d), $Z_i^* N_{T+1}$ is the monthly estimated zeta risk, and λ_{Z^*} is the monthly risk premium associated with zeta risk. Per unit zeta risk, the risk premium $\hat{\lambda}_{Z^*}$ is not changed by this rescaling. By rescaling Z_i^* to a monthly basis, we obtain λ_{Z^*} estimates that are comparable to monthly λ_β estimates associated with beta loadings.

A rolling regression approach was used in their analyses. Time-series regression (5.6) was estimated using one year of daily returns in the period from January 1964 to December 1964. Then cross-sectional regression (5.5) was estimated in January 1965. The analyses were rolled forward one month and repeated (e.g., the next estimation period is February 1964 to January 1965, and $T + 1$ is February 1965). This one-month-rolling process was continued until December 2018. A series of 636 monthly estimates are generated for the market prices of beta and zeta risk loadings denoted λ_a and λ_{RD}, respectively. Average market prices of λ_a and λ_{RD} are computed. Also, following Jagannathan and Wang (1996) and Lettau and Ludvigson (2001, footnote 17, p. 1254), to assess the goodness-of-fit of the cross-sectional regressions, an R^2 statistic is computed based on the single regression approach. Using

the 636 analysis period months, average one-month-ahead excess returns for N test assets for the entire analysis period from 1995 to 2018 are regressed on average monthly estimates of $\hat{\lambda}_k$ for the N test assets.

In an effort to validate the empirical ZCAPM, KLH compared its cross-sectional regression results to those for the following popular asset pricing models:

- Market model version of the CAPM with an excess market return factor ($MKT - RF$) defined as a value-weighted return of sample stocks in each country minus the 30-day U.S. Treasury bill rate;
- Fama and French (1992, 1993, 1995) three-factor model that augments the CAPM with a size factor (viz., small minus large firms' stock returns denoted *SMB*) and a value factor (viz., high value minus low value firms' stock returns denoted *HML*);
- Carhart (1997) four-factor model that augments the three-factor model with a momentum factor (viz., stocks with high past returns minus stocks with low past returns denoted *MOM*);
- Fama and French (2015) five-factor model[1] that augments the three-factor model with a profit factor (viz., high profit minus low profit firms' stock returns denoted *RMW*) and a capital investment factor (viz., high capital investment minus low capital investment firms' stock returns denoted *CMA*); and
- Fama and French (2018) six-factor model that augments the five-factor model with a momentum factor (denoted *MOM*).

The multifactors *SMB*, *HML*, *MOM*, *RMW*, and *CMA* for U.S stocks are downloaded from Kenneth French's website. These factors have become commonly used in asset pricing models.

5.3 Cross-Sectional Test Results

How does the ZCAPM perform in cross-sectional regression tests compared to other widely accepted asset pricing models? Table 5.1 shows the results for the CAPM and ZCAPM as well as multifactor models based on long/short portfolio factors. This table is exemplary of the extensive results published by KLH in their book. Panels A to D show the results for different test asset portfolios.

[1] This model is similar to the four-factor model of Hou et al. (2015), who do not include the value factor and modify somewhat the profit and capital investment factors

Let's review the evidence. Panel A of Table 5.1 reports the results using 25 size and book-to-market ratio portfolios. Fama and French (1992, 1993) created these portfolios to show that the CAPM could not price stock portfolios sorted on these firm characteristics (and therefore was declared *dead*). The table shows the following results: (1) average market prices of risk measured by λ_ks, (2) associated t-statistics (in parentheses), and (3) estimated R^2 values for each model. Recall that one-month-ahead excess returns for the portfolios are regressed on the previous year's factor loadings (i.e., betas and zeta coefficients estimated using one year of daily returns). Because this setup is an out-of-sample test, no possibility of cheating is possible. The cross-sectional regression seeks to determine if the factor loadings for portfolios are related to their future one-month returns. Many authors, including Simin (2008, p. 356), Ferson et al. (2013), Martin and Nagel (2021), and others, have recommended out-of-sample tests in comparisons of alternative asset pricing models to avoid data snooping issues and other problems in explaining stock returns.

It is clear from the results in Panel A that the ZCAPM dominates the other models. The goodness-of-fit is substantially higher with R^2 at 94% compared to at most 80% in the six-factor model. This finding suggests almost perfect goodness-of-fit for the ZCAPM—that is, future returns of the test asset portfolios line up almost perfectly with their previously estimated beta and zeta risks as measured by the empirical ZCAPM.

Notice from Panel A of the table that the multifactor models outperform the CAPM by boosting R^2 values above 48% CAPM. Regarding the CAPM, contrary to theory, the market price of the risk for the market factor is negative and insignificant with a t-statistic of only -0.35. For the multifactor models, the value factor is positively priced and significant with t-statistics in the range of 2.49 to 2.69. These t-statistics exceed the 5 percent significance level, which is a commonly used threshold to infer that the factor loadings are priced in the cross section of average stock returns.

More recently, due to the factor zoo in asset pricing and the possible problem of false discoveries related to data snooping, Harvey et al. (2016) and Chordia et al. (2020) have recommended a t-statistic of 3 as a minimum hurdle for the significance of factor loadings. Table 5.1 shows that the market price of zeta risk loadings denoted λ_{RD} in the ZCAPM is the only asset pricing factor that passes this hurdle. The t-statistic for average λ_{RD} is much higher than other factors at 4.22. Also, the magnitude of this market price of risk equals 0.46 percent per month. This market price translates to 5.52 percent per year which is close to the actual average market factor risk premium (i.e., CRSP index return minus Treasury bill rates) in the U.S. stock

Table 5.1 Out-of-sample Fama-MacBeth cross-sectional tests for ZCAPM regression factor loadings compared to other asset pricing models in the period from January 1965 to December 2018: 12-month rolling windows

Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{RD}$	$\hat{\lambda}_{SMB}$	$\hat{\lambda}_{HML}$	$\hat{\lambda}_{MOM}$	$\hat{\lambda}_{RMW}$	$\hat{\lambda}_{CMA}$	R^2
Panel A: 25 size-BM sorted portfolios									
CAPM	0.98	−0.35							0.48
	(4.03)	(−0.35)							
ZCAPM	0.78	−0.19	0.46						0.94
	(3.26)	(−0.77)	(4.22)						
Three-factor	0.93	−0.41		0.17	0.31				0.63
	(4.98)	(−1.96)		(1.27)	(2.64)				
Four-factor	1.12	−0.60		0.16	0.29	0.08			0.69
	(5.84)	(−2.83)		(1.29)	(2.57)	(0.35)			
Five-factor	0.98	−0.45		0.21	0.28		0.13	0.11	0.75
	(4.94)	(−2.06)		(1.73)	(2.49)		(1.10)	(0.84)	
Six-factor	1.07	−0.55		0.22	0.30	0.16	0.12	0.06	0.80
	(5.39)	(−2.52)		(1.80)	(2.69)	(0.68)	(1.03)	(0.45)	
Panel B: 47 industry portfolios									
CAPM	0.57	0.03							0.00
	(2.95)	(0.15)							
ZCAPM	0.49	0.11	0.32						0.70
	(2.71)	(0.53)	(4.07)						
Three-factor	0.43	0.11		0.02	0.14				0.15
	(2.21)	(0.48)		(0.14)	(1.17)				
Four-factor	0.46	0.08		0.04	0.08	0.43			0.39
	(2.31)	(0.37)		(0.31)	(0.73)	(2.09)			
Five-factor	0.36	0.19		0.07	0.09		0.20	0.24	0.18

(continued)

Table 5.1 (continued)

Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{RD}$	$\hat{\lambda}_{SMB}$	$\hat{\lambda}_{HML}$	$\hat{\lambda}_{MOM}$	$\hat{\lambda}_{RMW}$	$\hat{\lambda}_{CMA}$	R^2
	(1.78)	(0.83)		(0.51)	(0.76)		(2.03)	(2.48)	
Six-factor	0.51	0.04		0.04	0.06	0.39	0.20	0.17	0.50
	(2.53)	(0.16)		(0.32)	(0.52)	(1.92)	(2.05)	(1.78)	
Panel C: 25 beta-zeta sorted portfolios									
CAPM	0.49	−0.06							0.11
	(3.23)	(−0.26)							
ZCAPM	0.29	0.17	0.35						0.96
	(1.91)	(0.83)	(4.03)						
Three-factor	0.66	−0.05		−0.28	0.14				0.65
	(3.34)	(−0.24)		(−1.49)	(1.00)				
Four-factor	0.54	−0.03		−0.25	0.25	0.28			0.70
	(3.50)	(−0.15)		(−1.51)	(1.76)	(1.61)			
Five-factor	0.52	−0.03		−0.19	0.25		0.17	0.09	0.67
	(3.06)	(−0.13)		(−1.11)	(1.56)		(1.18)	(0.77)	
Six-factor	0.62	−0.11		−0.14	0.30	0.26	0.17	0.02	0.76
	(3.86)	(−0.54)		(−0.88)	(1.90)	(1.50)	(1.30)	(0.14)	
Panel D: 97 combined portfolios									
CAPM	0.64	−0.06							0.05
	(3.93)	(−0.31)							
ZCAPM	0.48	0.09	0.38						0.83
	(3.24)	(0.47)	(5.42)						
Three-factor	0.52	0.002		0.06	0.18				0.06
	(3.55)	(0.01)		(0.49)	(1.64)				
Four-factor	0.53	−0.01		0.09	0.17	0.43			0.28
	(3.75)	(−0.06)		(0.75)	(1.64)	(2.54)			

Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{RD}$	$\hat{\lambda}_{SMB}$	$\hat{\lambda}_{HML}$	$\hat{\lambda}_{MOM}$	$\hat{\lambda}_{RMW}$	$\hat{\lambda}_{CMA}$	R^2
Five-factor	0.44	0.09		0.12	0.14		0.21	0.21	0.22
	(2.97)	(0.45)		(1.02)	(1.35)		(1.99)	(2.36)	
Six-factor	0.57	−0.04		0.14	0.17	0.40	0.19	0.16	0.35
	(4.02)	(−0.24)		(1.20)	(1.68)	(2.40)	(1.94)	(1.84)	

In the ZCAPM, the price of beta risk associated with CRSP index excess returns is denoted $\hat{\lambda}_a$ rather than $\hat{\lambda}_m$ in the other models
This table reports selected results from Kolari et al. (2021). Out-of-sample (one-month-ahead) estimated prices of risk were estimated using the standard two-step Fama-MacBeth cross-sectional tests. Estimated prices of risk are denoted $\hat{\lambda}_k$ for the kth factor in monthly percent return terms (t-statistics in parentheses). Factors are denoted as *m* (CRSP index, see footnote), *RD* (return dispersion), *SMB* (size), *HML* (value), *MOM* (momentum), *RMW* (profit), and *CMA* (capital investment). Value-weighted returns were used in the period from January 1965 to December 2018. Different sets of test asset portfolios were employed as shown in Panels A to D. The results shown below are based on Kolari et al. (2021, Table 7.1, pp. 166–167)

market in the sample period from 1965 to 2018. Thus, the ZCAPM does a good job of explaining stock market returns over time.

Lewellen et al. (2010), Daniel and Titman (2012), and others have recommended that industry portfolios should be used in asset pricing model tests. The size and book-to-market ratio portfolios constructed by Fama and French are related to the size and value factors. This relation between independent variables (or factors) and dependent variables (or excess portfolio returns) may cause an endogeneity problem that unduly biases the regression equation results. Using exogenous test assets such as industry portfolios that have little or no relation to the factors in the model eliminate this problem. Panel B reports model comparisons using 47 industry portfolios downloaded from Kenneth French's website. Now the CAPM has no explanatory power at all with an R^2 of 0%. The multifactor models increase the R^2 values to as much as 50%. Note that the ZCAPM has the highest estimate at 70% and that the market price of risk of zeta risk loadings λ_{RD} is still highly significant with a *t*-statistic of 4.07. For the multifactor models, now the market prices of risk for momentum (*MOM*), profit (*RMW*), and capital investment (*CMA*) exhibit significance that at the 5 percent level but value (*HML*) is no longer significant. The fact that these factors' significance changes with different test assets is somewhat disconcerting. A factor that is priced should be priced (or significant) on a consistent basis across different samples of stock returns. The ZCAPM passes this consistency test.

Panel C gives the results for 25 portfolios formed by sorting stocks on their previous beta and zeta coefficient estimates in the previous year. The results in Panel C look similar to those in Panel A for the most part. Note that the R^2 estimate for the ZCAPM is even higher at 96%, an almost perfect fit with respect to one-step-ahead (one-month-ahead) stock returns.

Finally, Panel D contains the results for the 97 combined portfolios in the other panels. The multifactor models reach a maximum R^2 value of only 35%. The ZCAPM again performs well at 83%, and the *t*-statistic for the market price of zeta risk loadings λ_{RD} equals 5.42, which is very high by any standard in statistical analyses.

In sum, the results in Table 5.1 strongly support the empirical ZCAPM. The ZCAPM not only noticeably improves goodness-of-fit but has high factor loading significance with respect to zeta risk. Also, these results are consistent across different test asset portfolios. To the author's knowledge, no previous asset pricing studies have reported such convincing out-of-sample cross-sectional regression evidence. Moreover, multifactor models appear to suffer from two potential problems: (1) no factors have loadings that achieve *t*-statistics that surpass the 3 thresholds recommended by Harvey et al. (2016)

and Chordia et al. (2020), and (2) the factors tend to be significant at times and insignificant at other times depending on different test asset portfolios.

More recent work by Kolari et al. (2022a) extended the U.S. stock return analyses in KLH to a longer sample period of 1927 to 2020. The results corroborated those in KLH. Also, Kolari et al. (2022b) extended the analyses to Canada, Japan, Germany, and Japan. Again the results confirmed findings in KLH—namely, the ZCAPM consistently outperformed multifactor models for different test asset portfolios, especially when using exogenous industry portfolios. Furthermore, Kolari et al. (2023) showed that missing factors exist in popular multifactor models but not in the ZCAPM. Thus, the ZCAPM is a valid asset pricing model.

As discussed earlier, long/short portfolios used in multifactor models are themselves measures of return dispersion. They represent different slices of high/low returns within the cross-sectional distribution of stock returns. Given the significance of total return dispersion in the empirical ZCAPM, it is not surprising that these factors have some significance in cross-sectional regression tests. As more factors are added to these models, they demonstrate an increased ability to explain stock returns. A more parsimonious and comprehensive approach to modeling return dispersion is to simply use the total cross-sectional return dispersion of assets in the market as in the ZCAPM. Another advantage of the total return dispersion is that it can be readily computed as the cross-sectional standard deviation of returns in the market. There is no need to use researcher judgment and computer searches for significant long/short portfolios that can shift in terms of significance as the cross-sectional return distribution changes over time (i.e., rising and falling with average market returns as well as widening and narrowing with total return dispersion). We conclude that the empirical ZCAPM is a more powerful and stable asset pricing model than its multifactor model cousins.

5.4 Portfolio Implications of the ZCAPM

There are important portfolio management implications of beta risk and zeta risk estimates obtained from the empirical ZCAPM. For example, consider the relation between these risk loadings and future portfolio returns. Using the monthly rolling cross-sectional regression method discussed earlier with one-year estimation periods for the empirical ZCAPM, KLH computed a series of 636 monthly estimates of beta and zeta coefficients (or $\beta_{i,a}$ and $Z^*_{i,a}$) as well as 636 one-month-ahead returns for each of the 97 combined portfolios (i.e., 25 size and book-to-market ratio portfolios plus 25 beta and

zeta risk sorted portfolios plus 47 industry portfolios). For each portfolio, they computed the average betas, average zetas, and average one-month-ahead returns.

Figure 5.1 shows the relation between beta risk and one-month-ahead future returns for the 97 portfolios. It is clear that no relation is evident, which is consistent with prior research finding little or no relation between beta and returns.

Figure 5.2 graphically presents the results for zeta risk and average one-month-ahead returns. Casual inspection suggests a strong positive relation zeta risk and returns. These results mean that an investment manager could create stock portfolios ordered on zeta risk and control the future relative performance of stock portfolios. This portfolio implication was the earlier dream of the CAPM; however, as shown in Fig. 5.1, it is not possible in the real world. Nonetheless, by controlling zeta risk, portfolios can be constructed that earn predictable relative returns in the future—a powerful new tool in portfolio management!

So what if we use the empirical ZCAPM to order portfolios on their joint beta and zeta risks? What are the realized returns of these portfolios? To answer these questions, KLH compared the average one-month-ahead returns of portfolios to their predicted returns. The latter predicted returns are

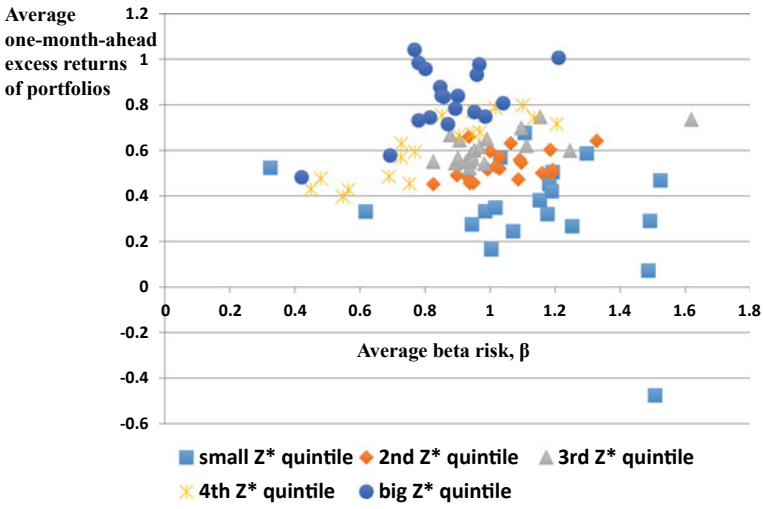


Fig. 5.1 Out-of-sample cross-sectional ZCAPM relationship between average one-month-ahead realized excess returns in percent (Y-axis) and average beta risk $\beta_{i,a}$ in the previous 12-month estimation period (X-axis). Results are shown for 97 U.S. stock portfolios. The analysis period is January 1965 to December 2018 (*Source* Kolari et al. 2021, p. 139)

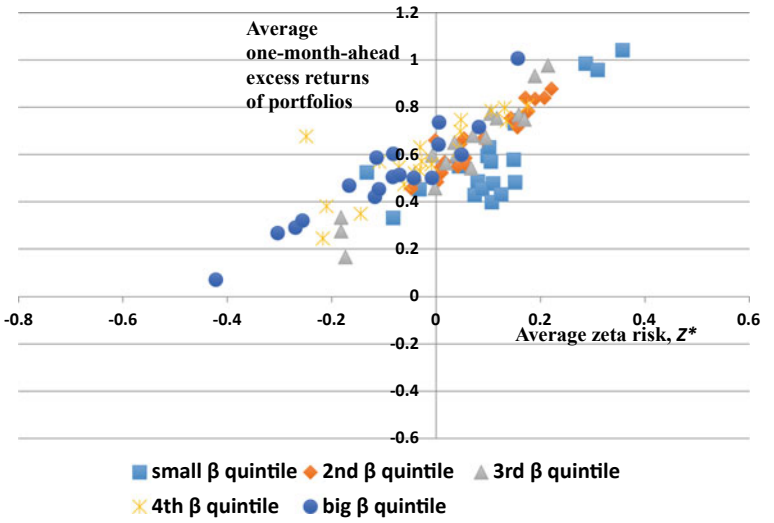


Fig. 5.2 Out-of-sample cross-sectional ZCAPM relationship between average one-month-ahead realized excess returns in percent (Y-axis) and average zeta risk $Z^*_{i,a}$ in the previous 12-month estimation period (X-axis). Results are shown for 97 U.S. stock portfolios. The analysis period is January 1965 to December 2018 (*Source* Kolari et al. 2021, p. 138)

computed by taking the beta and zeta coefficient estimates in a one-year period, and then plugging these risk estimates into the empirical ZCAPM in the next month using the values of average market excess returns $R_{at} - R_{ft}$ and market return dispersion σ_{at} in that month. Fama and MacBeth (1973) and many other authors have argued that normative models intended to help investors in their investment decisions should be evaluated by the extent to which past information is related to future returns. Here we use past estimates of beta and zeta risks to compute predicted future returns.

KLH found that the Fama and French three-factor model worked well on their endogenous 25 size and book-to-market ratio portfolios. In Panel A of Fig. 5.3, we see a close relation between realized and predicted returns for this model. However, the empirical ZCAPM works even better. An almost perfect relation is shown for the ZCAPM with the 25 portfolios that line up quite close to the 45-degree line.

When KLH added 47 exogenous portfolios to the 25 size and book-to-market ratio portfolios, the results change dramatically. Figure 5.4 illustrates their findings. Now the three-factor model is almost useless in predicting future returns. In stark contrast, the empirical ZCAPM continues to do a good job of predicting future returns. This graph highlights a potential problem with multifactor models. They are useful in predicting returns of

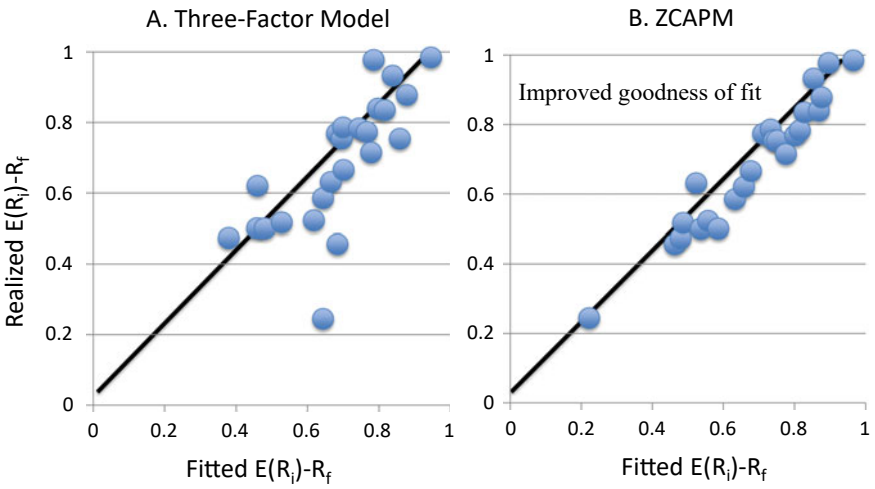


Fig. 5.3 Out-of-sample cross-sectional relationship between average one-month-ahead realized excess returns in percent (Y-axis) and average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 size-BM sorted portfolios: Fama and French Fama and French three-factor model in Panel A and empirical ZCAPM in Panel B. The analysis period is January 1965 to December 2018 (*Source* Kolari et al. 2021, p. 146)

portfolios sorted on firm characteristics related to the construction of long/short factors but can fall down when applied to other portfolios.

As we discussed in Chapter 3, multifactor models have evolved beyond the three-factor model. KLH compared the Fama and French six-factor model to the empirical ZCAPM using 25 size and profit-investment sorted portfolios plus 47 industry portfolios. Figure 5.5 demonstrates that the six-factor model improves upon the three-factor model using exogenous industry portfolios but still is not very good. However, except for a few portfolios, the empirical ZCAPM clearly outperforms the six-factor model.

In sum, the empirical ZCAPM generates zeta risk estimates for portfolios that are closely related to their future relative performance in the next month. Also, it does a good job of predicting future one-month-ahead returns of portfolios. This out-of-sample performance fulfills the dream of the CAPM—namely, investment portfolios can be formed by controlling risk in an asset pricing model so that they will earn future returns that line up with their relative risks. Customers preferring higher risk portfolios (e.g., younger investors) can be sold higher risk portfolios, and those desiring lower risk portfolios (e.g., older and retired investors) can buy lower risk portfolios. Now a close relation between risk and return can be achieved in investment portfolio management.

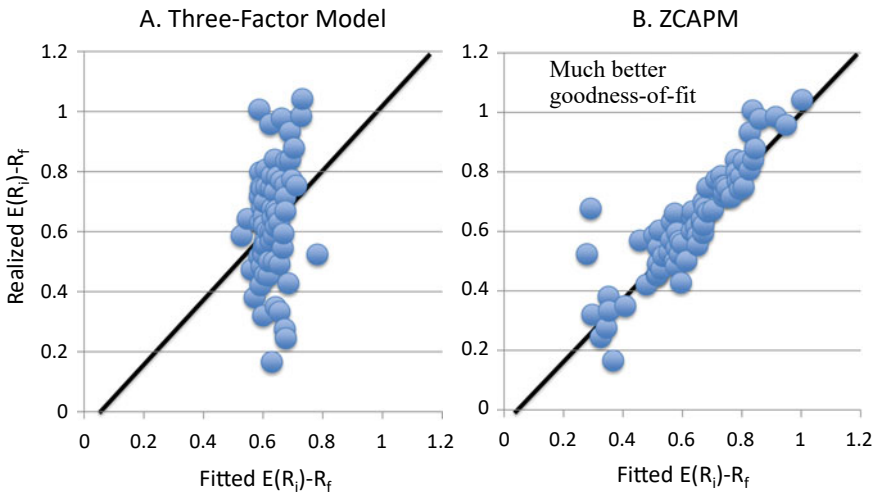


Fig. 5.4 Out-of-sample cross-sectional relationship between average one-month-ahead realized excess returns in percent (Y-axis) and average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 size-BM sorted plus 47 industry portfolios: Fama and French Fama and French three-factor model in Panel A and empirical ZCAPM in Panel B. The analysis period is January 1965 to December 2018 (*Source* Kolari et al. 2021, p. 147)

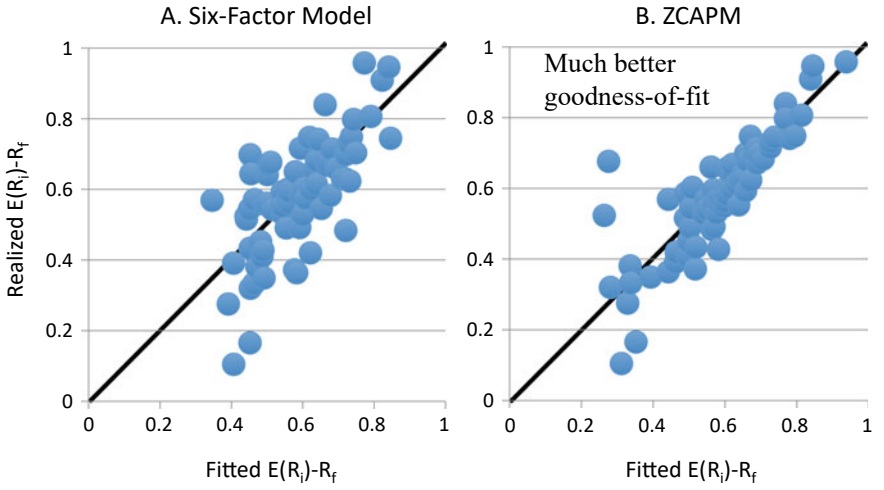


Fig. 5.5 Out-of-sample cross-sectional relationship between average one-month-ahead realized excess returns in percent (Y-axis) and average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 profit-investment sorted portfolios plus 47 industry portfolios: Fama and French six-factor model in Panel A and empirical ZCAPM in Panel B. The analysis period is January 1965 to December 2018 (*Source* Kolari et al. 2021, p. 151)

5.5 Recognition of the Empirical ZCAPM

Our initial experience with academic recognition was encouraging. We knew that the ZCAPM was a very special asset pricing model that broke new theoretical and empirical ground. The empirical results outdistanced anything that we had ever seen in published literature. Based on our working paper co-authored by Liu et al. (2012), the Financial Management Association International (FMA) awarded us the Best Paper Award in Investments at the 2012 conference in Atlanta, GA. Since the FMA is the largest finance conference in the world for academics and professionals in the world, we were excited to get the award. At the conference, a manager from the Texas Retirement System of Texas (TRS) approached us to possibly work jointly on implementing the ZCAPM in the real world to build diversified stock portfolios. After a series of meetings in Texas, an initial one-year period of paper trading was required to move forward with an actual stock portfolio. In the meantime, we sent the paper to a top tier journal in finance. We were promptly rejected due to only running in-sample Fama and MacBeth cross-sectional regression tests. Using Fama and French's often-applied 25 size and book-to-market ratio portfolios as test assets, we had obtained R^2 values as high as 98 that is impossible—no asset pricing models had ever reached that level of near perfect explanatory ability!

Some referees and the editor brushed aside the empirical ZCAPM outperformance by rejecting the theoretical ZCAPM. Since they could not understand the theory underlying the ZCAPM, they could not evaluate the incredibly strong empirical results in our paper. This criticism is seriously flawed. The theoretical ZCAPM was derived from the famous zero-beta CAPM of Black (1972). Everyone with a Ph.D. level finance education knows Black's asset pricing model. The innovation in the ZCAPM is discovering that the width or span of the mean-variance parabola is largely determined by the cross-sectional standard deviation of returns of all assets in the market (or market return dispersion). This new insight enables the derivation of the ZCAPM. It also says that the value-weighted average return of a general market index, such as the CRSP index or S&P 500 index, must lie near the axis of symmetry in the middle of the investment parabola. This latter implication goes against the grain of finance academics, many of whom persistently believe that these general market indexes are proxies for the market portfolio on the efficient frontier in the CAPM. Over time, more and more finance professors are conceding that these market indexes are inefficient, bad proxies for the market portfolio. Nonetheless, the idea that market indexes lie in the middle of the parabola is a far stretch for most academics.

Interestingly, when we present these ideas about Markowitz's investment parabola to undergraduate finance students in the classroom, they quickly understand that, if the width of the parabola is largely defined by market return dispersion, then it must be true that average general market returns must be located somewhere in the middle of the parabola. As a simple parallel example, if you have a normal distribution of people by their height, the dispersion of their heights gives the width of the distribution, and the average height is in the middle of the distribution. The average height and cross-sectional standard deviation of heights are the first and second moments of the distribution, respectively. Most finance students know these statistical facts by the time they reach their junior year of undergraduate studies. How can it be so difficult to accept these ideas with respect to Markowitz's mean-variance investment parabola? Perhaps the only answer we have is that finance PhDs are not trained up in these ideas. They reject them for the reason that, if it was this simple, someone would have taught them in their graduate classes or they would have read about it somewhere in the many research papers studied over the years.

Per the referee's suggestions from the first finance journal, we revised the paper by updating it to include only out-of-sample Fama and MacBeth cross-sectional tests and sent it to another top finance journal. Now our R^2 value for the 25 size and book-to-market ratios was 94%, which well exceeded previous evidence published for an existing asset pricing model. Not even the Fama and French multifactor models that contain size and book-to-value ratio factors came close to this high level of explanatory power. Nonetheless, we were rejected due to not including exogenous test assets, such as industry portfolios and individual stocks. Back to the drawing board, we added these test assets to our tests and found that only the ZCAPM worked for these exogenous test assets. So the referee must have known about the difficulty of pricing these exogenous test assets and thought that such tests would end our enthusiasm for the ZCAPM. By now the one-year paper trading tests had been completed with very positive results, and we made a business deal with TRS to manage $100 million in a stock portfolio. We signed legal papers to make the ZCAPM a proprietary technology that prohibited any publications, presentations, and outside discussions about the ZCAPM to anyone. Two years later, due to changes in management at TRS, we parted ways and began the process of publishing our accumulating knowledge and experience with the ZCAPM.

In our first attempt after TRS, it seemed that resistance from top tier academic finance papers had not changed. We were rejected for not including Gibbons et al. (1989) GRS tests for joint alphas equal to zero. These tests

focus on the alpha intercept term in the time-series regression asset pricing models. However, the GRS test is only applicable to linear asset pricing models, whereas the ZCAPM is a nonlinear model. Also, the empirical ZCAPM has no alpha intercept due to using expectation-maximization (EM) regression methods. Consequently, these criticisms seemed insurmountable.

Relevant to the GRS test criticism, Fama and French (2020) published a paper in which they proposed a way to estimate an alpha mispricing term for a time-series regression model even if it had no alpha intercept. As discussed in Chapter 3, they used the average of the residual error terms to obtain an estimate of the alpha intercept. Recently, in a working paper by Kolari et al. (2023), we implemented their alpha method. We obtained a positive alpha for the empirical ZCAPM, but the cross-sectional variance of our alphas across various sets of test assets was very small compared to multifactor models. Also, in new cross-sectional tests of alphas in models to determine if they contained a missing factor, we found that the empirical ZCAPM does not have any missing factors, but all of the well-known multifactor models had very significant missing factors. Repeating history, one referee rejected our empirical ZCAPM due to not using a large set of anomalous long-short portfolios as test assets that some authors had begun using to test models. By coincidence, we have been running tests of the empirical ZCAPM with over 130 anomalous portfolios (i.e., long-short stock portfolios) obtained from Andrew Chen at the Federal Reserve Board. These portfolios are showcased in a paper by Chen and Zimmermann (2019), who utilized them to test for publication biases in asset pricing tests and findings. We obtained another set of 153 anomaly portfolio returns from an online database provided by Jensen, Kelly, and Pederson (2023). Our ongoing research results show that only the empirical ZCAPM is able to price these test asset portfolios. All the multifactor models fail badly in our tests. We are now writing another version of our empirical ZCAPM paper that reports these exceptionally good anomalous portfolio results compared to other models. How will we be rejected at the next finance journal? Claiming ignorance, one top tier finance editor has already desk rejected us on grounds that they simply do not understand the ZCAPM theory and the empirical model. The editor happens to be a faculty member at the university where Kenneth French works. Oddly, in a previous rejection of the ZCAPM at a top finance journal, the editor was another faculty member at the same university. It seems possible that editor bias in favor of the Fama and French models and against contenders like the ZCAPM exists in top tier finance journals. In effect, the academic world of asset pricing appears to be so interconnected that entry is restricted.

The continued rejection of our ZCAPM by the top tier journals in finance is disappointing. We openly offer our software codes in Matlab, R, and Python on GitHub on the internet. Anyone can replicate our model, which is the gold standard in any science. Anyone can check its statistical significance with stock return data. But only one journal attempted to do this and was so amazed at the findings that the editor and referees just kept asking for more and more tests in round after round of referee comments. All our tests confirmed the fact that the empirical ZCAPM dominated all other asset pricing models in out-of-sample tests regardless of different test assets and time periods. After numerous referee rounds over a period of two and half years, with no decision in sight, we withdrew the paper from the journal. Was it another tactic to block us? That is, just hold up the paper in an endless review process!

What is motivating the repeated academic rejection of the ZCAPM? A major reason is that the editors and referees of the top finance journals themselves have their own multifactor models or research closely related to models. Acceptance of the ZCAPM would be the end of their model or research agenda. The models are important to academic researchers as they lead to publications in top journals, pay raises, promotions, and national and international recognition. Another reason is that the editors and referees are confederates of professors with outstanding asset pricing models. Professional and personal friendships could be affected, which would be deleterious to their academic careers. Another reason for rejection of the empirical ZCAPM is just laziness. If a referee or editor cannot believe the results, we recommend using our free software to conduct replication tests. Replication is the main process for verifying scientific discoveries. Rather than rely on opinions, judgments, and personal biases, obtain real world data and repeat our empirical tests. Run your own tests too. Subjective opinions should not be part of the process. Otherwise, the science of finance will be destroyed and the creation of new ideas will be shut down.

There should be a vibrant and open discourse of ideas that seeks to push forward our understanding of the natural finance world around us. New discoveries should be heard and scientific tools of logic, statistical analysis, and rigorous replication should be applied. The "euraka moment" manifested by the spark of imagination needs to be cherished and protected in our field of finance.

In an effort to fight back, we have been publishing our ZCAPM work in books like this one. Also, you can learn more about the ZCAPM in two other books: Kolari et al. (2021) and Kolari and Pynnonen (2023). Some finance journal publications on the ZCAPM are cited at the end of this chapter,

and we regularly post our working papers on the Social Science Research Network (SSRN). Not sure about our ZCAPM claims? Please feel free to use our software programs and check out the results for yourself.

5.6 Summary

As discussed in the previous chapter, the theoretical ZCAPM of Kolari, Liu, and Huang (KLH) (2021) proposed two factors: mean excess market returns and the cross-sectional standard deviation of returns for all assets in the market (return dispersion). Factor loadings on these two factors are defined as beta risk and zeta risk, respectively. The ZCAPM departs from the trend in asset pricing over the past 30 years to develop multifactor models based on research judgment and discretion. Instead, it has CAPM roots in the Markowitz (1959) mean-variance investment parabola, Sharpe's (1964) CAPM, and Black's zero-beta CAPM. Unlike other asset pricing models, the ZCAPM predicts that asset returns can move up for some assets and down for other assets in response to an increase in market return dispersion, and vice versa for a decrease in market return dispersion. Also, beta risk is related to sensitivity to average market returns rather than efficient market portfolio returns. Using the geometry of investment parabola, KLH located two unique orthogonal portfolios with the same total risk. The difference in returns between these two portfolios is closely related to the return dispersion in the cross-sectional distribution of the returns on all assets in the market on (for example) one day. By implication, the average market return lies approximately in the middle of the return distribution of all assets near the axis of symmetry of the parabola. Using these two portfolios, KLH mathematically derived the ZCAPM as a special case of Black's zero-beta CAPM.

KLH developed a novel empirical ZCAPM by introducing a dummy signal variable (equal to $+1$ or -1) to capture positive versus negative return dispersion effects on asset returns. Because this signal variable is unknown or not observable, they applied the well-known expectation-maximization (EM) algorithm to estimate the probability that the signal variable is positive or negative. In the regression model, this probability is multiplied by the zeta risk coefficient to estimate the zeta risk coefficient (denoted Z^*) for any asset. The empirical ZCAPM consists of a mixture of two-factor models—one model has positive zeta risk, and the second model has negative zeta risk. The probability estimate of the signal variable determines the operative model. The authors provide software programs on GitHub (https://github.com/zca

pm) to estimate the empirical ZCAPM in addition to their application in cross-sectional regression tests.

We reviewed a number of empirical tests conducted by KLH to assess the validity of the empirical ZCAPM compared to a number of prominent asset pricing models. Based on standard Fama and MacBeth (1973) cross-sectional regression analyses, the empirical ZCAPM tended to dominate the market model of the CAPM as well as popular multifactor models, in terms of both higher goodness-of-fit and higher significance of zeta risk loadings compared to multifactor loadings. The consistency of the strength of empirical ZCAPM evidence is remarkable in view of the tendency for multifactor models to have unstable results across different test asset portfolios. In endogenous test asset portfolios, the multifactor models perform fairly well, but their ability to explain out-of-sample returns collapses upon the use of exogenous portfolios (such as industry portfolios). As KLH have observed, long/short multifactors are rough measures of return dispersion and therefore have some significance in cross-sectional tests. However, total market return dispersion is a more parsimonious and comprehensive risk factor that outperforms component return dispersion measures such as the multifactors.

What are the implications of the empirical ZCAPM to portfolio management? KLH showed that portfolios sorted on zeta risk estimates earn out-of-sample returns in the next month that are closely related to their relative risk. However, beta risk does not work in this respect, which is consistent with previous evidence on the failure of the CAPM to demonstrate any relation between beta risk and stock returns. Further analyses of the relation between realized portfolio returns and predicted (one-month-ahead) portfolio returns showed that the empirical ZCAPM does a very good job on a normative level. Investors who buy portfolios using beta and zeta risk estimates from the empirical ZCAPM can earn future returns that line up with the relative risks of different portfolios. This investment strategy was the dream of the CAPM, a dream that now can be accomplished via the empirical ZCAPM.

The original justification of multifactor models over the CAPM was that they better explained stock returns. Now the tables have turned! The CAPM-based ZCAPM outperforms the multifactor models in terms of its empirical fit to out-of-sample returns in the cross section of stocks. In a surprising twist, it turns out that multifactor models are cousins of the ZCAPM due to the use of long/short factors that are essentially return dispersion measures. Thus, the empirical ZCAPM unifies the asset pricing models under the CAPM umbrella. While academic resistance to the ZCAPM has plagued us over the past decade, we have published a number of books that contain ZCAPM ideas and continue to make research publication progress.

In forthcoming chapters, we employ the empirical ZCAPM to produce a proxy for the minimum variance portfolio G that, in turn, is used to obtain better estimates of average market returns and market return dispersion. With improved market excess return and return dispersion factors available to estimate the empirical ZCAPM, we then proceed to construct empirically efficient portfolios based on beta and zeta risks that manifest the architecture of the mean-variance investment parabola. Real world applications of ZCAPM-based stock portfolios in collaboration with the investment community are sought by the authors.

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Carhart, M.M. 1997. On persistence in mutual fund performance. *Journal of Finance* 52: 57–82.

Chen, A.Y., and T. Zimmermann. 2019. Publication bias and the cross-section of stock returns. *The Review of Asset Pricing Studies* 10: 249–289.

Chordia, T., A. Goyal, and A. Saretto. 2020. Anomalies and false rejections. *Review of Financial Studies* 33: 2134–2179.

Daniel, K., and S. Titman. 2012. Testing factor-model explanations of market anomalies. *Critical Finance Review* 1: 103–139.

Dempster, A.P., N.M. Laird, and D.B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society* 39: 1–38.

Fama, E.F., and K.R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427–465.

Fama, E.F., and K.R. French. 1993. The cross-section of expected returns. *Journal of Financial Economics* 33: 3–56.

Fama, E.F., and K.R. French. 1995. Size and book-to-market factors in earnings and returns. *Journal of Finance* 50: 131–156.

Fama, E.F., and K.R. French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116: 1–22.

Fama, E.F., and K.R. French. 2018. Choosing factors. *Journal of Financial Economics* 128: 234–252.

Fama, E.F., and K.R. French. 2020. Comparing cross-section and time-series factor models. *Review of Financial Studies* 33: 1892–1926.

Fama, E.F., and J.D. MacBeth. 1973. Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy* 81: 607–636.

Ferson, W.E., S.K. Nallareddy, and B. Xie. 2013. The "out-of-sample" performance of long-run risk models. *Journal of Financial Economics* 107: 537–556.

Gibbons, M.R., S.A. Ross, and J. Shanken. 1989. A test of the efficiency of a given portfolio. *Econometrica* 57: 1121–1152.

Harvey, C.R., Y. Liu, and H. Zhu. 2016. ... and the cross-section of expected returns. *Review of Financial Studies* 29: 5–68.

Hou, K., C. Xue, and L. Zhang. 2015. Digesting anomalies: An investment approach. *Review of Financial Studies* 28: 650–705.

Jagannathan, R., and Z. Wang. 1996. The conditional CAPM and the cross-section of asset returns. *Journal of Finance* 51: 3–53.

Jensen, T. I., B. Kelly, and L. H. Pedersen. 2023. Is there a replication crisis in finance? *Journal of Finance* 78: 2465–2518.

Jones, P.N., and G.J. McLachlan. 1990. Algorithm AS 254: Maximum likelihood estimation from grouped and truncated data with finite normal mixture models. *Applied Statistics* 39: 273–282.

Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2022a. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*. Available online at https://www.mdpi.com/1911-8074/15/3/137. Reprinted in Kolari, J.W., and S. Pynnonen, eds. 2022. *Frontiers of asset pricing*. Basel, Switzerland: MDPI.

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022b. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. Testing for missing asset pricing factors. Paper presented at the Western Economic Association International, San Diego, CA.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Lettau, M., and S. Ludvigson. 2001. Consumption, aggregate wealth, and expected stock returns. *Journal of Finance* 56: 815–849.

Lewellen, J., S. Nagel, and J.A. Shanken. 2010. A skeptical appraisal of asset pricing tests. *Journal of Financial Economics* 96: 175–194.

Liu, W. 2013. A new asset pricing model based on the zero-beta CAPM: Theory and evidence. Doctoral dissertation, Texas A&M University.

Liu, W., J.W. Kolari, and J.Z. Huang. 2012. A new asset pricing model based on the zero-beta CAPM market model (CAPM). Presentation at the annual meetings of the Financial Management Association, Best Paper Award in Investments, October, Atlanta, GA.

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. Return dispersion and the cross-section of stock returns. Presentation at the annual meetings of the Southern Finance Association, October, Palm Springs, CA.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: Wiley.

Martin, I., and S. Nagel. 2021. Market efficiency in the age of big data. *Journal of Financial Economics* 45: 154–177.

McLachlan, G.J., and T. Krishnan. 2008. *The EM algorithm and extensions*, 2nd ed. New York, NY: Wiley.

McLachlan, G., and D. Peel. 2000. *Finite mixture models*. New York, NY: Wiley Interscience.

McLean, R.D., and J. Pontiff. 2016. Does academic publication destroy predictability? *Journal of Finance* 71: 5–32.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.

Simin, T. 2008. The poor performance of asset pricing models. *Journal of Financial and Quantitative Analysis* 43: 335–380.

Part IV

Portfolio Performance



6

Portfolio Performance Measures

6.1 Return Metrics

In this section, we revisit the two return concepts of *simple returns* and *continuously compounded returns* or *log returns*. It is important to recognize that simple returns can be aggregated cross-sectionally to portfolios, whereas log returns aggregate over time. Also, while simple holding period returns are the main focus of investors, log returns are often useful in academic research studies.

Asset returns are based on price changes between adjacent time periods $t-1$ to t. Using stocks as the asset, *simple returns* are defined as

$$R_t = (P_t + D_t - P_{t-1})/P_{t-1}, \tag{6.1}$$

where P_t is the stock price at time t, D_t is the dividend paid on the stock during the period from $t-1$ to t, and P_{t-1} is the stock price at previous time $t-1$. Assuming an infinite number of subperiods within period t, *continuously compounded returns* can computed using *log returns* as

$$r_t = \log(P_t + D_t) - \log(P_{t-1}), \tag{6.2}$$

where log is the natural logarithm. Simple and log returns are related to one another as follows:

$$r_t = \log(1 + R_t), \tag{6.3}$$

where $1 + R_t$ is the *gross return*, and $1 + r_t$ is the *gross log return*.

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It is notable that returns are scale free but not unitless, which means that they are always defined with respect to some time interval such as one day, month, etc. A "5% return per month" corresponds to $R_t = 0.05$ measured over a one-month period of time.

Mean returns can be either arithmetic or geometric. The arithmetic mean over the sample period of $t = 1, \ldots, T$ is

$$\bar{R} = \frac{1}{T} \sum_{t=1}^{T} R_t. \tag{6.4}$$

However, this mean return is always biased upward and therefore is too high. The main problem is that returns on investment develop in a compounded manner, not additive.[1]

The correct mean return in a multiperiod context is the *geometric mean*:

$$\bar{R}_g = \left[\prod_{t=1}^{T} (1 + R_t) \right]^{1/T} - 1. \tag{6.5}$$

This formula is more complex in mathematical terms but simplifies matters in the sense that, given the geometric mean, the gross return over the whole sample period becomes $1 + R(T) = \prod_{t=1}^{T} (1 + R_t) = (1 + \bar{R}_g)^T$.[2]

Using log returns, which are continuously compounded returns, the logarithm of $1 + \bar{R}_g$ in Eq. (6.5) transforms the product of the gross returns of securities to the following sum:

$$\bar{r} = \log(1 + \bar{R}_g) = \frac{1}{T} \sum_{t=1}^{T} \log(1 + R_t) = \frac{1}{T} \sum_{t=1}^{T} r_t \tag{6.6}$$

[1] A simple example illustrates the bias. If you start with \$200 and lose 20% in the first period, you have $(1 - 0.20) \times \$200 = \$160$ to invest in the second period. If the gain in this second period is 30%, the end value becomes $(1 + 0.30) \times \$160 = \$208$, such that the 2-period return of your investment is $(208 - 200)/200 = 4\%$, i.e., $(1 - 0.20) \times (1 + 0.30) - 1$. The additive return of 10% $(= -0.20 + 0.30)$ in the arithmetic mean formula is clearly too high and therefore yields an upward biased mean return equal to 5% $(= 10\%/2)$.

[2] Thus, in the simple example in footnote 1, the correct mean return is $[(1-0.2) \times (1+0.3)]^{1/2} - 1 = 1.98\%$ (rather than the arithmetic mean of $(-0.2 + 0.3)/2 = 5\%$). The 2-period return is then $(1 + 0.0198)^2 - 1 = 4\%$, which matches the correct 2-period return computed above.

with $r_t = \log(1 + R_t)$. Therefore, the arithmetic mean $\bar{r}$ correctly measures the average multiperiod log return. Also, it is useful to observe that $\bar{R}_g = e^{\bar{r}} - 1$ and $1 + R(T) = (1 + \bar{R}_g)^T = e^{T\bar{r}}$, where $e \approx 2.718$.[3]

As noted earlier, while returns are scale free, they are not unit free. That is, they depend on the time unit. Therefore, when discussing returns, it is important to indicate the unit, whether it is daily, weekly, monthly, yearly, etc. In particular, when making comparisons, it is important to transform returns to the same time unit. It is not useful to annualize single daily returns. A simple example demonstrates the problem. Assume 1% return on some day. If annualized by assuming 365 days, we get $(1 + 0.01)^{365} - 1 \approx 36.78$ or 3,678%, implying that the transformation explodes the numbers. However, average daily returns can be annualized as averaging smooths single day variation. The general annualization formula is

$$R_{\text{p.a}} = (1 + \bar{R}_g)^{1/\Delta t} - 1, \tag{6.7}$$

where Δt is the annualization (per annum, p.a) factor. For daily, weekly, and monthly returns, Δt equals 1/365, 1/52, and 1/12, respectively. For multiple year returns, Δt is the number of years, e.g., $\Delta t = 1$ for yearly returns and $\Delta t = 5$ for five years return. Thus, if for daily returns with $\bar{R}_g = 0.0002$, $R_{\text{p.a}} = (1.0002)^{365} - 1 \approx 7.6\%$, and for five years return of 20%, $R_{\text{p.a}} = 1.20^{1/5} - 1 \approx 3.7\%$.

Regarding simple versus log returns, in high frequency data based on daily or weekly returns, the numerical value differences between simple returns and log returns are usually negligible. However, there are fundamental differences between these return concepts. One difference is that the simple return is bounded below by -1 when the stock price falls to zero. In this case, log returns approach minus infinity. Therefore, while simple returns can take values from minus one upwards, log returns can be all real values.

A more important difference between these returns is the way they aggregate with multiple assets in a portfolio. We previously observed that, unlike log returns, summing simple returns over time does not aggregate correctly to total returns. The opposite is true when summing cross-sectionally stocks into portfolios at a point in time. Portfolio returns can be obtained as weighted sums of the simple returns, while log returns of a portfolio cannot be obtained as a weighted sum of log returns of multiple stocks. For this reason, it is common to use simple returns when the cross section of assets is measured in

[3] This value is the Neper or Napierian number after the Scottish mathematician John Napier. The term e is also known as Euler's number after the Swiss mathematician Leonard Euler.

some period of time and log returns when the temporal behavior of returns over time is measured.

Finally, in portfolio or fund return computations it is important to account for cash inflows and outflows during the investment period as these imply variation in amounts of funds invested in different time periods. A general practice is to use a compounded formula, called time-weighted return, in which returns are updated at the occurrences of cash flows. Formally this means that if $t_1, t_2, \ldots, t_n$ are time points for the cash flows within a period from initial investment at $t_0 = 1$ to end period $t_{n+1} = T$, the return on a fund with initial value V_0 is computed then as

$$R_T = \frac{V_{t_1^-}}{V_{t_0}} \times \frac{V_{t_2^-}}{V_{t_1^+}} \times \frac{V_{t_3^-}}{V_{t_2^+}} \times \cdots \times \frac{V_{t_n^-}}{V_{t_{n-1}^+}} \times \frac{V_{T^-}}{V_{t_n^+}} - 1, \tag{6.8}$$

where $V_{t_i^-}$ is the value of the portfolio before and $V_{t_i^+}$ the values after the inflow or outflow at the time point t_i, $i = 1, \ldots, n$. This formula, however, can be badly misleading if used for return computations of individual portfolios with cash inflows and outflows. An extreme example illustrates the problem. Suppose your initial investment at time t_0 is \$1 which appreciates ten times to \$10 by time point t_1 at which you invest \$90 more on the portfolio. At time point t_2, the value of the portfolio crashes to \$20, and you decide to cash out the portfolio. If you used the formula in Eq. (6.8) to compute the return on your portfolio, you would get $R_T = (\$10/\$1) \times (\$20/\$100) - 1 = 100\%$, even though the value of your total invested capital of \$91 $(= 1 + 90)$ has declined by \$71 to \$20, i.e., by $20/91 - 1 \approx -78\%$!

We should comment that the return of 100% is a correct measure for a managed fund when the fund manager does not have control over the timing and magnitudes of the cash flows. In such a case, Eq. (6.8) is well justified for computing returns. Also, it works for a buy-and-hold investor in the fund. In the above example, for a buy-and-hold investor, the return from t_0 to T would be the percentage change of the capital on the fund at t_0 (here \$1) which would increase tenfold (900%) by t_1 after which it would crash 80% by T. Thus, the time T value of the holdings would be $\$1 \times (1 + 9) \times (1 - 0.80) = \$2$ for a return of 100%.

To emphasize the importance of timing of the cash inflows and outflows, suppose that the above investor had switched the ordering of the cash inflows. Then again R_T by Eq. (6.8) would be 100%, but the end value of the

investor's holding would be \$180.2 (= (\$90 × 10 + \$1) × 0.20) with a return of 98% (= 180.2/91 − 1) on the 91 dollar total investment.[4]

As the earlier examples show, reliable return computation that controls interim cash flows needs precise tracking, book keeping, and timing of the invested capital. Therefore, even though variants of personal rates of returns have been developed, there is no all-purpose simple formula to compute returns.

One well-known solution is the internal rate of return (IRR). Denoted as Q, the IRR solves the present value of V_T:

$$\frac{V_T}{(1+Q)^T} = V_0 + \sum_{i=1}^{N_1} \frac{A_{t_i}}{(1+Q)^{t_i}} + \sum_{j=1}^{N_2} \frac{B_{s_j}}{(1+Q)^{s_j}}, \tag{6.9}$$

where V_0 and V_T are is the current and end values of the fund, $A_{t_i} > 0$ and $B_{s_j} < 0$ are the cash inflows and outflows at time points t_i and s_j within the holding period $t = 1, \ldots, T$, $i = 1, \ldots, N_1$, $j = 1, \ldots, N_2$ with N_1 and N_2 the number of instances of cash inflows and outflows, respectively. It is interesting that, unlike the time-weighted return, the IRR computation does not depend on the intermediate values of the fund. In the general case, the problem is solved numerically, and it may happen that the solution is not unique.[5]

6.2 Performance Comparison

Portfolio metrics seek to evaluate performance in terms of "reward per unit of risk." The most common performance measure is the Sharpe ratio. This measure scales the asset's excess return (return minus the riskfree rate) by the standard deviation of returns. Other commonly used measures are Jensen's alpha and the Treynor ratio. Additionally, most individual investors and fund managers are concerned with downside risk, wherein negative returns reduce wealth, as well as upside risk associated with lost positive returns. For a pension fund concerned about retirement savings, in which short-term losses can be compensated by long-term investment gains, market risk is the risk that matters. Given reasonably efficient capital markets, managers seek to outperform the general market. In this case, individual investors

[4] The timing issue of cash flows is dealt with in more detail by Fama (1972), Goetzmann et al. (2000), Ferson and Khang (2002), and others.

[5] Fischer and Wermers (2013) give an excellent discussion of IRR.

measure fund performance using a buy-and-hold strategy. Otherwise, timing, as demonstrated in the previous section, is a key dimension of investment performance.[6]

6.2.1 Sharpe Ratio

Sharpe (1966) proposed the following widely used performance metric:

$$\text{sr}_P = \frac{E(R_P) - R_f}{\sigma_P}, \tag{6.10}$$

where $E(R_P)$ is the expected return of the portfolio, R_f is the riskfree rate, and σ_P is the standard deviation of portfolio returns. The motivation for the Sharpe ratio stems from mean-variance analysis and the CAPM. As discussed in Sect. 2.2.3 in Chapter 2, a tangent portfolio on the Markowitz (1952, 1959) investment parabola has the maximum Sharpe ratio. Therefore, portfolios on the ray from the riskfree rate R_f through the tangent portfolio have the highest reward-to-risk ratios at any level of risk (as measured by the standard deviation of returns).

Importantly, the Sharpe ratio can be used to compare portfolios. The rule is the higher the better. Moreover, the ratio can be used to measure how far a particular portfolio is either from its intrinsic maximum potential efficiency or global efficiency. The distance between the tangent portfolio's Sharpe ratio and the Sharpe ratio of some portfolio P represents a measure of its performance. The tangent portfolio is formed from test assets plus portfolio P. The difference in Sharpe ratios can be interpreted in probabilistic terms using the famous Gibbons et al. (1989) (GRS) test:

$$F = \frac{T - N - 1}{N} \cdot \frac{\text{sr}_Q^2 - \text{sr}_P^2}{1 + \text{sr}_P^2}, \tag{6.11}$$

where sr_Q^2 is the square of the (estimated) Sharpe ratio of the tangent portfolio based on the N assets plus portfolio P, sr_P^2 is the square of the Sharpe ratio of portfolio P, T is the number of time-series observations used to estimate the parameters in the Sharpe ratios, and N is the number of test assets. If portfolio P is efficient, the stochastic behavior of F can be approximated by

[6] For more comprehensive discussions of portfolio performance evaluation, see Fischer and Wermers (2013), Ferson (2019, Part V), and citations therein.

the Fisher F-distribution with N and $N - T - 1$ degrees of freedom. Therefore, the p-value tells the distance of portfolio P from the efficient frontier in probabilistic terms. The smaller the p-value, the more inefficient portfolio P is, viz., relative to the efficient frontier (or tangent line). The difference is statistically significant if the p-value is less than 0.05.

The difference of squared Sharpe ratios, or $\text{sr}_Q^2 - \text{sr}_P^2$, has a number of other interesting characteristics. One attribute is the so-called *optimal orthogonal portfolio* (OOP)[7] The OOP with respect to portfolio P indicates the "most mispriced" portfolio that is orthogonal to P, i.e., OOP returns are not correlated with P returns. As such, the square of the Sharpe ratio of the OOP is the difference $\text{sr}_Q^2 - \text{sr}_P^2$. This result follows from the fact that, if we regress the excess returns for N test assets on the excess return of portfolio P (the N test assets need not be same in portfolio P), the optimal portfolio based on the intercepts (or alphas) of the regressions has the Sharpe ratio that equals the difference. Therefore, the GRS test statistic in Eq. (6.11) is a joint test statistic for zero alphas in the regressions. Finally, regarding the GRS statistic, we note that it readily generalizes to multifactor cases.

In spite of the attractive properties of the Sharpe ratio, its usefulness relies heavily on the symmetry of the return distribution. For example, assuming symmetry, ranking portfolios by their standard deviation of returns gives the same result as ranking by downside risk measures.[8] However, by utilizing derivatives in the portfolio, a manager can introduce asymmetry into the return distribution.[9]

6.2.2 Manipulation-Proof Performance Measure

The Sharpe ratio and essentially all portfolio performance measures are subject to artificial manipulation. To avoid this pitfall, Goetzmann et al. (2007) proposed the following manipulation-proof performance measure:

$$\hat{\Theta}(\rho) = \frac{1}{(1-\rho)\Delta t} \log\left(\frac{1}{T}\sum_{t=1}^{T}\left[(1+R_{Pt})/(1+R_{ft})\right]^{1-\rho}\right), \tag{6.13}$$

[7] See Chapter 21 in Ferson (2019).

[8] A popular downside risk measure is the semivariance:

$$sv = \frac{1}{T}\sum_{t=1}^{T}\min(0, R_t - \bar{R})^2, \tag{6.12}$$

where $\bar{R}$ is the sample mean of the returns.

[9] See Goetzmann et al. (2007) for an extreme example.

where log is the natural logarithm, R_{Pt} is the time period t return on the portfolio, T is the total number of time periods, Δt is the annualization factor (measuring the time length between observations in years), and ρ is a hyper parameter that should be selected to make holding of the benchmark optimal to an uninformed manager. The $\hat{\Theta}$ measure seeks to fulfill these requirements: (1) rank portfolios based on investor performances; (2) informationless trading not rewarded; (3) application to both small and large portfolios; and (4) consistent with standard market equilibrium models. The measure is based on power utility theory from which the parameter ρ derives. Typical values are between 2 and 4. A higher ρ penalizes risk more.

According to Goetzmann et al., another manipulation free performance measure is available from Morningstar—for example, the Morningstar Risk Adjusted Measure (MRAR) (Morningstar, 2021). The measure is derived from utility theory with emphasis on downside risk. The mathematical expression for MRAR is:

$$\text{MRAR}(\gamma) = \begin{cases} \left[\frac{1}{T}\sum_{t=1}^{T}(1+ER_{Pt})^{-\gamma}\right]^{-\frac{12}{\gamma}} - 1 \text{ for } \gamma > -1, \gamma \neq 0, \\ \left[\prod_{t=1}^{T}(1+ER_{Pt})\right]^{\frac{12}{T}} - 1 \qquad \text{for } \gamma = 0, \end{cases} \tag{6.14}$$

where $ER_{Pt} = (1+R_{Pt})/(1+R_{ft})-1$. Replacing 12 by $1/\Delta t$ in Eq. (6.14), the relation of $\hat{\Theta}$ and MRAR becomes

$$\hat{\Theta}(\rho) = \log[1 + \text{MRAR}(\rho - 1)]. \tag{6.15}$$

Morningstar uses $\gamma = 2$ ($\rho = 3$).

6.2.3 Treynor Measure

In the Sharpe ratio, the excess return is scaled by total risk measured by the standard deviation of the returns. Treynor (1965) scaled excess returns using systematic beta risk of the CAPM:

$$tr_P = \frac{E(R_P) - R_f}{\beta}. \tag{6.16}$$

This measure depends on CAPM beta, which has been shown to be unrelated to average stock returns by Fama and French (1992, 1993, 1995, 1996) and others. For this reason, the Sharpe ratio is recommended.

6.2.4 Jensen's Alpha

The performance measure, alpha, proposed by Jensen (1969) was originally motivated by the CAPM. It captures the deviation of actual returns from the return predicted by the CAPM. Along with the Sharpe ratio, this measure is a routinely reported performance statistic among institutional funds. Alpha can be estimated using the CAPM market model:

$$R_P - R_f = \alpha_P + \beta_P(R_m - R_f) + u_P, \tag{6.17}$$

where the α intercept is Jensen's alpha. Additionally, as multifactor models have appeared in the literature and practice, alpha intercepts are computed to evaluate their relative validity.

Researchers consider alpha to be a measure of mispricing error in an asset pricing model. The smaller the alpha for a model, the less mispricing error exists due to potential missing factors. If alphas are jointly zero for a large set of test asset portfolios as determined by GRS tests, a model is considered to be valid. In this case, the model can be used to do performance tests of investment portfolios. Investment funds that earn positive (negative) alphas earn positive (negative) abnormal returns and therefore outperform (underperform) the market on a risk-adjusted basis.

Related to GRS tests of alpha, recent work by Kolari et al. (2023) has found that, even if GRS tests indicate jointly zero alphas in test asset portfolios, missing factors can still exist. Novel statistical tests of alphas developed by the authors revealed that only the ZCAPM was found to have no missing factors. The ZCAPM is a new asset pricing model by Kolari et al. (2021) based on the zero-beta CAPM of Black (1972).[10] Thus, for purposes of evaluating investment portfolio performance, we can only recommend the ZCAPM for the estimation of alphas.[11]

6.2.5 Market Timing

As our simple example with fund returns in Eq. (6.8) demonstrated, the timing of cash inflows and outflows affects portfolio performance if the manager controls the cash flows. Treynor and Mazuy (1966) proposed a measure of the timing abilities of mutual funds. The measure is based on

[10] See also other publications by the authors on the ZCAPM, including Liu et al. (2012, 2020), Liu (2013), Kolari et al. (2022a, b, 2023), Kolari and Pynnonen (2023), and Kolari et al. (2024).

[11] The alpha in the ZCAPM can be estimated by computing the average of the residual error terms in the time-series EM regression model.

the quadratic time-series regression using the excess returns for the ith fund:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + \gamma_i(R_{mt} - R_{ft})^2 + u_{it}, \quad (6.18)$$

where γ_i indicates market timing ability if $\gamma_i > 0$, and other terms are as before. Henriksson and Merton (1981) proposed an alternative form as:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + \gamma_i D_t(R_{mt} - R_{ft}) + u_{it}, \quad (6.19)$$

where D_t is a dummy variable with $D_t = 1$ when $R_{mt} - R_{ft} \geq 0$ and zero otherwise. Thus, for up markets when $R_{mt} - R_{ft} \geq 0$, the regression is $R_{it} - R_{ft} = \alpha_i + (\beta_i + \gamma_i)(R_{mt} - R_{ft}) + u_{it}$. Again $\gamma_i > 0$ indicates timing ability. Because $D_t = 0$ when $R_{mt} - R_{ft} < 0$, β_i reflects the downward response of excess return $R_{it} - R_{ft}$ for the ith fund. Thus, ideal ranges for the parameters are $\beta_i < 0$ and $\gamma_i > |\beta_i|$.

6.2.6 Value at Risk

Value at risk (VaR) measures the investment risk of a large loss with low probability within a given time period, e.g., one day. Here we review VaR basics.[12] Typical risk probabilities are 1% and 5%. For example, a 5% VaR of $1 million for a portfolio means that there is 5% probability that the portfolio will lose $1 million or more in one day. In other words, in 1-out-of-20 days there is a risk that the portfolio will lose $1 million or more.

In probabilistic terms, for a long position, a VaR value is the left tail quantile corresponding to probability p of the probability distribution of the portfolio value. Similarly, for a short portfolio, a VaR value is the right tail quantile of probability $1 - p$ of the probability distribution of the portfolio value. Denoting portfolio value by a random variable X and its probability function by $F(x) = P(X \leq x)$, a p probability VaR can be defined for a long position as:

$$q_p = \inf\{q | P(X \leq q) \geq p\}, \quad (6.20)$$

where inf indicates the smallest value of q satisfying the probability bound. For a short position, p is replaced by $1 - p$. In both cases, VaR is a positive number for the loss. Therefore, if the probability distribution is symmetric, both cases can be solved by finding the right tail quantile of the distribution. Finally, rather than deriving the probability limits for the portfolio value

[12] See Holton (2013) and Jorion (2006) for further discussions of VaR.

quantiles, the initial values are derived for portfolio returns. Thereafter the monetary (dollar) value limits can be obtained by simply multiplying the return quantile. In the above example, a $1 million loss in a $10 million portfolio would equal a 10% loss in value, but only a 1% loss in a $100 million portfolio. Whether either of these corresponds to a 5% probability depends on the return distribution.

Starting with the simplest case by assuming that returns are normally (Gaussian) distributed with mean μ and standard deviation σ, the solution for one period (day) p% VaR is simply

$$\text{VaR}_p = V \times (\mu + z_{1-p}\sigma), \tag{6.21}$$

where V is the current value of the investment, and z_{1-p} is the standard normal quantile satisfying $\Phi(z_{1-p}) = 1 - p$ with $\Phi(\cdot)$ the standard normal probability function. As daily mean returns are negligible compared to the magnitude of the standard deviation, the key parameter is the standard deviation, and the less important mean is often ignored by assuming $\mu = 0$. This setup simplifies multiperiod VaR computations so that the T-day VaR becomes

$$\text{VaR}_p(T) = \sqrt{T}\,\text{VaR}_p, \tag{6.22}$$

i.e., the familiar square root rule of the volatility.

Since the assumption of a constant standard deviation (volatility) in Eq. (6.21) is unrealistic, substantial (downward) bias in the magnitude of potential losses is introduced. A stylized fact of financial returns is that volatility is clustered, which implies predictability of volatility. Therefore, rather than assuming a constant volatility, risk estimates can be improved by utilizing the predictability feature. The clustering volatility literature is vast and growing. For practical purposes, the most successful approach for now is the generalized autoregressive conditional heteroskedasticity (GARCH) approach by Engle (1982) and Bollerslev (1986).

Extending work by these researchers, the RiskMetrics variance model was developed by J. P. Morgan in 1992 at the request of Chairman Sir Dennis Weatherstone. The model adopts an exponentially weighted moving average (EWMA) approach that is a special GARCH model. Because of the negligible magnitude of the mean return compared to the volatility in return horizons of three months or less, the mean return is assumed to be zero. Given $m + 1$

one-day historical returns, the EWMA conditional variance becomes

$$\sigma_t^2 = \frac{1-\lambda}{1-\lambda^{m+1}} \sum_{i=0}^{m} \lambda^i r_{t-i}^2, \tag{6.23}$$

where $0 < \lambda < 1$ is the smoothing parameter, or *decay factor*. *RiskMetrics Classics* (RM1994) found that on average $\lambda = 0.94$ produces a very good forecast for 1-day volatility, and $\lambda = 0.97$ gives good estimates for 1-month volatility. Attractive characteristics of the EWMA variance is that it depends only on one parameter, or λ, and it preserves the square root rule in Eq. (6.22). Also, given a fixed common λ for each of n return series, the $n \times n$ (conditional) covariance matrix can be readily computed.[13]

Researchers have observed that EWMA weighting, which works reasonably well in short-term VaR computations, has not been useful in the long term. To improve the 1994 methodology, RiskMetrics 2006 (RM2006) was created.[14] Focusing on volatility, the underlying idea is similar to EWMA in Eq. (6.23) or

$$\sigma_t^2 = \sum_{i \geq 0} \lambda(i) r_{t-i}^2, \tag{6.24}$$

where the weights $\lambda(i) > 0$ with $\sum_{i \geq 0} \lambda(i) = 1$ designed such that they better match generic stylized facts of financial time series.

It is well known that volatility clustering implies long-range dependency (long memory) among absolute (and quadratic) returns. Empirically speaking, long-range dependency is manifested in autocorrelations that do not decay exponentially as implied by the standard EWMA approach in Eq. (6.23).

Figure 6.1 illustrates the case with semilog plots of absolute S&P 500 index and CRSP index daily log return autocorrelations up to the 252 lags (about one year of trading days) for the sample period from July 1, 1965 to December 30, 2022.[15] The fitted trend lines clearly show that memory structure measured in terms of autocorrelations does not decay exponentially (linear in the semilog scale), whereas the logarithmic well captures the structure in the overall decay. In this respect, RM2006 designers have conjectured that the RM1994 exponential weighting schemes in EWMA do not

[13] For technical details, see Mina and Xiao (2001).

[14] This methodology is described in Zumbach (2007).

[15] Data source: CRSP and https://finance.yahoo.com.

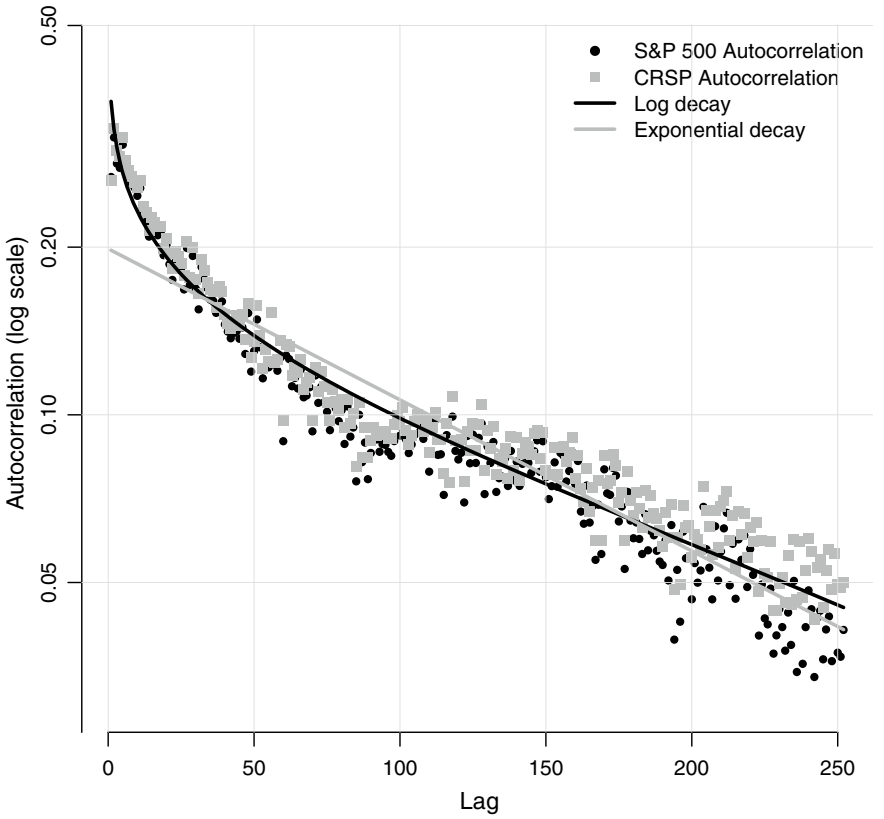


Fig. 6.1 S&P 500 index absolute return autocorrelations for lags 1–252 (The sample period is from July 1, 1965 to December 30, 2022. The black dots indicate the magnitudes of the autocorrelations, the gray straight line corresponds to a fitted exponential decaying trend, and the black line is a fitted logarithmic decaying trend of the autocorrelations. *Data sources* https://finance.yahoo.com and CRSP)

properly capture the long-range dependence structures in financial return volatilities. Therefore, especially long-term volatility forecasts may be systematically biased due to not reflecting systematic long memory in the historical data. Consequently, RM2006 improves the methodology by adjusting the weights, or $\lambda(i)$s, in Eq. (6.24) and weights, or $\lambda(n, i)$s, in the n period forecast variance as

$$\sigma_t^2(n) = n \times \sum_{i \geq 0} \lambda(n, i) r_{t-i}^2 \quad \text{with } \lambda(n, i) > 0 \text{ and } \sum_{i \geq 0} \lambda(n, i) = 1, \tag{6.25}$$

such that the resulting volatility forecasts take into account long memory.[16] Similar to RM1994, the advantage of the RM2006 is that it is a generic all-purpose solution.

Panels A, B, and C in Figs. 6.2 and 6.3 show daily returns and 1-day 5% VaR using RM1994 (Fig. 6.2) and RM2006 (Fig. 6.3) for the S&P 500

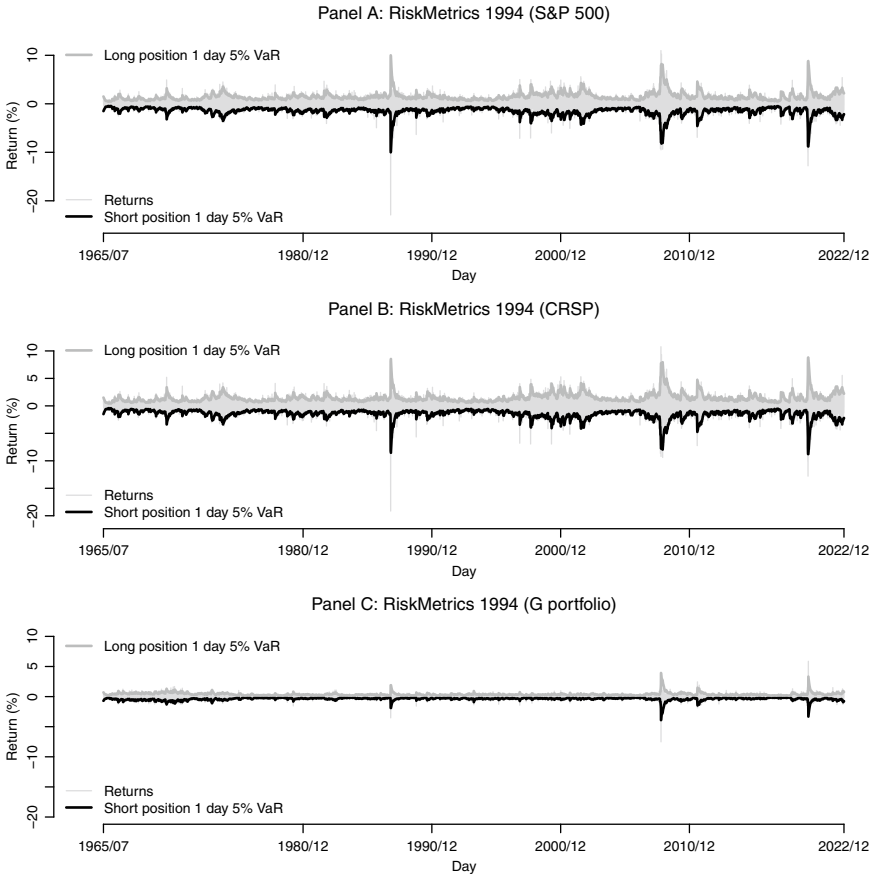


Fig. 6.2 S&P 500 index, CRSP index, and minimum variance portfolio (*G*) returns with 1-day 5% RM1994 VaR bounds for long and short positions (The sample period covers daily returns from July 1, 1965 to December 31, 2022 in which the first 252 days (until June 30, 1965) are used to estimate the first RiskMetrics variances for the first VaR on July 1, 1965. Thereafter the procedure is continued day by day until the end of the sample period. For the last VaR on December 30, 2022, the volatility is estimated from returns ending December 29, 2022. *Data sources* https://finance.yahoo.com and CRSP)

[16] For details on generic weighting schemes, see Appendix B in Zumbach (2007).

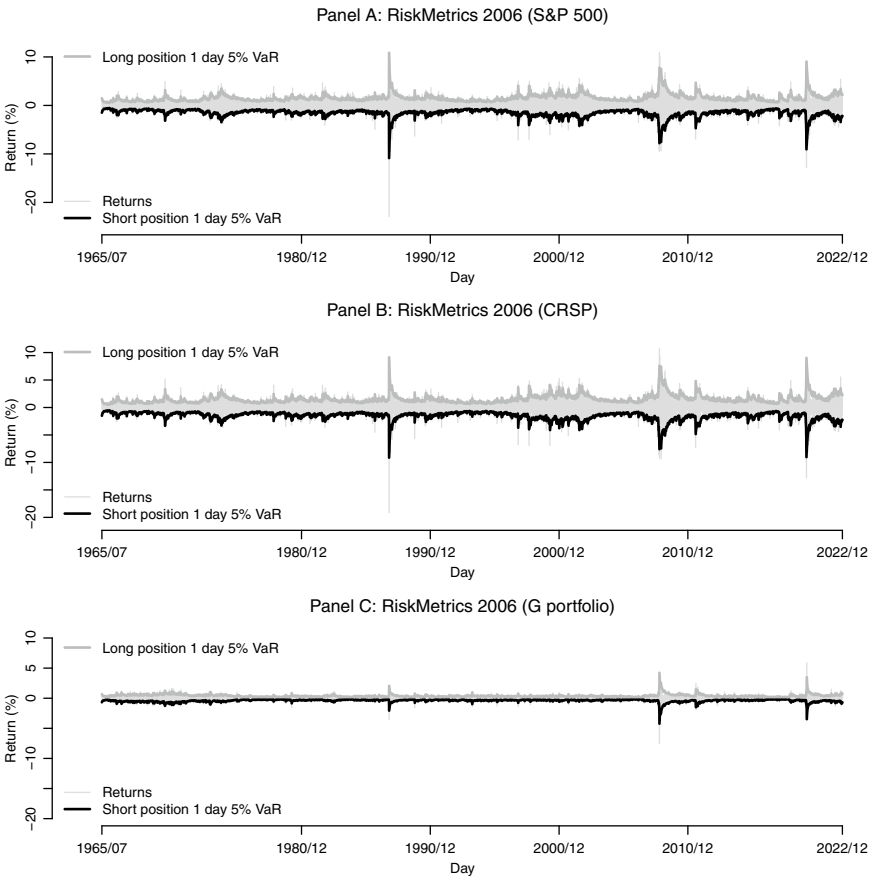


Fig. 6.3 Returns for the S&P 500 index, CRSP market index, and minimum variance portfolio (*G*) (see Chapter 7) with 1-day 5% RiskMetrics 2006 VaR bounds for long and short positions (The sample period covers daily returns from July 1, 1965 to December 30, 2022 of which the first 252 days (until June 30, 1965) are used to estimate the first RiskMetrics variances for the first VaR on July 1, 1965. Thereafter the procedure is continued day by day until the end of the sample period. For the last VaR on Dec 30, 2022, the volatility is estimated from returns ending December 29, 2022. *Data source* https://finance.yahoo.com and CRSP)

index, CRSP index, and daily updated minimum variance portfolio (*G* portfolio) with standard normal distribution quantiles. The latter *G* portfolio is constructed by the authors in forthcoming Chapter 7. The sample period covers the first trading day of July 1964 to last trading day of December 2022.

In terms of the S&P 500 and CRSP indexes, as expected, there are no discernible differences in the 1-day VaRs. By contrast, striking differences appear in the volatilities and related VaR values of the *G* portfolio in Panel

Table 6.1 Empirical exceedance percentages of RM1994 and RM2006 1-day 5% VaR limits for long and short positions in the S&P 500 index, CRSP index, and minimum variance *G* portfolio (see Chapter 7) in the holding period from July 1, 1964 to December 30, 2022

	S&P 500 index		CRSP index		*G* portfolio	
	Long position (lower tail) (%)	Short position (upper tail) (%)	Long position (lower tail) (%)	Short position (upper tail) (%)	Long position (lower tail) (%)	Short position (upper tail) (%)
Panel A: Normal distribution						
RM1994	4.61	4.41	4.71	4.21	3.58	4.88
RM2006	3.97	3.50	3.97	3.39	3.05	3.63
Panel B: *t*-distribution with 5 degrees of freedom						
RM1994	2.22	1.61	2.22	1.46	1.89	1.66
RM2006	1.59	1.07	1.69	0.97	1.40	1.04

C of Figs. 6.2 and 6.3. Since the *Y*-axes are in the same scale, these figures illustrate the extremely low riskiness of the *G* portfolio.

Table 6.1 provides realized 1-day 5% VaR values for each portfolio using the normal distribution in Panel A and the *t*-distribution with five degrees of freedom in Panel B.

The realized exceedance values in Panel A of Table 6.1 indicate that RM1994 based on the normal distribution is doing a fairly good job by matching a closer estimate of the 5% value than the RM2006 measure.[17] Both S&P 500 and CRSP market indexes tend to overstate severely the 1-day 5% VaR with $t(5)$ distribution (Panel B). Numerically, in both cases, it seems that RM2006 tends to be more conservative, thereby over exaggerating the VaR values. This result is particularly the case for the $t(5)$-distribution (Panel B).

Summary statistics in Table 6.2 further document differences in RM1994 and RM2006 VaR standards. In terms of these statistics, the differences between the S&P 500 index and the CRSP index are not as dramatic as might be expected by the thresholds in Table 6.1. For example, the average S&P 500 index realized 1-day 5% VaR measured by RM1994 with normal

[17] The 95% confidence interval for the 5% VaR here is (0.046%, 0.054%) which implies that, for the normal distribution RM1994 1-day 5% VaRs, short positions in the S&P 500 index and CRSP market index as well as long position in portfolio *G* significantly overstate the realized VaR threshold. That is, for example, a long position in the *G* portfolio with realized exceedance of 3.58% indicates that, for a $100 million investment, the realized 1-day 5% VaR would have been $3.58 million rather than $5 million stated by the VaR. However, these findings are not losses incurred, but rather the levels of the realized VaR values for different positions (long/short) of the portfolios.

Table 6.2 Summary statistics for 1-day 5% VaR estimates for a $100 investment (long or short) in the S&P 500 index, CRSP market index, and portfolio *G* in the holding period from July 1, 1965 to December 30, 2022

	Min	Q_{25}	Median	Mean	Q_{75}	Max
Panel A: RM1994, Normal distribution						
S&P 500	0.44	1.02	1.30	1.52	1.76	9.99
CRSP	0.42	0.99	1.25	1.49	1.72	8.77
G portfolio	0.13	0.26	0.33	0.39	0.46	3.90
Panel B: RM1994, *t(5) distribution*						
S&P 500	0.55	1.25	1.60	1.86	2.16	12.23
CRSP	0.51	1.22	1.53	1.82	2.11	10.75
G portfolio	0.16	0.32	0.41	0.48	0.56	4.78
Panel C: RM2006, Normal distribution						
S&P 500	0.54	1.08	1.37	1.56	1.81	10.85
GRSP	0.52	1.05	1.31	1.52	1.77	9.12
G portfolio	0.15	0.28	0.34	0.40	0.46	4.22
Panel D: RM2006, *t*(5) distribution						
S&P 500	0.66	1.32	1.68	1.91	2.22	13.29
CRSP	0.63	1.29	1.61	1.87	2.17	11.18
G portfolio	0.19	0.34	0.42	0.50	0.57	5.17

The VaR values are based on the (standard) normal distribution. By the symmetry of this distribution, the statistics are the same for both long and short positions. The values can be interpreted as 1-day 5% VaR dollar values per $100

distribution equals $1.52 per $100 investment, whereas based on RM2006 it is almost unchanged at $1.56 per $100 investment.

Lastly, the VaR statistics in Table 6.2 confirm the visual information conveyed in Fig. 6.3 about the low VaR risk of the *G* portfolio. Utilizing the normal distribution-based VaR, Panel C of the table shows that a 1-day 5% VaR risk for the *G* portfolio is on average $0.40 per $100 investments, while for the S&P 500 and CRSP indexes it is almost four times higher at $1.56 and $1.52, respectively.

6.2.7 Drawdown

Another downside risk concept that has gained popularity among practitioners is of the *drawdown* on an investment. This return concept can be mathematically defined as:

$$r_t = -\max_{s \le t}\left(1 - \frac{P_t}{P_s}\right), \tag{6.26}$$

where P_t is the time t price of the asset.[18] That is, drawdown is the maximum percentage loss (often expressed as a positive number) experienced by the investor on a buy-and-hold investment if bought at the worst time point s when the stock was at its peak during the investment horizon ending at t. Thus, similar to VaR, drawdown seeks to give the investor some idea about downside risk.

Important concepts related to the drawdown are: trough, maximum drawdown, duration, and recovery. Even though the meaning of these concepts varies somewhat among practitioners, we define them as follows:

- *Drawdown*: The return defined in Eq. (6.26) at time t.
- *Trough*: The bottom return (or value) during the drawdown.
- *Maximum drawdown*: (MDD) the return from the peak to the trough, i.e., the maximal negative return $P_{\text{trough}}/P_{\text{peak}} - 1$ during the drawdown. This is also called the *depth* of the drawdown.
- *Duration*: The lifetime of the drawdown, i.e., the time distance from the peak to the next time the peak is reached. This is also called *length* of the drawdown. See also *recovery* below.
- *Recovery*: The time until the price has recovered from the trough back to the peak value. Sometimes recovery is synonymous to *duration*.

Figure 6.4 illustrates the concepts.

Recovery from losses is an important performance metric. Given a bad investment with substantial losses, how long does it take to recover your funds? As an extreme example, suppose that you made a buy-and-hold investment in the Japanese stock market Nikkei index at 38,916 on October 1, 1989. At that time, the market was in a bubble. The bubble popped and the Nikkei steadily declined over time to only 7,972 on January 1, 2003. As of August 4, 2023, the Nikkei was 32,193 and had recovered most of the earlier losses. However, after more than 32 years, you would still be down 17.3%! Figure 6.5 tells the story. The dramatic crash of over 80% from the peak and ongoing duration of the recovery highlights the problems of worst case scenarios and risks lurking in even well-developed economies with large stock markets.

[18] Sometimes drawdown is defined in terms of log returns:

$$r_t = -\max_{s \le t} \log(P_s/P_t) = \min_{s \le t} \log(P_t/P_s),$$

where log is the natural logarithm. Also, drawdown is defined in monetary units at times.

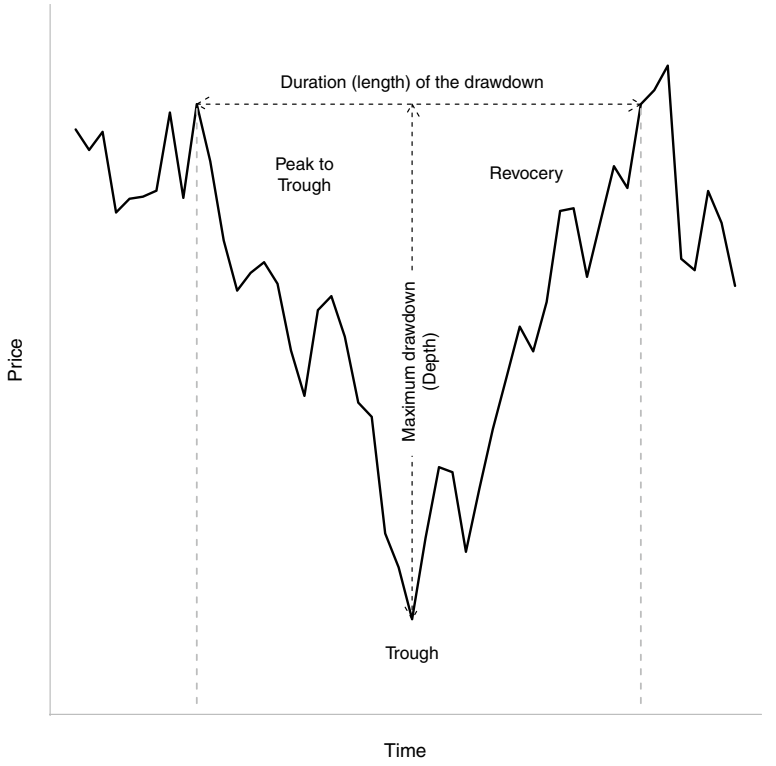


Fig. 6.4 Drawdown, duration (length), trough, maximum drawdown (depth), and recovery

Of course, nobody in the market is immune to major market downturns. History shows that serious drawdowns have occurred in global stock markets in response to the 2000 technology bubble, the 2008–2009 financial crisis, and the 2020 COVID-19 pandemic. From a professional investment management perspective, it is crucial to measure drawdowns, monitor their progress, and find ways to mitigate their losses and speed recovery. To measure drawdown characteristics of stock portfolios in forthcoming chapters, we utilize the R package PerformanceAnalytics.[19]

[19] https://cran.r-project.org/web/packages/PerformanceAnalytics/index.html. See also https://cran.r-project.org/web/packages/tidyquant/vignettes/TQ05-performance-analysis-with-tidyquant.html. R is a free software available at https://www.r-project.org.



Fig. 6.5 Nikkei 225 stock index with 1989 bubble, drawdown, and recovery from January 1985 to August 2023

6.3 Summary

Portfolio performance measurement is a complicated task. Multiple measures have been proposed by investment academics and professionals. In this chapter, we have reviewed some of the most popular performance measures. Of these, the Sharpe ratio is the most widely used. In view of possible artificial manipulation of Sharpe ratios, we reviewed manipulation-proof performance measures. Also, many investors use alpha to evaluate portfolio performance. Alpha indicates abnormal risk-adjusted return performance. However, as we discussed, a valid asset pricing model must be used to accurately estimate alpha. If missing factors exist in a model, alpha will incorporate mispricing error that contaminates it as a performance metric. Recent research has shown that the ZCAPM of Kolari et al. (2021) contains no missing factors and, therefore, is a valid asset pricing model for use in performance measurement. Forthcoming chapters rely upon the ZCAPM to build high-performing stock portfolios.

We discussed value at risk (VaR) measures that take into account downside risk. VaR estimates the potential loss of a portfolio at a given risk level and investment horizon. In this respect, the RiskMetrics variance model represents the state of the art. We demonstrated the usefulness of RM1994 and RM2006 to assess the riskiness of the S&P 500 index, CRSP index, and

minimum variance portfolio G (developed by the authors in forthcoming Chapter 7). Our results showed the portfolio G is much lower risk than these general market indexes.

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Bollerslev, T. 1986. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* 31: 307–327.

Engle, R.F. 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica* 50: 987–1007.

Fama, E.F. 1972. Components of investment performance. *Journal of Finance* 27: 551–567.

Fama, E.F., and K.R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427–465.

Fama, E.F., and K.R. French. 1993. The cross-section of expected returns. *Journal of Financial Economics* 33: 3–56.

Fama, E.F., and K.R. French. 1995. Size and book-to-market factors in earnings and returns. *Journal of Finance* 50: 131–156.

Fama, E.F., and K.R. French. 1996. The CAPM is wanted, dead or alive. *Journal of Finance* 51: 1947–1958.

Ferson, W. 2019. *Empirical asset pricing: Models and methods*. Cambridge MA: The MIT Press.

Ferson, W., and K. Khang. 2002. Conditional performance measurement using portfolio weights: Evidence for pension funds. *Journal of Financial Economics* 65: 249–282.

Fischer, B.R., and R. Wermers. 2013. *Performance evaluation and attribution of security portfolios*. San Diego, CA: Academic Press.

Gibbons, M.R., S.A. Ross, and Jay Shanken. 1989. A test of the efficiency of a given portfolio. *Econometrica* 57: 1121–1152.

Goetzmann, W.N., J. Ingersoll, and Z. Ivković. 2000. Monthly measurement of daily timers. *Journal of Financial and Quantitative Analysis* 35: 257–290.

Goetzmann, W., J. Ingersoll, M. Spiegel, and Ivo Welch. 2007. Portfolio performance manipulation and manipulation-proof performance measures. *Review of Financial Studies* 20: 1503–1546.

Henriksson, Roy D., and Robert C. Merton. 1981. On market timing and investment performance. II. Statistical procedures for evaluating forecasting skills. *Journal of Business* 54: 513–533.

Holton, G.A. 2013. *Value-at-risk: Theory and practice*, 2nd ed. Cambridge, MA: Academic Press.

Jensen, M.C. 1969. Risk, the pricing of capital assets, and the evaluation of investment portfolios. *Journal of Business* 42: 167–247.

Jorion, P. 2006. *Value at risk: The new benchmark for managing financial risk*, 3rd ed. New York, NY: McGraw-Hill Companies Inc.

Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham: Palgrave Macmillan.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2022a. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*. Available at: https://www.mdpi.com/1911-8074/15/3/137. Reprinted in Kolari, J.W., and S. Pynnonen, eds. 2022. *Frontiers of asset pricing*. Basel: MDPI.

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022b. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. Testing for missing asset pricing factors. Paper presented at the Western Economic Association International, San Diego, CA.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Liu, W. 2013. A new asset pricing model based on the zero-beta CAPM: Theory and evidence. Doctoral dissertation, Texas A &M University.

Liu, W., J.W. Kolari, and J.Z. Huang. 2012. A new asset pricing model based on the zero-beta CAPM market model (CAPM). Presentation at the annual meetings of the Financial Management Association, Best Paper Award in Investments, Atlanta, GA, October.

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. Return dispersion and the cross-section of stock returns. Presentation at the annual meetings of the Southern Finance Association, Palm Springs, CA, October.

Markowitz, H. M. 1952. Portfolio selection, *Journal of Finance* 7: 77–91.

Markowitz, H. M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: John Wiley & Sons.

Mina, J., and J.Y. Xiao. 2001. Return to RiskMetrics: The evolution of a standard. Available at: https://www.msci.com/www/research-report/return-to-riskmetrics-the/019088036, visited June 8, 2023.

Morningstar 2021. The Morningstar Rating^{TM} for funds. Morningstar, Inc., Morningstar Methodology, August 2021. Paper available online at https://www.morningstar.com/content/dam/marketing/shared/research/methodology/771945_Morningstar_Rating_for_Funds_Methodology.pdf, visited on January 4, 2024.

Sharpe, W.F. 1966. Mutual fund performance. *Journal of Business* 39: 119–138.

Treynor, Jack L. 1965. How to rate management of investment funds. *Harvard Business Review* 43: 63–75.

Treynor, Jack L., and M. Mazuy. 1966. Can mutual funds outguess the market? *Harvard Business Review* 44: 131–136.

Zumbach, G. 2007. The RiskMetrics 2006 methodology. Available at SSRN: https://ssrn.com/abstract=1420185 or https://doi.org/10.2139/ssrn.1420185, visited July 2023.

Part V

Building Stock Portfolios with the ZCAPM



7

Building the Global Minimum Variance Portfolio *G*

7.1 Previous Literature

A number of econometric problems have confronted researchers attempting to construct portfolio *G*. Previous studies built *G* using historical returns and their risks. The first problem that arises is estimation error in mean returns in the future. Historical returns provide poor proxies for expected returns. Estimation errors in the sample mean of stock returns are large, such that they are bad estimates of the population mean. For this reason, as well as the fact that they are not needed to estimate *G*, Jagannathan and Ma (2003) dropped mean returns from their global minimum variance portfolio analyses.

Focusing on the covariance structure of security returns (and excluding expected returns from the analyses) is not without its own problems, however. Researchers ran into difficulties estimating the variance-covariance matrix of Markowitz. In efforts to mitigate econometric challenges, many studies used shrinkage estimators and related non-Bayesian methods to improve covariance matrix estimations.[1] Even though these quantitative methods improved matters, they still tended to form optimized portfolios with larger (smaller)

[1] See Jobson and Korkie (1980, 1981), Jorion (1985, 1986), Frost and Savarino (1986), Haugen and Baker (2012), Best and Grauer (1991), Chan Karceski and Lakonishok (1999), Ledoit and Wolf (2004, 2020), Clarke et al. (2006, 2011), Maillet Tokpavi and Vaucher (2015), Ledoit and Wolf (2017, 2020), Reh et al. (2022), and citations therein. A good review of this extensive literature is contained in Ledoit and Wolf (2022). Tangentially related with the present analyses in this chapter, another study by De Nard et al. (2021) incorporated the factor structure of asset pricing models, such as the Fama and French (2015) five-factor model and principal component latent factors, in the estimation of large-dimensional covariance matrices of stock returns. Thus, asset pricing models have been used in an effort to improve covariance estimates.

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weights on securities with high (low) returns, low (high) return variances, and negative (positive) return correlations with other securities.

Michaud (1989) critically pointed out that the aforementioned securities tend to have the largest estimation errors. Citing stock return evidence by Jobson and Korkie (1981), he argued that mean-variance (MV) optimization tends to magnify the effects of errors in data inputs and thereby can result in portfolios that underperform simple equal-weighting schemes in out-of-sample portfolio tests.[2] For this and other reasons, he recommended that active equity managers take into account practical considerations in building investment portfolios, such as a benchmark portfolio, transaction costs, liquidity constraints, sector and industry constraints, and other directly controllable portfolio characteristics

Relevant to portfolio construction, Michaud observed that optimization could be performed in terms of residual returns defined as actual returns minus returns adjusted for systematic risk in an asset pricing model (e.g., beta risk in the CAPM). In this case, the efficient frontier minimizes residual return variance per unit residual return, not realized return variance per unit realized return. The latter residual return is computed as alpha or the intercept term in an asset pricing model.[3]

Black and Litterman (1992) discussed problems of optimized portfolios. When short positions are allowed, a large number of short positions is generated. Disallowing shorts, the algorithm generated corner solutions with zero weights for many assets and large weights for small capitalization stocks. Given problems in estimating expected returns from historical returns, they recommended that the optimal portfolio in equilibrium is market-value weighted (excluding potential currency risks from international investments). This benchmark market portfolio enables measurement of risk also. A CAPM investor would hold this portfolio as their optimal return per unit risk choice.

Pástor (2000) and Pástor and Stambaugh (2000) considered how different asset pricing models affect the portfolio choices of mean-variance-optimizing investors. They considered the CAPM, Fama and French's (1992, 1993) three-factor model based on betas of risk factors, and Daniel and Titman's (1997) characteristics model based on size and value variables (but not betas). Interestingly, given uncertainty about the validity of models to accurately

[2] See DeMiguel et al. (2009), who found that efficient portfolios on the frontier did not outperform naive equal-weighted portfolios in out-of-sample tests. Poor out-of-sample performance based on sample means and sample covariance matrices have been reported by Jobson and Korkie (1980, 1981), Frost and Savarino (1986), Best and Grauer (1991), and Black and Litterman (1992).

[3] He cited Rudd and Rosenberg (1979) for this asset pricing model approach to portfolio optimization.

price assets, they inferred that optimal portfolios are similar across models. In a world with a number of alternative models, they believed that investors should use some probabilistic weighted average of all the models to estimate expected returns of an optimal portfolio. How to compute such probabilities and combine different asset pricing models was left for future research.

Using the CAPM to build portfolios, Clarke et al. (2011) found that *G* portfolios containing long positions are entirely composed of low beta risk stocks as measured by the CAPM. They inferred that this emphasis on low beta stocks is due to the well-known empirical fact that the Security Market Line (SML) between stock returns and beta risk was flatter than expected by the CAPM (or even strictly flat). In effect, low beta risk stocks have higher than expected returns, thereby boosting their returns per unit systematic risk. This straightforward method of constructing *G* utilizes optimal security weights based on the CAPM in terms of the covariance matrix of stock returns and value-weighted market index returns. Only securities with relatively low betas are included in the optimal portfolio. No knowledge of expected returns is needed, thereby sidestepping issues related to estimating expected returns discussed above. Finally, they found that idiosyncratic risk was not generally useful in forming *G* portfolios, as systematic beta risk was the main determinant of minimum variance portfolio risk.

Extending previous literature, we next turn to our ZCAPM approach to optimally weight securities and build portfolio *G*. Our approach has some similarities to existing CAPM approaches but is different in many respects. Like Clarke, Silva, and Thorley, we emphasize low beta stocks in our *G* portfolio; however, our rationale is grounded in the theoretical ZCAPM, rather than the poor empirical results in researchers' tests of the CAPM wherein the SML was flatter than expected by theory. As we will show in forthcoming results, the performance of our *G* portfolio based on individual U.S. stocks outperforms the overall stock market in terms of returns per unit risk and, therefore, represents an optimized portfolio that contributes to our understanding of the global minimum variance portfolio in particular and the mean-variance investment parabola in general. Naturally, it should be of interest to securities companies building and selling minimum variance portfolios in the financial marketplace.

7.2 Global Minimum Variance Portfolio

We take advantage of the theoretical ZCAPM to develop an empirical approach for building portfolio G. As mentioned in the previous section, Clarke et al. (2011) formed portfolio G using only low beta stocks. To explain the inclusion of low beta stocks, the authors cited well-established empirical findings by researchers that the Security Market Line (SML) is much flatter than theory would predict. For this reason, low beta stocks have strong average return performance relative to other higher risk stocks. Like them, our portfolio G tends to emphasize low beta stocks but for different reasons. We explain our preference for low beta stocks using the theoretical ZCAPM.

Figure 7.1 shows a graph of the mean-variance investment parabola. Portfolio G is plotted at the global minimum variance point at the vertex of the parabola. According to the theoretical ZCAPM reviewed in Chapter 4, the parabola has an architecture with beta risk increasing from left to right. Orthogonal to beta risk, zeta risk increases or decreases above or below the dashed axis of symmetry (i.e., positive zeta risk securities lie above the axis and negative zeta risk assets lie below the axis). Here beta risk is not related to the market portfolio as in the CAPM; instead beta risk is associated with changes in average market returns. Zeta risk is related to the cross-sectional standard deviation of returns of all securities in the market (or return dispersion). Unlike beta risk, which generally is positive but can be negative in sign for some securities, the population of securities can be divided between those above or below the axis of symmetry that correspond to positive or negative zeta risk securities.

It is important to notice that portfolio G does not contain any zeta risk. This property of G is obvious from the diagram in Fig. 7.1. By the geometry of the parabola, it is clear that G has low beta risk. Higher beta risk stocks are located to the right of G in the parabola. What we seek is the minimum variance portfolio among those portfolios that lie somewhere along the axis of symmetry. To find portfolio G, a critical piece of information is the residual error variance from a valid asset pricing model. We want to over-(under-) weight those securities that have low (high) residual error volatility. High residual error volatility for a security suggests that it has high idiosyncratic volatility and, therefore, does not fit the asset pricing model for the most

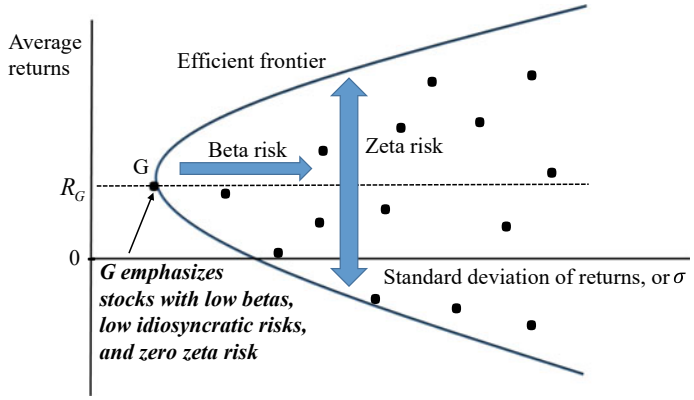


Fig. 7.1 The ZCAPM approach to finding the global minimum variance portfolio *G*

part. If a model is invalid, it will have high residual error variances for securities.[4] By contrast, a valid model will have lower residual errors and more accurately price securities.

We employ the empirical ZCAPM as the most valid asset pricing model among available models in the finance literature. Other models could be used in place of the ZCAPM but accurate estimates of portfolio G would be diminished.[5]

7.2.1 Mechanics of Building Portfolio *G*

We next turn to the basic mechanics for constructing portfolio G. To get started, some assumptions and definitions are needed concerning the global minimum variance portfolio G. Assume that there are N trading securities in the market (where N is large but finite) and no market frictions exist. Portfolios with aggregate weights $\sum_{i=1}^{N} w_i = 1$ are net long investment portfolios (but can contain short positions), and portfolios with $\sum_{i=1}^{N} w_i = 0$ are zero-investment (pure long-short) portfolios. The minimum variance portfolio G is the one among all net long portfolios with minimum variance.[6]

[4] In this case, residual error would include the alpha or intercept in the model that is commonly referred to as mispricing error. If a model has no alpha mispricing error, it is considered to be complete with no missing factors.

[5] Kolari et al. (2021) showed in Chapter 9 that multifactor risks in other asset pricing models are not useful in creating portfolios that exhibit out-of-sample return performance consistent with different levels of risk loadings.

[6] A net long position can contain short portfolios but total weights of securities sum to one.

We propose an empirically feasible way to construct an approximate minimum variance portfolio G. Of course, if the covariance matrix of N securities is ex ante known, the weights for these securities are fixed. Hence, building G is easy. However, in the real world, a gap exists between the in-sample estimated covariance matrix and the out-of-sample true matrix. Why does this happen?

First, for each individual security, idiosyncratic noise dominates its performance, as opposed to systematic risk factors. The situation worsens when data are collected for shorter intervals, e.g., from monthly to daily and intradaily returns. Higher frequency data causes the estimated in-sample covariance matrix for N individual securities to become unstable across time. Second, the number of securities N is large. To avoid singularity of the inverse of covariance matrix, the number of observations for each security should be larger than N. This constraint forces the use of long historical spans of data to estimate the covariance matrix. For example, if we use daily data, and assume the number of securities is 3,000, we need data going back as far as 12 years to invert the covariance matrix. It is unreasonable to believe that a stock's performance 10 years ago has a strong impact on its performance in the next month.

7.2.2 Second Stage Portfolios

Due to problems in estimating the covariance matrix of security returns, in order to build portfolio G, we propose the initial formation of what we call *second stage portfolios*. These portfolios are later combined to form the final G portfolio. Each of the second stage portfolios consists of a considerable number of securities. The securities in each portfolio share similar systematic risk properties, wherein idiosyncratic risk (noise) is actively de-emphasized and systematic risk is emphasized. To reach our empirical goal, we place the following constraint on G formed from a large number of securities:

> For a submarket with $M(M < N)$ securities, each of which shares a similar level of systematic risk, G is the minimum variance portfolio in the underlying submarket. It is constructed following some empirical process with weights for securities satisfying $\sum_{i=1}^{M} w_i^{ori} = 1$. If we randomly pick M-1 securities from the submarket, the normalized original weights $w_i^{ori}, i = 1, \cdots, M-1$ still form the minimum variance portfolio based on the same empirical process for these underlying $M-1$ securities, i.e., $\sum_{i=1}^{M-1} w_i^{nor} = 1$ with $w_i^{nor} = w_i^{ori} / \sum_{i}^{M-1} w^{ori}$. We refer to the original minimum variance portfolio with

M securities as the nontrivial portfolio. All other minimum variance portfolios, which violate the above constraints, are trivial portfolios.

We initially seek to identify the nontrivial minimum variance portfolio at any given systematic/common risk level. The nontrivial framework proposed here seeks to make the minimum variance portfolio more stable across time. In this regard, adding one more security to the underlying minimum variance portfolio to form a new minimum variance portfolio should not fundamentally change the original weight structure of the minimum variance portfolio. To construct the weight structure, we employ a valid asset pricing model.

Before proceeding, we need to clarify the requisite properties of the global minimum variance portfolio *G*. First, market risk is multi-dimensional in nature. Since higher risk means higher asset variance, construction of a portfolio with less variance requires reducing its risk. Second, given that systematic and idiosyncratic risks exist in the market, they both must be considered in risk minimization. Third, and last, we propose that *G* should only be correlated with the major component of market risk (e.g., a good proxy is the market index and associated beta risk) and neutral (or weakly correlated) with respect to other potential risk components, such as size, value, momentum, return dispersion, etc.[7] This last property of *G* implies that it should not consist of any other risk components than the major risk component.

The last property can be proved via contradiction. Assume that the original *G* portfolio consists of one other risk component. We can define the original *G* portfolio as the sum of the residual *G*' portfolio plus the long-short portfolios. These long-short portfolios mimic all other risk components. Since the long-short portfolios are not contained in the residual *G* portfolio, they should have zero correlation with residual *G* (i.e., residual portfolios are independent of the major component of market risk). As such, the residual *G* portfolio has smaller variance than the original *G* portfolio, which contradicts our original assumption. Thus, *G* contains only major market risk and no other risk components.

We introduce two additional assumptions to tie our derivation more closely to real world markets. First, we assume that the ex ante covariance matrix is unknown. It can be described as a *random matrix*, wherein the estimated covariance matrix for individual securities is not directly related to the out-of-sample matrix. This assumption is sensible as real markets have

[7] As a counterexample, an equal-weighted index is neutral to some risks (e.g., size) but is overweighted with high variance stocks; hence, it is not a viable candidate for the minimum variance portfolio.

thousands of securities with dynamically complex interactions. Second, we assume that investors only form portfolios based on their views about market risks. For example, if an investor holds a long-short portfolio, the portfolio should be formed based on one or more market risk components. Common examples are the well-known size, value, and momentum portfolios.

Under these assumptions, we argue that global minimum variance portfolio G is dependent on the asset pricing model. That is, the portfolio is constructed using the risk specification of the model. If there are no systematic risks, such that all assets move independently over time, portfolio G is a simple long only portfolio with the weight for each security equal to the inverse of its expected variance. Because the inverse of the estimated variance for each security is a nontrivial empirical estimate, formation of the minimum variance portfolio is nontrivial. It is noteworthy that the value of a portfolio's idiosyncratic volatility is at the level of $1/\sqrt{N}$, which decreases slowly as N increases. Consequently, when forming G, idiosyncratic volatility can never be ignored.

Suppose that the following general asset pricing model is valid:

$$R_{it} - R_{ft} = \alpha_i + b_{iM}(R_{Mt} - R_{ft}) + \sum_{k=1}^{K} b_{ik} F_{kt} + e_{it}, \tag{7.1}$$

where $R_{it} - R_{ft}$ is the realized excess return on the ith stock/portfolio for day or month t, R_{ft} is the riskless rate proxied by the Treasury bill rate, $(R_{Mt} - R_{ft})$ is the realized excess return on the market portfolio M, α_i is the intercept term or Jensen's alpha, F_{kt} are other asset pricing factors (or risk components) for $k = 1, \ldots, K$ factors, b_{ik}s are corresponding K beta risk loadings for the ith stock/portfolio, $i = 1, \ldots, N$ stocks/portfolios, $t = 1, \ldots, T$ is the estimation period, and $e_{it} \sim N(0, \sigma_i^2)$. The major market risk component M cannot be diversified away by adding more long position securities but can be diversified away by adding short positions. The F multifactors are related to other specific systematic market risks (or risk components).

We run the analyses on all stocks in the U.S. market within some estimation window. Here we will use a one-year estimation window with daily return data. Specifying Eq. (7.1) more precisely, we estimate the following empirical ZCAPM for each stock using one year of daily return starting in the period July 1963 to June 1964:

$$R_{it} - R_{ft} = \beta_{i,a}(R_{at} - R_{ft}) + Z^*_{i,a}\sigma_{at} + e_{it}, \quad t = 1, \ldots, T, \tag{7.2}$$

where beta risk coefficient $\beta_{i,a}$ is associated with average excess market returns, and zeta risk coefficient $Z^*_{i,a}$ is related to the cross-sectional standard deviation of returns across all stocks in the market (or return dispersion). We employ value-weighted CRSP market index returns for R_{at}. The return dispersion factor σ_{at} is computed using all CRSP stocks and value-weighting all stock returns. Departing from the general asset pricing model in Eq. (7.1), no intercept term is included in the ZCAPM. Also unlike other multifactor models that invariably use ordinary least squares (OLS) regression for their estimation, an expectation-maximization (EM) algorithm is used to estimate the ZCAPM regression model.[8]

After estimating empirical ZCAPM regression Eq. (7.2) for each stock $i = 1, \ldots, N$, stocks are sorted by their major systematic risk b_{iM}s. Stocks are then grouped into $B = 1, \ldots, n$ equal baskets of securities (where n is a fairly large number of basket portfolios such that $N >> n >> 1$). Each basket contains N/n securities. We choose the top N/n sorted basket of b_{iM} securities (with the highest beta risk) and seek to form a minimum variance portfolio using only stocks in this basket. To weigh each stock in this basket portfolio, we employ optimal weights based on proprietary methods.

Figure 7.2 illustrates the application of the empirical ZCAPM for some hypothetical sample of securities. Beta risk (or $\beta_{i,a}$) is represented by the slope of the line between excess security returns $R_{it} - R_{ft}$ on the Y-axis and $R_{at} - R_{ft}$ excess average market returns on the X-axis. Excess returns attributable to zeta risk, or $Z^*_{i,a}$, are shown as the vertical distance up or down from the beta risk line. The residual error term, or e_{it}, is shown for two securities as an additional component of excess security returns not explained by beta and zeta risks. If an asset pricing model does a poor job of pricing securities, two problems arise: (1) the overall sizes of the residual error terms for securities are large; and (2) residual error terms contain missing factors that change the relative sizes of residual error terms across securities.

Why use the above weighting scheme? Because N is large, the top N/n beta risk securities have b_{iM}s with similar values. Hence, their major risk is similar and the overall major risk of the formed portfolio cannot be reduced with any weighting scheme (i.e., their major risk is highly correlated and so cannot be reduced by diversifying). To reduce the variance of a particular basket of securities, we need to reduce the variance from other risks and idiosyncratic risk. For this purpose, we develop a proprietary optimal weighting scheme that mathematically reduces idiosyncratic risk.

[8] See Kolari et al. (2021) for details on step-by-step procedure to estimate the EM algorithm in the context of the empirical ZCAPM regression.

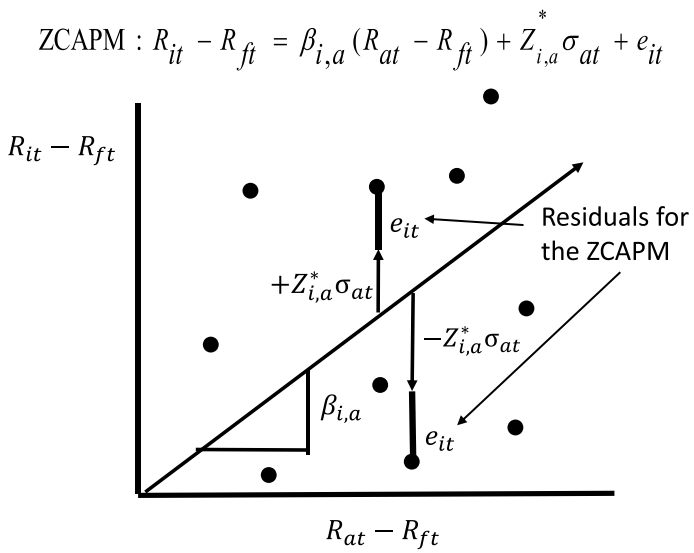


Fig. 7.2 Application of the empirical ZCAPM to securities

Next, we consider whether adding any short position can reduce the variance more than the above top b_{iM} formed portfolio. When short positions are added, we need to add the same amount of extra long positions to make the total weight equal one. To check the impact on the overall variance of the resultant portfolio, we know that adding extra long-short positions increases the idiosyncratic component variance and, therefore, potentially increases the overall variance of the resultant portfolio. Also, we know that the long-short portfolio bears small major risk as they cancel out due to the fact that all the top b_{iM} assets share similar major risk. For this reason, adding long-short positions will not increase the risk from major risk. Additionally, if long-short positions are randomly chosen, they will not bear any other risks and, subsequently, will not impact the variance of other risk components. If the long-short positions are picked to represent some of the other risk components, then they will increase the variance from other risks and will potentially increase the overall variance of the resultant portfolio. It is important to note that all of the systematic risk components are independent of one another (i.e., their correlation is very small). Based on the above conditions, adding short positions will only increase the overall variance of the resultant portfolio. Thus, it represents the nontrivial empirical scheme discussed above.

Repeating the same weighting scheme for other beta-sorted basket portfolios, we obtain the remaining $B - 1$ second stage portfolios. Each of these

portfolios has minimum variance for its major risk level. Now we have B portfolios wherein each security only bears major risk and idiosyncratic risk (as other risk components are diversified away). The key question is: How can we form a global minimum variance portfolio G using these B portfolios?

In this regard, it is worthwhile to consider the complexity of the G portfolio. We propose three possible scenarios that give conditions for when short positions are contained in G. Since idiosyncratic risks in principle are independent of any market risks, it is reasonable to assume that idiosyncratic risks are uniformly distributed across these B portfolios. Any B portfolio contains both systematic and idiosyncratic volatility. It is not known what these volatilities are in the real world. Portfolio G contains a complicated mix of systematic and idiosyncratic risks. We need to minimize both of these risks simultaneously to find G. Consider three possible weighting scenarios for combining the B portfolios to form G under different assumptions about idiosyncratic risk.

1. Among the $B = 1, \ldots, n$ beta-sorted basket portfolios, assume that $\sigma_M^1 < \sigma_M^2 < \ldots < \sigma_M^n << \sigma_I$. Here idiosyncratic volatility (denoted σ_I) is much larger than major risk for any of the B portfolios (denoted σ_M^B where $B = 1, \ldots, n$). We start with an equal-weight scheme as all the idiosyncratic variances are the same. To simplify the derivation with no loss of generality, we further assume that $\sigma_M^1 = \sigma_M^2 = \ldots = \sigma_M^{n-1} = \sigma_M^n/2 = \sigma_M$. As such, the variance of the combined portfolio containing the $B = 1$ to $B = 8$ ranked portfolios (assuming $n = 10$ portfolios) applying equal-weights to them yields $\sigma_M^2 + \sigma_I^2/(n-2)$. Note that, if using all n of the B portfolios, the portfolio variance can be smaller, but here we only use $n - 2$ portfolios to check whether adding any short positions can reduce the variance of the underlying portfolio. Next, consider long-short portfolio $R_{n-1} - R_n$, which has net major risk equal to $\sigma_M^{n-1} - \sigma_M^n = \sigma_M - 2\sigma_M = -\sigma_M$. Adding this long-short portfolio to the above pure long only portfolio will produce a combined portfolio with short positions and net major risk equal to zero. Now the variance of the resultant portfolio equals $\sigma_I^2/(n-2) + 2\sigma_I^2$, which is larger than the original pure long only portfolio. Notably, multiplying the long-short portfolio by an arbitrary constant will not change this conclusion. Summarizing this scenario, because the addition of short positions to diversify major risk does not reduce the variance of underlying pure long only portfolio, the global minimum variance portfolio should be pure long. This scenario is likely close to the real world as empirical tests show that idiosyncratic risk almost always dominates risk factors.

2. Assume that $\sigma_M^1 < \sigma_M^2 < \ldots < \sigma_M^n \approx \sigma_I$. Like the above scenario, to simplify the derivation, we further assume that $\sigma_M^1 = \sigma_M^2 = \ldots = \sigma_M^{n-1} = \sigma_M^n/2 = \sigma_M$. The variance for the combined portfolio consisting of 9 ranked portfolios, denoted as P_{n-1}, with equal-weights for the 9 portfolios equals $\sigma_M^2 + \sigma_I^2/(n-1)$. Now consider adding the Bth portfolio denoted as p_B to the P_{n-1}, portfolio, i.e., $P = x\widetilde{R}_{P_{n-1}} + (1-x)\widetilde{R}_{p_n}$, and minimizing its variance. Using the quantities $Var(R_{P_{n-1}}) = \sigma_M^2 + \sigma_I^2/(n-1)$, $Var(R_{p_n}) = 4\sigma_M^2 + \sigma_I^2$, and $Cov(R_{P_{n-1}}, R_{p_n}) = 2\sigma_M^2$, and applying the first order condition, we obtain $x = \frac{2\sigma_M^2 + \sigma_I^2}{\sigma_M^2 + n\sigma_I^2/(n-1)}$. Based on the assumption $\sigma_M^n \approx \sigma_I$, it is easy to see that $x > 1$ such that $1 - x < 0$, which says that portfolio B is a short position.
3. Assume that $\sigma_I << \sigma_M^1 < \sigma_M^2 < \ldots < \sigma_M^n$. In this case, idiosyncratic risk is negligible. By shorting some high β_M securities, we can form many portfolios that have zero major risk. To see this, consider a portfolio $\sum_{i=1}^n w_i R_i$ with $\sum_{i=1}^n w_i = 1$ and $w_i \geq 0$. Its total major risk is $\sqrt{\sum_{i=1}^n (\sigma_M^i)^2}$, as $Var(\sum w_i R_i) = \sum w_i^2 (\sigma_M^i)^2$ given no correlation between the portfolios. Now consider a long-short portfolio $R_1 - R_n$ with net major risk equal to $\sigma_M^1 - \sigma_M^n$. If we multiply the long-short portfolio by $\sum_{i=1}^n w_i \sigma_M^i/(\sigma_M^1 - \sigma_M^n)$ and add it to the original pure long only portfolio, it is easy to see that net major risk equals zero for the resultant portfolio. Now its variance is only determined by the combination of idiosyncratic variances and, therefore, should be smaller than those of portfolios with finite net major risk. There are many ways to form such portfolios. Thus, there exists one portfolio with the smallest variance, which contains short positions. This scenario is unrealistic as the number of securities N needs to be extremely large (e.g., one million or more securities), and the number of other risk components needs to be relatively small and not grow with increasing numbers of securities.

When idiosyncratic risk does not dominate, the above analyses reveal that the global minimum variance portfolio G formed from the $B = 1, \ldots, n$ beta-sorted basket or so-called *second stage* portfolios can have short positions. When adding short positions, the shorted securities are comprised of higher major risk securities. The above steps provide a framework to empirically build a *nontrivial* G portfolio by means of a valid asset pricing model. Using one year of daily returns for model estimation purposes, we can choose $B = 1, \ldots, n = 10$ or $n = 20$ and form these beta-sorted basket portfolios by means of the steps outlined above. After constructing 10 or 20 minimum

variance basket portfolios by optimally weighting securities in each portfolio, the final global minimum variance portfolio G can be formed.

We summarize the procedure for building portfolio G in Fig. 7.3. Again we estimate the empirical ZCAPM in regression Eq. (7.2) using one year of daily returns from July 1963 to June 1964. Three simple steps are:

1. create $B = 1, \ldots, n$ second stage portfolios sorted on beta risk;
2. form a minimum variance portfolio for each B portfolio using individual security weights that are optimized using our proprietary procedures; and
3. build the final portfolio G by using optimal weights for the $B = 1, \ldots, n$ portfolios to minimize the variance of the combined portfolio.

Critically, G contains only beta risk to the exclusion of zeta risk related to market return dispersion. Recall that market return dispersion largely determines the width of the parabola. Popular multifactors that are long-short portfolios (such as size, value, momentum, etc.) are themselves measures of return dispersion and therefore associated with the width of the parabola; hence, G should not contain these risk factors also.

The final steps to construct the daily return series for the G portfolio are as follows:

1. compute the return of G in the next out-of-sample day-ahead;
2. roll forward the above steps by one day and repeat to get the second day's return; and
3. continue rolling forward one day at a time until the end of December 2022.

This process generates a daily time series of G portfolio returns from July 1964 to December 2022. Our G return series is used in forthcoming chapters to estimate the empirical ZCAPM. In the above final steps, to reduce trading costs, the results would not change much if the process was rolled forward one month at a time as opposed to one day at a time.

Unlike Clark, Silva, and Thorley cited earlier, we do not use the SML in the CAPM to find G. The G portfolio itself has nothing to do with the derivation and theory of the CAPM that primarily focuses on the market portfolio M. By contrast, G pins the mean-variance parabola in return/risk space and has a pivotal role in both the theoretical and empirical ZCAPM. Taking advantage of the architecture of the parabola in the context of the ZCAPM with beta risk and zeta risk, we have proposed mechanical procedures for building an estimate of portfolio G. The remainder of this chapter

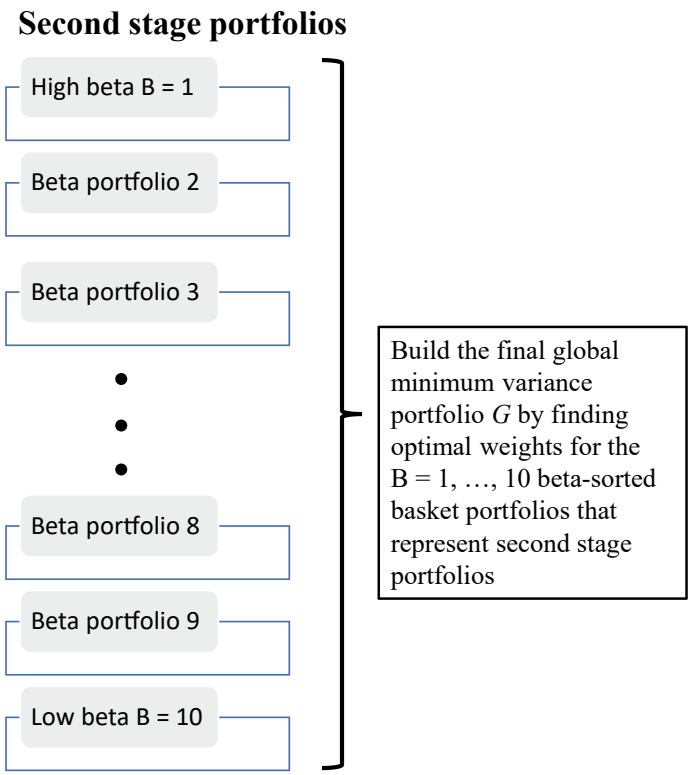


Fig. 7.3 Building the final portfolio *G* from the second stage portfolios which are beta-sorted portfolios. Each second stage portfolio is itself a minimum variance portfolio at its particular level of beta risk

gives the empirical *G* results upon implementing these procedures. As will be shown, the ZCAPM does an excellent job as evidenced by the very low return variance of *G* relative to common general market benchmarks such as the S&P 500 index and CRSP index.

7.3 Empirical Results for the *G* Portfolio

We downloaded all U.S. common stock returns from the Center for Research and Security Prices (CRSP) database provided by the University of Chicago. In each year we dropped very small stocks in the bottom 10% by market capitalization (hereafter the overall sample). Also, we formed a subset of relatively larger stocks in the top 3,000 by market capitalization (hereafter the top 3,000 sample). Daily stock returns from July 1963 to December 2022

are gathered for all individual stocks. The CRSP market index of all common stocks is used for comparison purposes in evaluating portfolio *G* results.

7.3.1 Overall Sample *G* Results

Using daily returns for the overall sample in selected years, Figs. 7.4 and 7.5 compare *G* portfolio returns (bold lines) to CRSP market index returns (thin lines). Again the daily returns for *G* are out-of-sample returns computed the next day after forming the *G* portfolio. As discussed above, the G portfolio formation process rolls forward one day at a time. Over the entire sample period from July 1963 to December 2022, the correlation of these two return series is fairly positive at 0.32. Casual inspection of the *G* portfolio and CRSP index returns in Figs. 7.4 and 7.5 clearly shows that *G* returns are similar to CRSP returns in terms of average magnitude over time. However, as shown in these figures, *G* portfolio returns are much less volatile than CRSP returns in any given year. Even during the 2008 financial crisis and stock market turmoil at that time, *G* returns are much less volatile than CRSP returns.

Based on the overall sample, Figs. 7.6 and 7.7 give more comparative details for *G* and CRSP daily returns in our sample period. Here we show average performance results for the following subperiods: July 1964–December 1969, January 1970–December 1979, January 1980–December 1989, January 1990–December 1999, January 2000–December 2010, and January 2011–December 2022.

In Fig. 7.6 we see the average daily returns in these different subperiods. The *G* portfolio and CRSP market index have similar average returns over time. For the entire sample period from July 1964 to December 2022, the average daily return of the *G* portfolio is 0.0332% per day versus 0.0401% per day for the CRSP market index. On an annualized basis, *G* earned 8.73% per year on average compared to 10.62% per year for the CRSP index. Thus, the CRSP market index earns higher average returns than the *G* portfolio but the difference is not large. If the CRSP index is an efficient portfolio that is a good proxy for the market portfolio in the CAPM, as many researchers have believed for the past 50 years, then *G* should underperform the CRSP index by a considerable amount. But this comparative outcome is not borne out by our evidence—namely, *G* underperforms *CRSP* but not by a large margin.

Figure 7.7 compares the average standard deviation of daily returns in different subperiods for the *G* portfolio and CRSP market index. These are time-series standard deviations of daily returns over time in each subperiod. It is obvious that the *G* portfolio consistently has lower volatility than the

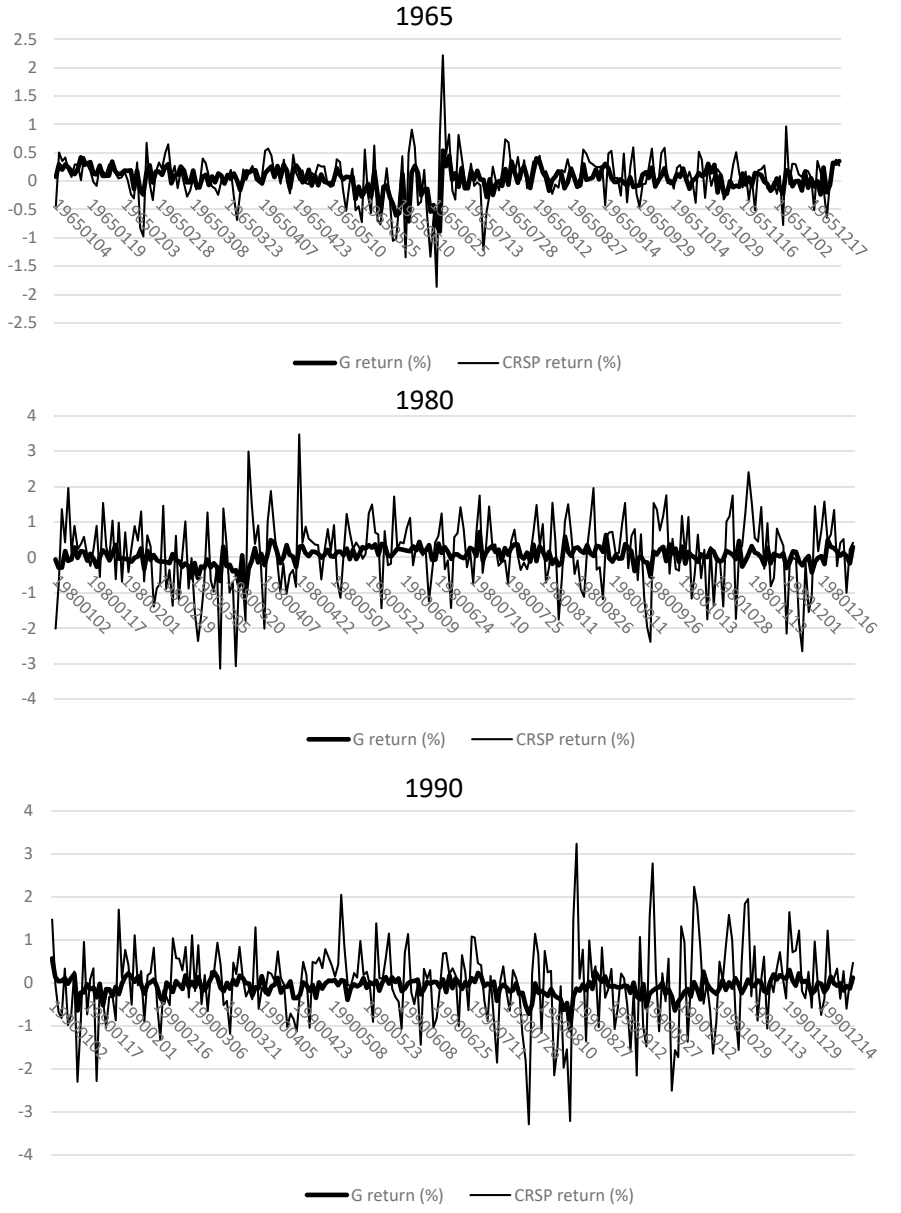


Fig. 7.4 *G* portfolio returns compared to CRSP index returns in selected years: 1965, 1980, and 1990

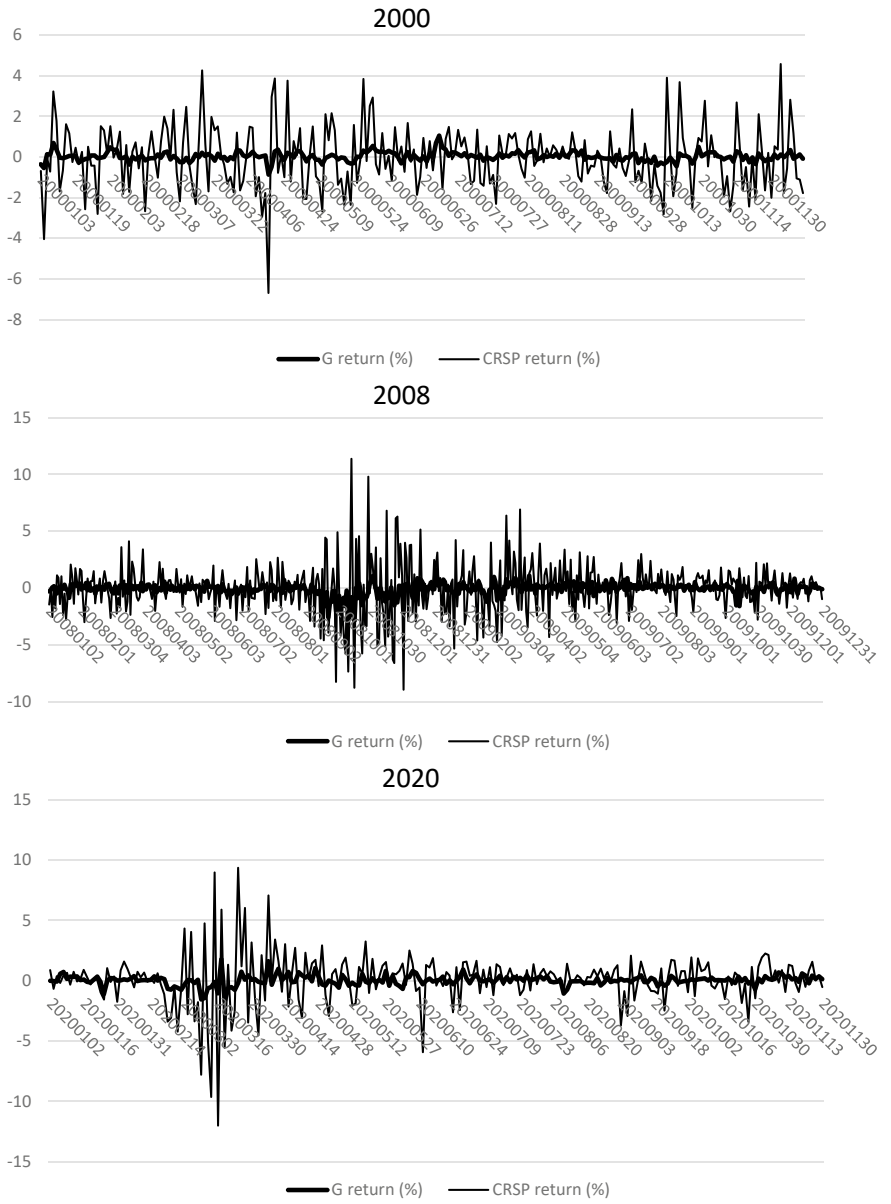
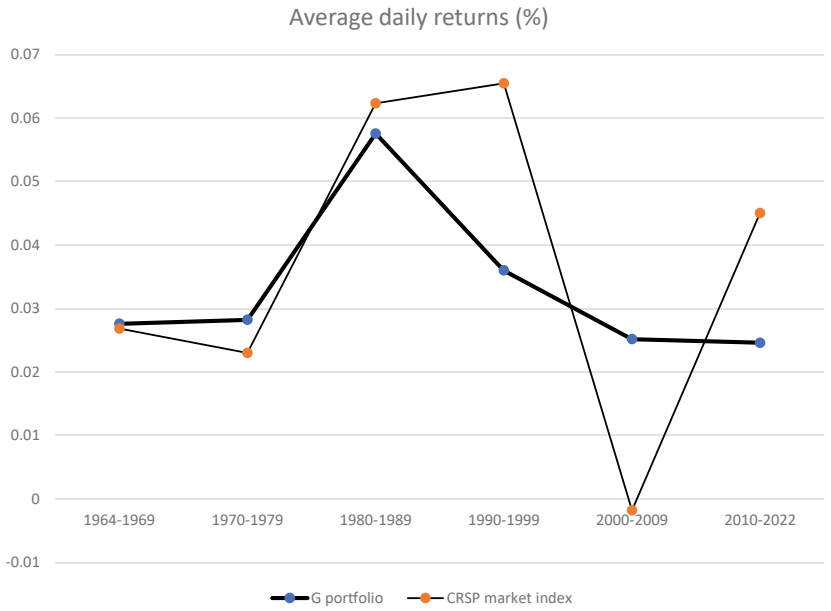


Fig. 7.5 *G* portfolio returns compared to CRSP index returns in selected years: 2000, 2008, and 2020

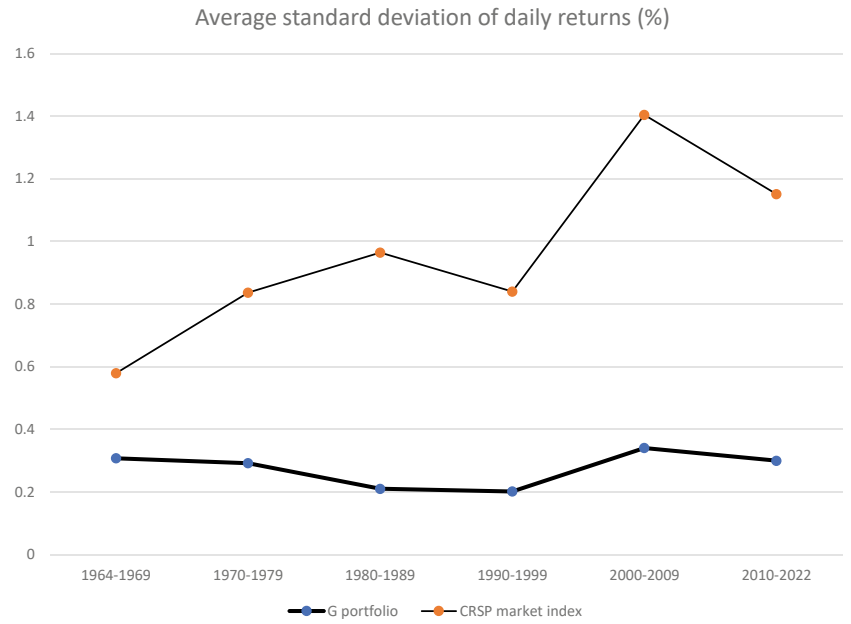


	1964-1969	1970-1979	1980-1989	1990-1999	2000-2009	2010-2022	1964-2022
G mean return %	0.0276	0.0282	0.0575	0.0360	0.0252	0.0246	0.033207
CRSP mean return %	0.0268	0.0230	0.0623	0.0655	-0.0018	0.0450	0.037894

Fig. 7.6 Average daily *G* portfolio returns compared to CRSP index returns in selected subperiods

CRSP index. Interestingly, the volatility of the CRSP index increases relative to the *G* portfolio over time from 1964 to 2022. However, the volatility of the *G* portfolio remains fairly constant over time. The overall average standard deviation of daily returns for *G* is 0.2802% versus 1.032% for the CRSP index. Hence, the CRSP market index is approximately 3.6 times more volatile than our *G* portfolio over more than 55 years. In terms of its location within the mean-variance parabola, taking into account the average daily returns in Fig. 7.6, we can infer that the CRSP index lies in the interior of the parabola to the right of portfolio *G* and somewhat above the axis of symmetry. Since volatility for the CRSP index trends upward over time, it is gradually moving further to the right within the parabola over time.

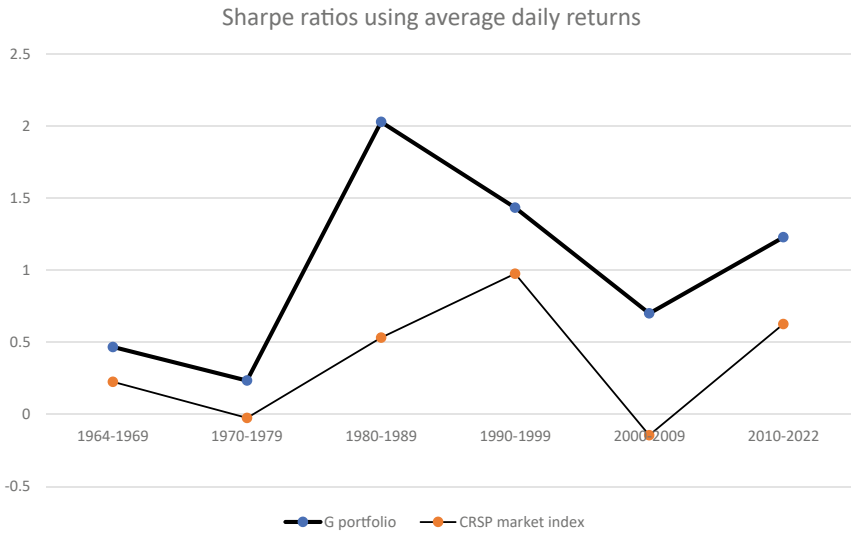
Now we can combine the average daily return and standard deviation of return evidence by means of the Sharpe ratio. Recall that the Sharpe ratio equals the average excess return of an asset over the riskless Treasury bill rate divided by the time-series standard deviation of asset returns over time.



	1964-1969	1970-1979	1980-1989	1990-1999	2000-2009	2010-2022	1964-2022
G standard deviation %	0.3077	0.2920	0.2102	0.2018	0.3406	0.2999	0.2802
CRSP standard deviation %	0.5795	0.8369	0.9649	0.8399	1.4049	1.1516	1.0315

Fig. 7.7 Average time-series standard deviations of *G* portfolio returns compared to CRSP index returns in selected subperiods

Figure 7.8 compares the Sharpe ratios in different subperiods for the *G* portfolio and CRSP market index. In all subperiods except 1964 to 1969, our *G* portfolio dominates the CRSP general stock market index. In the two subperiods 1970–1979 and 2000–2009, the CRSP index has negative Sharpe ratios due to its average daily market return falling below the average daily return of Treasury bills in these subperiods. By contrast, in all subperiods the *G* portfolio has positive Sharpe ratio performance. For the entire sample period, the average Sharpe ratio for the *G* portfolio is 0.97 versus 0.34 for the CRSP index, which represents an approximately 2.9-times higher return/risk performance! These results are indeed surprising. If the CRSP index is an efficient portfolio that is a good proxy for the market portfolio in the CAPM, as many researchers have believed, then the CRSP index should outperform *G* in terms of risk premiums per unit risk. But this comparative outcome is not borne out by our evidence—namely, *G* outperforms *CRSP*. We can conclude that the CRSP index is definitely not an efficient portfolio.



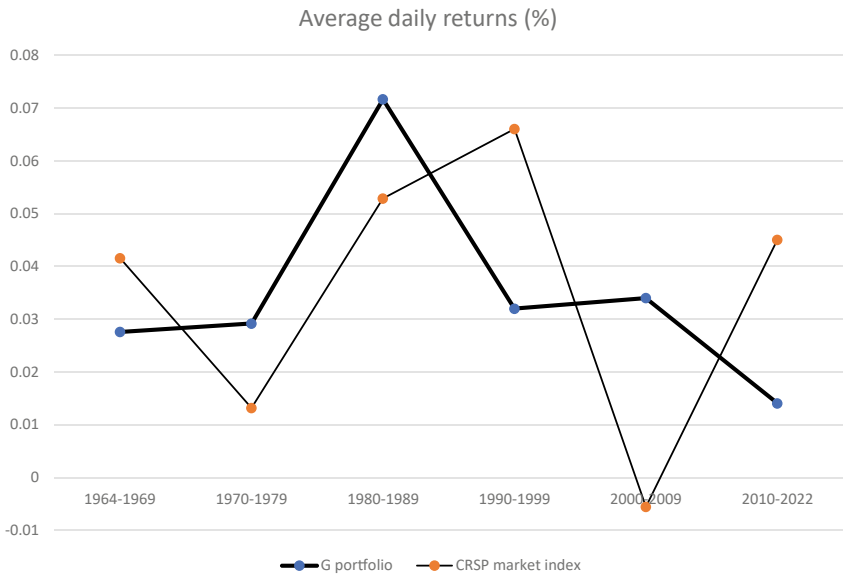
	1964-1969	1970-1979	1980-1989	1990-1999	2000-2009	2010-2022	1964-2022
G Sharpe ratio	0.4690	0.2358	2.0302	1.4349	0.7019	1.2305	0.9671
CRSP Sharpe ratio	0.2268	-0.0234	0.5338	0.9775	-0.1434	0.6276	0.3416

Fig. 7.8 Sharpe ratios for *G* portfolio returns compared to CRSP index returns in selected subperiods (all returns annualized)

7.3.2 Top 3,000 Sample *G* Results

The *G* portfolio results for the top 3,000 common stocks by market capitalization each year are shown in Figs. 7.9–7.11. In general, the results are very similar to those in Figs. 7.6–7.8, respectively. One difference in the results compared to the overall sample is that the Sharpe ratios for the CRSP index improved relative to the *G* portfolio. Now the *G* portfolio outperforms the CRSP index in 3-out-of-6 (rather than 5-out-of-6) subperiods. Nonetheless, for the entire sample period from July 1964 to December 2022, the Sharpe ratio for the *G* portfolio is still higher at 0.86 compared to 0.34 for the CRSP market index (i.e., 2.5-times higher than before).

These findings confirm the robustness of our empirical results for the *G* portfolio. Again, in view of the relative performance of *G* to the CRSP index, and contrary to the beliefs of many researchers, the market-valued weighted CRSP index cannot be an efficient portfolio.



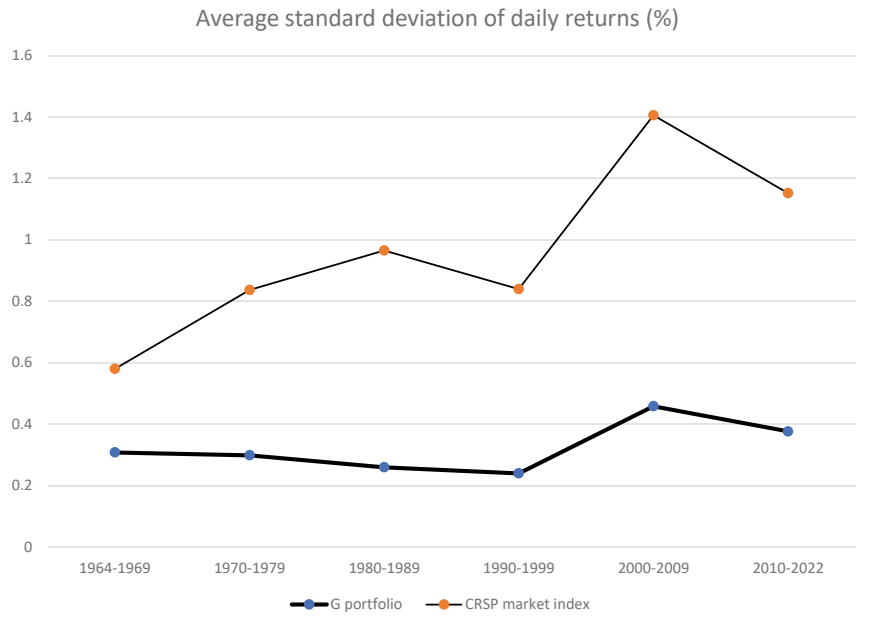
	1964-2022	1964-1969	1970-1979	1980-1989	1990-1999	2000-2009	2010-2022
G 3000 mean return %	0.0343	0.0276	0.0292	0.0717	0.0320	0.0340	0.0141
CRSP mean return %	0.0379	0.0416	0.0132	0.0529	0.0661	-0.0055	0.0450

Fig. 7.9 Average daily *G* portfolio returns for top 3,000 stocks compared to CRSP index returns in selected subperiods

7.4 Summary

Previous studies have found that, while it is not possible to develop efficient portfolios due to econometric problems, there has been some success in constructing global minimum variance portfolios denoted *G*. The *G* portfolio does not require any expected return estimates, which is a major source of estimation error in constructing efficient portfolios. Researchers have proposed a variety of methods for building *G* portfolios. Related to our ZCAPM approach, Clarke et al. (2011) created *G* portfolios by using only low beta stocks. Their rationale for this approach was that researchers found the Security Market Line (SML) of the CAPM to be flatter than theory predicted; as such, low beta stocks have higher expected returns per unit systematic risk than other stocks.

Like Clarke et al., we emphasize low beta stocks in the *G* portfolio but for different reasons. According to ZCAPM theory, *G* pins the parabola in return/risk space in the mean-variance landscape of Markowitz. As Kolari et al. (2021) have shown, the width of the mean-variance parabola is largely

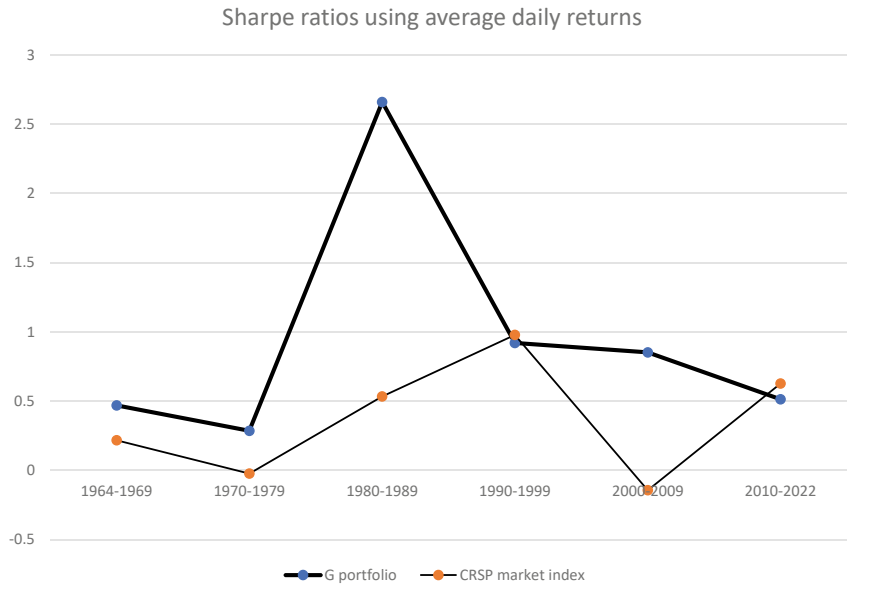


	1964-1969	1970-1979	1980-1989	1990-1999	2000-2009	2010-2022	1964-2022
G 3000 standard deviation %	0.3077	0.2985	0.2597	0.2398	0.4581	0.3764	0.3383
CRSP standard deviation %	0.5801	0.8369	0.9649	0.8399	1.4049	1.1516	1.0316

Fig. 7.10 Average time-series standard deviations of *G* portfolio returns for top 3,000 stocks compared to CRSP index returns in selected subperiods

determined by the cross-sectional standard deviation of returns of all assets in the market (or return dispersion). This insight implies that the average market return (e.g., the S&P 500 index or CRSP stock market index return) lies in the vicinity of the axis of symmetry in the middle of the parabola, which is equal to the return on portfolio *G*. The theoretical ZCAPM proposes an architecture for the parabola, with increasing beta risk to the right of portfolio *G* and increasingly positive (decreasingly negative) zeta risk above (below) the axis of symmetry. This architecture suggests that *G* contains low beta securities and zero zeta risk.

Given our theoretical framework, we proposed a three-step procedure for building portfolio *G*. First, we create a relatively small number of second stage portfolios based on sorting stocks on their betas estimated from the empirical ZCAPM. Second, stocks in each second stage portfolio are optimally weighted using a proprietary weighting scheme. Third, and last, the second stage portfolios are optimally weighted to form portfolio *G* with minimum return variance.



	1964-1969	1970-1979	1980-1989	1990-1999	2000-2009	2010-2022	1964-2022
G 3000 Sharpe ratio	0.4690	0.2854	2.6597	0.9198	0.8520	0.5134	0.8557
CRSP Sharpe ratio	0.2169	-0.0234	0.5338	0.9775	-0.1434	0.6276	0.3410

Fig. 7.11 Sharpe ratios for *G* portfolio returns for top 3,000 stocks compared to CRSP index returns in selected subperiods (all returns annualized)

To implement our *G* portfolio approach, we employed all U.S. common stocks from the CRSP database from July 1963 to December 2022. In our overall stock market sample, we exclude 10% of stocks in each year with the smallest market capitalization. We construct *G* using the remaining population of stocks. Another *G* portfolio is formed using only the top 3,000 stocks by market capitalization in any given year. All *G* returns are computed on an out-of-sample basis for all trading days in the next month after forming the portfolio. As such, out-of-sample daily returns for the *G* portfolio began in July 1964. We roll forward this process one month at a time to generate out-of-sample daily returns for *G* until December 2022. This investable strategy could be implemented by an actual investor.

Our empirical results for the *G* portfolio are impressive. In a number of subperiods over time, the overall sample (top 3,000 sample) of stocks had Sharpe ratios for the *G* portfolio that were approximately 2.9 (2.5) times those of the CRSP market index. While the average daily returns of the CRSP index was similar to those for the *G* portfolio, the CRSP index had much

higher time-series standard deviations of returns than G in different subsamples. It appeared that the volatility of the CRSP market index trended upward over the entire sample period, whereas the G portfolio's volatility was fairly constant over time. The rising volatility of the CRSP market index reduced its Sharpe ratios and likely explains its underperformance compared to the G portfolio. Of course, low and stable volatility are good traits for a global minimum variance portfolio.

In forthcoming chapters that cover the formation of different high-performing stock portfolios, we will employ the G portfolio in the empirical ZCAPM. Average returns per day for G are associated with beta risk, and the cross-sectional standard deviation of returns of stocks (or return dispersion) is related to zeta risk. We use our optimized weights for both factors in the ZCAPM: average G returns and their return dispersion.

References

Best, M.l J., and R. R. Grauer. 1991. Positively weighted minimum-variance portfolios and the structure of asset expected returns. *Journal of Financial and Quantitative Analysis* 26: 513–537.

Black, F., and R. Litterman. 1992. Global portfolio optimization. *Financial Analysts Journal* 48: 28–43.

Chan, L.K.C., J. Karceski, and J. Lakonishok. 1999. On portfolio optimization: Forecasting covariances and choosing the risk model. *Review of Financial Studies* 12: 937–974.

Clark, R., H. de Silva, and S. Thorley. 2006. Minimum-variance portfolios in the U.S. equity market. *Journal of Portfolio Management* 33: 10–24.

Clarke, R.G., H. De Silva, and S. Thorley. 2011. Minimum-variance portfolio composition. *Journal of Portfolio Management* 37: 31–45.

Daniel, K., and S. Titman. 1997. Evidence on the characteristics of cross sectional variation in stock returns. *Journal of Finance* 52: 1–33.

De Nard, G., O. Ledoit, and M. Wolf. 2021. Factor models for portfolio selection in large dimensions: The good, the better and the ugly. *Journal of Financial Econometrics* 19: 236–257.

DeMiguel, V., L. Garlappi, F.J. Nogales, and R. Uppal. 2009. A generalized approach to portfolio optimization: Improving performance by constraining portfolio norms. *Management Science* 55: 798–812.

Fama, E.F., and K.R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427–465.

Fama, E.F., and K.R. French. 1993. The cross-section of expected returns. *Journal of Financial Economics* 33: 3–56.

Fama, E.F., and K.R. French. 2015. A five-factor asset pricing model. *Journal of 31 Financial Economics* 116: 1–22.

Feldman, R. 2016. *Building minimum variance portfolios with low risk, low drawdowns and strong results*. Innovative Global Indices, Deutsche Börse Group: STOXX Ltd.

Frost, P.A., and J.E. Savarino. 1986. An empirical Bayes approach to efficient portfolio selection. *Journal of Financial and Quantitative Analysis* 21: 293–305.

Haugen, R. A., and N. L. Baker. 2012. Low risk assets outperform within all observable markets of the world. Working paper, SSIA and Haugen Custom Financial Systems. Available at SSRN 2055431.

Jagannathan, R., and T. Ma. 2003. Risk reduction in large portfolios: Why imposing the wrong constraints helps. *Journal of Finance* 58: 1651–1683.

Jobson, J.D., and B. Korkie. 1980. Estimation for Markowitz efficient portfolios. *Journal of the American Statistical Association* 75: 544–554.

Jobson, J.D., and B.M. Korkie. 1981. Putting Markowitz theory to work. *Journal of Portfolio Management* 7: 70–74.

Jorion, P. 1985. International portfolio diversification with estimation risk. *Journal of Business* 58: 259–278.

Jorion, P. 1986. Bayes-Stein estimation for portfolio analysis. *Journal of Financial and Quantitative Analysis* 21: 279–292.

Ledoit, O., and M. Wolf. 2017. Nonlinear shrinkage of the covariance matrix for portfolio selection: Markowitz meets Goldilocks. *Review of Financial Studies* 30: 4349–4388.

Ledoit, O., and M. Wolf. 2020. Analytical nonlinear shrinkage of large-dimensional covariance matrices. *Annals of Statistics* 48: 3043–3065.

Ledoit, O., and M. Wolf. 2022. The power of (non-)linear shrinking: A review and guide to covariance matrix estimation. *Journal of Financial Econometrics* 20: 187–218.

Kolari, J. W., J. Z. Huang, W. Liu, and H. Liao. 2022. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*. Available online at https://www.mdpi.com/1911-8074/15/3/137). Reprinted in Kolari, J. W., and S. Pynnonen (eds.). 2022. *Frontiers of Asset Pricing* (MDPI, Basel, Switzerland).

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. *Testing for missing asset pricing factors*. San Diego, CA: Paper presented at the Western Economic Association International.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham, Switzerland: Palgrave Macmillan.

Ledoit, O., and M. Wolf. 2004. Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management* 30: 110–119.

Ledoit, O., and M. Wolf. 2020. The power of (non-)linear shrinking: A review and guide to covariance matrix estimation. *Journal of Financial Econometrics* 18: 1–32.

Liu, W. 2013. *A new asset pricing model based on the Zero-Beta CAPM: Theory and evidence*. Doctoral dissertation, Texas A&M University.

Liu, W., J.W. Kolari, and J.Z. Huang. 2012. A new asset pricing model based on the zero-beta CAPM market model (CAPM), Presentation at the annual meetings of the Financial Management Association, Best Paper Award in Investments, Atlanta, GA (October).

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. *Return dispersion and the cross-section of stock returns*. Palm Springs, CA (October): Presentation at the annual meetings of the Southern Finance Association.

Maillet, B., S. Tokpavi, and B. Vaucher. 2015. Global minimum variance portfolio optimisation under some model risk: A robust regression-based approach. *European Journal of Operational Research* 244: 289–299.

Markowitz, H.M. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: Wiley.

Michaud, R.O. 1989. The Markowitz optimization enigma: Is "optimized" optimal? *Financial Analysts Journal* 45: 31–42.

Pástor, L. 2000. Portfolio selection and asset pricing models. *Journal of Finance* 55: 179–223.

Pástor, L., and R.F. Stambaugh. 2000. Comparing asset pricing models: An investment perspective. *Journal of Financial Economics* 56: 335–381.

Reh, L., F. Krüger, and R. Liesenfeld. 2022. Predicting the global minimum variance portfolio. *Journal of Business and Economic Statistics* 41: 440–452.

Rudd, A., and B. Rosenbert. 1979. Realistic portfolio optimization. In *Portfolio theory: 25 years after*, ed. N. Elton and M. Gruber. North Holland, New York, NY.



8

Net Long Portfolio Performance Analyses

8.1 Background Discussion

We begin with a brief review of the Markowitz efficient frontier and ZCAPM theory applied to efficient portfolios. As shown in Fig. 1.3 in Chapter 1, the efficient frontier of portfolios have the minimum total risk at any given expected return. While there is a distinct theoretical frontier defined by the curved outer boundary of the mean-variance parabola, in the real world we can think of the efficient frontier as a fuzzy boundary. Market noise causes expected returns and their standard deviations to randomly vary over time. At any snapshot of time, as illustrated in Fig. 8.1, the frontier actually looks like a blurry band of portfolios. Our goal is to find portfolios in this efficient region.

In Chapter 4 we discussed important new insights about the mean-variance parabola from the theoretical ZCAPM. First, market portfolio proxies, such as the S&P 500 or CRSP market indexes, have average returns that lie approximately on the axis of symmetry of the investment parabola. Contrary to widespread belief among many finance academics and researchers, it does **not** lie on the efficient frontier. Second, the span or width of the investment parabola is largely determined by the cross-sectional standard deviation of returns of all assets in the market (or return dispersion). Figure 8.2 illustrates these concepts.

The ZCAPM uses the geometry of the investment parabola to locate efficient portfolios. Figure 8.3 shows that the Capital Market Line (CML) of the CAPM is an infeasible approach due to the fact that we do not know how to find the tangent market portfolio. Rather than use the CML, the ZCAPM

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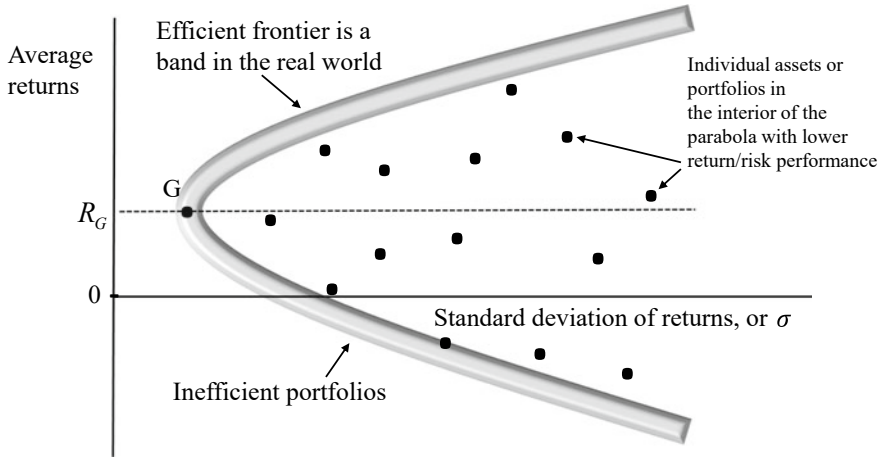


Fig. 8.1 The empirical efficient frontier of Markowitz

takes advantage of average market returns approximately equal to the return on global minimum variance portfolio G to move horizontally along the axis of symmetry of the parabola. From this axis that splits the parabola into two halves, we can move vertically upward using the cross-sectional standard deviation of returns of all assets in the market (or return dispersion). Different risk levels can be selected on the X-axis defined as the standard deviation of returns (or total risk). For each risk level, an efficient portfolio can be identified.

8.2 Empirical Methods

We initially estimate the empirical ZCAPM with one year of daily returns from July 1964 to June 1965 for individual stocks in the U.S. stock market. Individual stocks are formed into 25 portfolios based on their ranked zeta coefficient estimates. Portfolio 13 is dropped due to near zero zeta risk. The remaining 24 portfolios are used to construct 12 long/short portfolios that have different levels of zeta risk. Average stock returns in the next month (out-of-sample) of July 1965 are computed for these portfolios. This procedure is repeated month-by-month to obtain a time series of monthly returns for these portfolios from July 1965 to December 2022.

Our main goal is to form net long portfolios whose weights sum to one. To do this, we combine the G portfolio with our 12 long/short zeta risk portfolios to obtain 12 net long portfolios. Recall that G contains no zeta risk. Hence, we are using G as the zeta-riskfree portfolio and then adding

New investment insights from the ZCAPM:

(1) The market portfolio (e.g., the S&P 500 or CRSP market indexes) lies approximately on the axis of symmetry of the investment parabola, ***not*** on the efficient frontier as many believe.
(2) The cross-sectional standard deviation of returns of all assets in the market explains the span or width of the investment parabola.

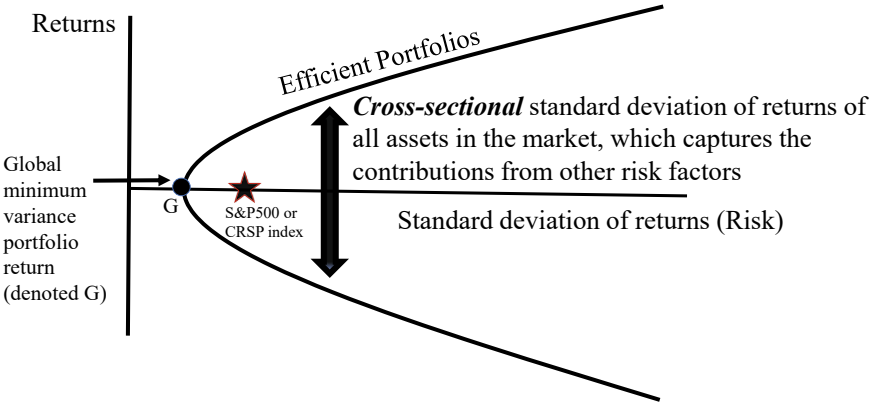


Fig. 8.2 The ZCAPM and mean-variance investment parabola

different levels of risk via the 12 long/short zeta risk portfolios. One-month-ahead returns are computed for these net long portfolios from July 1965 to December 2022. As we will see, when plotted in return/risk space, these portfolios trace out the shape of an upward sloping efficient frontier. Also, their average returns in our sample period well exceed those of the CRSP index.

We should note that transaction costs are fairly small due to only marginal changes in the holdings of any particular stock when portfolios are rebalanced each month. The authors have built long only portfolios in the real world and found that investment performance is not materially affected. Due to our professional experience in portfolio management using the ZCAPM, the authors actively seek joint investment activities and collaborations with securities firms to implement our technology for investment clients.

Lastly, it is important to recognize that our investment portfolio technology can be applied to the construction of a wide variety of different portfolios. Managers interested in building a specific type of portfolio, including style portfolios related to size, value, and other firm characteristics, international portfolios, exchange traded funds (ETFs), etc., can use the same basic methods. Also, applications are readily possible to portfolios comprised of other classes of assets, such as bonds, commodities, real estate assets, etc., with available daily market prices. The general applicability of ZCAPM-based

Creating efficient portfolios with ZCAPM technology:

(1) Move horizontally along the axis of symmetry at the average market return and then move vertically upward using the cross-sectional volatility of the market to reach efficient portfolios.
(2) Different efficient portfolios can be created using different risk levels.
(3) The CML of the CAPM is infeasible due to not knowing the location of the tangent portfolio.

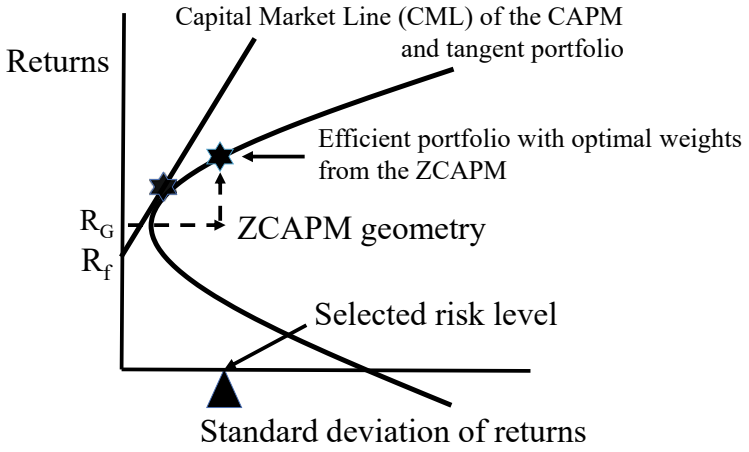


Fig. 8.3 The geometry of the ZCAPM and efficient portfolios

portfolios to many different kinds of portfolios is a major advantage of our investment technology.

8.2.1 Review of the ZCAPM

We are unaware of any other applications of asset pricing models to investment portfolio construction. The empirical ZCAPM enables the formation of high-performing portfolios that boost investment performance. In this section, we show how the ZCAPM can be used to create a series of stock portfolios that trace out an empirical efficient frontier (as opposed to theoretical frontier). Given portfolio *G* that defines the average market return, ZCAPM zeta risk associated with market return dispersion for individual stocks can be either positive or negative, which translates into higher (lower) returns than the average market return. At different levels of total risk, we can form portfolios with the highest (lowest) zeta risk that have relatively high (low)

returns relative to all other portfolios. These portfolios trace out the empirical boundary of the mean-variance parabola. Now for the first time we can see an empirical approximation of Markowitz's parabola in the real world.

In Chapter 4 we showed that the *theoretical ZCAPM* was derived by Kolari et al. (2021) as a special case of Black's (1972) zero-beta CAPM.[1] Incorporating portfolio G from Chapter 7, we can write the theoretical ZCAPM as follows:

$$E(R_{it}) - R_{ft} = \beta_{i,G}\left[E(R_{Gt}) - R_{ft}\right] + Z^*_{i,G}\sigma_{Gt}, \tag{8.1}$$

where $E(R_{Gt})$ is the expected market return based on the average return of all assets as measured by portfolio the global minimum variance portfolio G (rather than the theoretical market portfolio M in Sharpe's [1964] CAPM), R_{ft} is the riskless rate, $\beta_{i,G}$ is the beta risk coefficient measuring sensitivity to expected excess G portfolio returns, σ_{Gt} is the cross-sectional standard deviation of all asset returns in portfolio G (or return dispersion) at time t, and $Z^*_{i,G}$ is the zeta risk coefficient measuring dichotomous positive or negative sensitivity to G portfolio return dispersion. An asterisk is added to the zeta risk coefficient $Z^*_{i,G}$ to denote its theoretical relation to orthogonal portfolios I^* and ZI^*. These two portfolios lie on the upper and lower boundaries of the mean-variance investment parabola and have equal return variances.

Now we can re-write the empirical ZCAPM in Chapter 5 as follows:

$$\begin{aligned} R_{it} - R_{ft} = {} & \beta_{i,G}\left(R_{Gt} - R_{ft}\right) \\ & + Z_{i,G}D_{it}\sigma_{Gt} + u_{it}, \quad t = 1, \ldots, T \end{aligned} \tag{8.2}$$

where $R_{it} - R_{ft}$ is the excess return for the ith stock portfolio (or stock) over the Treasury bill rate, $\beta_{i,G}$ measures sensitivity of the ith stock portfolio (or stock) to excess average market returns based on portfolio G over the Treasury bill rate, or $R_{Gt} - R_{ft}$, $Z_{i,G}$ measures sensitivity of the ith stock portfolio (or stock) to return dispersion in portfolio G, or σ_{Gt}, D_{it} is a signal variable with values $+1$ and -1 associated with positive and negative return dispersion effects, respectively, on stock returns, and $u_{it} \sim$ iid $N(0, \sigma_i^2)$. Unlike all other empirical asset pricing models that utilize ordinary least squares (OLS) regression for estimation purposes, there is no intercept or alpha mispricing parameter in the empirical ZCAPM. The reason is that the

[1] See also related ZCAPM works by Liu et al. (2012), Liu (2013), Liu et al. (2020), Kolari et al. (2022), Kolari et al. (2022), Kolari et al. (2023), Kolari and Pynnonen (2023), and Kolari et al. (2024).

expectation-maximization (EM) algorithm is used to estimate the regression equation, which does not utilize an intercept term (see Dempster et al. 1977). The signal variable D_{it} is an independent random variable with a two-point distribution:

$$D_{it} = \begin{cases} +1 & \text{with probability } p_i \\ -1 & \text{with probability } 1 - p_i, \end{cases} \tag{8.3}$$

where p_i (or $1 - p_i$) is the probability of a positive (or negative) return dispersion effect, and D_{it} are independent of u_{it}.

Because $E(D_{it}) = 2p_i - 1$, we can specify the marginal form of the empirical ZCAPM as[2]:

$$R_{it} - R_{ft} = \beta_{i,G}\left(R_{Gt} - R_{ft}\right) + Z^*_{i,G}\sigma_{Gt} + u^*_{it}, \quad t = 1, \ldots, T \tag{8.4}$$

where $\beta_{i,G}$ is the beta risk coefficient, $Z^*_{i,G} = Z_{i,G}(2p_i - 1)$ is the zeta risk coefficient, u^*_{it} is the error term with variance related to return dispersion, and other notation is as before. It is noteworthy that the sign of zeta risk loading $Z^*_{i,G}$ is positive or negative depending on the probability p_i related to the expected value of signal variable D_{it} in period $t = 1, \ldots, T$.

Kolari et al. (2021) have provided Matlab, R, and Python software programs for estimating the empirical ZCAPM on the internet at GitHub (https://github.com/zcapm). R is a free software environment available on the internet. We have found that the R programs run at a faster turnaround speed than the other programs.

8.3 Building Net Long Portfolios Using the ZCAPM

We use all U.S. stocks in the intersection of the NYSE, AMEX, and NASDAQ return files from the Center for Research in Security Prices (CRSP) in the period July 1964 to December 2022. Relatively small firms with market capitalization in the bottom 10% at year-end are dropped in the next year. Daily CRSP value-weighted returns and Treasury bill rates are used to proxy mean G portfolio returns (R_{Gt}) and riskless returns (R_{ft}), respectively. To proxy the cross-sectional standard deviation of stocks' returns, returns for i

[2] See Kolari et al. (2021) for detailed step-by-step instructions on how to estimate the EM algorithm in the context of the empirical ZCAPM.

$= 1,\ldots, n$ individual stocks in portfolio G on day t (R_{it}) are used to compute

$$\sigma_{Gt} = \sqrt{\frac{n}{n-1}\sum_{i=1}^{n} w_{it-1}(R_{it} - R_{Gt})^2}, \tag{8.5}$$

where w_{it} = the weight for stock i in portfolio G, and $t = 1, \ldots, T$ spans one year. In our analysis period from July 1964 to December 2022, the mean daily $\bar{R}_G$ equals 0.03%, which is somewhat less than the mean daily CRSP return $\bar{R}_{CRSP}$ of 0.04%. The mean daily σ_{Gt} equals 2.55%. Thus, cross-sectional market return dispersion of G is quite large compared to its mean return. Given the magnitude of return dispersion, it is reasonable to believe that stock returns and risks are significantly impacted by σ_{Gt} movements over time.

We next construct long-short, zero-investment portfolios based on their zeta risk, or $Z^*_{i,G}$.[3] The step-by-step procedure for building zeta risk portfolios is provided below.

1. The empirical ZCAPM Eq. (8.4) is estimates using the expectation-maximization (EM) algorithm for all individual stocks with daily returns for the initial 12-month estimation period from July 1963 to June 1964.
2. Stocks are ranked in terms of their estimated zeta coefficient $\hat{Z}^*_{i,a}$.
3. A total of 25 portfolios are formed from positive to negative zeta risk. Optimized weights are used to compute each portfolio's return in the next month July 1965 (i.e., out-of-sample returns). As in the construction of the G portfolio, weights for each stock are used based on an optimized weighting scheme.
4. With these portfolios in hand, we form 12 zero-investment portfolios with one long and one short portfolio. Portfolio 13 is dropped as its zeta risk is near zero.
   (a) Portfolios 1 and 25 are combined into zero-investment portfolio (denoted $Z12$) that is long the highest positive zeta risk $\hat{Z}^*_{i,a}$ stocks and short the highest negative zeta risk $-\hat{Z}^*_{i,a}$ stocks.
   (b) Portfolios 2 and 24 form the next long-short zeta risk portfolio (denoted $Z11$).

[3] Somewhat related to these portfolios, Ang et al. (2006) constructed a portfolio (i.e., volatility mimicking factor) based on innovations in the *VIX* (or the Chicago Board Options Exchange's CBOE Volatility Index). The VIX seeks to measure the stock market's expectation of 30-day volatility, a time-series market volatility measure. Unlike their study, we have a volatility portfolio based on the cross-sectional standard deviation of returns, not time-series standard deviation.

(c) This process continues until the last zeta risk portfolio based on portfolios 14 and 12 (denoted $Z1$). Portfolios $Z1$ to $Z12$ form progressively higher long-short, zeta risk portfolios.

5. For each zeta risk portfolio, we compute the daily returns in the next month July 1964 (out-of-sample), or $R_{pT+1}, \ldots, R_{pT+21}$ where T is the last day in the estimation period $t = 1, \ldots, T$.[4]
6. We iterate this process by rolling the estimation period in step (1) forward one month at a time to rebalance the zero-investment, long-short zeta risk portfolios. The process ends in December 2022. The final daily time series of portfolio returns for each zeta risk portfolio (viz., $Z1$ to $Z12$) consists of 11,727 daily returns from July 1964 to December 2022.

As before for portfolio G in Chapter 7, our long only portfolios have returns based on their daily returns in the next month. This out-of-sample performance evaluation approach is consistent with actual investor behavior. Because our 12 long-short zeta risk portfolios span a wide range of systematic market risk related to cross-sectional market return dispersion (or σ_{Gt}), consistent with ZCAPM theory, we anticipate that out-of-sample returns will increase as zeta risk increases in these portfolios from $Z1$ to $Z12$.

The last step is to form net long portfolios by combining the G portfolio with the 12 zeta risk portfolios: $G + Z1, G + Z2, \ldots, G + Z12$. As an example, assume that you buy \$100 worth of portfolio G. Also, you buy \$100 stocks (long) and sell \$100 stocks (short) in a long-short zeta risk portfolio (i.e., net weight equals 0). In the next month, G earns \$2 and the zeta risk portfolio earns \$3. Your total return on this leveraged portfolio would be (\$2 + \$3)/\$100 = 5%.[5] By adding the returns of each respective long-short zeta risk portfolio $Z1$ to $Z12$ to G returns, the returns of our net long portfolios should increase with increasing zeta risk.

Why use short positions in our net long portfolios? The main reason is that they are needed to build Markowitz (1959) efficient portfolios. Many studies have found that short positions are essential in forming efficient portfolios—for example, see studies by Pulley (1981), Levy (1983), Kallberg and Ziemba (1983), Kroll et al. (1984), Green and Hollifield (1992), Jagannathan and Ma

[4] We should note that the 25 diversified zeta risk portfolios have beta risk near one that cancels out for the most part in the long-short $Z1$ to $Z12$ portfolios. Thus, the latter long-short portfolios have returns commensurate with their zeta risk levels.

[5] What if your brokerage firm requires cash collateral from you to hold a long-short, zero-investment portfolio? In this case, you can only buy \$50 of the G portfolio and another \$50 long-short, which changes your monthly return on this unleveraged portfolio to (\$1 + \$1.5)/\$100 = 2.5%. In our analyses, we assume the brokerage allows leveraged positions.

(2003), Brennan and Lo (2010), and Levy and Ritov (2010). Emphasizing this requirement, Brennan and Lo (2010) mathematically proved that it is almost impossible to create mean-variance efficient frontiers with long only portfolios.

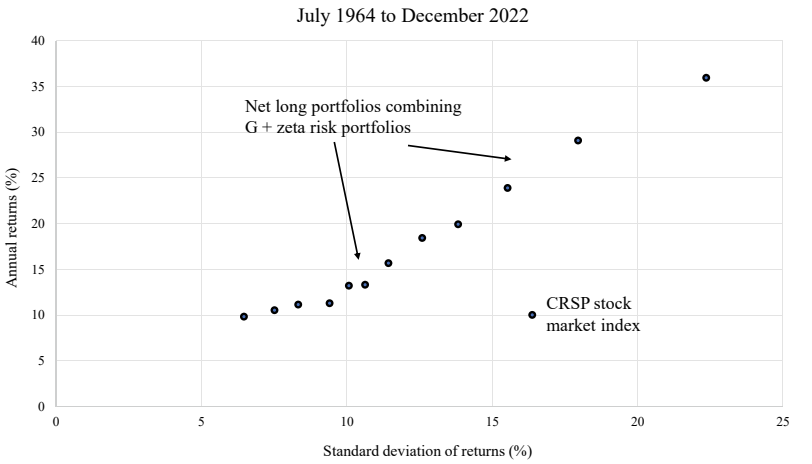
While we include short positions in our portfolios here, recognizing that institutional investors such as pension funds and mutual funds are restricted by law to long only portfolios, we employ ZCAPM-based methods to build long only portfolios in forthcoming Chapter 10, These portfolios can be readily constructed by focusing on positive zeta risk portfolios. As we will see, these portfolio analyses support earlier studies cited above that have found short positions boost the performance of portfolios in efficiency terms.

8.4 Empirical Results

We next report the performance of our net long portfolios. We employ some of our performance metrics discussed in Chapter 6. We compute the geometric mean of daily returns in our analysis period by means of Eq. (6.5). Subsequently, these geometric mean returns are annualized using Eq. (6.7). Daily standard deviations of returns in a portfolio are annualized as $\sigma_{pt} \times \sqrt{252}$. Lastly, Sharpe ratios are computed as shown in Eq. (6.10).

8.4.1 Net Long Portfolios in the Analysis Period

Figure 8.4 summarizes the results for our net long portfolios in the entire analysis period from July 1964 to December 2022. The graphical display of average annual returns (Y-axis) and their standard deviations (X-axis) are striking. Again, the 12 portfolios shown there are constructed by adding progressively higher long-short, zeta risk portfolios to the global minimum variance portfolio G. See Appendix for details on the average annual returns and total risks of these long-short zeta risk portfolios. As proposed by ZCAPM theory, the CRSP market index has returns that lie in the proximity of the axis of symmetry, which is well below the out-of-sample performance of the net long portfolios. At approximately the same total risk (or standard deviations of returns over time) as the CRSP index, while the CRSP index earns an average of 10.02% per year. Portfolio $G + Z10$ earns almost 24%, or about 2.4 times more than the CRSP index per year. For the most part, all of the net long portfolios earned the CRSP index return or more. Despite lower total risk than the CRSP index, portfolios $G + Z5$ to $G + Z9$ have



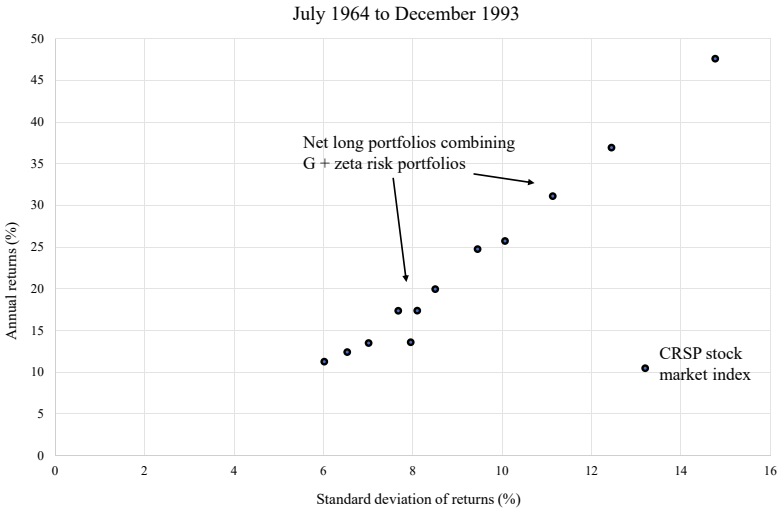
	G portfolio + zeta risk portfolio												
1964-2022: Annualized returns	G+Z1	G+Z2	G+Z3	G+Z4	G+Z5	G+Z6	G+Z7	G+Z8	G+Z9	G+Z10	G+Z11	G+Z12	CRSP
Net long portfolio returns %	9.83	10.53	11.15	11.30	13.21	13.32	15.67	18.43	19.93	23.90	29.08	35.94	10.02
Standard deviation of returns %	6.45	7.51	8.32	9.40	10.07	10.62	11.43	12.59	13.83	15.53	17.95	22.36	16.37

Fig. 8.4 Average daily net long portfolio returns compared to CRSP index returns in analysis period July 1964 to December 2022

average annual returns in the range of 13.21% to 19.93% that are materially above average CRSP returns.

To put the investment performance results in the context of investor wealth, we compute the compounded daily value of $1 invested on the first trading day of July 1964 to the last trading day of December 2022. The CRSP index generates a wealth accumulation of $242, which is quite good in terms of building wealth. Surprisingly, the $G + Z10$ portfolio with similar total risk as the CRSP index (as measured by the standard deviation of returns over this time period) yields about $224,700! This large wealth gain is 1,000 times greater than that of the CRSP index, which occurs due to persistently high returns over time that are compounded via the force of interest. These results amplify the fact that boosting portfolio performance per unit risk can pay major rewards to investors over time. Long-run portfolio performance is what matters for wealth creation. The world's most successful investor, Warren Buffet, no doubt would agree!

Returning to Fig. 8.4, notice that zeta risk controls the risk levels of the net long portfolios. As zeta risk increase, so does total risk on the X-axis. Also, as total risk increases, average annual returns rise almost linearly. These performance attributes demonstrate the power of ZCAPM-based machine methods to boost portfolio efficiency.



	G portfolio + zeta risk portfolio												
1964-1993: Annualized returns	G+Z1	G+Z2	G+Z3	G+Z4	G+Z5	G+Z6	G+Z7	G+Z8	G+Z9	G+Z10	G+Z11	G+Z12	CRSP
Net long portfolio returns %	11.25	12.41	13.49	13.58	17.36	17.37	19.94	24.74	25.71	31.10	36.91	47.58	10.46
Standard deviation of returns %	6.02	6.54	7.01	7.96	7.68	8.10	8.50	9.45	10.06	11.13	12.45	14.77	13.20

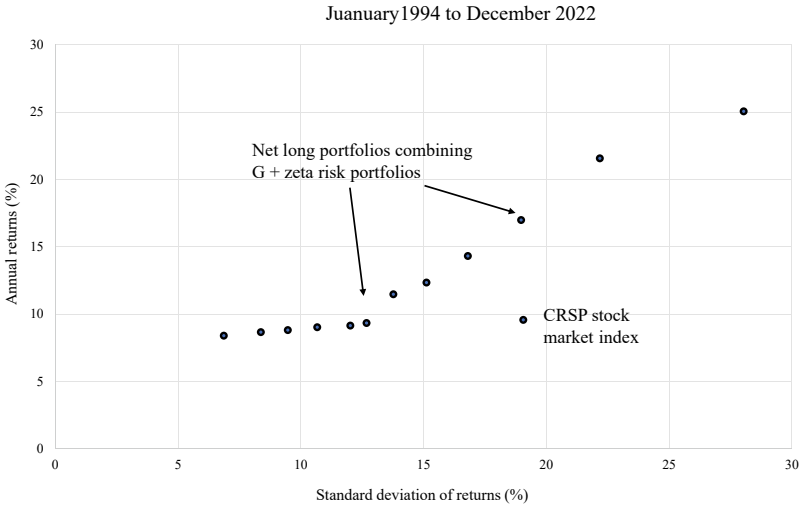
Fig. 8.5 Average daily net long portfolio returns compared to CRSP index returns in the subperiod July 1964 to December 1993

Portfolios $G+Z11$ and $G+Z12$, which lie at the high end of zeta risk, are very interesting. Assuming an investor is willing to accept relatively high total risk and accompanying volatility of returns over time, the long-run performance of these portfolios is exceptional. Portfolio $G + Z11$ earns on average 29.08% per year, and portfolio $G + Z12$ earns even more at 35.94%. These very high yields should be attractive to some investors with relatively less aversion to risk. Of course, some investors with risk loving preferences will be drawn to these portfolios also.

8.4.2 Net Long Portfolios for Subperiods

We split our analysis period into two subperiods of about 29 years each: July 1964 to December 1993, and January 1994 to December 2022. All analyses were repeated, net long portfolios formed, and out-of-sample (one-month-ahead) returns computed for portfolios. Figures 8.5 and 8.6 give the results for these two periods, respectively.

The results for the subperiod July 1964 to December 1993 are similar to those for the entire analysis period in Fig. 8.5. The value-weighted CRSP stock market index's performance is in the area of the axis of symmetry of



	G portfolio + zeta risk portfolio											
1994-2022: Annualized returns	G+Z1	G+Z2	G+Z3	G+Z4	G+Z5	G+Z6	G+Z7	G+Z8	G+Z9	G+Z10	G+Z11	G+Z12
Net long portfolio returns %	8.40	8.66	8.82	9.03	9.14	9.34	11.47	12.35	14.32	16.98	21.57	25.05
Standard deviation of returns %	6.87	8.38	9.47	10.67	12.02	12.68	13.77	15.12	16.81	18.97	22.18	28.04

Fig. 8.6 Average daily net long portfolio returns compared to CRSP index returns in subperiod January 1994 to December 2022

the parabola. Additionally, our net long portfolios substantially boost performance above that of the CRSP index, which lies far below these portfolios in the graph. One difference from the entire analysis period is that all of the net long portfolios have noticeably higher annual returns per unit total risk (i.e., standard deviation of returns on the X-axis) than before. Portfolios $G + Z7$ to $G + Z12$ have average annual returns at 20% or more compared to the average CRSP return of 10.46% per year. Remarkably, portfolios $G + Z11$ and $G + Z12$ have average annual returns of 36.91% and 47.58%, respectively. These returns are outstanding by any investment metric! For investors with a risk tolerance for market volatility, these high zeta risk portfolios should be of keen interest!

Again the results in the subperiod January 1994 to December 2022 parallel those for the entire analysis period. As shown in Fig. 8.6, the CRSP index is visibly outperformed by our net long portfolios. Holding total risk approximately the same for the CRSP index and portfolio G_Z10, the latter net long portfolio earns an average out-of-sample return of 16.98% per year compared to 9.57% for CRSP. Like the results for the entire analysis period, low zeta risk portfolios $G + Z1$ to $G + Z6$ have relatively lower returns in the range of the CRSP index with only small incremental return gains as zeta risk increases.

Thereafter returns tend to increase more rapidly as zeta risk (and concomitantly total risk) increases. Like before, portfolios $G+Z11$ and $G+Z12$ have relatively high annual returns equal to 21.57% and 25.05%, respectively.

We computed Sharpe ratios for the entire analysis period and two subperiods. Table 8.1 displays the results. All net long portfolios have Sharpe ratios that exceed the CRSP index by considerable margins. In the three different periods, the CRSP index has Sharpe ratios in the fairly narrow range of 0.29 to 0.34. In sharp contrast, the net long portfolios have Sharpe ratios in the range of 0.47 to 2.73. Even the low zeta risk portfolio $G + Z1$ had much higher Sharpe ratios between 0.74 and 0.84. The results are generally stronger for the net long portfolios in the entire analysis period and July 1964 to December 1993 subperiod. Sharpe ratios are normally lower for net long portfolios in the more recent subperiod January 1994 to December 2022. Even so, they well outperform the CRSP index on a consistent basis across different levels of zeta risk.

As alluded to earlier, high risk net long portfolios exhibit exceptional performance. In Table 8.1, we see that the Sharpe ratios of portfolios $G + Z8$ to $G + Z12$ dominate those of the CRSP index in the overall analysis period as well as subperiods. Among these high risk portfolios, the $G + Z11$ and $G + Z12$ portfolios perform head and shoulders above the other portfolios. Figures 8.4–8.6 show that these high zeta risk portfolios have outsized average returns per year. We infer from these findings that the tails of the cross-sectional distribution of returns closest to the upper and lower boundaries of the mean-variance parabola contain valuable market information. The ZCAPM is able to exploit this information by means of its ability to estimate zeta risk associated with cross-sectional return dispersion.

In general, the subperiod results provide robustness checks that corroborate our earlier entire analysis period results. Our asset pricing model approach to forming investment portfolios using the empirical ZCAPM to control zeta risk associated with market return dispersion yields extraordinary performance results. Investment performance is substantially boosted relative to the CRSP market index. We infer that the CRSP index is most certainly not an efficient portfolio as believed by many academic researchers in the past.

Table 8.1 Sharpe ratios of net long portfolios compared to the CRSP index in selected subperiods

Period	*G*+Z1	*G*+Z2	*G*+Z3	*G*+Z4	*G*+Z5	*G*+Z6	*G*+Z7	*G*+Z8	*G*+Z9	*G*+Z10	*G*+Z11	*G*+Z12	CRSP
7/1964–12/2022	0.84	0.81	0.81	0.73	0.87	0.84	0.98	1.11	1.12	1.25	1.37	1.41	0.34
7/1964–12/1993	0.76	0.87	0.97	0.87	1.39	1.32	1.56	1.91	1.9	2.19	2.43	2.73	0.29
1/1994–12/2022	0.74	0.64	0.58	0.53	0.48	0.47	0.59	0.60	0.65	0.72	0.82	0.77	0.33

8.5 Summary

This chapter proposed a novel asset pricing model approach to building high-performing stock portfolios. To the authors' knowledge, despite many academic articles over the past 60 years on asset pricing models, their application to the creation of high-performing, well-diversified portfolios in real world investment practice has been limited. Utilizing research published in Kolari et al. (2021), we employ ZCAPM theory to financially engineer diversified investment portfolios with different risk levels. First, building on Chapter 7, the global minimum variance portfolio G is used as a foundational portfolio that contains no zeta risk associated with cross-sectional market return dispersion. The G portfolio earns an average return that splits the mean-variance Markowitz parabola into two halves (see Fig. 8.1). The width of the parabola to the right of G in return/risk space is largely determined by zeta risk and return dispersion (see Fig. 8.2). As zeta risk increases above zero (decreases below zero) for portfolios, their returns increase (decrease). We use this geometry to propose zero-investment, long-short zeta risk portfolios that go long high positive zeta risk portfolios and short high negative zeta risk portfolios. By varying these long-short zeta risk positions, a series of long-short portfolios are conceived that have increasing levels of zeta risk. Lastly, the G portfolio are combined with the long-short zeta risk portfolios to establish a lower bound return in the middle of the parabola and then move upward in the investment parabola via different levels of zeta risk (see Fig. 8.3).

We implement ZCAPM theory by estimating the empirical ZCAPM with U.S. stock returns. Our analysis period was July 1964 to December 2022. Data was collected for U.S. common stocks in the CRSP database. Stocks in the bottom 10% by market capitalization per year were dropped. Our ZCAPM-based approach to building investment portfolios can be summarized as follows:

1. Estimate the empirical ZCAPM using one year of daily returns. Drawing upon our analyses and results in Chapter 7, as factors in the empirical ZCAPM, we employ G portfolio returns and their cross-sectional standard deviation of returns. The global minimum variance portfolio G is formed by optimally weighting individual stocks.
2. Stocks are sorted into 25 zeta risk portfolios using the estimated $Z^*_{i,G}$ zeta risk parameter for each individual stock (viz., $1, \ldots, 25$). Zeta risk portfolio 1 has the highest positive zeta risk, and portfolio 25 has the highest

negative zeta risk. Portfolio 13 near zero zeta risk is dropped. Like portfolio G, individual stock returns are weighted by means of an optimized weighting scheme.

3. We formed 12 long-short zeta risk portfolios using zeta portfolios $1, \ldots, 25$. Zeta risk portfolios 1 and 25 are used in long and short positions, respectively, to form $Z12$. These 12 portfolios have increasing levels of zeta risk.
4. Portfolio G is added to each long-short zeta risk portfolio to build 12 net long portfolios $G + Z1, \ldots, G + Z12$.
5. In the next month, out-of-sample daily returns for these 12 net long portfolios are computed to imitate an investable strategy that a real world investor could implement.
6. The above process is rolled forward one month and repeated until December 2022. As such, portfolios are rebalanced monthly.

Our investment performance results are striking. The net long portfolios significantly boost investment performance relative to the well-known CRSP market index, especially as zeta risk levels increase in these portfolios. At the lowest zeta risk level, the CRSP index had higher average returns than the net long portfolio, but its total risk was substantially higher, which lowered its Sharpe ratio relative to the net long portfolio. As zeta risk increased, net long portfolios demonstrated superior returns. Indeed, for the highest zeta risk portfolios, their outperformance relative to the CRSP index was nothing short of remarkable. For example, in the entire analysis period, the CRSP index earned about 10% per year compared to from 29% to 36% per year for the two net long portfolios with the highest zeta risk. This outperformance should get the attention of practicing professional investment managers! Investment clients willing to accept relatively high total risk can earn exceptionally high average returns over long investment horizons by means of our ZCAPM-based approach.

In closing, it is worthwhile to highlight two key findings. First, the CRSP index is clearly not an efficient portfolio. Our evidence should put the often-accepted notion that the CRSP index is a valid proxy for the efficient market portfolio to rest. Also, the CRSP index is a close cousin of the S&P 500 index, one of the most popular investment portfolios in the world, which is likewise not an efficient portfolio. Since most investment professionals cannot consistently beat the S&P 500 index over three-to-five-year horizons (e.g., see often-cited evidence by Malkiel [1995, 2005]), our net long portfolio results are worthy of attention in the investment world. Second, our results prove that the ZCAPM asset pricing model can be applied to stock

portfolio analyses to build high-performing portfolios for investors. In this regard, the authors are interested in partnering with the investment community to help bring these high-performing portfolios to investors in the real world.

Appendix: Long-Short Portfolios Based on Zeta Risk Levels

This appendix provides performance results for the 12 long-short portfolios formed with respect to zeta risk in the empirical ZCAPM. Here we compare the average annual returns and standard deviation of returns of our long-short zeta risk portfolios to the following portfolios: CRSP index, *G* portfolio, *SMB* portfolio (long small stocks and short big stocks), *HML* portfolio (long value stocks and short growth stocks), *RMW* portfolio (long high profit firms and short low profit firms), and *CMA* portfolio (long conservative capital investment firms and short aggressive capital investment firms). As discussed earlier in Chapter 3, the *SMB*, *HML*, *RMW*, and *CMA* multifactors were proposed in a series of papers by Fama and French (1992, 1993, 1995, 1996, 1998, 2004, 2015, 2018, 2020). Since they are long-short portfolios, it is appropriate to compare them to our long-short zeta risk portfolios.

Figure 8.7 gives the results for our analysis period from July 1964 to December 2020. The Fama and French multifactors are similar to the lower zeta risk portfolios. They have relatively low annual returns and total risks relative to a number of long-short zeta risk portfolios. Compared to the multifactor portfolios, the CRSP index has much higher total risk and higher average annual returns too. The *G* portfolio has similar total risk as the lowest zeta risk portfolios but much higher average annual return than these portfolios as well as the multifactor portfolios.

The five zeta risk portfolios with higher risk in Fig. 8.7 lie along an almost linear upward sloping line. At similar total risk as the CRSP index, the zeta risk portfolios earn about 6% higher annual returns than the CRSP index—that is, annual returns are substantially boosted from about 10% per year for the CRSP index to 16% per year for the zeta risk portfolios. Finally, the long-short zeta risk portfolio with highest zeta risk at the far upper right of the graph has outsized returns of about 24% per year.

In sum, average annual returns increase with zeta risk in long-short portfolios, especially as zeta risk rises above initially lower levels. In our analysis period, average returns are boosted above average general stock market returns per unit total risk. For those investors interested in earning relatively high

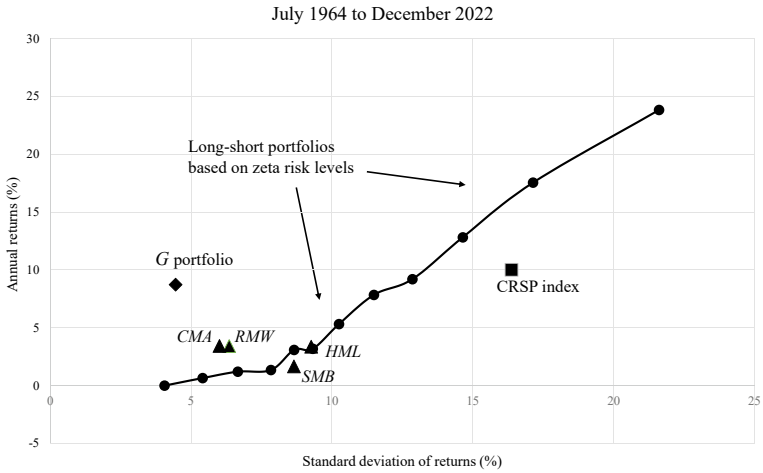


Fig. 8.7 Long-short zeta risk portfolio returns compared to CRSP index returns as well as well-known multifactors in the analysis period July 1964 to December 2022

returns, zeta risk enables the construction of such portfolios by controlling their risk levels as estimated by the empirical ZCAPM. Despite its higher total risk than the CRSP index, the Sharpe ratio of the highest zeta risk portfolio 12 with average annual returns of about 24 percent equals 0.90, which is considerably larger than the Sharpe ratio of the CRSP index at 0.34.

References

Ang, A., R.J. Hodrick, Y. Xing, and X. Zhang. 2006. The cross-section of volatility and expected returns. *Journal of Finance* 61: 259–299.

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Brennan, T.J., and W.W. Lo. 2010. Impossible frontiers. *Management Science* 56: 905–923.

Dempster, A.P., N.M. Laird, and D.B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society* 39: 1–38.

Fama, E.F., and K.R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427–465.

Fama, E.F., and K.R. French. 1993. The cross-section of expected returns. *Journal of Financial Economics* 33: 3–56.

Fama, E.F., and K.R. French. 1995. Size and book-to-market factors in earnings and returns. *Journal of Finance* 50: 131–156.

Fama, E.F., and K.R. French. 1996. The CAPM is wanted, dead or alive. *Journal of Finance* 51: 1947–1958.

Fama, E.F., and K.R. French. 1998. Value versus growth: The international evidence. *Journal of Finance* 53: 1975–1999.

Fama, E.F., and K.R. French. 2004. The capital asset pricing model: Theory and evidence. *Journal of Economic Perspectives* 18: 25–46.

Fama, E.F., and K.R. French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116: 1–22.

Fama, E.F., and K.R. French. 2018. Choosing factors. *Journal of Financial Economics* 128: 234–252.

Fama, E.F., and K.R. French. 2020. Comparing cross-section and time-series factor models. *Review of Financial Studies* 33: 1892–1926.

Green, R.C., and B. Hollifield. 1992. When will mean-variance efficient portfolios be well diversified? *Journal of Finance* 47: 1785–1809.

Jagannathan, R., and T. Ma. 2003. Risk reduction in large portfolios: A role for portfolio weight constraints. *Journal of Finance* 58: 1651–1684.

Kallberg, J.G., and W.T. Ziemba. 1983. Comparison of alternative utility functions in portfolio selection problems. *Management Science* 29: 1257–1276.

Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2022. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*, available online at https://www.mdpi.com/1911-8074/15/3/137. Reprinted in Kolari, J.W., and S. Pynnonen, eds. 2022. *Frontiers of asset pricing*. Basel, Switzerland: MDPI.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. Testing for missing asset pricing factors. Paper presented at the Western Economic Association International, San Diego, CA.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Kroll, Y., H. Levy, and H. Markowitz. 1984. Mean-variance versus direct utility maximization. *Journal of Finance* 39: 47–61.

Levy, H. 1983. The capital asset pricing model: Theory and empiricism. *Economic Journal* 93: 145–165.

Levy, M., and Y. Ritov. 2010. Mean-variance efficient portfolios with many assets: 50% short. *Quantitative Finance* 11: 1461–1471.

Liu, W. 2013. A new asset pricing model based on the zero-beta CAPM: Theory and evidence. Doctoral dissertation, Texas A&M University, Chapter 8.

Liu, W., J.W. Kolari, and J.Z. Huang. 2012. A new asset pricing model based on the zero-beta CAPM market model (CAPM). Presentation at the annual meetings of the Financial Management Association. Best Paper Award in Investments, Atlanta, GA (October).

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. Return dispersion and the cross-section of stock returns. Presentation at the annual meetings of the Southern Finance Association, Palm Springs, CA (October).

Malkiel, B.G. 1995. Returns from investing in equity mutual funds 1971 to 1991. *Journal of Finance* 50: 549–572.

Malkiel, B.G. 2005. Reflections on the efficient market hypotheses: 30 years later. *The Financial Review* 40: 1–9.

Markowitz, H.M. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York: Wiley.

Pulley, L.B. 1981. General mean-variance approximation to expected utility for short holding periods. *Journal of Financial and Quantitative Analysis* 16: 361–373.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.



9

Net Long Portfolio Risk Analyses

9.1 GRS Risk Metrics

This section contains Gibbons, Ross, and Shanken (GRS) (1989) tests of the net long portfolios. All returns for the portfolios under study are out-of-sample in the next month. Hence, the portfolios are investable in the sense that investors buy stocks in the portfolios and then observe their returns and risks in the next month. However, we should comment that some analyses (e.g., the theoretical efficient frontier) are essentially in-sample analyses as discussed shortly.

Table 9.1 reports the Sharpe ratios and GRS statistics for the net long portfolios. The GRS test is computed as defined in Eq. (6.11) of Chapter 6. Figure 9.1 depicts the theoretical (in-sample) parabola generated from monthly returns on the CRSP value-weighted index and the 12 net long $G + Z1,\ G + Z2,\ \ldots,\ G + Z12$ portfolios. It is obvious that the CRSP market return is well below the theoretical efficient frontier as represented by the upper curve of the parabola.

The ZCAPM is derived as a special case of Black's (1972) zero-beta CAPM. The authors have published a number of related works, including Liu, Kolari and Huang (2012), Liu (2013), Liu, Kolari, and Huang (2020), Kolari, Huang, Liu, and Liao (2022), Kolari, Huang, Butt, and Liao (2022), Kolari, Huang, Liu, and Liao (2023), Kolari and Pynnonen (2023), and Kolari, Huang, Liu, and Liao (2024).

The average excess return of the portfolio per unit of total risk as measured by the standard deviation of portfolio returns over time.

See citations to this literature in Chapter 6.

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We refer to the efficient frontier as theoretical due to the fact that it is computed using out-of-sample returns for the net long portfolios and, therefore, does not represent an investable strategy. From a practical standpoint, as already mentioned, investors buy a portfolio and then compute its performance in say the next out-of-sample month. Optimal weights for the 12 net long portfolios would need to be computed prior to this next month. By contrast, in Fig. 9.1 the optimal weights are computed ex post in the out-of-sample next month. Nonetheless, the graph shows that the net long portfolios mimic the shape of the efficient frontier. Also, we should note that optimal weighting of these 12 portfolios could be done prior to the ex post month to create combined portfolios that come closer to the theoretical efficient frontier. Certainly, the graph in Fig. 9.1 provides a glimpse of the possibility with some additional investment portfolio effort.

Referring to Table 9.1 with $N = 12$ test assets and time in months $T = 702$, the GRS statistic for the CRSP index is highly significant at $F = 6.93$ with p-value near zero at 0.0000. That is, for any reasonable confidence level, the observed deviation of the CRSP market portfolio from the efficient frontier cannot be considered to be due to random noise. The same result holds in the two subperiods: (1) July 1964 to December 1993 and (2) January 1994 to December 2022. Therefore, we infer that the CRSP market portfolio is not mean-variance efficient.

Compared to the theoretical frontier, according to GRS tests in Table 9.1, most of the net long portfolios are inefficient also. One exception is that portfolio G+Z12 is efficient in the first subperiod from 7/1964 to 12/1993. In this regard, notice that the net long portfolios approach theoretical efficiency as the zeta risk increases. Also their F-statistics decrease in the more recent subperiod from 1/1994 to 12/2022. This finding suggests that the net long portfolios became more efficient in the later subperiod. As already mentioned, optimal construction of combined portfolios comprised of the 12 net long portfolios holds promise as near theoretically efficient portfolios.

The GRS statistics are statistical measures of the deviation of the given portfolios from the theoretical (abstract) frontier. Practical interest, however, is in the relative performance of the portfolios with respect to the CRSP index of the general market. Table 9.2 reports the results for evaluating whether the observed Sharpe ratio outperformance in Table 8.1 of Chapter 8 concerning the net long portfolios $G + \text{Z1}$, $G + \text{Z2}$, $\ldots$, $G + \text{Z12}$ over the CRSP index are statistically significant.

Looking at the results for the entire analysis period from 7/1964 to 12/2022 in Table 9.2, all net long portfolios outperform the CRSP index as evidenced by the statistically significant Sharpe ratio differences. Of these,

Table 9.1 GRS *F*-tests of efficiency for the net long portfolios *G*+Z1, *G*+Z2, ..., *G*+Z12 and the CRSP index (*Data source* CRSP database from the University of Chicago)

	Sample period								
	7/1964 to 12/2022			7/1964 to 12/1993			1/1994 to 12/2022		
Portf	Sharpe	*F*-stat	*p*-value	Sharpe	*F*-stat	*p*-value	Sharpe	*F*-stat	*p*-value
G+Z1	0.84	6.07	0.0000	0.68	7.88	0.0000	0.74	2.37	0.0061
G+Z2	0.81	5.95	0.0000	0.87	7.71	0.0000	0.64	2.60	0.0025
G+Z3	0.81	5.80	0.0000	0.97	7.44	0.0000	0.58	2.94	0.0007
G+Z4	0.73	6.02	0.0000	0.87	7.55	0.0000	0.53	3.08	0.0004
G+Z5	0.87	5.10	0.0000	1.39	6.54	0.0000	0.48	3.10	0.0004
G+Z6	0.84	5.41	0.0000	1.32	6.75	0.0000	0.47	3.23	0.0002
G+Z7	0.98	4.41	0.0000	1.56	5.91	0.0000	0.59	2.91	0.0007
G+Z8	1.11	3.76	0.0000	1.91	4.87	0.0000	0.60	2.98	0.0006
G+Z9	1.12	3.57	0.0000	1.89	4.70	0.0000	0.65	2.85	0.0010
G+Z10	1.25	3.10	0.0003	2.19	3.62	0.0000	0.72	2.85	0.0010
G+Z11	1.37	2.12	0.0138	2.43	2.60	0.0025	0.82	2.51	0.0035
G+Z12	1.41	1.95	0.0263	2.77	1.12	0.3450	0.77	2.66	0.0020
CRSP	0.34	6.93	0.0000	0.29	8.17	0.0000	0.33	3.44	0.0001

The sample period covers daily returns from July 1, 1964 to December 30, 2022. Construction of the portfolios $G+Z1,\ G+Z2,\ \ldots,\ G+Z12$ are described in Chapter 8

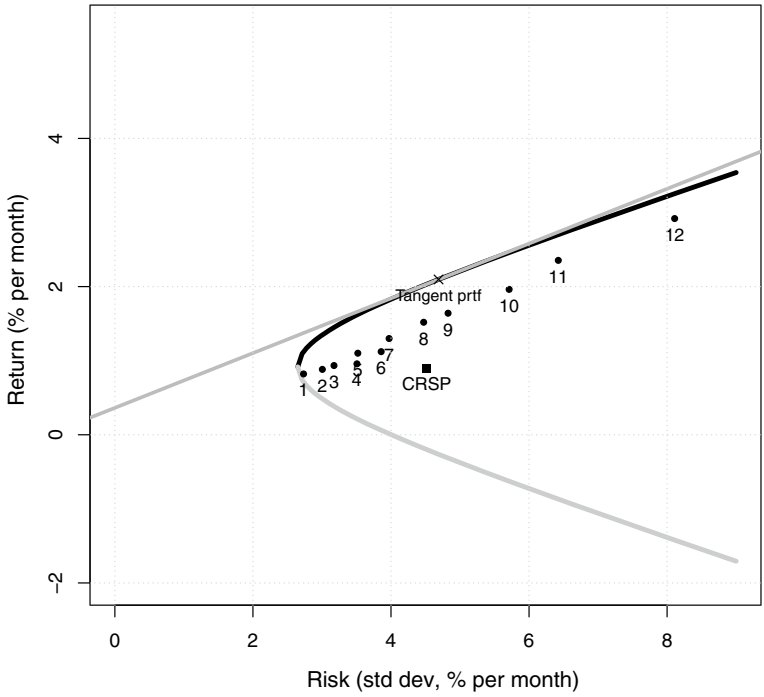


Fig. 9.1 Theoretical efficient frontier based on the *G*+Z1, *G*+Z2, ..., *G*+Z12 portfolios constructed in Chapter 8 and the CRSP value-weighted market index. The sample period covers daily returns from July 1, 1964 to December 30, 2022. The numbers $1, \ldots, 12$ refer to the *G*+Z1, *G*+Z2, ..., *G*+Z12 net long portfolios, respectively (*Data source* CRSP database from the University of Chicago)

G+Z4 is on the borderline with a *p*-value equaling 0.047. The subperiods in the table show that total period differences stem mainly from the initial subperiod from 7/1964 to 12/1993. In the later period from 1/1994 to 12/2022, only *G*+Z1 statistically outperforms the CRSP index. Nonetheless, by inspecting the Sharpe ratios in the subperiod from 1/1994 to 12/2022, we observe that the numerical Sharpe ratios of the net long portfolios are well above that of the CRSP index. Furthermore, the majority of them are well above 0.50 (based on annualized returns) versus 0.33 of the CRSP index. Practitioners typically use a 0.50 Sharpe ratio as a high watermark or threshold for high-performing portfolios.

Table 9.2 Statistical significance of the mean difference between Sharpe ratios of the net long portfolios *G*+Z1, *G*+Z2, ..., *G*+Z12 constructed in Chapter 8 and the Sharpe ratio of the CRSP market index

	Sample period								
	7/1964 to 12/2022			1/1964 to 12/1993			1/1994 to 12/2022		
Portf	Sharpe	*t*-val	*p*-val	Sharpe	*t*-val	*p*-val	Sharpe	*t*-val	*p*-val
G+Z1	0.84	2.87	0.004	0.76	1.97	0.049	0.74	2.12	0.034
G+Z2	0.81	2.65	0.008	0.87	2.50	0.013	0.64	1.50	0.133
G+Z3	0.81	2.50	0.013	0.97	2.86	0.004	0.58	1.16	0.247
G+Z4	0.73	1.99	0.047	0.87	2.02	0.043	0.53	0.94	0.350
G+Z5	0.87	2.72	0.006	1.39	4.54	0.000	0.48	0.68	0.494
G+Z6	0.84	2.48	0.013	1.32	4.18	0.000	0.47	0.62	0.538
G+Z7	0.98	3.08	0.002	1.56	4.93	0.000	0.59	0.95	0.342
G+Z8	1.11	3.65	0.000	1.91	6.11	0.000	0.60	0.93	0.351
G+Z9	1.12	3.64	0.000	1.89	5.99	0.000	0.65	1.08	0.282
G+Z10	1.25	4.15	0.000	2.19	6.79	0.000	0.72	1.24	0.214
G+Z11	1.37	4.62	0.000	2.43	7.36	0.000	0.82	1.53	0.127
G+Z12	1.41	4.70	0.000	2.77	8.19	0.000	0.77	1.33	0.183
CRSP	0.34	–	–	0.29	–	–	0.33	–	–

The Sharpe ratios are computed from annualized returns and standard deviations (see Table 8.1 in Chapter 8) by employing 14,727, 7,425, and 7,302 daily returns from the full sample and the two subperiods, respectively. The test statistics are computed with R package SharpeR (Pav, S. E., 2021, SharpeR: Statistical significance of the Sharpe ratio. R package version 1.3.0, https://github.com/shabbychef/SharpeR)

9.2 Value at Risk Metrics

Table 9.3 reports 1-day 5% VaRs (RM2006, see Chapter 6) computed for net long portfolios $G + Z1$, $G + Z2$, ..., $G + Z12$ based on the ZCAPM. Figure 9.2 graphically displays the VaR results in the table. As shown there, VaR values increase along with the zeta riskiness of the portfolios.

In the entire analysis period as well as subperiods, the CRSP VaR in Table 9.3 exceeds those for the net long portfolios with the exceptions of two or three of the most risky net long portfolios. For the lowest risk portfolio G+Z1, the 1-day 5% VaR is approximately 0.622, i.e., with 95% probability one can expect to lose at most 0.622%, or 62 basis points the next day. For the highest risk portfolio G+Z12, VaR is 2.005 for the entire analysis period compared to the CRSP index at 1.509. It is obvious from the graph that the later subperiod from 1/1994 to 12/2022 is riskier than the initial subperiod from 7/1964 to 12/1993. Finally, only portfolios G+Z11 and G+Z12 are riskier than the CRSP index.

9.3 Drawdown Risk Metrics

Chapter 6 introduced the concept of *drawdowns* that are an important measure of downside risk. For the net long portfolios $G + Z1$, $G + Z2$, ..., $G + Z12$ and the CRSP index during the analysis period from July 1964 to December 2022, Table 9.4 displays frequency distributions of drawdowns, and Table 9.5 reports descriptive statistics for *depth*, *duration*, *peak-to-trough*, and *recovery* measures. Readers can refer back to Fig. 6.4 in Chapter 6 and related discussion for details of these drawdown concepts. The tables utilize the R package Performance Analytics[1] in the analyses.

Similar to VaR results in the previous section, the clear pattern of drawdowns documented in Table 9.4 is the increasing riskiness of net long portfolios from G+Z1 toward G+12. The frequency and percentage distributions of the drawdown depths in Panels A and B, respectively, are informative. The total row in Panel A and first row in Panel B tell the story. Drawdowns for the low risk portfolios of G+Z1, G+Z2, and G+Z3 tend to be a much lower than those for the high risk portfolios G+Z10, G+Z11, and G+Z12. For the former, at least 78% of the time, drawdowns are less than one percent, whereas for the high risk portfolios it is 65% or lower. For the CRSP index, we get 61.1%. Additionally, the likelihood for more than 5%

[1] https://www.r-project.org and https://cran.r-project.org/web/packages/PerformanceAnalytics/index.html

Table 9.3 One-day 5% VaR per $100 investment in each of the net long portfolios *G*+Z1, *G*+Z2, ..., *G*+Z12 and the CRSP index portfolio with respect to the entire analysis period as well as two subperiods during the analysis period from July 1, 1964 to December 30, 2022

Period	*G*+Z1	*G*+Z2	*G*+Z3	*G*+Z4	*G*+Z5	*G*+Z6	*G*+Z7	*G*+Z8	*G*+Z9	*G*+Z10	*G*+Z11	*G*+Z12	CRSP
7/1964–12/2022	0.622	0.709	0.792	0.883	0.929	0.988	1.052	1.149	1.250	1.403	1.614	2.005	1.509
7/1964–12/1993	0.602	0.640	0.690	0.760	0.753	0.802	0.843	0.925	0.986	1.089	1.225	1.461	1.269
1/1994–12/2022	0.644	0.783	0.899	1.012	1.112	1.183	1.270	1.383	1.529	1.734	2.023	2.577	1.762

The VaRs are based on the normal distribution. The *G* portfolios are rebalanced at the end of each calendar month

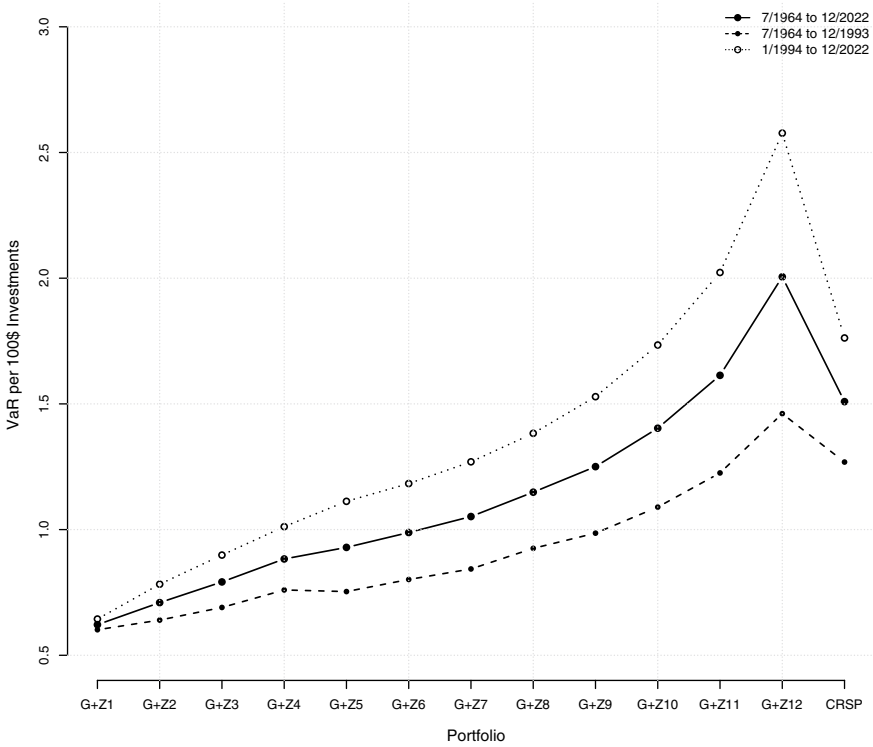


Fig. 9.2 One-day 5% Value at Risk for net long portfolios *G*+Z1, *G*+Z2, ..., *G*+Z12 and CRSP index. The sample period covers daily returns from July 1, 1964 to Dec 30, 2022. The first 252 days are used to estimate RiskMetrics variances for the first VaR on July 1, 1965. Thereafter the procedure is continued day by day until the end of the analysis period. For the last VaR on December 30, 2022, the volatility is estimated from returns ending December 29, 2022 (*Data source* CRSP database from the University of Chicago)

drawdowns ranges for the low risk porfolios of *G*+Z1, *G*+Z2, and *G*+Z3 from 3.4% to 5.2%, which is less than the high risk portfolios of *G*+Z10, *G*+Z11, and *G*+Z12 with likelihood ranges from 9.6% to 12.2%. For the CRSP index, the likelihood is 10%.

Panels A and B in Table 9.5 provide summary statistics for the depth and duration measures, respectively. Because the arithmetic mean is sensitive to outliers, the rare extreme drawdowns acutely influence the statistic. Median drawdown depth values, as already seen from the relative frequency distributions in Panel B of Table 9.4, are well below 1% (viz., only 0.3% for the low risk portfolios and 0.8% for the high risk portfolio). The CRSP index belongs to this high risk end with a median depth equal to 0.6%. The maximum drawdowns for low risk portfolios are below 45%, whereas for the high risk

portfolios the maximum drawdowns are 58.4%, 64.1%, and 81.9% for the G+10, G+11, and G+12 portfolios, respectively. For the CRSP index, the maximum drawdown is 54.7%, which is slightly lower than the high risk portfolios but much higher than those for the low to medium risk net long portfolios.

Panel B of Table 9.5 displays summary statistics for the total duration of the drawdowns. Again the means are contaminated by a few long-lasting drawdowns. Therefore, while the median durations are three or four days for all portfolios, the means range from 16.6 days for portfolio G+Z1 to 22.1 days for the CRSP index. Otherwise there is no clear pattern in the means with respect to the risk of the portfolios. However, the riskiness is clearly visible in the maximum durations—namely, higher risk portfolios have longer maximum drawdown duration. The CRSP index experienced the longest drawdown. The longest drawdowns are related to technology bubble in 2000, financial crisis in 2008–2010, and pandemic in 2020–2021.

Panel C of Table 9.5 shows the peak to trough time spans. Again, because the majority of drawdown depths are less than 1%, the median time spans from a peak to trough are short with a maximum of only 2 days. More interesting statistics are the maximums in Panel C. A clear decreasing trend is apparent—that is, lower risk net long portfolios decline more slowly to the trough than high risk portfolios. In other words, higher risk portfolios tend to crash faster in catastrophic market conditions. The CRSP index appears to be an exception with the longest-lasting duration before reaching the bottom.

Finally, Panel D of Table 9.5 summarizes some interesting features of the recovery from a trough. Again the median recoveries are one or two days due to typically small drawdowns as shown in Table 9.4. Also, similar to peak-to-trough patterns, recoveries tend to be faster for less risky portfolios. The longest recovery of 1,282 trading days, or about 5 years, occurred in G+Z12 portfolio after the 1,609 trading days (over 6 years) drawdown related to the 2008–2010 financial crisis (i.e., the large drawdown was almost 82% as shown in Panel A). For the CRSP index, the financial crisis drawdown was also the deepest one reaching almost 55%. The recovery took 760 trading days, but it was not the longest recovery in Panel D. The longest recovery was associated with the drawdown in response to the post-2000 crash after the technology bubble. This drawdown lasted 1,654 trading days (or over 6 and half years) as shown in Panel B.

For net long portfolios G+Z1, G+Z4, G+Z8, and G+Z12, in the upper panel of each graph, Figs. 9.3 to 9.6 depict the cumulative values of a \$1 investment in July 1964 that is held to the end of December 2022. The lower panels exhibit the drawdowns of the investment over time. The Y-axis in each

Table 9.4 Drawdown frequency distributions for net long portfolios *G*+Z1, *G*+Z2, ..., *G*+Z12 and the CRSP index using daily returns (*Data source* CRSP database from the University of Chicago)

Drawdown	*G*+Z1	*G*+Z2	*G*+Z3	*G*+Z4	*G*+Z5	*G*+Z6	*G*+Z7	*G*+Z8	*G*+Z9	*G*+Z10	*G*+Z11	*G*+Z12	CRSP
Panel A. Frequency distributions of drawdowns													
$-1<0$	663	571	538	465	536	455	507	509	473	475	478	404	385
−2 to −1	68	58	66	76	84	79	81	83	93	107	114	130	101
−3 to −2	23	20	23	23	34	31	37	45	35	48	49	64	37
−4 to −3	12	14	8	11	12	9	18	21	22	18	23	28	24
−5 to −4	8	8	14	17	14	17	11	14	14	11	19	22	20
−6 to −5	4	8	6	5	11	6	6	3	17	17	21	14	20
−7 to −6	8	7	5	3	4	7	6	12	5	8	8	14	5
−8 to −7	2	1	4	6	4	4	10	3	5	7	8	6	5
−9 to −8	0	3	2	7	4	3	4	4	6	5	9	7	1
−10 to −9	1	0	1	4	4	3	2	2	2	3	6	4	11
−15 to −10	3	6	11	7	11	12	11	10	9	9	9	19	6
−20 to −15	4	4	0	3	2	3	5	8	7	7	13	7	3
−25 to −20	3	2	3	3	5	3	5	4	2	5	2	5	5
−50 to−25	2	3	4	4	3	6	5	6	7	8	9	11	6
<−50	0	0	0	0	0	0	0	0	1	1	1	3	1
Total	801	705	685	634	728	638	708	724	698	729	769	738	630
Panel B. Percentage distributions of drawdowns													
$-1<0$	82.8	81.0	78.5	73.3	73.6	71.3	71.6	70.3	67.8	65.2	62.2	54.7	61.1
−2 to −1	8.5	8.2	9.6	12.0	11.5	12.4	11.4	11.5	13.3	14.7	14.8	17.6	16.0
−3 to −2	2.9	2.8	3.4	3.6	4.7	4.9	5.2	6.2	5.0	6.6	6.4	8.7	5.9
−4 to −3	1.5	2.0	1.2	1.7	1.6	1.4	2.5	2.9	3.2	2.5	3.0	3.8	3.8

Drawdown	*G*+Z1	*G*+Z2	*G*+Z3	*G*+Z4	*G*+Z5	*G*+Z6	*G*+Z7	*G*+Z8	*G*+Z9	*G*+Z10	*G*+Z11	*G*+Z12	CRSP
−5 to −4	1.0	1.1	2.0	2.7	1.9	2.7	1.6	1.9	2.0	1.5	2.5	3.0	3.2
−6 to −5	0.5	1.1	0.9	0.8	1.5	0.9	0.8	0.4	2.4	2.3	2.7	1.9	3.2
−7 to −6	1.0	1.0	0.7	0.5	0.5	1.1	0.8	1.7	0.7	1.1	1.0	1.9	0.8
−8 to −7	0.2	0.1	0.6	0.9	0.5	0.6	1.4	0.4	0.7	1.0	1.0	0.8	0.8
−9 to −8	0.0	0.4	0.3	1.1	0.5	0.5	0.6	0.6	0.9	0.7	1.2	0.9	0.2
−10 to −9	0.1	0.0	0.1	0.6	0.5	0.5	0.3	0.3	0.3	0.4	0.8	0.5	1.7
−15 to −10	0.4	0.9	1.6	1.1	1.5	1.9	1.6	1.4	1.3	1.2	1.2	2.6	1.0
−20 to −15	0.5	0.6	0.0	0.5	0.3	0.5	0.7	1.1	1.0	1.0	1.7	0.9	0.5
−25 to −20	0.4	0.3	0.4	0.5	0.7	0.5	0.7	0.6	0.3	0.7	0.3	0.7	0.8
−50 to −25	0.2	0.4	0.6	0.6	0.4	0.9	0.7	0.8	1.0	1.1	1.2	1.5	1.0
<−50	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.4	0.2

The table reports both frequency (Panel A) and percentage (Panel B) distributions of drawdowns for the net long portfolios *G*+Z1, *G*+Z2, ..., *G*+Z12 and the CRSP index with drawdown percentages indicated in the first column in the table. The analysis period covers daily returns from July 1964 to December 2022

Table 9.5 Drawdown summary statistics for net long portfolios *G*+Z1, *G*+Z2, ..., *G*+Z12 and the CRSP index computed from daily returns (*Data source* CRSP database from the University of Chicago)

Drawdown	*G*+Z1	*G*+Z2	*G*+Z3	*G*+Z4	*G*+Z5	*G*+Z6	*G*+Z7	*G*+Z8	*G*+Z9	*G*+Z10	*G*+Z11	*G*+Z12	CRSP
Panel A. Drawdown (depth) summary statistics													
Mean	−0.9	−1.1	−1.2	−1.5	−1.4	−1.6	−1.6	−1.7	−1.9	−2.0	−2.2	−2.7	−2.1
Median	−0.3	−0.3	−0.3	−0.4	−0.4	−0.4	−0.4	−0.4	−0.5	−0.5	−0.6	−0.8	−0.6
Q1	−0.1	−0.1	−0.1	−0.1	−0.1	−0.2	−0.2	−0.2	−0.2	−0.2	−0.2	−0.3	−0.2
Q3	−0.6	−0.7	−0.8	−1.1	−1.1	−1.1	−1.2	−1.2	−1.4	−1.4	−1.8	−2.3	−1.8
Min	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Max	−43.0	−44.4	−41.0	−45.5	−39.4	−44.8	−42.0	−38.3	−51.0	−58.3	−64.1	−81.9	−54.7
Panel B. Summary statistics: Drawdown durations													
Mean	16.6	18.9	19.5	21.1	18.2	20.9	18.7	18.1	18.8	17.9	16.9	17.6	22.1
Median	3.0	3.0	3.0	3.0	4.0	4.0	4.0	3.0	4.0	3.0	3.0	4.0	4.0
Q1	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Q3	7.0	8.0	7.0	9.0	9.0	9.0	8.0	8.0	8.0	8.0	8.0	9.0	9.8
Min	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Max	1082.0	1020.0	1020.0	1270.0	844.0	908.0	1027.0	1024.0	1034.0	1110.0	1188.0	1609.0	1654.0
Panel C. Summary statistics: Peak-to-trough durations													
Mean	7.7	8.5	8.6	9.3	7.5	8.5	7.6	7.4	7.2	7.4	6.3	6.6	10.3
Median	1.0	1.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
Q1	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Q3	3.0	4.0	3.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	5.0
Min	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Max	502.0	497.0	493.0	487.0	493.0	450.0	408.0	374.0	368.0	328.0	231.0	327.0	637.0
Panel D. Summary statistics: Recoveries from trough-to-peak													
Mean	8.8	10.4	10.9	11.9	10.7	12.4	11.1	10.7	11.7	10.5	10.6	10.9	11.7
Median	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

Drawdown	*G*+Z1	*G*+Z2	*G*+Z3	*G*+Z4	*G*+Z5	*G*+Z6	*G*+Z7	*G*+Z8	*G*+Z9	*G*+Z10	*G*+Z11	*G*+Z12	CRSP
Q1	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Q3	4.0	4.0	4.0	4.0	5.0	5.0	4.0	4.0	4.0	4.0	4.0	5.0	5.0
Min	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Max	610.0	703.0	849.0	783.0	704.0	703.0	866.0	860.0	878.0	947.0	1024.0	1282.0	1017.0

The table reports summary statistics for drawdown depths in Panel A, duration (in trading days) in Panel B, peak-to-trough in Panel C, and recoveries (duration from trough back to previous peak level) in Panel D. The analysis period covers daily returns from July 1964 to December 2022

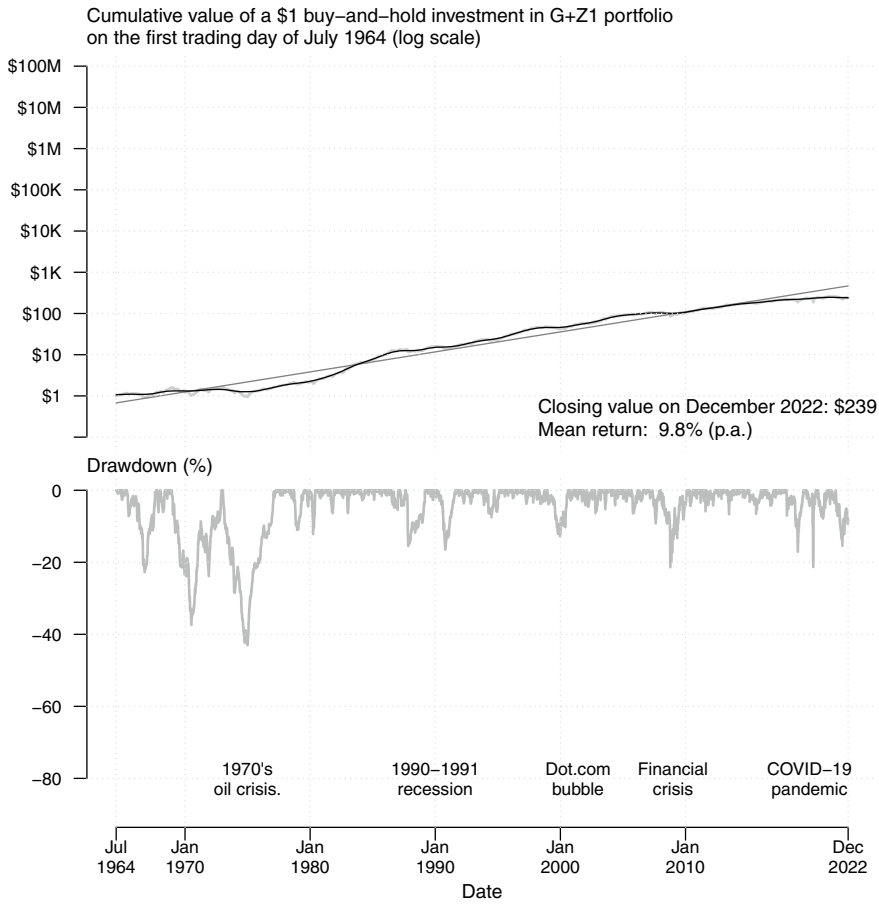


Fig. 9.3 Cumulative value and drawdowns of a $1 buy-and-hold investment in net long portfolio *G*+Z1 from July 1964 to December 2022. The cumulative value graph shows the compounded value of a buy-and-hold $1 investment from July 1964 to December 2022. The *Y*-axis of the value graph is in log-scale. Thus, the linear trend line indicates the average daily returns of the portfolio. When the smoothed value line is steeper than the linear trend, the portfolio return exceeds the mean return, and vice versa. The lower line depicts the drawdown in percentages (*Data source* CRSP database from the University of Chicago)

cumulative value panel is in log-scale, such that the slope of the linear trend represents the average daily return of the portfolio. The steepness of the slope gives a quick indication of the performance of the portfolio over time.

For comparative purposes, Fig. 9.7 gives the overall return performance of the CRSP index and its drawdowns over time. The much lower slope of the CRSP index versus *G*+Z12 portfolio highlights the superior return

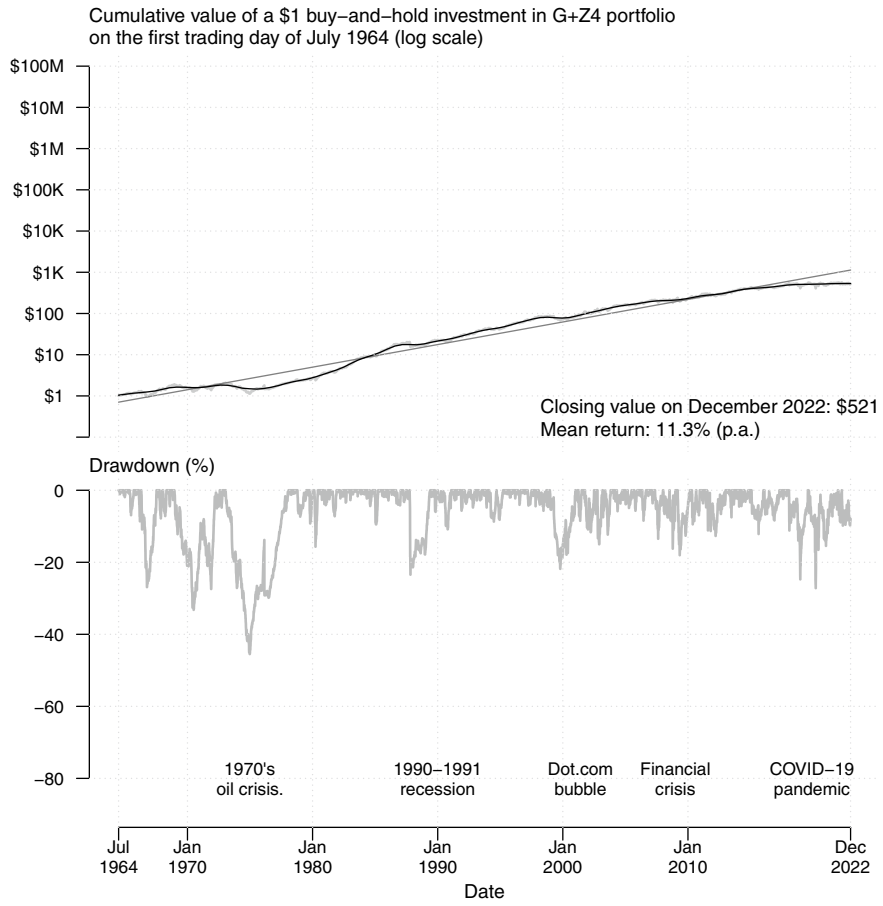


Fig. 9.4 Cumulative value and drawdowns of a $1 buy-and-hold investment in net long portfolio *G*+Z4 from July 1964 to December 2022. The cumulative value graph shows the compounded value of a buy-and-hold $1 investment from July 1964 to December 2022. The *Y*-axis of the value graph is in log-scale. Thus, the linear trend line indicates the average daily returns of the portfolio. When the smoothed value line is steeper than the linear trend, the portfolio return exceeds the mean return, and vice versa. The lower line depicts the drawdown in percentages (*Data source* CRSP database from the University of Chicago)

performance of *G*+Z12. Also, increasing steepness of the slopes and associated increases in cumulative values can be observed as zeta risk increases from *G*+Z1 to *G*+Z12 in Figs. 9.3 to 9.6.

Remarkably, the *G*+Z12 portfolio's cumulative value of $1 invested in 1964 exceeds $62 million by the end of 2022, which far exceeds the CRSP index at $263. Of course, in the real world, it would be difficult to reap these large gains due to impact costs from trading that are not included

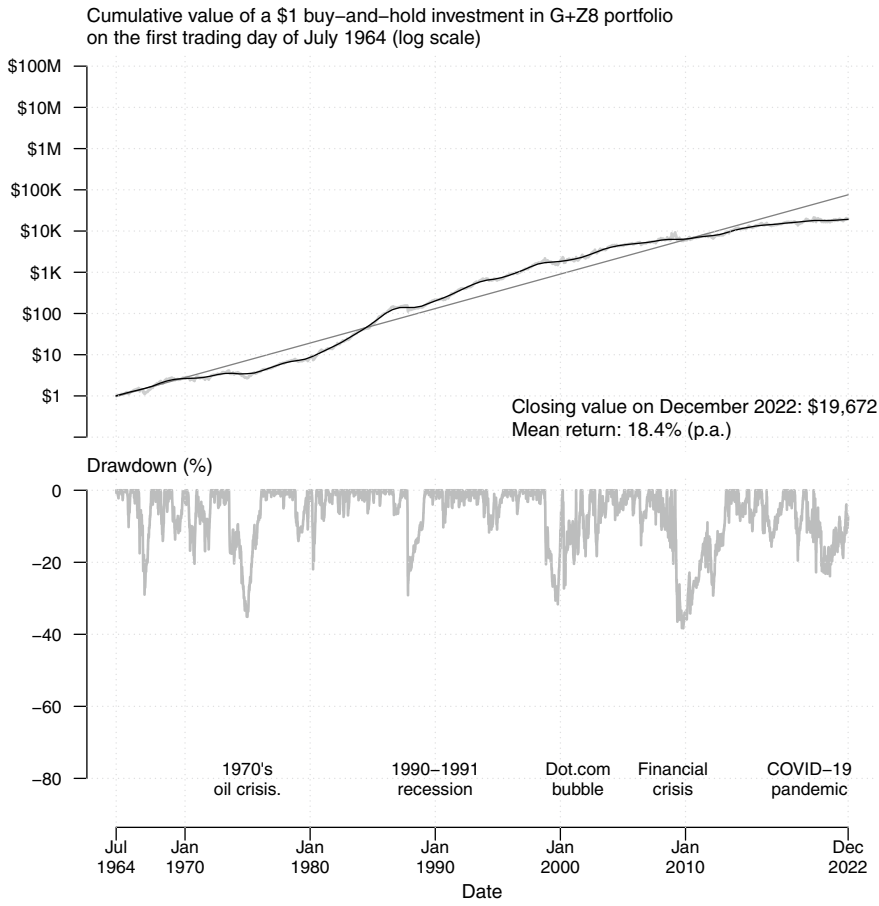


Fig. 9.5 Cumulative value and drawdowns of a $1 buy-and-hold investment in net long portfolio *G*+Z8 from July 1964 to December 2022. The cumulative value graph shows the compounded value of a buy-and-hold $1 investment from July 1964 to December 2022. The *Y*-axis of the value graph is in log-scale. Thus, the linear trend line indicates the average daily returns of the portfolio. When the smoothed value line is steeper than the linear trend, the portfolio return exceeds the mean return, and vice versa. The lower line depicts the drawdown in percentages (*Data source* CRSP database from the University of Chicago)

in the cumulative value calculations. Despite this practical constraint, even after subtracting reasonable trading costs, the value creation potential of the *G*+Z12 portfolio outweighs the larger drawdowns for this investment strategy. While the large gains computed for this high zeta risk portfolios seem unrealistic, to put things in context, Warren Buffett started with a fairly small amount of investment funds, invested in stocks, and today is estimated to

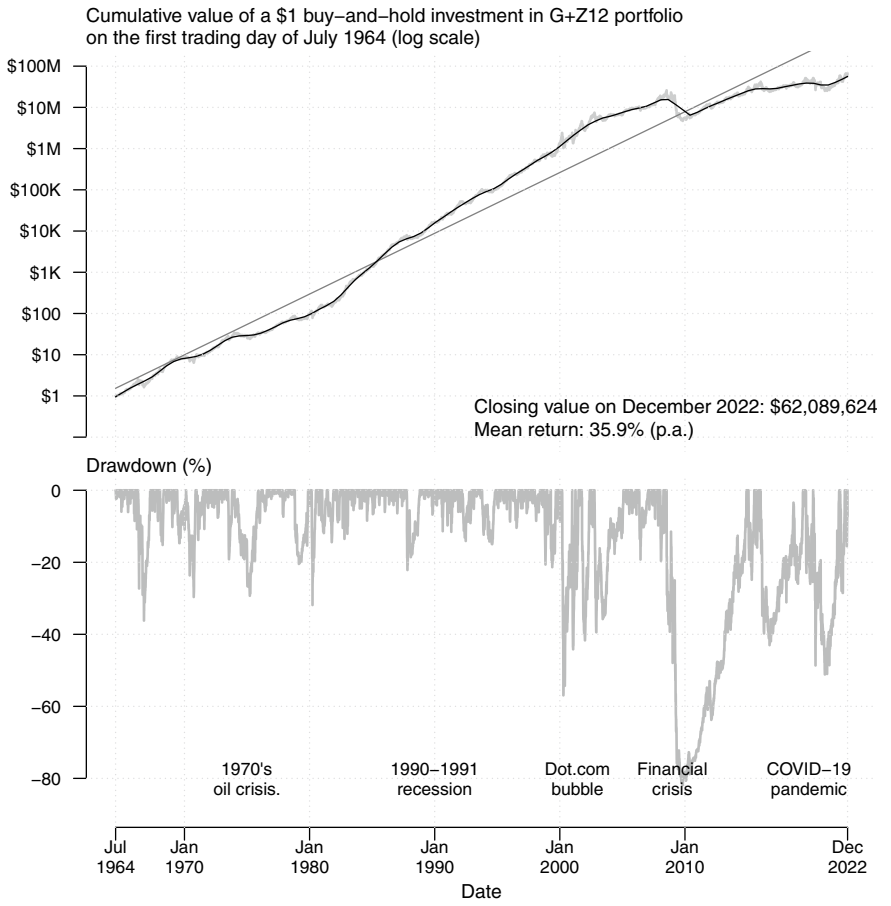


Fig. 9.6 Cumulative value and drawdowns of a $1 buy-and-hold investment in net long portfolio *G*+Z12 from July 1964 to December 2022. The cumulative value graph shows the compounded value of a buy-and-hold $1 investment from July 1964 to December 2022. The *Y*-axis of the value graph is in log-scale. Thus, the linear trend line indicates the average daily returns of the portfolio. When the smoothed value line is steeper than the linear trend, the portfolio return exceeds the mean return, and vice versa. The lower line depicts the drawdown in percentages (*Data source* CRSP database from the University of Chicago)

have a net worth of $120 billion. His investment philosophy includes fundamental analyses of businesses, using market timing to buy (sell) when prices are relatively low (high), and sticking with good quality companies for the long run. Our machine-made net long portfolios are constructed for different levels of zeta risk that are controlled over time so that investors can choose portfolios that fit their risk preferences.

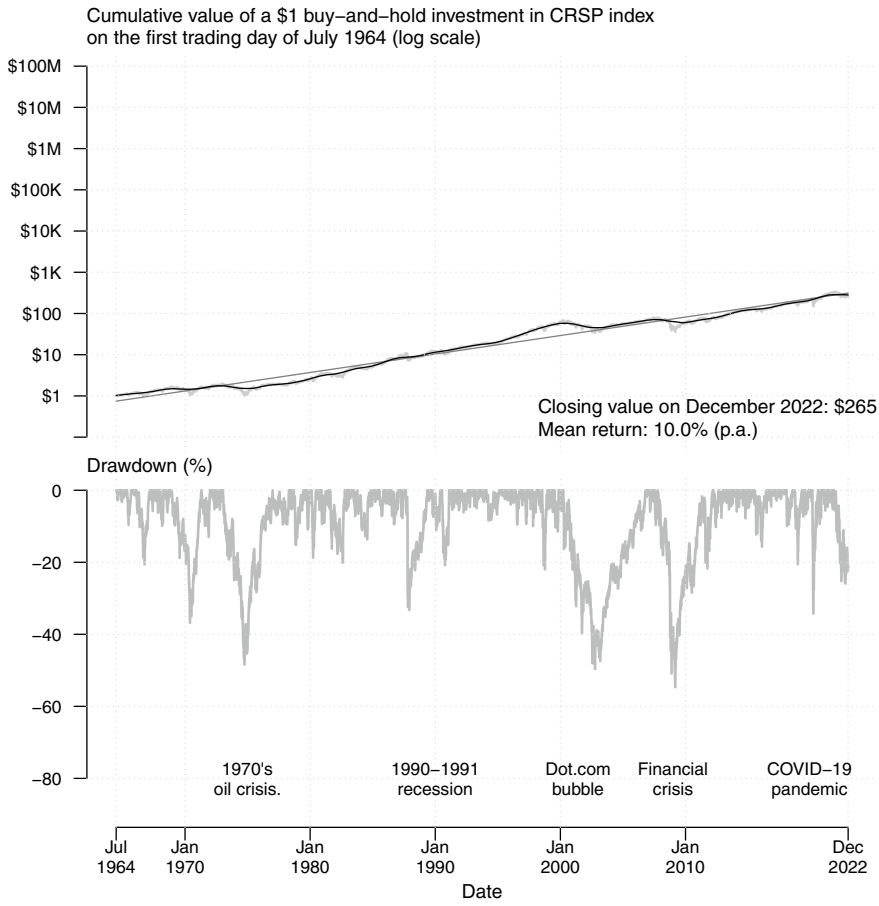


Fig. 9.7 Cumulative value and drawdowns of a $1 buy-and-hold investment in the CRSP index from July 1964 to December 2022. The cumulative value graph shows the compounded value of a buy-and-hold $1 investment from July 1964 to December 2022. The *Y*-axis of the value graph is in log-scale. Thus, the linear trend line indicates the average daily returns of the portfolio. When the smoothed value line is steeper than the linear trend, the portfolio return exceeds the mean return, and vice versa. The lower line depicts the drawdown in percentages (*Data source* CRSP database from the University of Chicago)

The drawdowns of the portfolios have some interesting shared features. First, as seen from the drawdown panels in Figs. 9.3 to 9.6, portfolios, G+Z1 and G+Z4 suffered most from the 1970's oil crisis with similar drawdowns of 40%. Second, portfolios G+Z8 and G+Z12 experienced their worst drawdowns during the 2008–2010 financial crisis. Third, the 2000 technology (dot.com) bubble had fairly mild drawdowns in the lower risk G+Z1 and G+Z4 portfolios. Fourth, all of the major crisis periods, including the 1970's

oil crisis, 1990–1991 recession, 2000 dot.com bubble, 2008–2010 financial crisis, and COVID-19 pandemic are clearly visible in the higher risk portfolios G+Z8 and G+Z12 as well as in the CRSP index in Fig. 9.7. Fifth, and last, the depth of the drawdown due to the 2008–2010 financial crisis increases as zeta risk increases, which is especially obvious in Fig. 9.6 with G+Z12 collapsing by about 80% of its value from the previous peak. In this regard, the CRSP index experienced a large loss of about 55% at this time too. We infer that major market downturns seriously impact the values of most stock market investors' portfolios. In times of extreme market turbulence, it is appropriate to rotate funds out of the stock market to the safety and liquidity of the money market. Alternatively, taking short positions in major market indexes such as the S&P 500 index can help to mitigate drawdowns or possibly even profit from large price declines in the market.

9.4 Summary

This chapter reported the risk analyses of our net long portfolios constructed in Chapter 8. Constructed using the ZCAPM asset pricing model of Kolari et al. (2021), these portfolios combine a proxy for the global minimum variance portfolio G developed in Chapter 7 and long-short portfolios with increasing levels of risk. The latter portfolios are formed based on zeta risk associated with market return dispersion in the ZCAPM. The inclusion of the G portfolio in the net long portfolios tends to decrease their total risk relative to the CRSP market index. On the other hand, as higher zeta risk portfolios are added to G, the total risk of the net long portfolios increases. While their risk increases in zeta risk, average annual returns increases also. In Chapter 8, we found that some net long portfolios with relatively high zeta risk had extraordinary positive returns that well exceeded the CRSP index. Also, after adjusting for total risk, their Sharpe ratios were much greater than that of the CRSP index. These outstanding performance results motivated the current chapter on risk metrics.

We conducted three risk metric analyses: (1) GRS tests of portfolio efficiency, (2) value at risk (VaR) tests, and (3) drawdown tests. Our analysis period was from July 1964 to December 2022. Also, subperiod analyses were performed by splitting the analysis period into earlier and later subperiods. The GRS tests revealed that the net long portfolios form an empirical efficient frontier that mimics the shape of the theoretical efficient frontier. The results showed that the net long portfolios could be combined into optimally

weighted aggregate portfolios that theoretically lie on a higher efficient frontier. Surprisingly, as zeta risk increased in the net long portfolios, they came closer to the theoretical efficient frontier; indeed, the net long portfolio with the highest zeta risk was theoretically efficient according to the GRS test.

The VaR and drawdown tests looked at potential losses in the net long portfolios. Not surprisingly, these losses increased as zeta risk increased in the net long portfolios. While the highest zeta risk portfolio had a VaR value greater than the CRSP market index, the other zeta risk portfolios had lower or equal VaR values. Thus, higher zeta risk portfolios with average returns superior to the CRSP index had lower or equal VaR characteristics. Another finding was that VaR values were higher in the later subperiod compared to the earlier subperiod. No doubt these findings are attributable to the market turmoil in the later subperiod, including the 2000 technology bubble, 2008–2010 financial crisis, and 2020–2021 COVID-19 pandemic. Finally, according to various drawdown test metrics, including frequency, depth, duration, peak-to-trough, and recoveries from trough-to-peak, as zeta risk increased in the net long portfolios, their drawdown risks increased. To assuage skeptics of our high zeta risk portfolios, we reported the cumulative values of $1 invested from 1964 to December 2022. These results showed that the high risk portfolios can generate extremely high cumulative wealth that more than offsets their potential drawdown characteristics.

Professional managers and investors can use the return and risk information in Chapter 8 as well as this chapter to better understand the returns and risks of the net long portfolios and, in turn, make informed investment decisions. The authors encourage cooperative research to implement in the real world of these innovative stock portfolios based on the ZCAPM asset pricing model.

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Gibbons, M.R., S.A. Ross, and J. Shanken. 1989. A test of the efficiency of a given portfolio. *Econometrica* 57: 1121–1152.

Kolari, J. W., J. Z. Huang, W. Liu, and H. Liao, 2022, Further tests of the ZCAPM asset pricing model, *Journal of Risk and Financial Management*, available online at https://www.mdpi.com/1911-8074/15/3/137) Reprinted in Kolari, J. W., and S. Pynnonen, 2022, eds., *Frontiers of Asset Pricing* (MDPI, Basel, Switzerland).

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A New Model of Capital Asset Prices: Theory and Evidence*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. *Testing for missing asset pricing factors*. San Diego, CA: Paper presented at the Western Economic Association International.

Kolari, J.W., and S. Pynnonen. 2023. *Investment Valuation and Asset Pricing: Models and Methods*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Liu, W., 2013, *A New Asset Pricing Model Based on the Zero-Beta CAPM: Theory and Evidence*, Doctoral dissertation, Texas A&M University.

Liu, W., J. W. Kolari, and J. Z. Huang, 2012, A new asset pricing model based on the zero-beta CAPM market model (CAPM), Presentation at the annual meetings of the Financial Management Association, Best Paper Award in Investments, Atlanta, GA (October).

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. *Return dispersion and the cross-section of stock returns*. Palm Springs, CA (October): Presentation at the annual meetings of the Southern Finance Association.

Markowitz, H.M. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio Selection: Efficient Diversification of Investments*. New York, NY: John Wiley & Sons.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.



10

Long Only Efficient Portfolios

10.1 Empirical Methods

As in Chapters 8 and 9 for net long portfolios, we begin by estimating the empirical ZCAPM with one year of daily returns from July 1964 to June 1965 for individual stocks in the U.S. stock market. We drop the bottom 10 percent of stocks in each year that have the smallest market capitalization. These stocks tend to be very small firms with high idiosyncratic risks that most investors, especially institutional investors, tend to avoid.

Using the estimated zeta risk coefficients of individual stocks, we rank them and then form 25 portfolios. Out-of-sample returns are computed for each portfolio in the next month starting with July 1965. Rolling forward one month at a time, this procedure is repeated to generate monthly returns for each portfolio over time from July 1965 to December 2022. The same procedures are implemented to create 10 beta sorted portfolios and their out-of-sample returns over time.

Chapter 4 covered the theoretical ZCAPM of Kolari et al. (2021), a special case of Black's (1972) zero-beta CAPM.[1] According to the ZCAPM, the ith asset's expected return at time t can be specified as follows:

$$E(R_{it}) - R_{ft} = \beta_{i,a}[E(R_{at}) - R_{ft}] + Z^*_{i,a}\sigma_{at}, \tag{10.1}$$

where $E(R_{at})$ is the expected market return equal to the average return of all assets (rather than a market proxy m for the theoretical market portfolio M

[1] See also other ZCAPM publications by Liu et al. (2012, 2020), Liu (2013), Kolari et al. (2022a, b, 2023a), Kolari and Pynnonen (2023b), and Kolari et al. (2024).

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in Sharpe's [1964] CAPM), R_{ft} is the riskless rate, $\beta_{i,a}$ is the beta risk coefficient measuring sensitivity to expected excess market returns, σ_{at} is the cross-sectional standard deviation of all asset returns (or return dispersion) at time t, and $Z^*_{i,a}$ is the zeta risk coefficient measuring either positive or negative sensitivity to return dispersion. $Z^*_{i,a}$ is developed in the ZCAPM with respect to orthogonal portfolios I^* and ZI^* on the boundary of the mean-variance investment parabola. Unlike other pairs of efficient and zero-beta portfolios on the parabola, these unique efficient and zero-beta portfolios have equal return variances.

To estimate the theoretical ZCAPM, we employ the novel empirical ZCAPM proposed by Kolari et al. (2021):

$$R_{it} - R_{ft} = \beta_{i,G}(R_{Gt} - R_{ft}) + Z_{i,G} D_{it} \sigma_{Gt} + u_{it}, \quad t = 1, \ldots, T \tag{10.2}$$

where $R_{it} - R_{ft}$ is the excess return for the ith stock over the Treasury bill rate, $\beta_{i,G}$ measures sensitivity of the ith stock to excess average market returns based on the optimized G portfolio (see Chapter 7) minus the Treasury bill rate R_{ft}, $Z_{i,G}$ measures sensitivity of the ith stock portfolio (or stock) to the cross-sectional standard deviation of returnsσ_{Gt} (return dispersion), D_{it} is a signal variable with values $+1$ and -1 representing positive and negative return dispersion effects, respectively, on stock returns, and $u_{it} \sim$ iid N(0, σ_i^2).[2] Signal variable D_{it} is an independent random variable with a two-point distribution:

$$D_{it} = \begin{cases} +1 \text{ with probability } p_i \\ -1 \text{ with probability } 1 - p_i, \end{cases} \tag{10.3}$$

where p_i (or $1 - p_i$) is the probability of a positive (or negative) return dispersion effect, and D_{it} are independent of u_{it}.

Unlike other asset pricing models that invariably employ ordinary least squares (OLS) regression models, the empirical ZCAPM utilizes the expectation-maximization (EM) algorithm of Dempster et al. (1977) for regression parameter estimation (including the probability parameter p_i).[3]

[2] As discussed in Chapter 5, the intercept parameter α_i in the empirical ZCAPM is assumed to be zero. All other empirical asset pricing models based on regression analyses contain an intercept term. For this and other reasons, the empirical ZCAPM is quite different from other asset pricing models with no close parallels in the asset pricing literature.

[3] Kolari et al. (2021) detailed step-by-step instructions on how to implement the EM algorithm to estimate the empirical ZCAPM. Also, they provided citations to other finance and statistics studies that have employed the EM algorithm.

Since $E(D_{it}) = 2p_i - 1$, the marginal form of the empirical ZCAPM can be stated as:

$$R_{it} - R_{ft} = \beta_{i,G}(R_{Gt} - R_{ft}) + Z^*_{i,G}\sigma_{Gt} + u^*_{it}, \quad t = 1, \ldots, T \quad (10.4)$$

where $\beta_{i,G}$ is the beta risk coefficient, $Z^*_{i,G} = Z_{i,G}(2p_i - 1)$ is the zeta risk coefficient, u^*_{it} is the error term with variance related to return dispersion, and other notation is as before. The sign of zeta risk loading $Z^*_{i,G}$ is either positive or negative as determined by the probability p_i associated with the expected value of signal variableD_{it} in period $t = 1, \ldots, T$.

Kolari, Huang, and Liu provide free software to estimate the empirical ZCAPM on GitHub (https://github.com/zcapm). There you can find Matlab, Python, and R programs. We utilize their R programs, which tend to run at a faster estimation speed than the other programs.

10.1.1 Building Long Only Portfolios

Here we describe the steps for building long only portfolios sorted on different zeta risk levels.

1. Times-series regressions are used to estimate empirical ZCAPM Eq. (10.4) via the expectation-maximization (EM) algorithm for each individual stock with daily returns for the initial 12-month estimation period July 1963 to June 1964. These regression models utilized our proxy for the global minimum variance portfolio G (see Chapter 7) to compute average market returns and the cross-sectional standard deviation of all stocks' returns (or market return dispersion) related to beta risk and zeta risk, respectively.[4]
2. All stocks are ranked in terms of their estimated zeta coefficient $\hat{Z}^*_{i,G}$.
3. A total of 25 portfolios are formed from low (negative) zeta risk to high (positive) zeta risk, viz., $Z1, \ldots, Z25$.

[4] In empirical asset pricing tests based on the often-used Fama and MacBeth (1973) two-step test procedure, in unreported results for the analysis period July 1964 to December 2022, we find that beta risk is not significantly priced in the cross-sectional regression but zeta risk is very significant. The latter t-statistic on the market price of risk associated with estimated zeta risk coefficients in the cross-sectional regression is 6.17. This significance level far exceeds all known tests for the significance of various multifactors, including size, value, profit, capital investment, momentum, and other factors (see Fama and French 2015, 2018, 2020), which typically reach maximum t-statistics in the range of 2–3. It is clear that zeta risk related to market return dispersion is a powerful systematic risk factor that should be taken into account in asset pricing models.

4. For all stocks in each zeta risk portfolio, we compute their one-month-ahead (out-of-sample) return R_{pt+1} in July 1964. Using these returns, we compute the zeta risk portfolio's optimally-weighted, one-month-ahead holding period return. Our weights are based on optimaized methods. This out-of-sample approach is consistent with real world investor behavior in that portfolios are formed at time $t = T$ and performance is observed in the next month $T + 1$.
5. The entire process is repeated by rolling the estimation period in step (1) forward one month to rebalance each of the 25 zeta risk portfolios in August 1964. By rolling forward month-by-month to the end of the sample period, we generate out-of-sample returns for each portfolio from July 1964 to December 2022.

We also build 10 beta sorted portfolios using the same rolling procedures. These portfolios are intended to test whether there is a relation between beta risk and average stock returns. As we will see, CAPM beta may well be dead but not ZCAPM beta. Both beta risk and zeta risk are important systematic risk measures related to average stock returns. It is critical that investors take into account these powerful risk metrics in their investment decisions. As Markowitz, the founder of modern investment portfolio analyses established, maximizing returns is a wrong-headed investment strategy. Instead, investors should consider the returns per unit risk in diversified investment portfolios. Without good measures of market risk, investors are flying blind. We next review our results that document important relationships between average stock returns and beta risk as well as zeta risk.

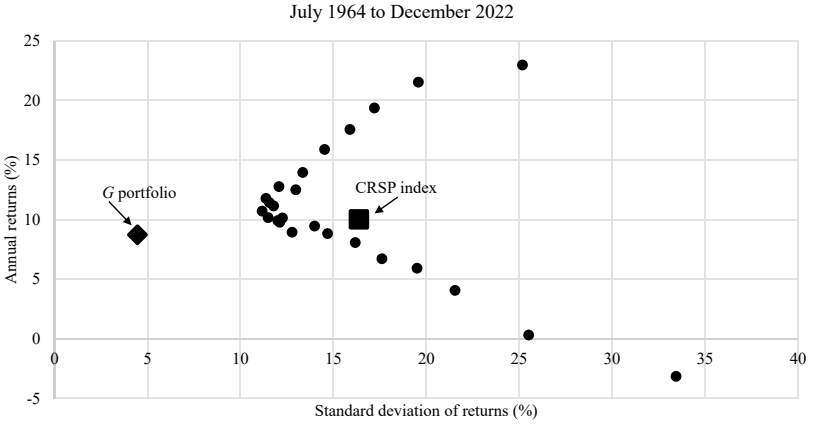
10.2 Empirical Results

This section reviews our empirical evidence for long only portfolios. Again it is important to note that all portfolios are formed once a month and then returns in the following month are computed for the portfolios. This out-of-sample empirical design is consistent with investor behavior—that is, select the stocks in a portfolio and then observe the returns in the next month. We rebalance the portfolios once a month. Results cover the analysis period July 1964 to December 2022.

10.2.1 Long Only Zeta Risk Portfolios

The out-of-sample performance of the 25 long only zeta risk portfolios are shown in Fig. 10.1. Average annual returns and the standard deviations of these returns from July 1964 to December 2022 are used to plot each portfolio. Portfolio Z1 has the lowest zeta risk, and Z25 has the highest zeta risk as estimated by the empirical ZCAPM. A number of results are readily seen there. First, the 25 zeta risk portfolios trace out the shape of the theoretical Markowitz parabola. However, this curve is not theoretical, as it is generated from the actual returns of U.S. stocks; hence, we refer to it as the *empirical parabola*. Second, consistent with ZCAPM theory, the CRSP market index lies in the interior of the empirical parabola approximately on its axis of the symmetry (i.e., the average market return is located in the middle of the parabola, not on an upper efficient frontier as many finance academics and practitioners have believed in the past). Third, and last, the global minimum variance portfolio *G* has very low return volatility and has an average return that approximates the average market return (or CRSP index return).

The empirical parabola formed by the zeta risk portfolios in Figure 10.1 has a return width or span from –3.15% to 22.98% per year. Focusing on



	Zeta risk portfolios (low to high zeta risk coefficients)												
7/1964-12/2022 Annualized returns	Z1	Z2	Z3	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13
Long only portfolio returns %	-3.15	0.31	4.05	5.92	6.71	8.06	8.83	9.46	8.93	10.14	9.78	9.91	10.17
Standard deviation of returns %	33.45	25.52	21.55	19.51	17.62	16.18	14.69	13.99	12.78	12.26	12.12	12.01	11.49

	Zeta risk portfolios (low to high zeta risk coefficients) plus CRSP index and *G* portfolio													
	Z14	Z15	Z16	Z17	Z18	Z19	Z20	Z21	Z22	Z23	Z24	Z25	CRSP	*G*
Long only portfolio returns %	10.71	11.77	11.42	11.15	12.77	12.50	13.95	15.88	17.57	19.36	21.54	22.98	10.02	8.73
Standard deviation of returns %	11.17	11.38	11.56	11.79	12.07	12.98	13.35	14.53	15.89	17.21	19.58	25.18	16.37	4.45

Fig. 10.1 Average returns and their standard deviation of returns for 25 long only zeta risk portfolios plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is July 1964 to December 2022

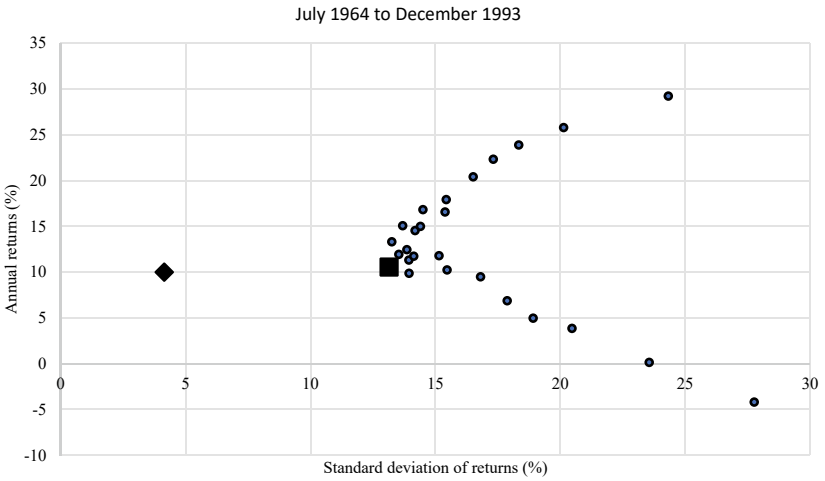
portfolios Z21–Z25, these high positive zeta risk portfolios earn outstanding average returns that range from 17.57% to 22.98% per year. This performance is roughly two times the annual average returns of the CRSP market index at 10.02% per year. While they have higher total risk (as measured by the standard deviation of returns), their Sharpe ratios—defined as excess returns over the Treasury bill rate divided by total risk—range from 0.73 to 0.87 which far outperforms the CRSP index at 0.34 in the analysis period. Hence, it is clear that high performing long only portfolios can be built based on zeta risk in the empirical ZCAPM.

We repeat these analyses for the following two subperiods: (1) July 1964 to December 1993, and (2) January 1994 to December 2022. Figures 10.2 and 10.3 show the results. In general, these subperiod graphs of the 25 long only zeta risk portfolios are very similar to Fig. 10.1 encompassing the entire analysis period from July 1964 to December 2022. Indeed, the subperiod July 1964 to December 1993 results in Fig. 10.2 are unchanged for the most part. One exception is that high positive zeta risk portfolios Z21–Z25 now have even higher Sharpe ratios ranging from 0.83 to 0.95 compared to the CRSP index at only 0.29. The results for the subperiod January 1994 to December 2022 exhibit a lower upper (efficient) boundary of the empirical parabola than the entire analysis period in Fig. 10.1 with a maximum average return per year of 16.39% for the Z24 portfolio. The zeta risk portfolios Z23–Z25 earn average returns in the range of 14.45–16.26%, which is still around two times higher than the CRSP market index at 8.73% per year in this subperiod. Moreover, the Sharpe ratio for the CRSP of 0.48 underperforms portfolios Z10–Z25 with ratios in the range of 0.52–0.75 (i.e., portfolios Z13–Z16 have relatively high Sharpe ratios from 0.64 to 0.71).

Thus, we infer from these results that the shape of the Markowitz parabola is largely determined by zeta risk in the ZCAPM. Zeta risk can be used to earn returns above or below the average market return (viz., the CRSP market index).

10.2.2 Long Only Beta Risk Portfolios

Here we review the empirical results for portfolios sorted by beta risk. As mentioned earlier, beta risk in our empirical ZCAPM Eq. (10.2) relates to excess returns on the G portfolio, not the CRSP index or S&P 500 index as typically used in other asset pricing models. We formed 10 beta risk portfolios and plot their average returns in the analysis period July 1964 to December 2022. B1 has the lowest beta risk, and B10 has the highest beta risk. Figure 10.4 summarizes the results.

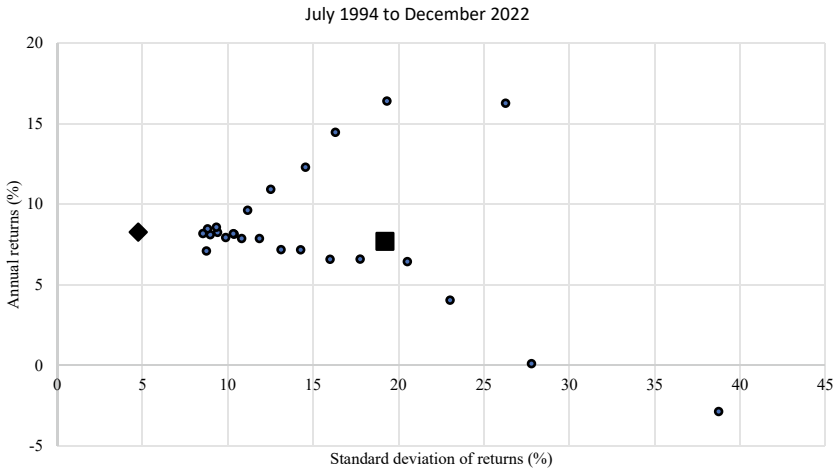


	Zeta risk portfolios (low to high zeta risk coefficients)													
7/1964 -12/1993 Annualized returns	Z1	Z2	Z3	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13	
Long only portfolio returns %	-3.15	0.31	4.05	5.92	6.71	8.06	8.83	9.46	8.93	10.14	9.78	9.91	10.17	
Standard deviation of returns %	33.45	25.52	21.55	19.51	17.62	16.18	14.69	13.99	12.78	12.26	12.12	12.01	11.49	
	Zeta risk portfolios (low to high zeta risk coefficients) plus CRSP index and *G* portfolio													
	Z14	Z15	Z16	Z17	Z18	Z19	Z20	Z21	Z22	Z23	Z24	Z25	CRSP	*G*
Long only portfolio returns %	10.71	11.77	11.42	11.15	12.77	12.50	13.95	15.88	17.57	19.36	21.54	22.98	10.02	8.73
Standard deviation of returns %	11.17	11.38	11.56	11.79	12.07	12.98	13.35	14.53	15.89	17.21	19.58	25.18	16.37	4.45

Fig. 10.2 Average returns and their standard deviation of returns for 25 long only zeta risk portfolios plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis subperiod is July 1964 to December 1994

As seen there, average portfolio returns increase for portfolios B1–B8 from 7.68% to 12.70% per month. While returns increase almost linearly from portfolios B1–B4, returns thereafter only marginally increase for portfolios B5–B8 from 12.62% to 12.97% per year. Subsequently, average returns decline for portfolios B9 and B10. The latter portfolios likely contain many small stocks with very high idiosyncratic volatility. In forthcoming results, we drop high idiosyncratic volatility stocks and re-compute the return performances of beta sorted stocks. The results confirm our conjecture that high beta portfolios contain high idiosyncratic risk stocks that are not amenable to portfolio analyses using asset pricing models (that are based on systematic market risks).

Notice that the return/risk performance CRSP market index in Figure 10.4 lies below a number of beta sorted portfolios. Portfolios B1–B8 have Sharpe ratios (viz., excess returns over the Treasury bill rate divided by the standard deviation of returns) in the range of 0.43 for B8 to 0.66 for B3, whereas the CRSP index has a lower Sharpe ratio of 0.34. Also, the *G* portfolio itself has relatively lower standard deviation of returns as well as average returns than most beta sorted portfolios.



	Zeta risk portfolios (low to high zeta risk coefficients)												
1/1994-12/2022 Annualized returns	Z1	Z2	Z3	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13
Long only portfolio returns %	-2.88	0.10	4.04	6.42	6.58	6.57	7.15	7.16	7.86	7.86	8.13	7.93	8.24
Standard deviation of returns %	38.77	27.80	23.02	20.52	17.75	15.99	14.27	13.12	11.86	10.81	10.38	9.88	9.40

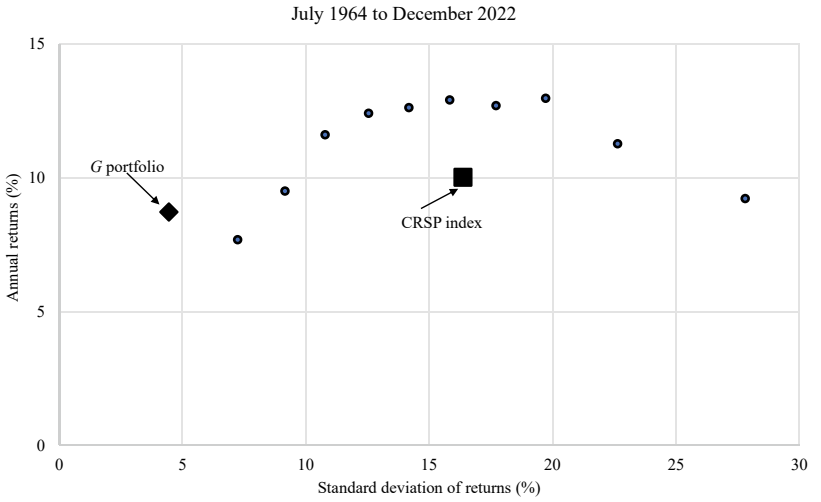
	Zeta risk portfolios (low to high zeta risk coefficients) plus CRSP index and *G* portfolio													
	Z14	Z15	Z16	Z17	Z18	Z19	Z20	Z21	Z22	Z23	Z24	Z25	CRSP	*G*
Long only portfolio returns %	8.10	8.45	8.16	7.08	8.56	8.16	9.61	10.91	12.28	14.45	16.39	16.26	7.69	8.26
Standard deviation of returns %	8.98	8.82	8.55	8.75	9.33	10.33	11.17	12.52	14.55	16.30	19.32	26.28	19.21	4.74

Fig. 10.3 Average returns and their standard deviation of returns for 25 long only zeta risk portfolios plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis subperiod is January 1994 to December 2022

It is interesting that we do not obtain a random pattern between beta risk and average returns. It is well known that beta as measured with respect to CRSP index or S&P 500 index excess returns has little or no relation to average stock returns—in other words, a random pattern is found. Fama and French (1992, 1993, 1995) have demonstrated in numerous papers that CRSP-based CAPM betas fail in the cross section of average stock returns due to a lack of any relation to average stock returns. They concluded that CAPM beta is dead from their extensive tests. In sharp contrast to the CAPM beta evidence, with the exception of very high beta risk portfolios, we see an obvious positive relation between *G*-based ZCAPM beta risk average stock returns. We infer that, excluding very high beta risk portfolios, average returns tend to increase with *G*-based beta risk. Apparently beta is not dead when measured using the *G* portfolio!

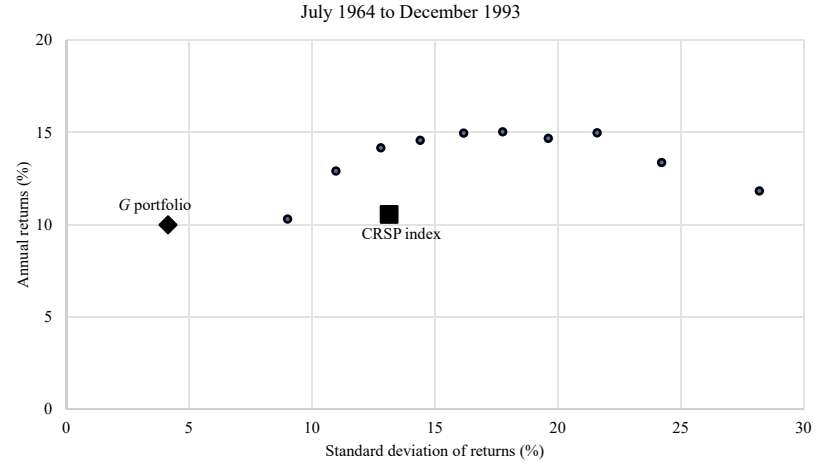
As before for the 25 zeta risk portfolios, we repeat the above analyses for beta sorted portfolios in two subperiods: (1) July 1964 to December 1993, and (2) January 1994 to December 2022. Figures 10.5 and 10.6 give the out-of-sample portfolio performance results, respectively.

Figure 10.5 for the earlier subperiod looks very similar to the entire analysis period in Fig. 10.4. Portfolios B1–B8 have Sharpe ratios in the range from



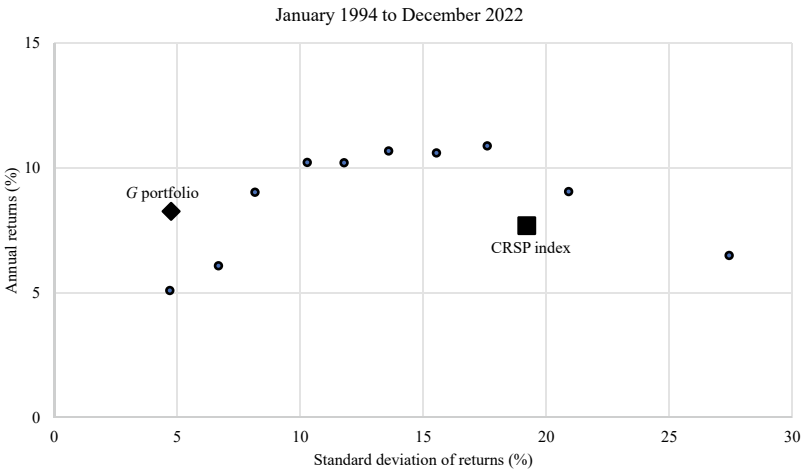
	Beta risk portfolios (low to high beta risk coefficients) plus CRSP index and *G* portfolio											
7/1964-12/2022 Annualized returns	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	CRSP	*G*
Long only portfolio returns %	7.68	9.50	11.61	12.41	12.62	12.91	12.70	12.97	11.27	9.22	10.02	8.73
Standard deviation of returns %	7.24	9.15	10.78	12.54	14.18	15.83	17.71	19.72	22.64	27.81	16.37	4.45

Fig. 10.4 Average returns and their standard deviation of returns for 10 long only beta risk portfolios plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is July 1964 to December 2022



	Beta risk portfolios (low to high beta risk coefficients) plus CRSP index and *G* portfolio											
7/1964-12/1993 Annualized returns	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	CRSP	*G*
Long only portfolio returns %	10.30	12.91	14.16	14.57	14.96	15.03	14.68	14.98	13.36	11.82	10.54	9.99
Standard deviation of returns %	9.01	10.98	12.81	14.41	16.18	17.76	19.62	21.61	24.23	28.20	13.15	4.14

Fig. 10.5 Average returns and their standard deviation of returns for 10 long only beta risk portfolios plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis subperiod is July 1964 to December 1994



	Beta risk portfolios (low to high beta risk coefficients) plus CRSP index and *G* portfolio											
1994-2022 Annualized returns	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	CRSP	*G*
Long only portfolio returns %	5.54	6.62	9.41	10.55	10.56	11.13	11.20	11.40	9.85	7.64	7.69	8.26
Standard deviation of returns %	4.74	6.63	8.05	10.09	11.56	13.34	15.24	17.32	20.57	26.96	19.21	4.74

Fig. 10.6 Average returns and their standard deviation of returns for 25 long only zeta risk portfolios plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis subperiod is January 1994 to December 2022

0.38 for B8 to 0.58 for B3 compared to 0.29 for the CRSP index. Thus, portfolio B8 has a Sharpe ratio two times higher than the CRSP index.

In the later subperiod, Fig. 10.6 shows that the results change to some degree. The *G* portfolio has return volatility similar to the low beta risk B1 portfolio, rather than being lower volatility. Also, the CRSP index has relatively higher return volatility than before. Only high beta risk portfolios B9 and B10 have return volatility exceeding the CRSP index. The other beta portfolios B1–B8 have lower total risk than the CRSP index. We infer that an increasing number of small stocks in the CRSP index over time has increased its total risk relative to past decades. Due to increasing total risk in the CRSP index, it is a particularly bad proxy to use for the market portfolio in estimating CAPM beta risk for stocks in this subperiod.

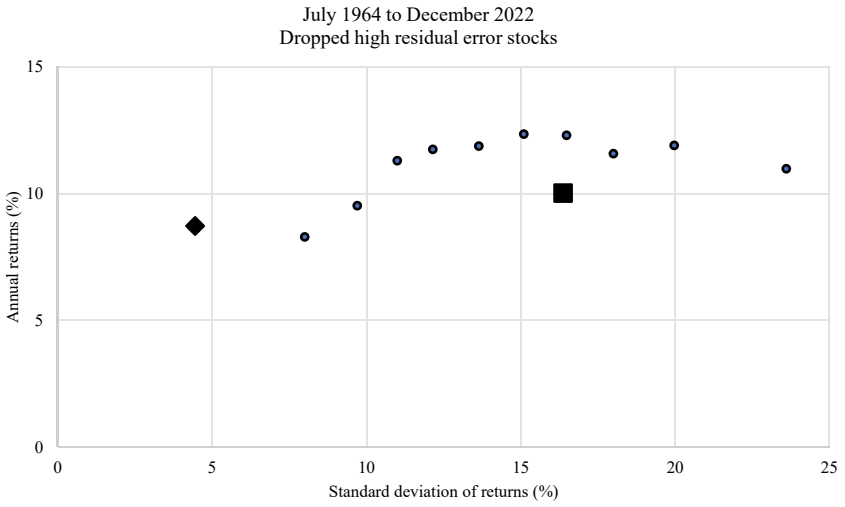
A number of beta risk portfolios stand out in Fig. 10.6. In this later subperiod, portfolios B3–B7 all have higher average returns than the CRSP index. These portfolios' returns range from 9.41% to 11.40% compared to 7.69% for the CRSP index. Also, they have lower total risk (as measured by the standard deviation of returns over time) than the CRSP index. Taking into account both excess returns and total risk, their Sharpe ratios range from 0.54 to 0.84, which exceed the CRSP index at 0.48 in this subperiod.

As discussed earlier, high beta stocks experienced decreased rates of return compared to other beta sorted portfolios. We next drop stocks that have residual error variances in the empirical ZCAPM in the top 20% of sample stocks. These stocks' returns have high idiosyncratic risk and, therefore, are difficult (or even impossible) to fit well with an asset pricing model containing systematic risk factors. It appears that previous empirical tests of the CAPM would have been advised to drop these high beta risk stocks from the analyses.

Figures 10.7 and 10.8 repeat the above analyses after dropping high idiosyncratic risk stocks. As we expected, now the high beta portfolios do not exhibit a marked drop in average returns. For the entire sample period July 1963 to December 2022, Fig. 10.7 shows that portfolios B3–B10 have average returns greater than the that of the CRSP index. Also, only portfolio B10 has a lower Sharpe ratio than the CRSP index. Portfolio B3 has the highest Sharpe ratio at 0.58 compared to the CRSP index at 0.29. Both subperiod analyses are contained in Fig. 10.8. The earlier subperiod July 1964 to December 1993 results are similar to the entire sample period in Fig. 10.7. However, the later subperiod results in January 1994 to December 2022 is somewhat changed, with a higher CRSP index return indicating that its beta risk had increased over time. Even so, the Sharpe ratios for B1–B9 range from 0.37 to 0.70 compared to 0.48 for the CRSP index. Especially for lower to middle ranges of beta risk, after dropping high idiosyncratic risk stocks, these results confirm that stock returns tend to be linearly increasing with beta risk.

We do find that, even after dropping high idiosyncratic risk stocks, average stock returns tend to level out for higher beta risk portfolios. It is possible that these portfolios are affected by other risks that tend to impact their return performance, such as risk related to high financial leverage that increases beta. In a well-known theoretical CAPM study, Hamada (1972) proved that levered betas are greater than unlevered betas. He combined the CAPM of Sharpe (1964) with the corporate valuation and capital structure theories of Modigliani and Miller (MM) (1958, 1963) to derive this result. According to MM, as financial leverage (as measured by debt to equity ratios) increases in a firm, it value increases due to tax gains on debt interest payments. However, at some point financial leverage can have negative effects due to higher interest rates demanded by debtholders as well as rising probabilities of firm bankruptcy. For these reasons, firm profits will decline at relatively high levels of financial leverage and thereby decrease the value of the firm and (relevant to the present discussion) the rate of return on its stock.[5]

[5] See an excellent review of theory and evidence by Modigliani (1988), who received the 1985 Nobel Prize in Economic Sciences in large part due these corporate finance ideas.



Dropped 20% high residual error stocks	Beta risk portfolios (low to high beta risk coefficients) plus CRSP index and *G* portfolio											
7/1964-12/2022 Annualized returns	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	CRSP	*G*
Long only portfolio returns %	8.29	9.52	11.29	11.74	11.87	12.34	12.30	11.57	11.90	10.98	10.02	8.73
Standard deviation of returns %	8.00	9.70	11.00	12.15	13.64	15.10	16.48	18.00	19.97	23.60	16.37	4.45

Fig. 10.7 Average returns and their standard deviation of returns for 10 long only beta risk portfolios plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is July 1964 to December 2022

In Figs. 10.6, 10.7 and 10.8, we see empirical evidence to support the notion that rates of return decline for stock portfolios with relatively beta risk that is likely associated with high levels of financial leverage. Since this evidence is important to famous corporate finance theories, the authors have started to conduct more detailed studies of the links between betas, stock returns, financial leverage, and taxes.

10.3 Summary

In this chapter we demonstrated the application of the ZCAPM asset pricing model to building high-performing long only portfolios. We built two kinds of portfolios: (1) zeta risk sorted portfolios, and (2) beta risk sorted portfolios. The empirical ZCAPM is estimated using one year of daily returns for individual stocks, minimum variance portfolio *G* returns, and the cross-sectional standard deviation of *G* portfolio returns. Resultant parameter estimates for beta risk coefficients and zeta risk coefficients are used to sort individual stocks into beta risk and zeta risk portfolios.

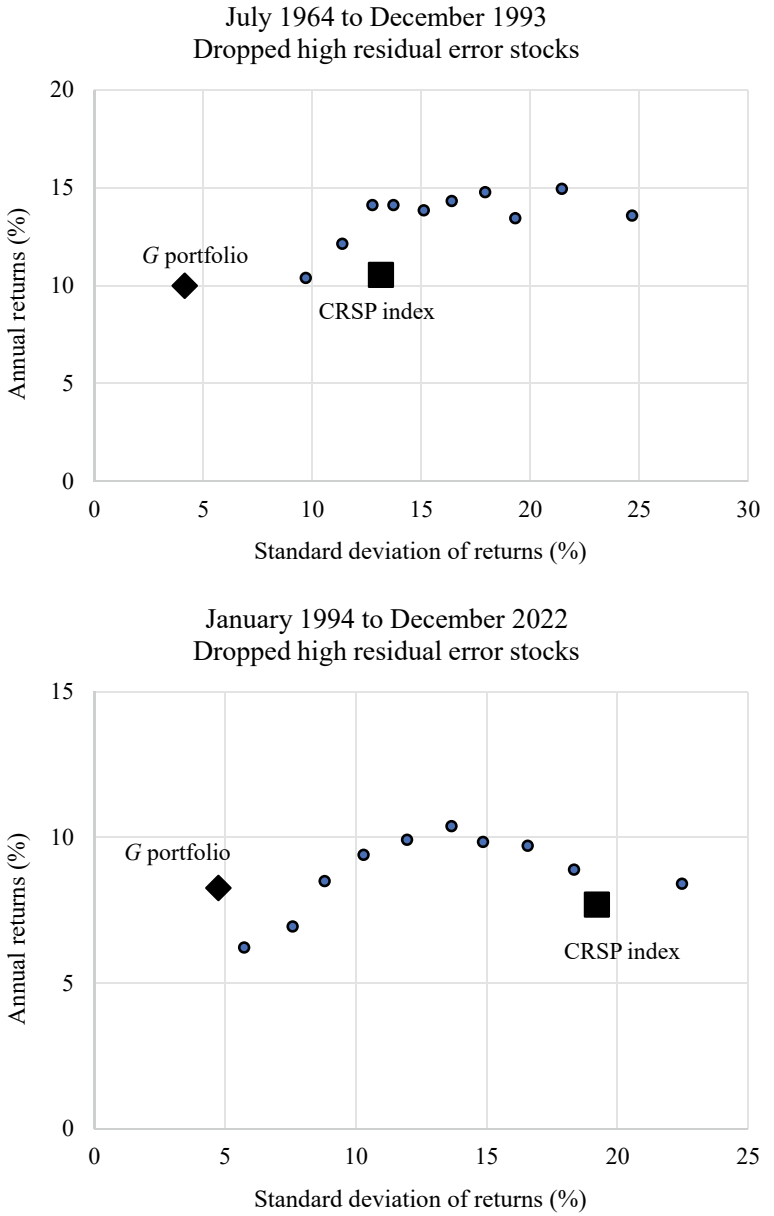


Fig. 10.8 Average returns and their standard deviation of returns for 10 long only beta risk portfolios plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). In the upper panel, the analysis subperiod is July 1964 to December 1994. In the lower panel, the analysis subperiod is January 1994 to December 2022

Our results using U.S. stock returns for the entire analysis period July 1964 to December 2022 were impressive. Some zeta risk portfolios earned average stock returns two times or more greater than the CRSP market index (i.e., 20% or more per year)! The CRSP index is closely related to the S&P 500 index, which is a very popular investment portfolio of many investors in the market. To beat the market by large margins is extraordinary. Over 90 percent of professional managers cannot beat the S&P 500 index over extended periods of time from three to five years or more. These high-performing portfolio should be of keen interest among institutional money managers and their client investors. The authors would like to work with managers to implement these portfolios. For example, zeta risk exchange traded funds (ETFs) could be developed for the marketplace.

Consistent with ZCAPM theory, the G portfolio had clearly lower total risk than the zeta and beta sorted portfolios. Also, the CRSP market index was located near the axis of symmetry that splits the mean-variance empirical parabola into two halves at an interior location therein. Upon splitting the return series into two subperiods (viz., July 1964 to December 1993 and January 1994 to December 2022), our empirical results indicated that the CRSP index increased in total risk over time in our analysis period. This finding is likely related to periodic bouts of substantial market volatility that occurred during and after 2000 technology bubble, the 2008–2009 financial crisis, and 2020–2022 COVID-19 pandemic. Also, the number of small stocks in the CRSP index has increased over time. Even with these turbulent events, the zeta and beta sorted portfolios tended to follow similar patterns to the entire analysis period.

Unlike previous studies that find no relation between beta risk and average stock returns, our results using the G portfolio to estimate beta coefficients (instead of the CRSP market index) revealed a definite positive relation between beta and returns that gradually leveled off as beta increased and then turned negative at relatively high beta risk levels. These results overturn the notion that beta is dead. ZCAPM beta risk is consistent with CAPM theory but is based on the G portfolio rather than the market portfolio M. From a practical investment perspective, a number of our beta risk portfolios had average stock returns that well exceeded the CRSP market index. However, the gains from beta risk were not as large as those for zeta risk.

An unexpected finding was the concave shape of the stock return and beta risk. We believe that this relation can be explained (in part at least) by famous corporate valuation theories by Modigliani and Miller (MM) (1958, 1963). MM hypothesized that, as financial leverage increases, firm value increases but eventually reverses and then decreases as excessive levels

of financial leverage are reached that significantly increase the probability of firm bankruptcy. Hamada (1972) showed that levered betas are greater than unlevered betas due to financial leverage. Hence, consistent with our evidence for beta sorted portfolios, increasing beta risk is associated with financial leverage that initially tends to boost firm profits and its stock returns but eventually lowers profits and returns. The authors are exploring this important corporate finance issue in more detail at the present time.

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Dempster, A.P., N.M. Laird, and D.B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society* 39: 1–38.

Fama, E.F., and K.R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427–465.

Fama, E.F., and K.R. French. 1993. The cross-section of expected returns. *Journal of Financial Economics* 33: 3–56.

Fama, E.F., and K.R. French. 1995. Size and book-to-market factors in earnings and returns. *Journal of Finance* 50: 131–156.

Fama, E.F., and K.R. French. 1996. The CAPM is wanted, dead or alive. *Journal of Finance* 51: 1947–1958.

Fama, E.F., and K.R. French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116: 1–22.

Fama, E.F., and K.R. French. 2018. Choosing factors. *Journal of Financial Economics* 128: 234–252.

Fama, E.F., and K.R. French. 2020. Comparing cross-section and time-series factor models. *Review of Financial Studies* 33: 1892–1926.

Fama, E.F., and J.D. MacBeth. 1973. Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy* 81: 607–636.

Hamada, R.S. 1972. The effect of the firm's capital structure on the systematic risk of common stocks. *Journal of Finance* 27: 435–452.

Kolari, J. W., J. Z. Huang, W. Liu, and H. Liao. 2022a. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*, available online at https://www.mdpi.com/1911-8074/15/3/137) Reprinted in ed. J.W. Kolari, and S. Pynnonen, 2022. *Frontiers of asset pricing* (MDPI, Basel, Switzerland).

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022b. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. *Testing for missing asset pricing factors*. San Diego, CA: Paper presented at the Western Economic Association International.

Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Liu, W., 2013, *A new asset pricing model based on the zero-beta CAPM: Theory and evidence*. Doctoral dissertation, Texas A &M University.

Liu, W., J.W. Kolari, and J.Z. Huang. 2012. A new asset pricing model based on the zero-beta CAPM market model (CAPM). Presentation at the annual meetings of the Financial Management Association, Best Paper Award in Investments, Atlanta, GA, October.

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. Return dispersion and the cross-section of stock returns. Presentation at the annual meetings of the Southern Finance Association, Palm Springs, CA, October.

Markowitz, H. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: Wiley.

Modigliani, F. 1988. MM—Past, present, and future. *Journal of Economic Perspectives* 2: 149–158.

Modigliani, F., and M.H. Miller. 1958. The cost of capital, corporation finance and the theory of investment. *American Economic Review* 48: 261–297.

Modigliani, F., and M.H. Miller. 1963. Corporation income taxes and the cost of capital: A correction. *American Economic Review* 53: 433–443.

Roll, R. 1977. A critique of the asset pricing theory's tests, part I: On past and potential future testability of the theory. *Journal of Financial Economics* 4: 129–176.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.



11

The Beta-Zeta Risk Architecture of the Mean-Variance Parabola

11.1 Empirical Methods

As in Chapter 8 for net long portfolios, we begin by estimating the empirical ZCAPM with one year of daily returns from July 1964 to June 1965 for individual stocks in the U.S. stock market. We drop the bottom 10% of stocks each year that have the smallest market capitalization. These stocks tend to be very small firms with high idiosyncratic risks that most investors, especially institutional investors, tend to avoid.

Using the estimated zeta risk coefficients of individual stocks, we rank them and then form 5 portfolios. Out-of-sample returns are computed for each portfolio in the next month starting with July 1964. Within each zeta risk quintile, we form 5 beta risk sorted portfolios. Rolling forward one month at a time, this procedure is repeated to generate monthly returns for each portfolio over time from July 1964 to December 2022. The same procedures are implemented to create 5 beta-sorted portfolios and their out-of-sample returns over time. Subsequently, within each beta risk quintile, we form 5 zeta risk portfolios.

Chapter 4 covered the theoretical ZCAPM of Kolari et al. (2021), a special case of Black's (1972) zero-beta CAPM.[1] According to the ZCAPM, the ith asset's expected return at time t can be specified as follows:

$$E(R_{it}) - R_{ft} = \beta_{i,a}[E(R_{at}) - R_{ft}] + Z^*_{i,a}\sigma_{at}, \tag{11.1}$$

[1] See also other ZCAPM publications by Liu et al. (2012, 2020), Liu (2013), Kolari et al. (2022a, 2023), Kolari et al. (2022b), Kolari and Pynnonen (2023) and Kolari et al. (2024).

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J. W. Kolari et al., *Professional Investment Portfolio Management*,
https://doi.org/10.1007/978-3-031-48169-7_11

where $E(R_{at})$ is the expected market return equal to the average return of all assets (rather than a market proxy m for the theoretical market portfolio M in Sharpe's (1964) CAPM), R_{ft} is the riskless rate, $\beta_{i,a}$ is the beta risk coefficient measuring sensitivity to expected excess market returns, σ_{at} is the cross-sectional standard deviation of all asset returns (or return dispersion denoted RD) at time t, and $Z^*_{i,a}$ is the zeta risk coefficient measuring either positive or negative sensitivity to return dispersion. $Z^*_{i,a}$ is developed in the ZCAPM with respect to orthogonal portfolios I^* and ZI^* on the boundary of the mean-variance investment parabola. Unlike other pairs of efficient and zero-beta portfolios on the parabola, these unique efficient and zero-beta portfolios have equal return variances.

To estimate the theoretical ZCAPM, the novel empirical ZCAPM proposed by Kolari et al. (2021) is employed:

$$R_{it} - R_{ft} = \beta_{i,G}(R_{Gt} - R_{ft}) + Z_{i,G} D_{it} \sigma_{Gt} + u_{it}, \quad t = 1, \ldots, T \tag{11.2}$$

where $R_{it} - R_{ft}$ is the excess return for the ith stock over the Treasury bill rate, $\beta_{i,G}$ measures sensitivity of the ith stock to excess average market returns based on the optimized G portfolio (see Chapter 7) minus the Treasury bill rate R_{ft}, $Z_{i,G}$ measures sensitivity of the ith stock portfolio (or stock) to the cross-sectional standard deviation of returns σ_{Gt} (return dispersion), D_{it} is a signal variable with values $+1$ and -1 representing positive and negative return dispersion effects, respectively, on stock returns, and $u_{it} \sim$ iid $\mathrm{N}(0, \sigma_i^2)$.[2] Signal variable D_{it} is an independent random variable with a two-point distribution:

$$D_{it} = \begin{cases} +1 \text{ with probability } p_i \\ -1 \text{ with probability } 1 - p_i, \end{cases} \tag{11.3}$$

where p_i (or $1 - p_i$) is the probability of a positive (or negative) return dispersion effect, and D_{it} are independent of u_{it}.

Unlike other asset pricing models that invariably utilize ordinary least squares (OLS) regression models, the empirical ZCAPM makes use of the expectation-maximization (EM) algorithm of Dempster et al. (1977) for

[2] As discussed in Chapter 5, the intercept parameter α_i in the empirical ZCAPM is assumed to be zero. All other empirical asset pricing models based on regression analyses contain an intercept term. For this and other reasons, the empirical ZCAPM is quite different from other asset pricing models with no close parallels in the asset pricing literature.

regression parameter estimation (including the probability parameter p_i).[3] Since $E(D_{it}) = 2p_i - 1$, the marginal form of the empirical ZCAPM can be stated as:

$$R_{it} - R_{ft} = \beta_{i,G}(R_{Gt} - R_{ft}) + Z^*_{i,G}\sigma_{Gt} + u^*_{it}, \quad t = 1, \ldots, T \quad (11.4)$$

where $\beta_{i,G}$ is the beta risk coefficient, $Z^*_{i,G} = Z_{i,G}(2p_i - 1)$ is the zeta risk coefficient, u^*_{it} is the error term with variance related to return dispersion, and other notation is as before. The sign of zeta risk loading $Z^*_{i,G}$ is either positive or negative as determined by the probability p_i associated with the expected value of signal variable D_{it} in period $t = 1, \ldots, T$.

Kolari et al. provide free software to estimate the empirical ZCAPM on GitHub (https://github.com/zcapm). There you can find Matlab, Python, and R programs. We utilize their R programs, which tend to run at a faster estimation speed than the other programs.

11.1.1 Building Long Only Portfolios

Here we describe the steps for building long only portfolios sorted on different beta risk and zeta risk levels.

1. Times-series regressions are used to estimate empirical ZCAPM equation (11.4) via the expectation-maximization (EM) algorithm for each individual stock with daily returns for the initial 12-month estimation period from July 1963 to June 1964. As market factors in the empirical ZCAPM, we employ our estimate of the global minimum variance portfolio G (see Chapter 7) to compute average stock market returns and the cross-sectional standard deviation of all stocks' returns (or market return dispersion), which are associated with beta risk and zeta risk, respectively.[4]
2. All stocks are ranked in terms of their estimated zeta coefficients $\hat{Z}^*_{i,G}$.

[3] Kolari et al. (2021) have provided step-by-step instructions on how to implement the EM algorithm to estimate the empirical ZCAPM. Also, they reference citations to other finance and statistics studies that have employed the EM algorithm.

[4] In empirical asset pricing tests based on the often-used Fama and MacBeth (1973) two-step test procedure, in unreported results, we find that beta risk is not significantly priced in the cross-sectional regression but zeta risk is very significant. The latter t-statistic on the market price of risk associated with estimated zeta risk coefficients in the cross-sectional regression is 6.17. This significance level far exceeds all known tests for the significance of various multifactors, including size, value, profit, capital investment, momentum, and other factors (see Fama and French [1992, 1993, 1995, 1996 2015, 2018, 2020]), which typically reach maximum t-statistics in the range of 2 to 3. It is clear that zeta risk related to market return dispersion is a powerful systematic risk factor that should be taken into account in asset pricing models.

3. A total of five portfolios are formed from low (negative) zeta risk to high (positive) zeta risk, viz., $Z1, \ldots, Z5$.
4. Within each zeta risk quintile, we form five beta risk sorted portfolios. Based on optimization methods, we weight the returns of each portfolio in the next month July 1964 (i.e., out-of-sample returns). This out-of-sample approach is consistent with real world investor behavior in that portfolios are formed at time $t = T$ and performance is observed in the next month $T + 1$.
5. The entire process is repeated by rolling the estimation period in step (1) forward one month to rebalance each of the 25 beta-zeta risk portfolios in August 1964. By rolling forward month-by-month to the end of the sample period, we generate out-of-sample returns for each portfolio from July 1964 to December 2022.

We repeat the above steps by initially sorting on beta risk and then zeta risk within each beta risk quintile to create 25 beta-zeta risk portfolios. Now stocks are initially sorted into five beta risk portfolios, and then within each of these quintiles, we form five zeta risk portfolios. Out-of-sample returns are computed for each portfolio.

11.2 Empirical Results

We next report and discuss the empirical results for stock portfolios sorted into beta risk and zeta risk quintiles. As we will see, these portfolios enable us to map out a grid for the mean-variance empirical parabola with different levels of beta and zeta risks. Not only portfolios on the outer boundary of the parabola are defined by beta and zeta risks but the structure of the interior of the parabola is revealed also. We provide results for the entire analysis period July 1964 to December 2022 in addition to the subperiods July 1964 to December 1993 and January 1994 to December 2022.

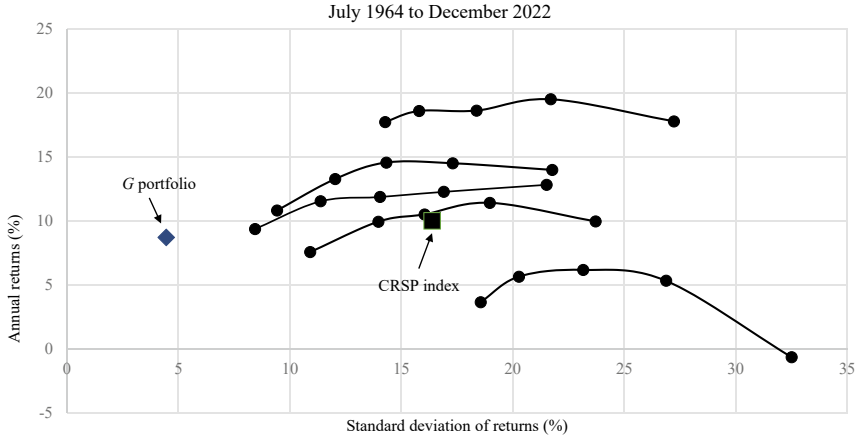
11.2.1 Zeta-Beta Risk Portfolios

We begin with the zeta-beta risk sorted portfolio results. Figure 11.1 summarizes the findings for the entire analysis period July 1964 to December 2022. The curves in the graph indicate different levels of zeta risk from low to high. It is obvious that average stock returns increase as zeta risk increases. Looking at Z5 portfolios that are used to draw the highest zeta risk curve, average stock returns range from 17.75% to 18.64%. By comparison, the CRSP market

index earns 10.02% in this analysis period, which is far below this high zeta risk curve. Clearly, the CRSP index is not an efficient portfolio! It would be ill advised to use this portfolio as a proxy for the market portfolio in tests of the CAPM as in so many empirical asset pricing tests in finance.

Within each zeta risk quintile are five beta risk portfolios. With the exception of the highest beta portfolio in each zeta risk curve, average stock returns tend to increase as beta risk increases. This trend for all but high beta risk portfolios suggests that beta risk related to the *G* portfolio is positively related to average stock returns. Recall that previously published tests of beta risk associated with the CRSP index as proxy for the market portfolio yield no relation to average stock returns.

Casual observation of the zeta-beta risk portfolios in Fig. 11.1 reveals important information on the architecture of the mean-variance empirical parabola. Referring back to Fig. 7.1 in Chapter 7, we see that zeta risk defines the width or span of the parabola, and beta risk increases returns within any given zeta risk level. Consistent with ZCAPM theory, the *G* portfolio and CRSP market index are located within the empirical parabola as expected—namely *G* is the leftmost minimum variance portfolio, and the CRSP



	Zeta-beta risk sorted portfolios (by quintiles)													
	Low Z1					Z2					Z3			
7/1964-12/2022 Annualized returns	Low B1	B2	B3	B4	High B5	Low B1	B2	B3	B4	High B5	Low B1	B2	B3	
Long only portfolio returns %	3.66	5.66	6.19	5.34	-0.63	7.58	9.96	10.53	11.43	9.98	9.38	11.55	11.89	
Standard deviation of returns %	18.55	20.27	23.15	26.87	32.50	10.90	13.96	16.03	18.96	23.70	8.43	11.37	14.03	

	Zeta-beta risk sorted portfolios (by quintiles)													
	Z3		Z4					High Z5						
	B4	High B5	Low B1	B2	B3	B4	High B5	Low B1	B2	B3	B4	High B5	CRSP	*G*
Long only portfolio returns %	12.29	12.85	10.83	13.29	14.58	14.53	14.00	17.75	18.61	18.64	19.53	17.80	10.02	8.73
Standard deviation of returns %	16.90	21.51	9.42	12.02	14.32	17.30	21.76	14.26	15.79	18.37	21.70	27.22	16.37	4.45

Fig. 11.1 Average returns and their standard deviation of returns for 25 portfolios sorted into zeta risk quintiles and then beta risk quintiles (within each zeta risk quintile) plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is July 1964 to December 2022

index is located in the interior of the parabola near its axis of symmetry in the middle. Finally, as in Chapter 10 on long only beta risk portfolios in Figures 10.4–10.6, average annual returns decrease somewhat for high beta risk portfolios in each zeta risk quintile. As conjectured there, consistent with well-known corporate theory by Hamada (1972), this pattern is likely related to high financial leverage that reduces equity returns due to high interest rates demanded by debtholders as well as increasing risks of bankruptcy.

11.2.2 Beta-Zeta Risk Portfolios

The results from sorting stocks into beta-zeta risk portfolios are shown in Fig. 11.2. The beta risk curves show that average stock returns increase with beta risk. Within each beta risk curve, average stock returns increase with zeta risk for the most part.

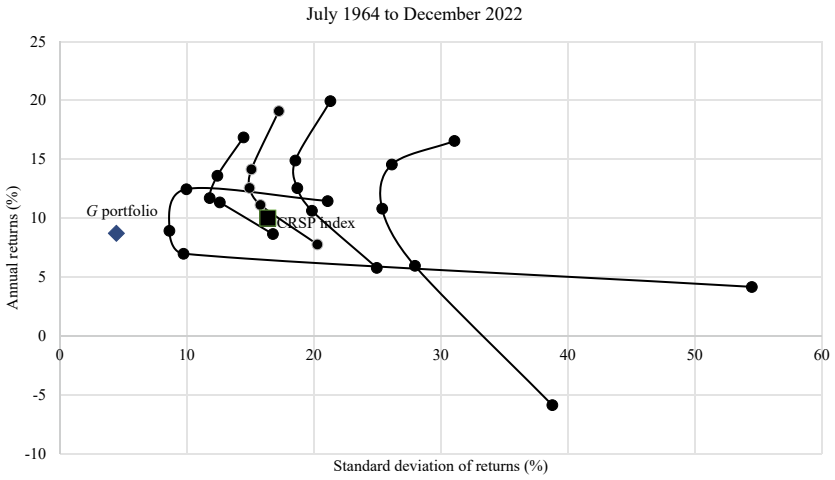
We do see that the lowest beta risk curve is somewhat different with high total risk in the highest and lowest zeta risk portfolios on this curve. Especially stocks that have low beta risk and low zeta risk have very high total risk. These stocks likely do not fit the empirical ZCAPM very well. However, excluding high and low zeta risk portfolios in the low beta quintile portfolios, higher returns are evident as zeta risk increases.

From Fig. 11.2, we infer that zeta risk is critically important to mapping out the risk structure of the mean-variance parabola. The combination of beta risk and zeta risk can be used to locate portfolios within and on the outer boundary of the parabola.

11.2.3 Subperiod Results for Beta Risk and Zeta Risk Portfolios

In this Subsection, we report that beta risk and zeta risk sorted portfolio results for the subperiods July 1964 to December 1994 and January 1994 to December 2022.

Concerning the earlier subperiod, the upper panel of Fig. 11.3 shows the results for portfolios sorted on zeta risk and then beta risk. The zeta risk curves increase returns from a range of 0 to 5% for the lowest zeta risk quintile to a range of 20 to 25% for the highest zeta risk range. The latter high zeta risk portfolio average returns are very high relative to the CRSP market index at 10% per year! As predicted by ZCAPM theory, the G portfolio and CRSP index lie on the axis of symmetry with G at the minimum variance point and CRSP in the interior of the empirical parabola. Again the CRSP index is



	Beta-zeta risk sorted portfolios (by quintiles)												
	Low B1					B2					B3		
7/1964-12/2022 Annualized returns	Low Z1	Z2	Z3	Z4	High Z5	Low Z1	Z2	Z3	Z4	High Z5	Low Z1	Z2	Z3
Long only portfolio returns %	4.17	6.99	8.94	12.47	11.47	8.66	11.34	11.72	13.61	16.87	7.78	11.13	12.58
Standard deviation of returns %	54.49	9.73	8.62	9.95	21.08	16.77	12.58	11.78	12.40	14.46	20.27	15.79	14.92

	Beta-zeta risk sorted portfolios (by quintiles)													
	B3		B4					High B5						
	Z4	High Z5	Low Z1	Z2	Z3	Z4	High Z5	Low Z1	Z2	Z3	Z4	High Z5	CRSP	*G*
Long only portfolio returns %	14.15	19.10	5.79	10.64	12.56	14.90	19.94	-5.86	5.98	10.82	14.56	16.56	10.02	8.73
Standard deviation of returns %	15.07	17.24	24.95	19.84	18.69	18.54	21.30	38.78	27.95	25.37	26.13	31.07	16.37	4.45

Fig. 11.2 Average returns and their standard deviation of returns for 25 portfolios sorted into beta risk quintiles and then zeta risk quintiles (within each beta risk quintile) plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis subperiod is July 1964 to December 1994

far from the empirical efficient frontier; hence, it is inappropriate to use this market index as a proxy for the market portfolio in empirical tests of the CAPM as done by numerous researchers over the years.

Turning to the later subperiod in the lower panel of Fig. 11.3, average returns for the highest zeta risk portfolios are reduced considerably to a range of 13.5 to 14%. Thus, the width or span of the mean-variance parabola decreased substantially over time. What could cause this major shift in the shape of the investment parabola? Over time the number of small stocks in the market has increased dramatically to over 8,000 stocks in recent years. However, as these small stocks have entered the market, we would have expected return dispersion across all stocks to have increased, not decreased. We conjecture that the multiple periods of stock market crashes after the 2000 technology bubble, during the financial crisis of 2008–2009, and in response to the COVID-19 pandemic of 2020 are responsible for declining market return dispersion. When the market suffers a strong downturn and then later rebounds recovering large losses, all stocks move together in a more synchronous fashion than during normal stock market conditions. Hence, we

conjecture that the dispersion of stock returns was diminished by these major marketwide swings in returns across individual stocks. Even so, the average returns of the highest zeta risk quintile curve are very high compared to the CRSP market index at about 7.7% in this subperiod (i.e., almost two times higher than the CRSP index). Certainly these portfolios should be of interest to stock market investors!

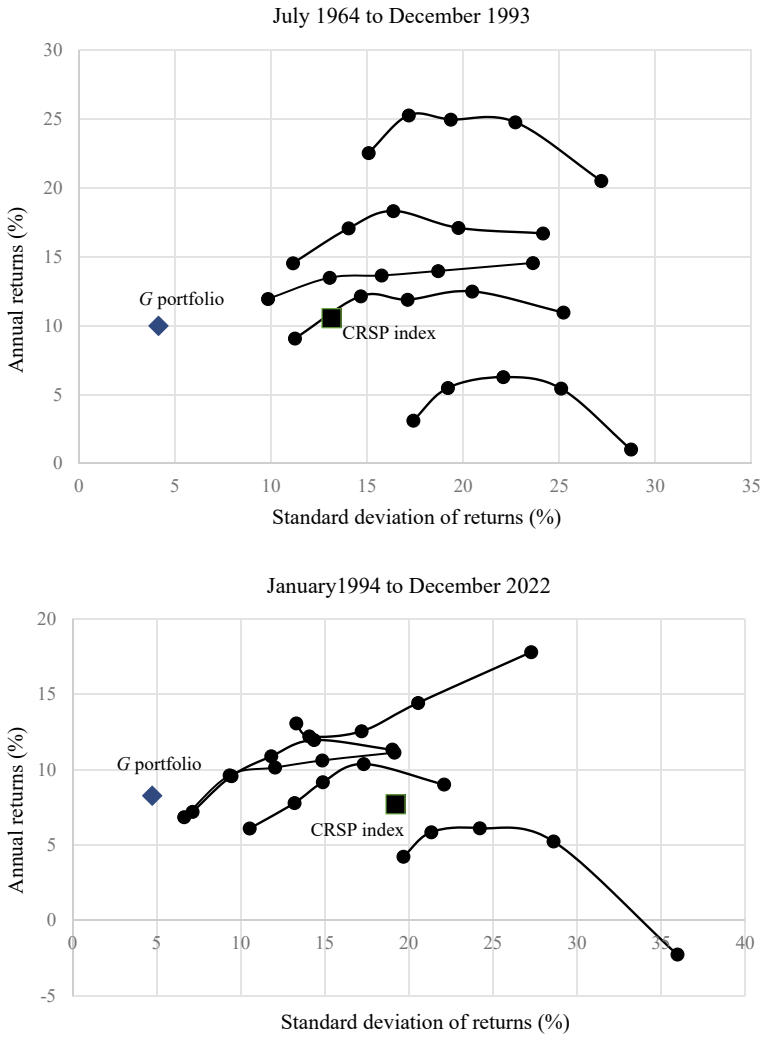


Fig. 11.3 Average returns and their standard deviation of returns for 25 portfolios sorted into zeta risk quintiles and then beta risk quintiles (within each zeta risk quintile) plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). In the upper panel, the analysis period is July 1964 to December 1993. In the lower panel, the analysis period is January 1994 to December 2022

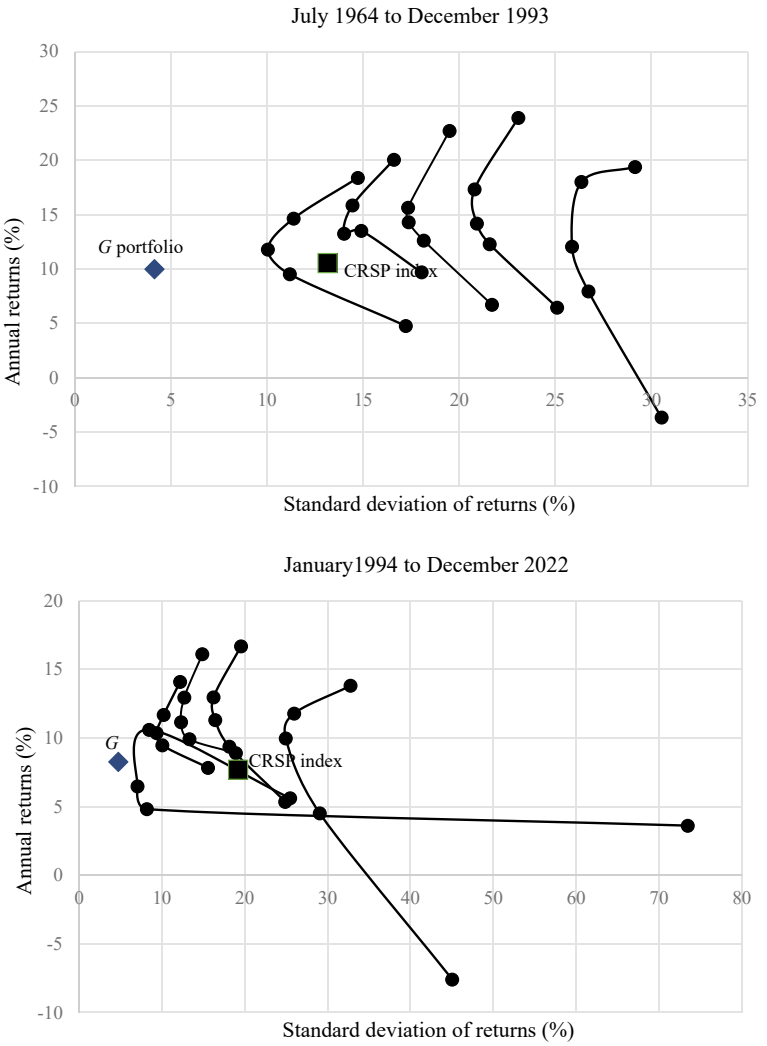


Fig. 11.4 Average returns and their standard deviation of returns for 25 portfolios sorted into beta risk quintiles and then zeta risk quintiles (within each beta risk quintile) plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). In the upper panel, the analysis period is July 1964 to December 1993. In the lower panel, the analysis period is January 1994 to December 2022

Figure 11.4 illustrates the results for portfolios sorted into beta risk quintiles and then zeta risk quintiles within each beta risk quintile. The earlier subperiod in the upper panel shows that average returns increase markedly as beta risk increases. Also, within each beta risk quintile curve, zeta risk is compensated by higher returns in all cases. The later subperiod in the lower

panel exhibits the same general risk/return patterns; however, in the lowest beta quintile, the low zeta risk portfolio has very high total risk at over 70% standard deviation of returns. These stocks no doubt have very high idiosyncratic risks, which we address in the next subsection. In general, beta risk increases the average returns of stocks. and within each beta risk quintile curve, zeta risk normally increases average returns.

11.2.4 Results After Dropping High Idiosyncratic Risk Stocks

Due to our suspicion that high volatility stocks have high idiosyncratic (firm-specific) risk, we dropped 20% of stocks with the highest residual error variance in empirical ZCAPM relation (10.4). The new results for zeta-beta risk sorted stocks in Fig. 11.5 are similar to those in Fig. 11.1 without dropping stocks, but the maximum time-series standard deviations of returns (or total risk) of the zeta risk curves are reduced. Hence, our earlier conjecture is supported—namely, total risk can be reduced in the zeta-beta portfolios by dropping high idiosyncratic risk stocks.

Referring to Fig. 11.5, exceptional return/risk performance is achieved by the high zeta risk Z5 portfolios. In our analysis period July 1964 to December 2022, their average returns are in the range of 17.20% to 18.75% compared to the value-weighted market average return of the CRSP index equal to 10.02%. Also, with few exceptions, average returns tend to increase as beta risk increases in any given zeta risk quintile portfolio curve. Again this beta risk is related to the G portfolio, not the CRSP index used in most all published academic studies that find little or no relation between beta risk and average stock returns.

Figure 11.6 shows the beta-zeta sorted portfolio results after dropping high idiosyncratic risk stocks. Unlike Fig. 11.2 without dropping these stocks, the total risk of the portfolios is noticeably reduced. Now it is easier to see the effects of zeta risk within each beta risk quintile portfolio curve. The width of the parabola is clearly observable in this diagram. Also, by visually looking at the middle Z3 portfolio in each beta risk curve, we see that average stock returns gradually increase as beta risk is increased from the low B1 portfolio to the high B5 portfolio.

As predicted by ZCAPM theory, the CRSP index is located in the vicinity of the axis of symmetry of the mean-variance parabola at an interior location. It is not near the empirical efficient frontier defined by the high zeta risk Z5 portfolios in each beta risk quintile curve. Also, the G portfolio has much

lower total risk than the beta-zeta risk sorted portfolios and, therefore, is a good proxy for the global minimum variance stock portfolio.

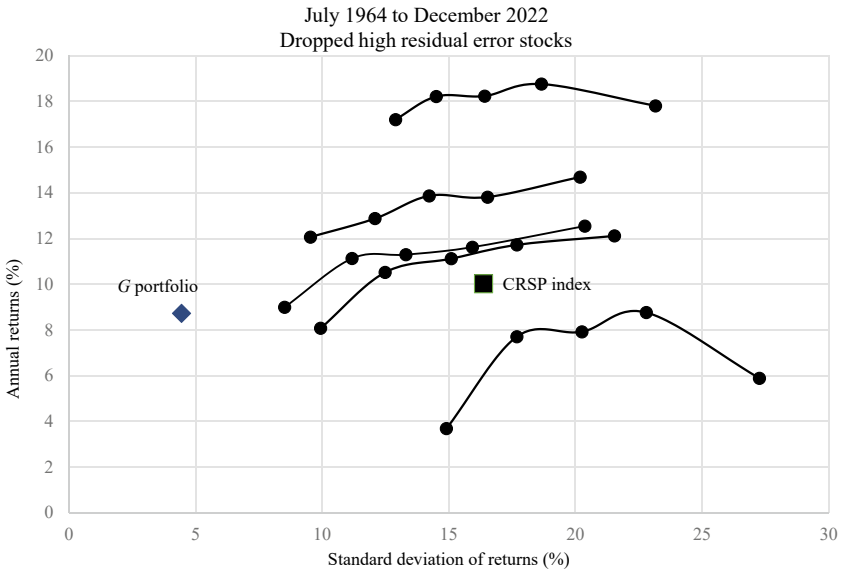
Upon repeating these analyses using subperiods July 1964 to December 1993 and January 1994 to December 2022, the average stock return results in Figs. 11.7 and 11.8 give a clearer picture of the relation between beta and zeta risks and stock returns than Figs. 11.3 and 11.4, respectively, without dropping these stocks. Comparing these two subperiods, average stock returns for the stock portfolios decrease over time. As mentioned earlier, major market downturns after the 2000 technology bubble, the 2008–2009 financial crisis, and the 2020 COVID-19 pandemic negatively impacted average long-run stock returns. Even so, on a relative basis, high zeta risk portfolios strongly outperformed the CRSP market index in both subperiods. Additionally, the CRSP index experienced an increase in total risk over time in these two subperiods; notably, in the later subperiod January 1994 to December 2022, its risk is higher than most of our zeta-beta and beta-zeta sorted quintile portfolios.

11.3 Summary

This chapter presented empirical evidence on the architecture of the mean-variance parabola of Markowitz (1952, 1959). We showed for U.S. stock returns in the analysis period July 1964 to December 2022 that beta risk and zeta risk in the empirical ZCAPM are not only important in defining the outer boundary of an empirical efficient frontier but portfolios in the interior of the empirical parabola also.

In our tests, we used the G portfolio (see Chapter 7) to estimate beta and zeta risk coefficients in empirical ZCAPM relation (10.4) developed by Kolari et al. (2021). Stock portfolio returns were computed in the next month after estimating beta and zeta risks for stocks. Portfolios were formed based on quintiles of beta risk and zeta risk, and their optimally weighted returns were computed in the next out-of-sample month consistent with an investable stock strategy.

We found that average stocks returns increased strongly with increases in zeta risk in quintile portfolios and increased somewhat more with beta risk in any given zeta risk quintile. Also, high idiosyncratic risk stocks tended to increase the total risk of zeta-beta and beta-zeta sorted quintile portfolios. By dropping these stocks, a clearer picture of the mean-variance parabola was possible.



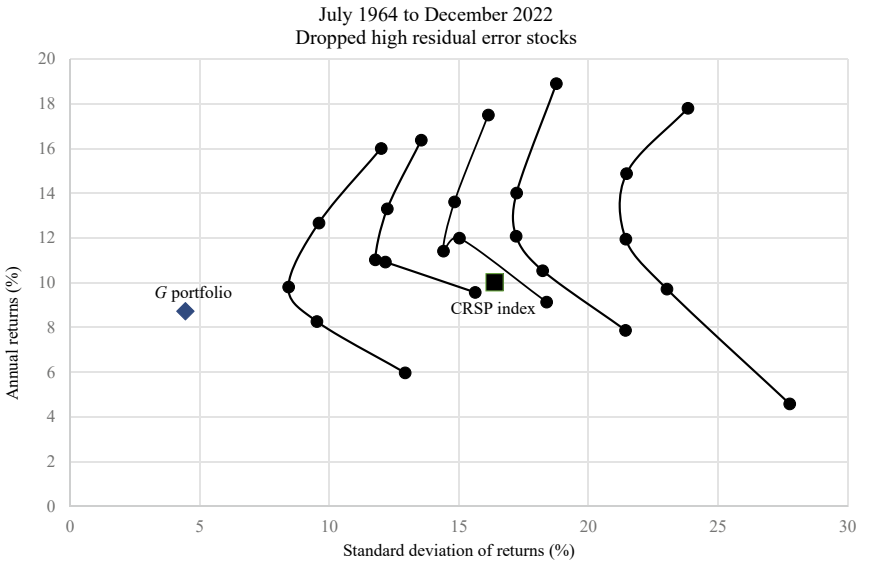
Dropped high residual error stocks												
Zeta-beta risk sorted portfolios (by quintiles)												
		Low Z1					Z2				Z3	
Low B1	B2	B3	B4	High B5	Low B1	B2	B3	B4	High B5	Low B1	B2	B3
3.69	7.71	7.92	8.76	5.89	8.07	10.51	11.12	11.72	12.11	8.99	11.12	11.29
14.91	17.69	20.25	22.79	27.26	9.94	12.48	15.09	17.69	21.54	8.52	11.18	13.31

Zeta-beta risk sorted portfolios (by quintiles)														
Z3		Z4					High Z5							
B4	High B5	Low B1	B2	B3	B4	High B5	Low B1	B2	B3	B4	High B5	CRSP	*G*
11.62	12.54	12.06	12.87	13.86	13.81	14.69	17.20	18.21	18.23	18.75	17.80	10.02	8.73
15.93	20.37	9.54	12.09	14.22	16.53	20.18	12.90	14.50	16.41	18.66	23.16	16.37	4.45

Fig. 11.5 Average returns and their standard deviation of returns for 25 portfolios sorted into zeta risk quintiles and then beta risk quintiles (within each zeta risk quintile) plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). In this graph, 20% of stocks are dropped that have high residual error variance in the empirical ZCAPM (i.e., high idiosyncratic risk). The analysis period is July 1964 to December 2022

To our knowledge, with the exception of Kolari, Liu, and Huang, no previous research studies or books on investment analysis have used an empirical asset model to build relatively efficient stock portfolios. Our results showed that beta risk and zeta risk portfolios outperform the value-weighted returns of the overall stock market as measured by the CRSP index by large margins. Remarkably, high zeta risk portfolios have average stock returns per year that are 75% to 250% higher than the CRSP index.

Compounding our high zeta risk returns over time periods of 10 to 20 years would substantially boost cumulative portfolio values. For example, in the 1994 to 2022 period, let's assume that you invest $1 at the CRSP average return of 8% for 10 years. Your portfolio value would more than double to $2.16. However, by investing in our high zeta risk portfolios, you could have



Dropped 20% high residual error stocks	Beta-zeta risk sorted portfolios (by quintiles)													
	Low B1					B2					B3			
7/1964-12/2022 Annualized returns	Low Z1	Z2	Z3	Z4	High Z5	Low Z1	Z2	Z3	Z4	High Z5	Low Z1	Z2	Z3	
Long only portfolio returns %	5.97	8.27	9.81	12.67	16.01	9.57	10.92	11.03	13.31	16.38	9.14	11.99	11.41	
Standard deviation of returns %	12.92	9.51	8.42	9.59	12.00	15.62	12.16	11.77	12.23	13.54	18.38	15.01	14.40	
	Zeta-beta risk sorted portfolios (by quintiles)													
	Z3		B4					High B5						
	B4	High Z5	Low Z1	Z2	Z3	Z4	High Z5	Low Z1	Z2	Z3	Z4	High Z5	CRSP	*G*
Long only portfolio returns %	13.62	17.50	7.87	10.54	12.08	14.01	18.90	4.58	9.72	11.95	14.88	17.80	10.02	8.73
Standard deviation of returns %	14.83	16.13	21.42	18.23	17.20	17.22	18.76	27.76	23.02	21.44	21.47	23.83	16.37	4.45

Fig. 11.6 Average returns and their standard deviation of returns for 25 portfolios sorted into beta risk quintiles and then zeta risk quintiles (within each beta risk quintile) plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). In this graph, 20% of stocks are dropped that have high residual error variance in the empirical ZCAPM (i.e., high idiosyncratic risk). The analysis period is July 1964 to December 2022

earned 14% with an ending portfolio value after 10 years equal to $3.71. Your ending valuation would have increased by a factor of 1.72 times compared to that of the CRSP index. Over a long investment horizon of 20 years, your ending portfolio values would be $4.66 for the CRSP index versus $13.74 for our zeta risk portfolio. Thus, your portfolio value would increase by a factor of 2.94 times the CRSP index, or almost 3 times more investment valuation. Such large differences in long-run valuation would have significant impacts on personal wealth over time. We invite investment practitioners to work with us to develop such high return per unit risk portfolios in the real world.

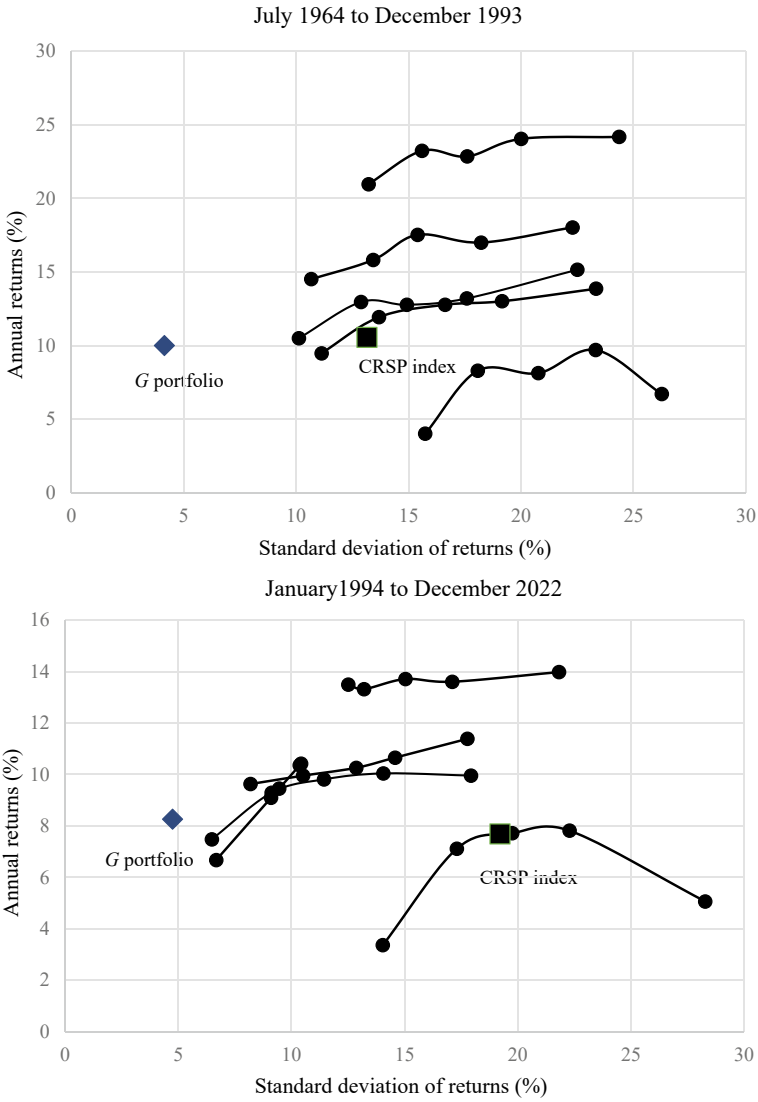


Fig. 11.7 Average returns and their standard deviation of returns for 25 portfolios sorted into zeta risk quintiles and then beta risk quintiles (within each zeta risk quintile) plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). In this graph, 20% of stocks are dropped that have high residual error variance in the empirical ZCAPM (i.e., high idiosyncratic risk). In the upper panel, the analysis period is July 1964 to December 1993. In the lower panel, the analysis period is January 1994 to December 2022

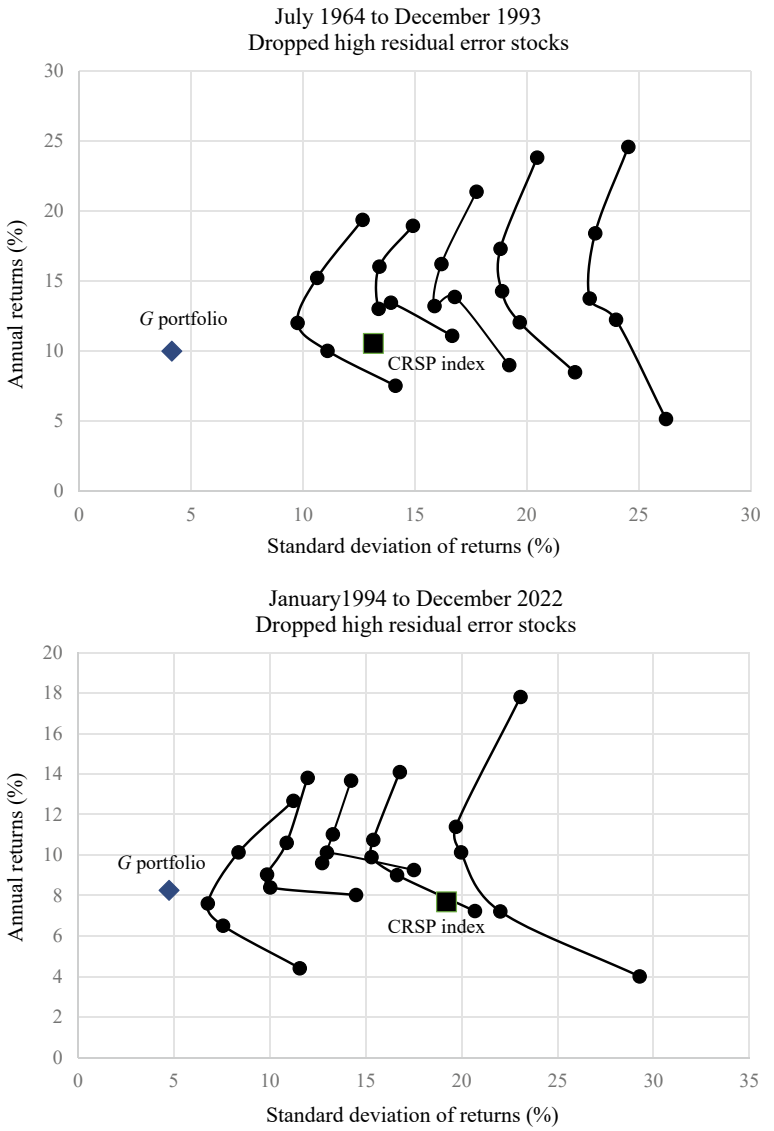


Fig. 11.8 Average returns and their standard deviation of returns for 25 portfolios sorted into beta risk quintiles and then zeta risk quintiles (within each beta risk quintile) plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). In this graph, 20% of stocks are dropped that have high residual error variance in the empirical ZCAPM (i.e., high idiosyncratic risk). In the upper panel, the analysis period is July 1964 to December 1993. In the lower panel, the analysis period is January 1994 to December 2022

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Dempster, A.P., N.M. Laird, and D.B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society* 39: 1–38.

Fama, E.F., and K.R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427–465.

Fama, E.F., and K.R. French. 1993. The cross-section of expected returns. *Journal of Financial Economics* 33: 3–56.

Fama, E.F., and K.R. French. 1995. Size and book-to-market factors in earnings and returns. *Journal of Finance* 50: 131–156.

Fama, E.F., and K.R. French. 1996. The CAPM is wanted, dead or alive. *Journal of Finance* 51: 1947–1958.

Fama, E.F., and K.R. French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116: 1–22.

Fama, E.F., and K.R. French. 2018. Choosing factors. *Journal of Financial Economics* 128: 234–252.

Fama, E.F., and K.R. French. 2020. Comparing cross-section and time-series factor models. *Review of Financial Studies* 33: 1892–1926.

Fama, E.F., and J.D. MacBeth. 1973. Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy* 81: 607–636.

Hamada, R.S. 1972. The effect of the firm's capital structure on the systematic risk of common stocks. *Journal of Finance* 27: 435–452.

Kolari, J. W., J. Z. Huang, W. Liu, and H. Liao, 2022a. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*, available online at https://www.mdpi.com/1911-8074/15/3/137) Reprinted in Kolari, J. W., and S. Pynnonen, 2022, eds., *Frontiers of Asset Pricing* (MDPI, Basel, Switzerland).

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022b. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. *Testing for missing asset pricing factors*. San Diego, CA: Paper presented at the Western Economic Association International.

Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Liu, W., 2013. *A new asset pricing model based on the zero-beta CAPM: Theory and Evidence*, Doctoral dissertation, Texas A &M University.

Liu, W., J. W. Kolari, and J. Z. Huang, 2012. *A new asset pricing model based on the zero-beta CAPM market model (CAPM)*, Presentation at the annual meetings of the Financial Management Association, Best Paper Award in Investments, Atlanta, GA (October).

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. *Return dispersion and the cross-section of stock returns*. Palm Springs, CA (October): Presentation at the annual meetings of the Southern Finance Association.

Markowitz, H. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: John Wiley & Sons.

Roll, R. 1977. A critique of the asset pricing theory's tests, part I: On past and potential future testability of the theory. *Journal of Financial Economics* 4: 129–176.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.



12

Mutual Fund Portfolios

12.1 Empirical Methods

U.S. mutual fund daily shareholder returns were downloaded from the CRSP database. To apply the ZCAPM to the problem of investing in mutual funds, we estimated the empirical ZCAPM for individual mutual funds using one year of daily returns. A total of 1,000 mutual funds with the smallest residual error volatility in the empirical ZCAPM were retained for subsequent analyses. These funds daily returns respond to beta and zeta risks associated with average market returns and the cross-sectional standard deviation of returns (or market return dispersion).

The 1,000 mutual funds are grouped into 100 portfolios sorted on their zeta risk coefficients, each containing 10 mutual funds. Portfolios of 10 mutual funds each are held for one month, and their returns are computed in this out-of-sample month. All portfolios are rebalanced every month with optimized weights. This process is repeated to build 100 portfolios sorted on beta risk coefficients, each again containing 10 mutual funds. The analysis period is from January 2000 to June 2022.

Previously in Chapter 4 we provided discussion of the theoretical ZCAPM by Kolari et al. (2021). The ZCAPM is a special case of Black's (1972) renowned zero-beta CAPM.[1] The zero-beta CAPM is based on an efficient portfolio and its orthogonal zero-beta portfolio. In the ZCAPM, the

[1] A number of ZCAPM publications have been written by the authors, including Liu et al. (2012, 2020), Liu (2013), Kolari, Huang, Liu et al. (2022, 2023), Kolari, Huang, Butt et al. (2022), Kolari and Pynnonen (2023), and Kolari et al. (2024).

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unique pair of efficient and zero-beta portfolios (denoted I^* and ZI^*, respectively) have the same total risk as measured by the variance of returns. The general equilibrium asset pricing ZCAPM proposes that the expected excess return for the ith asset can be specified as:

$$E(R_{it}) - R_{ft} = \beta_{i,a}\left[E(R_{at}) - R_{ft}\right] + Z^*_{i,a}\sigma_{at}, \tag{12.1}$$

where $E(R_{at})$ is the expected market return equal to the average return of all assets (rather than a market proxy m for the theoretical market portfolio M in Sharpe's (1964) CAPM), R_{ft} is the riskless rate, $\beta_{i,a}$ is the beta risk coefficient measuring sensitivity to expected excess market returns, σ_{at} is the cross-sectional standard deviation of all asset returns (or market return dispersion denoted RD) at time t, and $Z^*_{i,a}$ is the zeta risk coefficient measuring either positive or negative sensitivity to return dispersion. The systematic risk term $Z^*_{i,a}$ in the ZCAPM takes into account efficient portfolio I^* and orthogonal, inefficient portfolio ZI^*, both of which lie on the boundary of the mean-variance investment parabola and have equal return variances. The former (latter) portfolio has positive (negative) sensitivity to zeta risk coefficient $Z^*_{i,a}$. As such, depending on whether portfolio I^* or portfolio ZI^* is used, the expected excess returns for the ith asset can be positively or negatively affected by market return dispersion σ_{at}.

A challenge for specifying the empirical form of the ZCAPM is to allow for both positive and negative sensitivity to market return dispersion σ_{at} for any given asset. How can we determine the direction of the zeta risk coefficient for an asset? To solve this problem, as discussed in Chapter 5, Kolari, Liu, and Huang proposed a novel empirical ZCAPM:

$$R_{it} - R_{ft} = \beta_{i,G}\left(R_{Gt} - R_{ft}\right) + Z_{i,G}D_{it}\sigma_{Gt} + u_{it}, \quad t = 1, \ldots, T \tag{12.2}$$

where $R_{it} - R_{ft}$ is the excess return for the ith stock over the Treasury bill rate, $\beta_{i,G}$ measures sensitivity of the ith stock to excess average market returns based on the optimized G portfolio (see Chapter 7) minus the Treasury bill rate R_{ft}, $Z_{i,G}$ measures sensitivity of the ith stock portfolio (or stock) to the cross-sectional standard deviation of returnsσ_{Gt} (return dispersion), D_{it} is a signal variable with values $+1$ and -1 representing positive and negative return dispersion effects, respectively, on stock returns, and $u_{it} \sim$ iid $\mathrm{N}(0, \sigma_i^2)$. Notice that there is no intercept parameter α_i in the empirical ZCAPM. The lack of an α_i parameter is unlike all other empirical asset pricing models that use ordinary least squares (OLS) regression methods for estimation purposes.

In the above equation, signal variable D_{it} is an independent random variable with a two-point distribution:

$$D_{it} = \begin{cases} +1 & \text{with probability } p_i \\ -1 & \text{with probability } 1 - p_i, \end{cases} \tag{12.3}$$

where p_i (or $1-p_i$) is the probability of a positive (or negative) return dispersion effect, and D_{it} are independent of u_{it}. The empirical ZCAPM utilizes the expectation-maximization (EM) algorithm of Dempster, Laird, and Rubin (1977) for regression parameter estimation including the probability parameter p_i.[2] Since $E(D_{it}) = 2p_i - 1$, the marginal form of the empirical ZCAPM can be stated as:

$$R_{it} - R_{ft} = \beta_{i,G}\left(R_{Gt} - R_{ft}\right) + Z^*_{i,G}\sigma_{Gt} + u^*_{it}, \quad t = 1, \ldots, T \tag{12.4}$$

where $\beta_{i,G}$ is the beta risk coefficient, $Z^*_{i,G} = Z_{i,G}(2p_i - 1)$ is the zeta risk coefficient, u^*_{it} is the error term with variance related to return dispersion, and other notation is as before. The probability p_i associated with the expected value of signal variable D_{it} in period $t = 1, \ldots, T$ determines the sign of zeta risk loading $Z^*_{i,G}$ as either positive or negative.

Free software to estimate the empirical ZCAPM are provided by Kolari, Huang, and Liu on GitHub (https://github.com/zcapm). Matlab, Python, and R programs are available. In forthcoming analyses, we implement their R programs, which tend to run faster than the other programs.

12.1.1 Building Mutual Fund Portfolios

As mentioned earlier, we use the CRSP database to download daily stock returns for mutual funds. Mutual funds hold long portfolios with no short positions. Only actively managed mutual funds specializing in publicly traded equities are selected. Daily returns for the period from January 1999 to June 2021 are downloaded. Also, we drop mutual funds with relatively large residual error terms in the empirical ZCAPM. These funds' returns are sensitive to beta and zeta risks in the empirical ZCAPM. We retain 1,000 mutual funds for our analyses.

The following steps are implemented to build mutual fund portfolios sorted on their zeta risk and beta risk characteristics.

[2] See Kolari et al. (2021) for step-by-step instructions on how to implement the EM algorithm to estimate the empirical ZCAPM. Also, they provide citations to other finance and statistics studies that have employed the EM algorithm.

1. Times-series regressions are used to estimate empirical ZCAPM Eq. (12.4) via the expectation-maximization (EM) algorithm for each individual mutual fund with daily returns for the initial 12-month estimation period from January 1999 to December 1999. As market factors in the empirical ZCAPM, we employ our estimate of the global minimum variance portfolio G (see Chapter 7) to compute average stock market returns and the cross-sectional standard deviation of all stocks' returns (or market return dispersion), which are associated with beta risk and zeta risk, respectively.
2. All stocks are ranked on their residual error volatility. The 1,000 mutual funds with the lowest residual volatility are retained, and other mutual funds are dropped.
3. All stocks are ranked in terms of their estimated zeta coefficient $\hat{Z}^*_{i,G}$.
4. A total of 100 portfolios are formed with 10 mutual funds in each zeta risk portfolio.
5. For all stocks in each zeta risk portfolio, we compute their one-month-ahead (out-of-sample) return R_{pt+1} in January 2000. Using these returns, we compute the zeta risk portfolio's optimally weighted, one-month-ahead holding period return. This out-of-sample approach is consistent with real world investor behavior in that portfolios are formed at time $t = T$ and performance is observed in the next month $T + 1$.
6. The entire process is repeated by rolling the estimation period in step (1) forward one month to rebalance each of the 100 mutual fund portfolios sorted on zeta risk in February 2000. By rolling forward month-by-month to the end of the sample period, we generate out-of-sample returns for each portfolio from January 2000 to June 2022.

Employing these same steps, we also build 100 mutual fund portfolios sorted on beta risk. Is beta risk related to average mutual fund returns? Numerous studies have shown that CAPM beta risk measured using the CRSP index (as a proxy for the market portfolio) is not related to average stock returns. As we will see, average stock returns of mutual fund portfolios are related to both beta risk and zeta risk as estimated by the empirical ZCAPM. Hence, mutual fund investors can use the empirical ZCAPM to earn higher returns and better manage their risk too.

12.2 Empirical Results

This section reports the results for our mutual fund portfolios. All portfolios consist of 10 mutual funds that are rebalanced at the end of each month. Optimized weights are used to form each portfolio in any given month. Portfolio returns are computed out-of-sample in the next month after formation. The analysis period is from January 2000 to June 2022, or 22.5 years.

12.2.1 Mutual Fund Portfolios Sorted on Zeta Risk

The out-of-sample performance in the next month for the 100 mutual fund portfolios sorted on zeta risk in the empirical ZCAPM is shown in Fig. 12.1. We graph the G portfolio and CRSP market index also. The results clearly show a positive relation between zeta risk and average annual returns as well as total risk as measured by the standard deviation of returns over the analysis period. Two mutual fund portfolios are outliers with very high total risk on the right side of the graph. We checked these portfolios and found that one mutual fund in each portfolio had exceptionally large daily returns at times which biased the findings. After dropping them from the analyses, we created Fig. 12.2, which provides a more accurate picture of mutual fund portfolio performance.

The CRSP market index has greater total risk than our optimized zeta risk portfolios that each comprise of 10 mutual funds. Also, as expected, portfolio G has much lower total risk. From an investor standpoint, the three points at the top of the cluster of mutual fund portfolios are particularly interesting. These portfolios have average returns around 10% per year compared to the CRSP at 6.45% per year, which are approximately 55% higher in terms of average returns in the analysis period. This outperformance bodes well for recommending our ZCAPM-based approach to investing in mutual funds.

Figure 12.3 shows the performance of the mutual fund portfolios sorted on zeta risk in terms of Sharpe ratios. Looking at the top graph there, most portfolios have higher Sharpe ratios than the CRSP index; however, some portfolios fall below this market benchmark. The G portfolio well outperforms all of the mutual fund portfolios. In general, these results demonstrate that zeta risk measurement is useful for choosing high performing mutual fund portfolios. In the bottom graph, we dropped the G portfolio to get a better look at the mutual fund portfolios. Now we can see more clearly where the CRSP index lies on the spectrum of Sharpe ratios. The mutual fund portfolios at the far right of the cluster of mutual fund portfolios are most interesting to investors. Their Sharpe ratios are the in range of 0.45 to

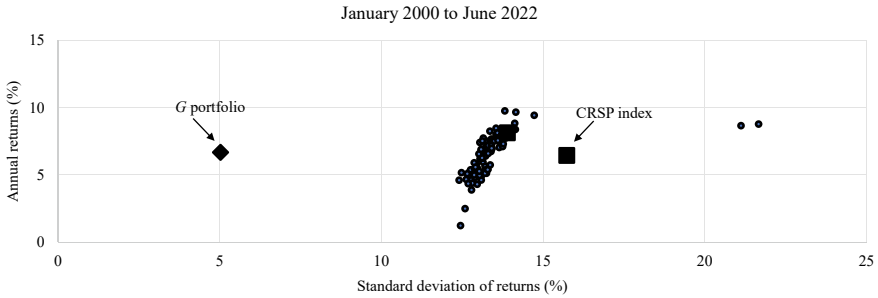


Fig. 12.1 Average returns and their standard deviation of returns for 100 mutual fund portfolios sorted on zeta risk plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is January 2000 to June 2022

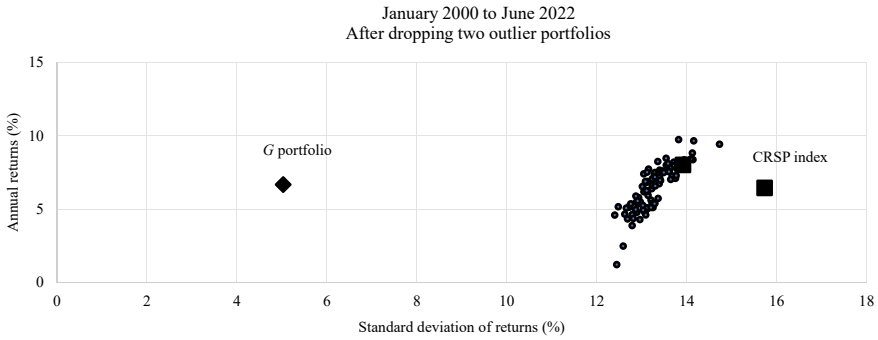


Fig. 12.2 After dropping two outlier portfolios, average returns and their standard deviation of returns for 100 mutual fund portfolios sorted on zeta risk plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is January 2000 to June 2022

0.48 compared to 0.32 for the CRSP index. Hence, these mutual fund portfolios can "beat the market" in the sense of outperforming the CRSP market index.

12.2.2 Mutual Fund Portfolios Sorted on Beta Risk

We next review the evidence for mutual fund portfolios sorted on beta risk in the empirical ZCAPM. Visual inspection of Fig. 12.4 suggests a clear pattern with higher average annual returns as beta risk increases among the mutual funds. Like findings in Chapter 10, where long only beta portfolios depicted a concave function with returns decreasing at the highest beta risk levels, we again see average portfolio returns increase as total risk increases but then level out and even reverse somewhat at the highest beta risk levels. As conjectured

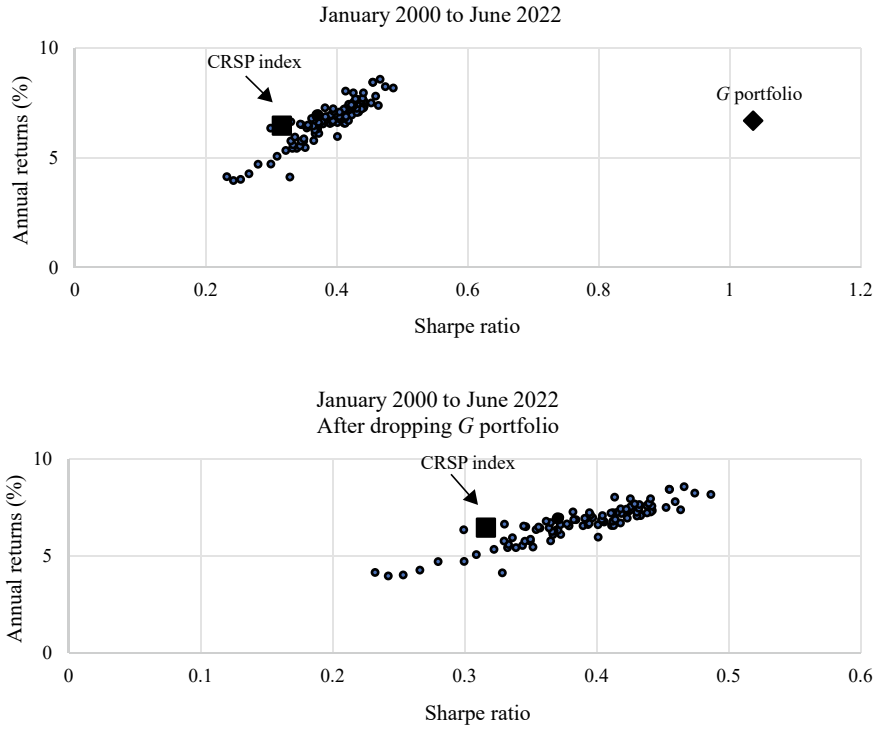


Fig. 12.3 Average returns and their Sharpe ratios for 100 mutual fund portfolios sorted on zeta risk plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is January 2000 to June 2022

earlier, high best risk levels are likely due to high financial leverage that diminishes firm profits and stock returns at some point. Moreover, there is one outlier mutual fund portfolio with negative average returns due to extremely poor performance.

We graph the Sharpe ratios of beta-sorted mutual fund portfolios in Fig. 12.5. In the top graph, again the *G* portfolio is much more efficient than the mutual fund portfolios with a distinctly higher Sharpe ratio in the analysis period. Dropping the *G* portfolio, the graph in the bottom Fig. 12.5 shows the dispersion of mutual fund portfolios' performance. The CRSP index falls about in the middle of the portfolios. Hence, to find high-performing mutual funds, it is useful to use beta risk in the empirical ZCAPM.

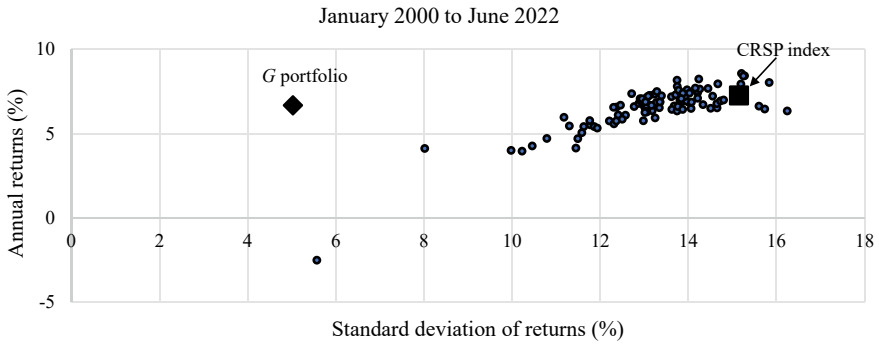


Fig. 12.4 Average returns and their standard deviation of returns for 100 mutual fund portfolios sorted on beta risk plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is January 2000 to June 2022

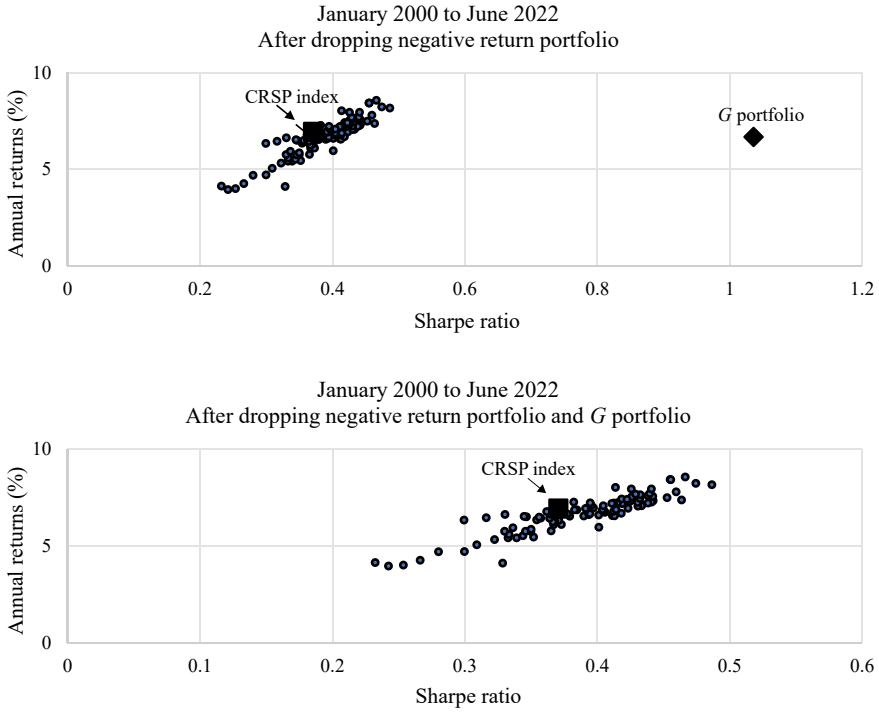


Fig. 12.5 Average returns and their Sharpe ratios for 100 mutual fund portfolios sorted on beta risk plus the CRSP market index and minimum variance portfolio *G* (see Chapter 7). The analysis period is January 2000 to June 2022

12.3 Summary

This chapter sought to answer the question of whether the ZCAPM can be applied to mutual funds in the same way as individual stocks in earlier chapters to build portfolios that outperform general stock market indexes. In our analyses, we employ 1,000 equity mutual funds with daily returns available on the CRSP database. As before, we estimated the empirical ZCAPM with one year of daily returns. Mutual funds are sorted into 100 portfolios with 10 mutual funds in each portfolio. We sort the 1,000 mutual funds in two ways: (1) funds are sorted on their zeta risks and formed into 100 portfolios; and (2) funds are sorted on their beta risks into 100 portfolios. Using optimized weights for the 10 mutual funds in a portfolio, out-of-sample returns in the next month are computed for each of these portfolios. The analysis is rolled forward one month at a time, and the portfolios are rebalanced each month. A time series of out-of-sample monthly returns is generated from January 2000 to December 2022.

The results for zeta risk sorted mutual fund portfolios showed that, as zeta risk increases, average annual returns increase. Some of the zeta risk portfolios outperformed that CRSP market index by large margins. For example, three portfolios earned approximately 10% per year that well outperformed the CRSP index at 6.45%. Our optimized mutual fund portfolios had lower total risk than the CRSP index and normally higher Sharpe ratios too.

Turning to the results for the beta-sorted portfolios, a positive relation between average returns and beta risk was found. Thus, even though beta as computed using the CRSP index returns in past tests of Sharpe's (1964) CAPM in the finance literature has little or no relation to average stock returns, beta risk estimated from G portfolio returns reveals a positive relation as predicted by the ZCAPM. Beta risk portfolios had lower Sharpe ratios than the G portfolio but similar ratios to that of the CRSP index. Furthermore, as in Chapter 10, we found that average returns for very high beta risk portfolios tended to decline, which we conjecture is due to high levels of debt or financial risk in these companies.

In general, the main takeaway from our mutual fund results is that ZCAPM methods can be applied to building high-performing portfolios of mutual funds. We recommend that mutual funds offer ZCAPM portfolio services to their investors to help them increase their personal wealth and pension savings over time. Such an investment service would benefit millions of people who are saving for retirement. The authors are open to work with the mutual fund industry to achieve this investment goal.

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Dempster, A.P., N.M. Laird, and D.B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society* 39: 1–38.

Hamada, R.S. 1972. The effect of the firm's capital structure on the systematic risk of common stocks. *Journal of Finance* 27: 435–452.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2022. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*, available online at https://www.mdpi.com/1911-8074/15/3/137. Reprinted in Kolari, J.W., and S. Pynnonen, eds. 2022. *Frontiers of asset pricing*. Basel, Switzerland: MDPI.

Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.

Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. Testing for missing asset pricing factors. Paper presented at the Western Economic Association International, San Diego, CA.

Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J. Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Liu, W. 2013. A new asset pricing model based on the zero-beta CAPM: Theory and evidence. Doctoral dissertation, Texas A&M University.

Liu, W., J.W. Kolari, and J.Z. Huang. 2012. A new asset pricing model based on the zero-beta CAPM market model (CAPM). Presentation at the annual meetings of the Financial Management Association. Best Paper in Investments, Atlanta, GA (October).

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. Return dispersion and the cross-section of stock returns. Presentation at the annual meetings of the Southern Finance Association, Palm Springs, CA (October).

Markowitz, H. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York: Wiley.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.

Part VI

Conclusion



13

The Future of Investment Practice, Artificial Intelligence, and Machine Learning

13.1 Asset Pricing Discussion

Based on the simple assumption that a riskless rate exists (e.g., U.S. Treasury bill rates), Sharpe (1964) and others took advantage of Markowitz's mean-variance parabola to derive the now famous Capital Asset Pricing Model (CAPM). According to the CAPM, diversified investors should buy the market portfolio on Markowitz's efficient frontier that is located at a tangent point from a ray extending from the riskless rate to the efficient frontier. In so doing, the only priced risk for assets (including portfolios) is systematic beta risk related to market portfolio returns. The CAPM was the first asset pricing model derived in a general equilibrium market setting with risk averse investors, perfect capital markets, and other market assumptions. It immediately became the centerpiece of valuation in finance and earned Sharpe the Nobel Memorial Prize in Economic Sciences in 1990. Researchers proposed a number of variations on the CAPM that took into account conditional, consumption, international, intertemporal, and production conditions. Interestingly, Sharpe was Markowitz's student, and they both received the Nobel Prize at the same time in 1990 for their modern portfolio and asset pricing discoveries.

In its early years, empirical evidence found that beta risk in the CAPM only weakly explained the cross section of average stock returns. Then in the 1990s, the bottom fell out. A series of top tier finance publications by Fama and French (1992, 1993, 1995, 1996, 1998, 2004) proved using stock returns that beta risk in the CAPM was not related to average stock returns in the cross section. They concluded that the beloved CAPM was *dead*. In its

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place, the authors proposed an innovative three-factor model by augmenting the market factor with size and value factors. These so-called *multifactors* were long-short portfolios akin to Ross' (1976) arbitrage pricing theory (APT). The empirical success of their multifactors motivated many researchers to propose alternative multifactors. Academic asset pricing research became embroiled in a battle between multifactor models that continues today. Which factors are the best ones? Which models are the most valid? Fama and French (2015, 2018, 2020) proposed five- and six-factor models in attempts to improve upon their earlier three-factor model. Other researchers, such as Hou, Xue, and Zhang (2015) and Stambaugh and Yuan (2017), proposed other multifactor models. Nowadays there are hundreds of long-short factors that have been suggested by researchers. Recognizing this fact, Cochrane (2011) has humorously referred to the proliferation of factors in asset pricing as a "factor zoo."

A major problem with multifactor models is that they have little theoretical foundation. They are not based on Markowitz's mean-variance parabola or general equilibrium theory as in Sharpe's CAPM. In an attempt to overcome this problem, filling a gap in the literature, Kolari, Liu, and Huang (2021) proposed a new CAPM dubbed the ZCAPM. This asset pricing model is derived as a special case of Black's (1972) zero-beta CAPM, which includes an efficient portfolio and an inefficient, orthogonal zero-beta portfolio that lie on Markowitz's mean-variance parabola. The authors chose a unique pair of efficient and inefficient, orthogonal portfolios on the parabola that have the same total risk as measured by their variance of returns. This pair of portfolios enabled them to derive the theoretical ZCAPM with two factors: (1) average market returns, and (2) the cross-sectional standard deviation of returns of all assets in the market (or market return dispersion). These two factors correspond to beta risk and zeta risk, respectively. Notice here that beta risk is not related to the market portfolio on the efficient frontier as in the CAPM but average market returns that lie somewhere along the axis of symmetry in the middle of the parabola. Additionally, notice that market return dispersion is not a long-short portfolio or multifactor. In another innovation, to estimate the theoretical ZCAPM in the real world, Kolari, Liu, and Huang developed an empirical ZCAPM that utilizes an expectation-maximization (EM) algorithm in the regression equation. Virtually all previous asset pricing models employ ordinary least squares (OLS) regression methods; hence, the empirical ZCAPM is an entirely new methodological approach to estimating an asset pricing model. The authors make available free Matlab, Python and R coded software to estimate the empirical ZCAPM via the EM algorithm on the internet at GitHub (https://github.com/zcapm).

In their book as well as numerous research studies, including Liu, Kolari and Huang (2012), Liu (2013), Liu, Kolari, and Huang (2020), Kolari, Huang, Liu, and Liao (2022), Kolari, Huang, Butt, and Liao (2022), Kolari, Huang, Liu, and Liao (2023), Kolari and Pynnonen (2023) and Kolari, Huang, Liu, and Liao (2024), the empirical ZCAPM proved to dominate well-known multifactor models. Note that the main argument for multifactor models by Fama and French and others is that they better fit real world stock return data in empirical tests than the CAPM. However, the ZCAPM consistently outperforms multifactor models in cross-sectional regression tests with different test assets and analyses periods. The repeated outperformance of the ZCAPM in terms of better fitting average stock returns in the cross section says that Markowitz, Sharpe, and others with general equilibrium theories were right after all. The CAPM is not dead as Fama and French have pronounced; instead, it is *alive and well* in the new form of the ZCAPM!

The authors have found that academic finance journals and even conferences tend to resist and deny the ZCAPM. Accepting the ZCAPM means to abandon multifactor models that have become so popular over the last 30 years. All of the long-short factors in the multifactor models are subsumed into the market return dispersion factor of the ZCAPM. The reason is that long-short factors are themselves rough measures of return dispersion. Each factor measures a return dispersion slice in the span or width of the mean-variance investment parabola. The ZCAPM incorporates the total return dispersion as measured by the cross-sectional return dispersion of returns of all assets in the market (i.e., the second moment of the return distribution) and, therefore, all long-short factors are contained therein. There is no longer any need for long-short factors as they are all taken into account within the market return dispersion factor of the ZCAPM. To paraphrase Fama and French, multifactor models are *dead*. We recommend that researchers turn their attention to the new frontier in asset pricing—the ZCAPM. Replicate our empirical results, run new tests, explore our new approach to asset pricing, etc. As research grows on the ZCAPM, it is likely that new discoveries will be made. We look forward to the day when even the top finance journals recognize the theoretical advances and empirical successes of the ZCAPM.

13.2 The ZCAPM and Investment Practice

In this book, we have sought to answer a simple but practical question: Can an asset pricing model be used to boost performance in machine-made portfolios?

Our first challenge was to develop a proxy for the global minimum variance portfolio *G*. This portfolio "pins" the mean-variance investment parabola in the sense that it determines the position of the entire parabola in return/risk space. The *G* portfolio has no zeta risk and only beta risk. Consequently, it is the most suitable portfolio to compute average market returns and market return dispersion for use in the empirical ZCAPM.

To build *G*, we employed all U.S. common stocks downloaded from the Chicago Research in Security Prices (CRSP) database from July 1963 to December 2022. Very small stocks that are illiquid were dropped from the analyses. Our empirical results for the *G* portfolio were impressive. The portfolio had average returns similar to the CRSP market index but very low total risk (or return variance) compared to the CRSP index. In turn, its Sharpe ratio (defined as excess returns over the Treasury bill rate divided by the standard deviation of returns) was about two and half times larger than the CRSP index. In our analysis period from July 1964 to December 2022, the volatility of the CRSP index trended upward over time due to periodic market turmoil, but the volatility of our *G* portfolio was fairly constant. The low and stable volatility of our *G* portfolio are good characteristics that recommend it for application in the empirical ZCAPM.

With portfolio *G* in hand, we estimated the empirical ZCAPM using this portfolio to compute average market returns and market return dispersion related to beta risk and zeta risk, respectively. Individual stocks were sorted into portfolios with low to high zeta risk as measured by the estimated zeta risk coefficient in the empirical ZCAPM. Optimal weights were utilized to form the final zeta risk portfolios. Using these portfolios, long-short zeta risk portfolios were formed that had increasing levels of zeta risk. Finally, net long portfolios were constructed by adding low variance *G* (with zero zeta risk) to each of the long-short zeta risk portfolios. These portfolios had increasing levels of zeta risk. Out-of-sample returns for each portfolio were computed in the next month similar to an investable strategy implemented by a real world investor.

Based on an analysis period from July 1964 to December 2022, our results were extraordinary—more specifically, the 12 net long portfolios traced out an empirical efficient frontier. Confirming earlier work by Kolari, Liu, and Huang, the empirical shape of the efficient frontier was mapped in return/risk space as estimated from actual stock returns over a long period of time. Consistent with ZCAPM theory, the CRSP market index was located in the interior of the mean-variance parabola approximately on the axis of symmetry of the parabola (i.e., it represents an average market return). It clearly is not a good proxy for the efficient market portfolio of the CAPM as

so many academic researchers have supposed over the years. Strikingly, some of the net long portfolios earned average annual returns in the range of 29 to 36% per year compared to about 10% for the CRSP index. Also, their Sharpe ratios were considerably higher than the CRSP index.

Motivated by the excellent performance of our net long portfolios, we conducted a number of different risk analyses of the portfolios. Gibbons, Ross, and Shanken (GRS) (1989) tests of portfolio efficiency showed that our empirical efficient frontier of net long portfolios increasingly approximates the theoretical efficient frontier of Markowitz as zeta risk increases. Also, the results suggested that, by combining the net long portfolios, we could move the empirical efficient frontier even higher and closer to the theoretical efficient frontier.

Value at risk (VaR) tests showed that, with the exception of the highest zeta risk portfolio, our net long portfolios have lower VaRs than the CRSP market index. Additionally, VaRs tended to increase for portfolios over time in our analysis period due to market turmoil surrounding the 2000 technology (dot.com) bubble, the 2008–2010 financial crisis, and the 2020–2021 COVID-19 pandemic.

Drawdown tests corroborated these findings. Also, they documented that various dimensions of drawdown risk (i.e., frequency, depth, duration, peak-to-trough, and recoveries from trough-to-peak) tended to increase as zeta risk increases. Even so, these increasing risks were counterbalanced by relatively high return performance that boosted accumulated wealth as well as Sharpe ratios for the net long portfolios in comparison to the CRSP index.

Many investors do not employ short positions in their stock portfolios. Pension investments in mutual funds is a good example as well as many exchange traded funds (ETFs). Relevant to these investors, we examined the performance and risk of long only stock portfolios constructed using our ZCAPM-based methods. Our empirical results for the analysis period from July 1964 to December 2022 were compelling. We found that 25 long only portfolios sorted on zeta risk in the empirical ZCAPM traced out the shape of Markowitz's mean-variance parabola. As hypothesized by ZCAPM theory, the CRSP market index was located in the interior of the parabola approximately on the axis of symmetry that splits the parabola into upper and lower halves. The *G* portfolio had average returns similar to the CRSP index and very low total risk compared to the zeta risk and CRSP portfolios (i.e., it has no zeta risk as expected by ZCAPM theory). Of particular interest to investors, some of the higher zeta risk portfolios earned average returns equal to 20% per year or more, which well outperformed the CRSP index at about 10%. Some of these portfolios had similar total risk as the CRSP index. Thus, their Sharpe

ratios were much greater than the CRSP index. Again, over time in our analysis period, average returns for all portfolios declined due to periodic bouts with stock market turmoil due to various economic and other shocks.

We also sorted the long only portfolios by their beta risk estimated in the empirical ZCAPM. Beta risk here is related to average market returns and not a proxy for the market portfolio as in the CAPM.

Unlike CAPM beta risk, wherein little or no relation exists between average stock returns and beta, average annual returns for the long only portfolios tended to increase as beta risk increased.

For portfolios with relatively high beta, we saw a flattening and even reversal of average stock returns. We conjectured that this pattern was consistent with the famous corporate capital structure and firm valuation theories of Modigliani and Miller (1958, 1963) as well as Hamada (1972). These authors posited that financial leverage due to debt usage by firms would initially increase value due to interest tax deductions on debt interest payments. However, eventually rising debt levels in the firm would lead to an optimal capital structure and thereafter a decline in value due to rising risks of bankruptcy and coincident costs to the firm. These findings are intriguing and deserving of further investigation.

Adding to our analyses of beta risk and zeta risk in long only portfolios, we sorted stocks into zeta risk quintiles and then beta risk quintiles within each zeta risk quintile. These 25 zeta-beta risk portfolios sorted on zeta-beta risk quintiles confirmed ZCAPM theory (see Figure 7.1 in Chapter 7). That is, average annual stock returns were higher in each zeta risk quintile slice of portfolios. Also, average returns increased with beta risk but to a lesser extent than zeta risk. Similar results were found by sorting stocks on beta risk and then zeta risk to form 25 beta-zeta risk portfolios. Our results revealed that Markowitz's mean-variance parabola has a definite structure or architecture that is determined by the beta risk and zeta risk characteristics of portfolios. It is apparent that portfolios within the parabola are located in different interior positions due to their beta and zeta risks. Higher zeta risk is needed to reach the empirical efficient frontier or upper boundary of the empirical parabola. Given zeta risk, beta risk moves portfolios toward higher average returns and higher total risk within the empirical parabola. Professional managers and investors can use this beta-zeta risk architecture to build portfolios for clients, better understand the performance of their stock portfolios, and make informed decisions about future investments.

While Markowitz invented the mean-variance investment parabola in the 1950s, efforts by researchers and academics to create efficient portfolios in the real world have failed. Studies have shown that optimally-weighted portfolios

do not outperform equally-weighted portfolios. The illusory nature of the efficient frontier is a major anomaly in the field of finance. A Nobel prize was awarded to Markowitz for ideas that do not work in the real world? The prize is supposed to recognize ideas that help to solve practical problems facing the society. Of course, if the mean-variance parabola does not exist, then the CAPM does not exist either as there is no tangent market portfolio on the efficient frontier of the parabola. We believe that our work on the architecture of the mean-variance parabola is the first strong evidence that the parabola indeed exists on an out-of-sample basis and is not merely an interesting theoretical concept. Importantly, given these findings about the investment parabola, we can safely infer that the CAPM of Sharpe and other researchers may well exist, thereby providing support for the zero-beta CAPM of Black and ZCAPM of Kolari, Liu, and Huang. Using the ZCAPM, as already mentioned, both investment managers and individual investors can build portfolios that are located at different positions within the parabola. For example, they can build portfolios that outperform general market indexes, such as the S&P 500 index, as an out-of-sample investable strategy. Given that over 90 percent of professional managers cannot beat the S&P 500 index over periods of 3-to-5 years, this prospect is exciting indeed!

Lastly, we applied ZCAPM investment analyses to actively managed equity mutual funds. Our analysis period was January 2000 to December 2022. We grouped 1,000 U.S. equity mutual funds into 10 portfolios sorted on zeta risk and 10 portfolios sorted on beta risk. As zeta risk increased, average annual returns for optimized portfolios of mutual funds increased. In our analysis period, three of these mutual fund portfolios earned approximately 10% per year compared to 6.45% for the CRSP index. Additionally, our mutual fund portfolios had lower total risk and higher Sharpe ratios than the CRSP index. Beta risk portfolios exhibited higher average returns as beta increased, but Sharpe ratios were similar to the CRSP index. In general, our mutual fund experiments demonstrated that the ZCAPM can be used to substantially boost pension savings invested in mutual funds. Certainly long-run savers concerned about having enough funds to maintain their lifestyle in retirement should be informed about these promising mutual fund results.

13.3 Implications of Artificial Intelligence and Machine Learning

Nowadays there is a lot of excitement about artificial intelligence (AI) and the application of machine learning (ML) in many areas of business. Relevant to our focus, machine learning is already here in investment management. A recent study by Grobys, Kolari, and Niang (2022) found that hedge funds with the highest level of automation outperformed those with more reliance on human decision-making. AIML models are particularly well suited to forecasting in addition to uncovering hidden patterns in data that human traders would not be able to discern. They can identify new trading strategies, are able to learn and adapt, and do not have human behavioral biases. The authors examined 826 U.S. hedge funds from 2006 to 2021. They grouped these funds into four clusters: discretionary, systematic, combined, and AIML. After risk-adjusted returns using the CAPM and Fama and French (1992, 1993, 1995, 1996, 1998) three-factor model, the AIML funds outperformed the other funds in terms of average returns. Also, they found that a man-versus-machine investment portfolio that is long AIML funds and short discretionary funds generated significant average returns ranging from 20 to 26 basis points per month. They concluded that machine decision-making beat human decision-making as well as mixed automated-human strategies.

Another study by Cao, Jiang, Wang, and Yang (2022) conducted an experiment involving a hypothetical AI analyst that collects a variety of information, including corporate disclosures, industry trends, and macroeconomic indicators. When the firm is more complex and has intangible assets, humans tended to outperform machines in terms of stock forecasts. However, when information was readily available and voluminous, machines beat humans. A man-versus-machine investment portfolio that is long AI forecasts and short human analyst forecasts produced monthly abnormal returns of more than 75 basis points. In these tests, they adjusted returns for risk using the Fama and French (1993, 2015, 2018, 2020) three-, five-, and six-factor models as well as the Carhart (1997) four-factor model. A hybrid strategy that combined AI and human judgment proved to outperform the machine only strategy. They concluded that man plus machine is a superior investment approach. Thus, the future of AIML in professional investment management is complementary to traditional human decision-making approach. We need both to produce the highest performing investment portfolios.

These and other studies cited therein make a strong case for AIML models in investment management. Asset pricing models can be employed to

help construct superior performing portfolios. We believe the ZCAPM can play a role in this investment technology movement. Managers need statistical and computer software tools to create new investment strategies to boost returns and control risk in portfolios. This book has shown that the ZCAPM can be used to achieve these investment goals. No doubt ZCAPM technology will advance in the years ahead as its applications increase in the real world. We have only scratched the surface to demonstrate its feasibility. Real world implementation and portfolio track records are needed to further develop and prove the concepts. In this regard, we actively seek partners in an effort to employ ZCAPM-based AIML models in professional practice.

As a final comment, our ZCAPM-based investment technology would not be possible without the landmark contributions of previous researchers, including Black (1972), Dempster, Laird, and Rubin (1977), Fama and French (1993, 2018), Fama and MacBeth (1973), Hou, Xue, and Zhang (2015), Gibbons, Ross, and Shanken (1989), Lintner (1965), Markowitz (1952, 1959), Merton (1973), Mossin (1966), Ross (1976), Sharpe (1964), Stambaugh and Yuan (2017), Treynor (1961, 1962), among many others that have made important contributions to asset pricing. Most certainly, we stand on the "shoulders of giants" that enabled the derivation of the theoretical ZCAPM, specification of the empirical ZCAPM, and applications of beta and zeta risks in the ZCAPM to the problem of constructing high-performing investment portfolios. We hope that academic researchers and practitioners will further develop our ideas in the future.

References

Black, F. 1972. Capital market equilibrium with restricted borrowing. *Journal of Business* 45: 444–454.

Carhart, M.M. 1997. On persistence in mutual fund performance. *Journal of Finance* 52: 57–82.

Cao, S., W. Jiang, J. L. Wang, and B. Yang. 2022. From man vs. machine to man + machine: The airt and AI of stock analyses. Columbia Business School Research Paper, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3840538.

Cochrane, J.H. 2011. Presidential address: Discount rates. *Journal of Finance* 56: 1047–1108.

Dempster, A.P., N.M. Laird, and D.B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society* 39: 1–38.

Fama, E.F., and K.R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427–465.
Fama, E.F., and K.R. French. 1993. The cross-section of expected returns. *Journal of Financial Economics* 33: 3–56.
Fama, E.F., and K.R. French. 1995. Size and book-to-market factors in earnings and returns. *Journal of Finance* 50: 131–156.
Fama, E.F., and K.R. French. 1996. The CAPM is wanted, dead or alive. *Journal of Finance* 51: 1947–1958.
Fama, E.F., and K.R. French. 1998. Value versus growth: The international evidence. *Journal of Finance* 53: 1975–1999.
Fama, E.F., and K.R. French. 2004. The Capital Asset Pricing Model: Theory and evidence. *Journal of Economic Perspectives* 18: 25–46.
Fama, E.F., and K.R. French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116: 1–22.
Fama, E.F., and K.R. French. 2018. Choosing factors. *Journal of Financial Economics* 128: 234–252.
Fama, E.F., and K.R. French. 2020. Comparing cross-section and time-series factor models. *Review of Financial Studies* 33: 1892–1926.
Fama, E.F., and J.D. MacBeth. 1973. Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy* 81: 607–636.
Gibbons, M.R., S.A. Ross, and J. Shanken. 1989. A test of the efficiency of a given portfolio. *Econometrica* 57: 1121–1152.
Grobys, K., J.W. Kolari, and J. Niang. 2022. Man versus machine: On artificial intelligence and hedge funds performance. *Applied Economics* 54: 4632–4646.
Hamada, R.S. 1972. The effect of the firm's capital structure on the systematic risk of common stocks. *Journal of Finance* 27: 435–452.
Hou, K., C. Xue, and L. Zhang. 2015. Digesting anomalies: An investment approach. *Review of Financial Studies* 28: 650–705.
Kolari, J. W., J. Z. Huang, W. Liu, and H. Liao. 2022. Further tests of the ZCAPM asset pricing model. *Journal of Risk and Financial Management*. https://www.mdpi.com/1911-8074/15/3/137). Reprinted in Kolari, J. W., and S. Pynnonen. 2022. eds., *Frontiers of Asset Pricing* (MDPI, Basel, Switzerland).
Kolari, J.W., J.Z. Huang, H.A. Butt, and H. Liao. 2022. International tests of the ZCAPM asset pricing model. *Journal of International Financial Markets, Institutions, and Money* 79: 101607.
Kolari, J.W., W. Liu, and J.Z. Huang. 2021. *A new model of capital asset prices: Theory and evidence*. Cham, Switzerland: Palgrave Macmillan.
Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2023. *Testing for missing asset pricing factors*. San Diego, CA: Paper presented at the Western Economic Association International.
Kolari, J.W., and S. Pynnonen. 2023. *Investment valuation and asset pricing: Models and methods*. Cham, Switzerland: Palgrave Macmillan.

Kolari, J.W., J.Z. Huang, W. Liu, and H. Liao. 2024. A quantum leap in asset pricing: Explaining anomalous returns. Working paper, Texas A&M University, available on SSRN at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4591779.

Lintner, J. 1965. The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics* 47: 13–37.

Liu, W. 2013 *A new asset pricing model based on the Zero-Beta CAPM: Theory and evidence*. Doctoral dissertation, Texas A&M University.

Liu, W., J. W. Kolari, and J. Z. Huang. 2012. A new asset pricing model based on the zero-beta CAPM market model (CAPM). Presentation at the annual meetings of the Financial Management Association, Best Paper Award in Investments, Atlanta, GA (October).

Liu, W., J.W. Kolari, and J.Z. Huang. 2020. *Return dispersion and the cross-section of stock returns*. Palm Springs, CA (October): Presentation at the annual meetings of the Southern Finance Association.

Markowitz, H.M. 1952. Portfolio selection. *Journal of Finance* 7: 77–91.

Markowitz, H.M. 1959. *Portfolio selection: Efficient diversification of investments*. New York, NY: John Wiley & Sons.

Merton, R.C. 1973. An intertemporal capital asset pricing model. *Econometrica* 41: 867–887.

Modigliani, F., and M.H. Miller. 1958. The cost of capital, corporation finance and the theory of investment. *American Economic Review* 48: 261–297.

Modigliani, F., and M.H. Miller. 1963. Corporation income taxes and the cost of apital: A correction. *American Economic Review* 53: 433–443.

Mossin, J. 1966. Equilibrium in a capital asset market. *Econometrica* 34: 768–783.

Ross, S.A. 1976. The arbitrage theory of capital asset pricing. *Journal of Economic Theory* 13: 341–360.

Sharpe, W.F. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 19: 425–442.

Stambaugh, R.F., and Y. Yuan. 2017. Mispricing factors. *Review of Financial Studies* 30: 1270–1315.

Treynor, J.L. 1961. *Market value, time, and risk*. Unpublished manuscript.

Treynor, J.L. 1962. *Toward a theory of market value of risky assets*. Unpublished manuscript.

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